

Disturbance feedforward control for vibration isolation systems : analysis, design, and implementation

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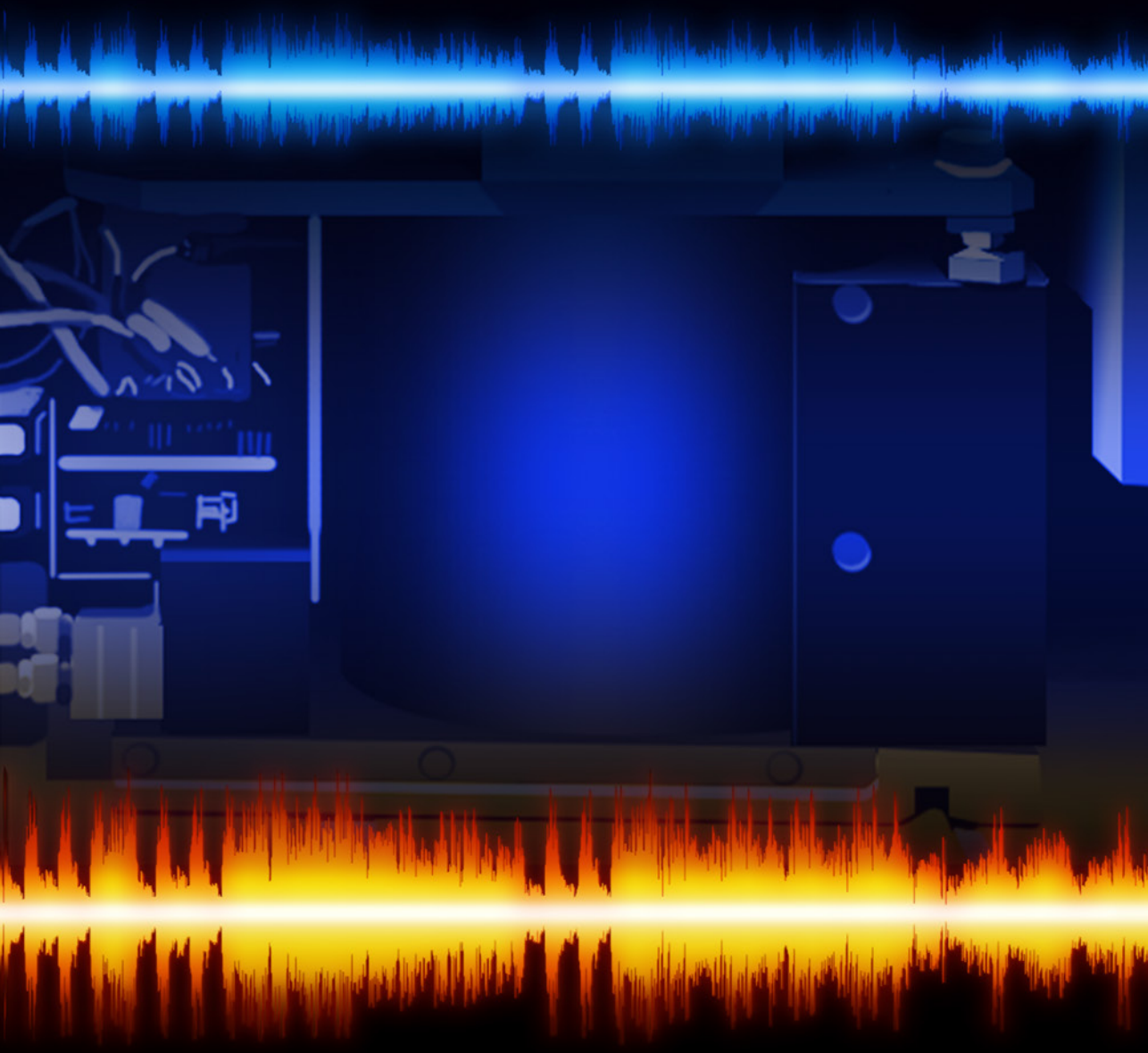
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DISTURBANCE FEEDFORWARD CONTROL FOR VIBRATION ISOLATION SYSTEMS

ANALYSIS, DESIGN, AND IMPLEMENTATION



Michiel Beijen

Disturbance feedforward control
for vibration isolation systems:
analysis, design, and implementation

Michiel Beijen

disc

The research reported in this thesis is part of the research program of the Dutch Institute of Systems and Control (DISC). The author has successfully completed the educational program of the Graduate School DISC.

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Disturbance feedforward control for vibration isolation systems: analysis, design, and implementation

PROEFSCHRIFT

ter verkrijging van de graad van doctor aan de
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commissie aangewezen door het College voor
Promoties, in het openbaar te verdedigen
op donderdag 5 april 2018 om 16:00 uur

door

Michiel Alexander Beijen

geboren te Borne

Dit proefschrift is goedgekeurd door de promotoren en de samenstelling van de promotiecommissie is als volgt:

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Het onderzoek dat in dit proefschrift wordt beschreven is uitgevoerd in overeenstemming met de TU/e Gedragscode Wetenschapsbeoefening.

Publiekssamenvatting

Onze maatschappij vraagt in toenemende mate om extra functionaliteit in draagbare elektronische apparaten zoals smartphones, laptops, enzovoort. Computer chips, ook wel “integrated circuits” of IC’s genoemd, vormen het hart van zulke apparaten. De hoeveelheid functies op een IC wordt bepaald door de kleinst mogelijke structuren die op een IC aangebracht kunnen worden. Verdere verhoging van de functionaliteit van IC’s is alleen mogelijk door verdere verkleining van deze structuren. Momenteel zijn wafer scanners in staat om IC’s met structuren tot 13 nanometer te produceren, waarbij 1 nanometer een miljoenste millimeter is. Echter, op zulke kleine schalen worden machines zoals wafer scanners extreem gevoelig voor trillingen vanuit de omgeving (bijvoorbeeld vloertrillingen) of trillingen veroorzaakt door de machine zelf (bijvoorbeeld omdat deze agressief bewegende componenten bevat). Voor een optimale werking van de machine dienen deze trillingen dan ook te worden tegengegaan door plaatsing van trillingsisolatie systemen tussen de trillende omgeving en de machine-onderdelen die het meest gevoelig zijn voor deze trillingen.

Passieve trillingsisolatie systemen zoals mechanische veren en dempers kunnen slechts een beperkt deel van alle trillingen isoleren. Om de prestaties van deze passieve systemen te verbeteren kunnen sensoren, regelsystemen, en actuatoren worden toegevoegd waardoor een actief trillingsisolatie systeem ontstaat. Een veelgebruikte regelstrategie voor actieve systemen is de feedback regeling, waarbij een sensor direct de nog resterende en ongewenste trillingen in de machine terugkoppelt. Hoewel het ontwerp van feedback regelingen relatief simpel is, levert het slechts een beperkte prestatiewinst op. Een veelbelovend alternatief is de verstoringsfeedforward regeling (DFC), waarbij sensoren niet in de machine maar in de trillende omgeving worden geplaatst. Dit maakt verdere prestatiewinst mogelijk omdat inkomende trillingen eerder gemeten en dus gecompenseerd kunnen worden. Echter, er is momenteel te weinig bekend over DFC voor grootschalige industriële toepasbaarheid.

Het hoofddoel van dit proefschrift is het zetten van een nieuwe stap richting

het analyseren, ontwerpen en implementeren van DFC in toekomstige actieve trillingsisolatiesystemen voor industriële toepassingen zoals wafer scanners en elektronenmicroscopen. Dit proefschrift bevat vier kernbijdragen. Ten eerste wordt nauwkeurig beschreven wat de mogelijkheden en beperkingen van DFC zijn. Ten tweede is een systematische methode ontwikkeld voor het daadwerkelijk meten van de trillingsisolatie prestaties in industriële systemen. Ten derde wordt een nieuwe ontwerp methode voor DFC regelingen gepresenteerd, waarbij de regelaar zelflerend is gemaakt om het ontwerpproces te vereenvoudigen. Ten vierde is beschreven hoe DFC slim kan worden gecombineerd met feedback regelingen om tot optimale prestaties te komen op het gebied van actieve trillingsisolatie. Al deze kernbijdragen zijn getoetst op meerdere industriële systemen en laten een sterke verbetering van de trillingsisolatie prestaties zien.

Public summary

There is an ever-increasing demand from our society to put more functionality in portable electronic devices such as smartphones and laptops. The heart of these devices is formed by computer chips, so-called “integrated circuits” (ICs). The amount of functionality that can be handled by one IC is determined by the feature size, which indicates the size of the smallest component that can be manufactured. To include more functionality in ICs, the minimum feature size has to be further decreased. Current ICs produced with so-called wafer scanners have a feature size of about 13 nanometers, with 1 nanometer being a millionth of a millimeter. At such small scales, wafer scanners become very sensitive for undesired vibrations caused by the environment (e.g. floor vibrations) or vibrations induced by the machine itself (e.g. due to aggressively moving machine parts). These vibrations strongly limit the smallest feature size and, therefore, must be counteracted using vibration isolation systems that are installed between a vibrating environment and the equipment-to-be-isolated.

Passive vibration isolation systems such as mechanical springs and dampers can only achieve a limited vibration isolation performance. To further improve performance, these systems are equipped with sensors, controllers, and actuators resulting in active vibration isolation systems. Many of these active systems use feedback control, where sensors directly measure any undesired movements of the sensitive equipment. Although feedback control design is relatively straightforward, it only induces moderate performance improvements. A promising addition is given by disturbance feedforward control (DFC), where the sensors are placed in the vibrating environment instead. This enables the possibility to further improve performance because incoming vibrations can be measured and compensated earlier. However, proper control design is fairly complex, which currently prevents DFC from being widely used in industry.

The main goal of this thesis is to make a step towards the design and implementation of DFC in next-generation active vibration isolation systems for wafer scanners, electron microscopes, and other industrial high-precision systems. This

thesis has four main contributions. First, it accurately describes the possibilities and limitations of DFC. Second, a systematic approach is developed to measure the actual performance of industrial vibration isolation systems. Third, a novel design method for DFC is presented, using a self-tuning algorithm to simplify the design process. Fourth, it is described how DFC should be combined with feedback control strategies to obtain optimal vibration isolation performance. All four contributions have been assessed at multiple industrial systems and show a strong improvement in vibration isolation performance.

Abstract

Performance of high-precision machines in terms of accuracy and throughput is often limited by undesired vibrations. To deal with such vibrations, active vibration isolation systems are used. Undesired vibrations can enter the machine either as floor vibrations transmitted via the vibration isolator, or as direct disturbances acting on the isolated payload. Various feedback control methods have been developed in the past to further improve the vibration isolation performance. However, high-gain feedback can lead to stability problems and undesired amplification of sensor and actuator noise, which poses a problem for vibration isolation performance. In view of this problem, disturbance feedforward control (DFC) seems a promising alternative as it does not affect the closed-loop stability properties, and generally induces less amplification of sensor noise. The aim of this thesis is to develop a theoretical framework to show the possibilities and limitations of DFC, and to provide the design tools for identification and control toward an implementation of DFC in next-generation active vibration isolation systems, and has four main contributions.

The first contribution consists of a detailed analysis of design and performance tradeoffs in DFC for Multi-Input Multi-Output (MIMO) systems. These tradeoffs occur in the feedforward sensitivity matrix due to causality aspects, i.e. disturbances must be measured before any compensation can be applied. This leads to performance limitations similar to the waterbed effect in feedback control. Given these tradeoffs, the cost of disturbance rejection can be minimized by only reducing the principal gains of the feedforward sensitivity matrix that correspond to the disturbance directions, and only at frequencies that are relevant for disturbance rejection. This generally results in a centralized controller being the optimal solution if the disturbances are correlated, even in cases where the system itself is perfectly decoupled.

The second contribution consists of an identification approach to estimate the frequency response function (FRF) of the transmissibility matrix. This matrix is a performance measure describing the ability to suppress floor vibrations

of vibration isolation systems. Identification of this matrix is often limited in practice due to the heavy weight of high-precision machines and the inability to sufficiently excite the floor. A non-parametric system identification method is presented that combines floor and shaker excitations to improve the signal-to-noise ratio. Furthermore, a method is presented to analyze the input power spectrum of the floor excitations, both in terms of magnitude and direction. This method is also used for input design of the shaker excitation signals to obtain sufficient excitation power in all directions with minimum cost of experiment.

The third contribution consists of a self-tuning feedforward controller that has the structure of a generalized FIR filter. The structure of the feedforward controller follows from modeling, but to achieve robust performance the tuning of the controller parameters will be based on measurement data. More specifically, the controller poles, which correspond to internal isolator dynamics, are obtained offline from system identification experiments and are fixed a priori in basis functions; the controller zeros are obtained from online self-tuning using a Filtered-error Least Mean Squares (FeLMS) algorithm. Residual noise shaping is used to reduce bias, and to improve convergence speed. This results in a convex optimization algorithm with a unique global minimum. The proposed strategy is applicable to both hard-mounted and soft-mounted vibration isolators to improve the rejection of floor vibrations.

The fourth contribution consists of a combined feedforward-feedback control strategy based on physical insights. In this strategy, a combination of relative position feedback, acceleration feedback, and disturbance feedforward control is designed and evaluated. By mixing the sensor signals, amplification of sensor noise is minimized while vibration isolation and leveling performance are optimized. It is shown that feedback control greatly enhances the rejection of disturbances that act directly on the isolated metrology frame. In addition, feedforward control improves the rejection of floor vibrations that are transmitted via the vibration isolator.

The proposed identification and control strategies are successfully implemented and tested on several hard-mounted and soft-mounted industrial vibration isolation systems. It is shown that the transmissibility of floor vibrations is reduced up to 40 dB in a broad frequency range without compromising the ability to reject disturbances that act directly on the isolated machine parts.

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Chapter 1

Introduction

1.1 Vibration isolation in high-precision systems

There are ever-increasing demands on both the accuracy and the throughput of high-precision machines. Examples of such machines are lithographic wafer scanners for the semi-conductor industry [19, 45], scanning microscopes [112, 53], and metrology systems like coordinate measuring machines (CMMs) [48, 28]. An increase of throughput for such machines generally comes at the cost of accuracy, because the moving parts require more aggressive setpoints. For example, moving stages within wafer scanners nowadays typically accelerate with 160 m/s^2 . This leads to large reaction forces exerted on the machine frames and, subsequently, heavy vibrations induced by the machine itself. Besides, environmental disturbance sources such as floor vibrations contribute to machine vibrations, leading to a decrease in performance in terms of accuracy.

In order to achieve the extreme accuracy requirements in such a vibrating environment, sensors and other sensitive equipment are placed on isolated metrology frames with an approximately ‘zero’ stiffness connection to the environment. By doing so, the isolated metrology frame serves as a fixed reference for all other parts of the machine. Figure 1.1 shows an example of such a machine architecture for wafer scanners, but similar architectures are proposed for other mentioned applications as well [31].

Passive vibration isolators have been developed to isolate the metrology frame [77, 67], but such passive solutions have several limitations. First, a perfect zero-stiffness is never obtained in practice, as this would lead to leveling problems and induces resonances caused by internal isolator dynamics. Second, it is practically impossible to isolate the metrology frame from all incoming disturbances, so there will always be residual disturbances that act directly on the isolated

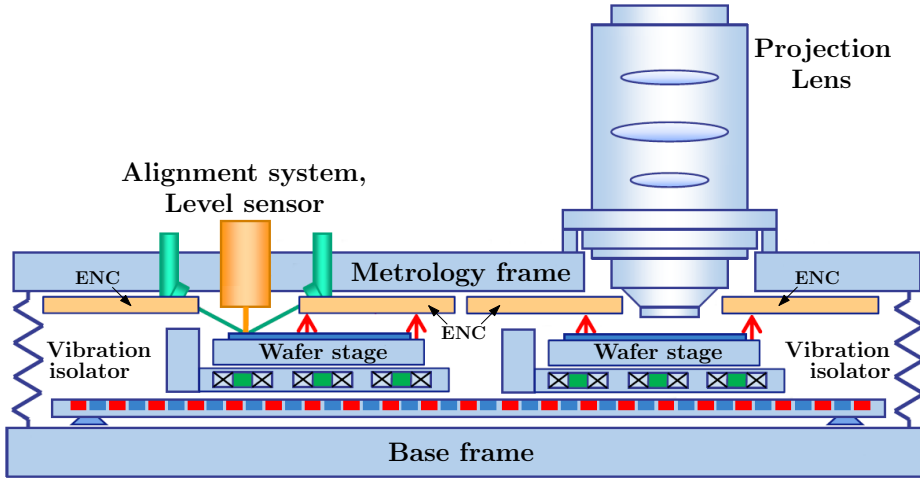


Figure 1.1. Dual-stage wafer scanner layout; vibration isolators are used to isolate the metrology frame from the base frame supporting moving wafer stages with aggressive setpoints causing machine vibrations; the wafer stages are positioned with respect to the isolated metrology frame using encoder (ENC) measurements with nanometer accuracy. [18]

metrology frame and that need to be rejected. Examples of such direct disturbances are acoustic waves, band cables and tubes with a parasitic stiffness, and flow-induced vibrations used for cooling or immersion technology [64]. For a perfect suppression of such direct disturbances, an infinitely high support stiffness is desired, but this conflicts with the goal of zero-stiffness to suppress indirect disturbances that enter via the isolator.

Control strategies for active vibration isolation systems have been developed to overcome the design tradeoffs existing in passive systems. These strategies are further discussed in the next section.

1.2 Next-generation active vibration isolation

An example of a modern vibration isolation system is shown in Figure 1.2, and consists of the following parts: (a) a 6 degree-of-freedom movable metrology frame, (b) a base frame that is connected to the floor, and (c) three vibration isolators that are installed between the base frame and the metrology frame. Each vibration isolator contains a pneumatically controlled air mount that provides gravity compensation, damping, and a low suspension stiffness. In addition, the vibration isolator is equipped with Lorentz motors to enable active vibration isolation using feedback or feedforward control.

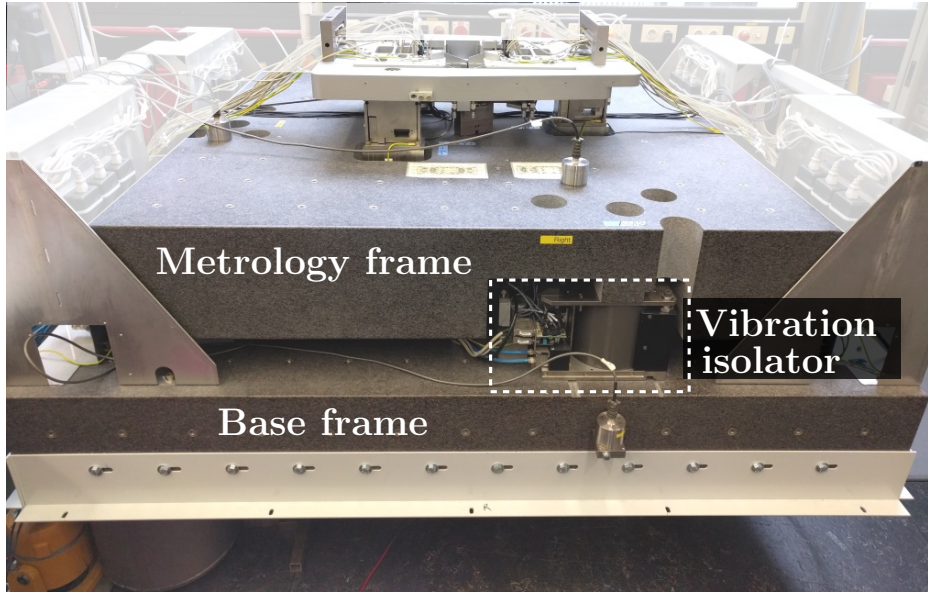


Figure 1.2. Modern active vibration isolation system; the picture shows the base frame, the isolated metrology frame, and one of the vibration isolators. Photo: floating reticle stage test rig at the Eindhoven University of Technology; the setup contains three air mount vibration isolators of the type *ASML-XTII* developed by the Integrated Dynamics Engineering (IDE) Group.

A classic example of feedback control is the addition of skyhook damping to damp the suspension modes of the isolated system. In fact, skyhook damping is a form of inertial feedback, which exploits measurements of the absolute velocity or acceleration of the metrology frame. A second example is force feedback [73, 90], where a load cell is installed to measure the forces transmitted through the isolator. A third example is relative position feedback, exploiting measurements of the relative distance between the metrology frame and the base frame. In addition, combinations of feedback control strategies have been proposed to further improve the vibration isolation performance, see for example [22, 90, 116]. However, performance improvements obtained with feedback control are generally limited. Inertial feedback improves the suppression of both direct and indirect disturbances, but cannot be used at low frequencies because of sensor noise which induces drift [21]. Force feedback is suitable to suppress indirect disturbances at high frequencies, but this reduces the ability to reject direct disturbances at low frequencies [90]. A similar limitation holds for relative position feedback, where enhanced suppression of direct disturbances comes at the cost of reduced suppression of indirect disturbances, or vice versa [10].

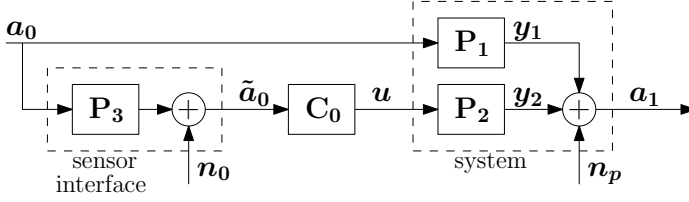


Figure 1.3. Block diagram of a vibration isolation system with disturbance feedforward control.

A promising alternative is disturbance feedforward control (DFC), which exploits measurements of indirect disturbances using sensors located at the base frame [95, 58]. Subsequently, it applies a control force that mimics the disturbance transmitted by the isolator but with an opposite sign. This technique is closely related to DFC in active noise cancellation [29, 7, 13], in which a microphone picks up sound waves from the environment. A controller and a loudspeaker are then used to generate a counteracting sound wave that ideally cancels the incoming sound wave, which can be seen as the disturbance. An example of DFC from advanced motion control systems is found in the synchronization of positioning stages of wafer scanners [46, 8, 79]. In [46], the tracking error of a low-bandwidth wafer stage is forwarded as a setpoint to a high-bandwidth reticle stage, to minimize the synchronization error between the two stages. Compared to feedback control, application of DFC has two main advantages. First, it does not affect the stability properties of the system because no control loop is closed. This enables disturbance suppression also at high frequencies, i.e. far beyond the bandwidth achievable by any feedback controller. Second, it uses a direct measurement of the disturbance, which generally improves the signal-to-noise ratio because the disturbance signal to be measured is not (mechanically) low-pass filtered by the vibration isolator itself.

1.3 Disturbance feedforward control

The general setting of disturbance feedforward control used throughout this thesis is represented by the block diagram shown in Figure 1.3. The base frame acceleration $\mathbf{a}_0(t) \in \mathbb{R}^{n_r}$ is considered as an indirect disturbance (we will come back to this assumption in the next section). The metrology frame acceleration $\mathbf{a}_1(t) \in \mathbb{R}^{n_y}$ is considered as the signal to be minimized. The latter is formed by three signals, i.e. $\mathbf{y}_1$ that is caused by $\mathbf{a}_0$ via the isolator described by the primary path $\mathbf{P}_1$, $\mathbf{y}_2$ that represents the impact of the control signal $\mathbf{u}(t) \in \mathbb{R}^{n_u}$ transferred via the secondary path $\mathbf{P}_2$, and process noise $\mathbf{n}_p(t) \in \mathbb{R}^{n_y}$ that represents metrology frame accelerations caused by direct disturbance forces. The feedforward controller $\mathbf{C}_0$ provides disturbance feedforward control based

on input $\mathbf{a}_0$. Note that instead of $\mathbf{a}_0$, the signal $\tilde{\mathbf{a}}_0$ is measured, the difference being the additive measurement noise represented by $\mathbf{n}_0$. Using the Laplace transform $\mathcal{L}\{\mathbf{x}(t)\} = \mathbf{X}(s)$, with s the Laplace variable, the signal $\mathbf{a}_1$ can be written as

$$\mathbf{A}_1(s) = \mathcal{T}(s)\mathbf{A}_0(s) + \mathcal{S}_0(s)\mathbf{N}_0(s) + \mathbf{N}_p(s), \quad (1.1)$$

with closed-loop transmissibility matrix $\mathcal{T}$ and noise sensitivity matrix $\mathcal{S}_0$ defined as

$$\mathcal{T}(s) = \mathbf{P}_1(s) + \mathbf{P}_2(s)\mathbf{C}_0(s)\mathbf{P}_3(s), \quad \mathcal{S}_0(s) = \mathbf{P}_2(s)\mathbf{C}_0(s). \quad (1.2)$$

The transmissibility matrix is a measure for vibration isolation performance, whereas the noise sensitivity matrix is a measure for noise amplification.

Under the assumptions that $n_y = n_u = n_r$ such that $\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3$ are square systems, $\mathbf{a}_0$ and $\mathbf{n}_p$ are uncorrelated, and in the absence of $\mathbf{n}_0$, perfect cancellation of base frame vibrations is obtained if $\mathcal{T} = \mathbf{0}$ in (1.2), which is obtained if

$$\mathbf{C}_0(s) = -\mathbf{P}_2^{-1}(s)\mathbf{P}_1(s)\mathbf{P}_3^{-1}(s). \quad (1.3)$$

This follows from substitution of (1.3) in (1.2). However, a discrete-time implementation of the design in (1.3) is practically infeasible and does not lead to perfect cancellation due to the following limitations:

- I. Causality. Disturbance feedforward controllers are generally implemented in discrete-time. Since the disturbance has to be measured before compensation can be applied, the compensation inherently comes too late when a discrete-time compensation is used to reject a continuous-time disturbance.
- II. Bounded signals. The design in (1.3) needs the inverse of $\mathbf{P}_2$ and $\mathbf{P}_3$. In general, these terms can contain non-minimum-phase (transmission) zeros. Inversion of these zeros leads to an unbounded control signal.
- III. Model uncertainty. Eq. (1.3) needs accurate parametric models for $\mathbf{P}_1$, $\mathbf{P}_2$, and $\mathbf{P}_3$ to achieve sufficiently good performance. However, estimating $\mathbf{P}_1$ in general is only possible with a limited accuracy, because in identification experiments it is often difficult (if not impossible) to sufficiently excite the base frame.
- IV. Sensor noise. Minimization of $\mathbf{A}_1$ in (1.1) implies finding a balanced trade-off between minimizing $\mathcal{T}\mathbf{A}_0$ and $\mathcal{S}_0\mathbf{N}_0$, while perfect cancellation using (1.3) is only obtained in the absence of $\mathbf{n}_0$.

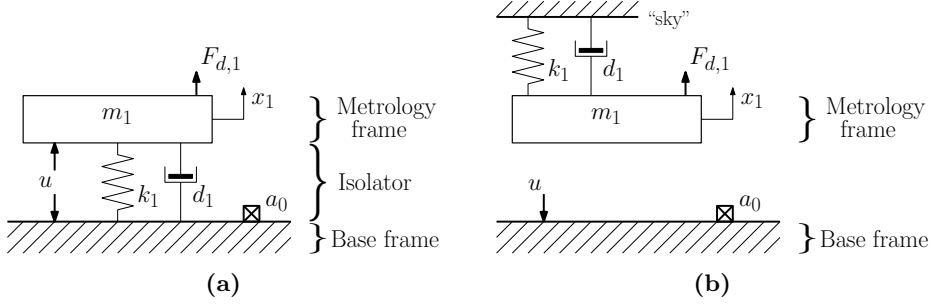


Figure 1.4. (a): Simple model of a vibration isolation system; base frame (BF) motion is represented by x_0 , and metrology frame (MF) motion is represented by x_1 ; the MF has rigid-body mass m_1 ; the vibration isolator is modeled as a spring k_1 , a damper d_1 , and an actuator force u ; $F_{d,1}$ represents a direct disturbance force on the metrology frame; the BF acceleration is measured by the sensor a_0 ; (b) physical interpretation of a vibration isolation system with perfect disturbance feedforward control, which effectively reconnects the isolator to a virtual point in the “sky”.

1.3.1 Illustrative example

Consider a simple vibration isolation model given by the mass-spring-damper system in Figure 1.4a. The equation of motion (in Laplace domain) for this system is given by

$$(m_1 s^2 + d_1 s + k_1)X_1(s) = (d_1 s + k_1)X_0(s) + F_{d,1}(s) + U(s), \quad (1.4)$$

where m_1 denotes the isolated metrology frame mass and d_1, k_1 represent the isolator damping and stiffness, respectively. X_0 and X_1 represent respectively the base frame and metrology frame displacement, whereas the corresponding accelerations are given by the second derivatives $A_0(s) = s^2 X_0(s)$ and $A_1(s) = s^2 X_1(s)$, respectively. U and $F_{d,1}$ represent the actuator and disturbance forces, respectively. The goal is to minimize A_1 . Using (1.4), the input-output relations for $\mathbf{P}_1$ and $\mathbf{P}_2$ in (1.3) are straightforwardly derived as

$$\mathbf{P}_1(s) := \frac{A_1(s)}{A_0(s)} = \frac{d_1 s + k_1}{m_1 s^2 + d_1 s + k_1}, \quad \mathbf{P}_2(s) := \frac{A_1(s)}{U(s)} = \frac{s^2}{m_1 s^2 + d_1 s + k_1}.$$

Measurements of the base frame accelerations are obtained with an accelerometer and denoted by a_0 which is assumed to be a perfect sensor without any dynamics, i.e. $\mathbf{P}_3 = 1$ in Figure 1.3. The perfect disturbance feedforward controller then reads

$$\mathbf{C}_0(s) = -\mathbf{P}_2^{-1}(s)\mathbf{P}_1(s)\mathbf{P}_3^{-1}(s) = -\frac{d_1 s + k_1}{s^2}, \quad (1.5)$$

and the resulting actuator signal becomes

$$U(s) = -\frac{d_1 s + k_1}{s^2} A_0(s) = -(d_1 s + k_1) X_0(s). \quad (1.6)$$

Substitution of (1.6) in (1.4) gives

$$(m_1 s^2 + d_1 s + k_1) X_1(s) = F_{d,1}(s). \quad (1.7)$$

Eq. (1.7) shows that X_0 no longer appears in the equation of motion for m_1 , hence it is perfectly canceled. A physical interpretation of (1.7) is that the control signal virtually removes the isolator's spring and damper from Figure 1.4a and replaces it with a virtual skyhook spring and damper, see Figure 1.4b. Note that the actuator signal in (1.6) only depends on X_0 while it is intended to control X_1 , hence there is no feedback loop. In this regard, $\mathbf{C}_0$ in (1.6) is called a disturbance feedforward controller.

1.4 Problem statement

Although DFC is a promising technique for suppressing indirect disturbances, there is a lack of both theoretical knowledge and a systematic design approach which currently prevents DFC from being used in industrial high-precision machines like wafer scanners. Besides the limitations already given regarding an implementation of the perfect controller in (1.3), several other limitations exist. For example, it is not clear what the possibilities and limitations of DFC are in terms of performance improvement, there is no systematic method to measure vibration isolation performance in terms of transmissibility matrices for heavy-weight industrial systems, nor is there a systematic design procedure addressing both DFC and feedback control. The aim of this thesis is to tackle these and several other problems to make a step toward the implementation of DFC in next-generation vibration isolation systems. The general research goal of this thesis can therefore be formulated as:

Develop a theoretical framework to show the possibilities and limitations of DFC, and provide the design tools for identification and control toward an implementation of DFC in next-generation active vibration isolation systems.

In this thesis, such a framework is developed under the following assumptions.

Assumption 1.1 (Disturbances). *Each disturbance entering the system can be classified as one of the following disturbances:*

- Indirect disturbances, denoted by $\mathbf{F}_{d,0}$, are for example base frame vibrations which can only excite the metrology frame via the isolation modules; there are no other parasitic disturbance transmission paths between the base frame and the metrology frame;
- Direct disturbances, denoted by $\mathbf{F}_{d,1}$, act directly on the metrology frame but do not induce base frame vibrations.

There is no cross-correlation between the elements of $\mathbf{F}_{d,0}$ and $\mathbf{F}_{d,1}$.

Assumption 1.1 is often satisfied in practice given two conditions. First, the total stiffness of all parasitic disturbance transmission paths between the base frame and the metrology frame other than the isolator itself, such as band cables or acoustic waves, must be small compared to the stiffness of the isolator. Second, the base frame must be much stiffer and heavier than the isolator stiffness and the mass of the isolated metrology frame, such that $\mathbf{F}_{d,1}$ does not influence $\mathbf{a}_0$ significantly. When Assumption 1.1 does not hold, $\mathbf{F}_{d,0}$ and $\mathbf{F}_{d,1}$ might become cross-correlated. In that case, application of DFC might still improve performance, but optimality can no longer be guaranteed.

Assumption 1.2 (Base frame). *The base frame is considered rigid and infinitely heavy such that base frame motion can be considered as an exogenous input representing an indirect disturbance that is not affected by the isolator or actuator forces. This assumption has two implications:*

- (i) *there is no need to check stability of the intrinsic feedback loop formed by the actuator (reaction) forces and the base frame motion sensors;*
- (ii) *the transfer functions $\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3$ do not contain structural dynamics related to the base frame.*

If Assumption 1.2 does not hold, an intrinsic feedback loop is created from the actuator force back to the measured base frame motion. This is not a problem as long as the intrinsic feedback loop is stable, which can be checked using the Nyquist stability criterion discussed in the appendix of this chapter, see pp. 14. Moreover, if Assumption 1.2 does not hold, structural dynamics occur in the base frame. These dynamics do not influence $\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3$ as long as the base frame sensors are co-located with the isolators, see the appendix of this chapter on pp. 14. However, for the several implementations considered in this thesis, instability of the intrinsic feedback loop and base frame dynamics have never been an issue, and are therefore not further considered.

1.5 Research challenges and contributions

This section discusses four academic research challenges toward the desired framework for DFC. For each of the four challenges, the discussion starts with a literature overview giving the state-of-the-art and the current limitations encountered. The discussions end with showing the contributions of this thesis to each specific research challenge.

1.5.1 Inherent design limitations in DFC

DFC generally comes too late, hence a causality problem is present. This inherently leads to the fact that perfect disturbance cancellation is not possible in most practical applications. From a theoretical point of view, the best solution is then given by an optimal controller that minimizes vibrations of the metrology frame in some norm-based sense. An often used norm in this regard is the L_2 -norm, for which the so-called causal Wiener filter is shown to be the optimal solution [32, 94]. In the same works, scalar expressions for the steady-state error are given in the time domain considering the L_2 -norm, but do not provide a more generic description of the fundamental performance limitations of DFC in the frequency domain.

For single-input single-output (SISO) systems, such a frequency domain description is found in terms of the Bode integral constraint [81], which leads to performance tradeoffs along the frequency axis and which is known as the waterbed effect. For multi-input multi-output (MIMO) systems, however, also the directions of the disturbances play an important role in performance optimization. In [16], design and performance tradeoffs based on the MIMO sensitivity matrix are extensively discussed for feedback control systems, but not within the context of feedforward control. In [81], fundamental limitations are derived for the MIMO filtering sensitivity matrix, which is very similar to the feedforward sensitivity matrix, but only in terms of a Poisson integral constraint and not in terms of a Bode integral constraint. The latter, however, is preferred in exploiting the waterbed effect for controller design.

Contribution I: *A novel Bode integral constraint to describe the fundamental performance limitations of DFC in MIMO systems, together with an H_2 control design method to optimize performance under these limitations by adequate modeling of disturbances both in terms of magnitudes and directions.*

1.5.2 Measurement of the transmissibility matrix

The transmissibility matrix $\mathcal{T}$ is of key interest, because it serves as a measure for vibration isolation system performance, and it is needed to enable model-based controller design as proposed in (1.3). However, an accurate estimation of

$\mathcal{T}$ from identification experiments is often extremely difficult due to the inability to sufficiently excite the base frame (or the floor).

Amongst others, [74, 62] present methods for transmissibility matrix measurements where external shaker solutions are used to ensure that the base frame is sufficiently excited. However, these methods are generally not applicable because industrial machines are often too heavy for commercially available shakers to support and excite the system. Therefore, estimation methods that allow the machine to perform under operating conditions are highly preferred. Such methods rely on environmental base frame excitations, e.g., floor vibrations [82]. These methods can result in a reasonable estimation, as long as the environmental vibrations provide sufficient spectral power in all relevant directions to obtain a sufficient signal-to-noise ratio (SNR). Available methods are restricted to those that can deal with random excitations, because the environmental vibrations are not known beforehand and are typically non-periodic. In this regard, spectral analysis is often used [70, 59], but this assumes that either the base frame or the metrology frame excitation is measured free of noise to prevent biased estimates [2, 71].

Several methods are proposed in literature to improve the estimation results for systems that can only be partly identified due to an insufficient SNR, such as operational modal analysis [76, 104], finite-element modeling [26], and frequency-based substructuring [100, 65]. However, none of these methods provides a validation regarding the actual performance of the assembled system in closed-loop. The latter is important, since control loops might deteriorate performance due to changing system dynamics and noise amplification [115, 90].

Contribution II: *A novel systematic approach to identify the transmissibility matrix for heavy-weight MIMO vibration isolation systems using a combination of environmental vibrations and shaker excitations by appropriate input design.*

1.5.3 Data-based self-tuning

Vibration isolation performance of the model-based feedforward controller as proposed in (1.3) is very sensitive to model uncertainties. An accurate estimation of the passive transmissibility matrix $\mathbf{P}_1$ is part of the model-based solution, but extremely difficult to obtain. Although the approach from Contribution II enables measurement of $\mathbf{P}_1$ with sufficient accuracy to evaluate vibration isolation performance, the approach is generally insufficient to realize the required high performance obtained from model-based design. As an alternative approach, data-based self-tuning can be used because it circumvents the necessity of an accurate model for $\mathbf{P}_1$ and it automatically makes a tradeoff between maximizing vibration isolation performance and minimizing noise amplification.

Many self-tuning controllers consist of adaptive finiteimpulse response (FIR)

filters [94, 30, 56], but these filters are inappropriate for active vibration isolation systems in industrial high-precision machines. This is because FIR filters need many parameters to approximate lightly damped resonances and low-frequency poles which typically occur in this class of systems. In [32, 58] adaptive infinite impulse response (IIR) filters are proposed which are able to describe lightly damped resonances and low-frequency poles with fewer filter parameters. However, adaptive IIR filtering generally leads to a non-convex optimization problem suffering from local minima or resulting in an unstable feedforward controller. As an intermediate solution, generalized FIR filters are proposed in [113, 23], which are IIR filters with fixed poles. This solution needs pre-estimation of the poles which, opposed to the zeros, can reasonably be done using the approach from Contribution II.

For self-tuning filters several recursive update algorithms exist. Typical examples are least mean squares (LMS) and recursive least squares (RLS) algorithms. LMS-type algorithms are attractive because they perform relatively simple computations compared to RLS algorithms [80]. This is important for practical implementation. There are two main classes of LMS algorithms that can deal with plant dynamics in the secondary path $\mathbf{P}_2$, namely, filtered-reference LMS (FxLMS) and filtered-error LMS (FeLMS), which involve filtering of $\mathbf{a}_0$ and $\mathbf{a}_1$ with a model of $\mathbf{P}_2$, respectively. Compared to FxLMS, FeLMS algorithms require less computational power in case of MIMO systems [101], but with the disadvantage of having additional requirements on the model of $\mathbf{P}_2$. Hybrid methods based on FeLMS [24] or Filtered-reference Filtered-error LMS (Fx-Fe-LMS) [63, 86] are proposed in an attempt to have the best of both worlds. However, none of the above works apply self-tuning of generalized FIR filters.

Contribution III: *A novel self-tuning controller for MIMO vibration isolation systems. The self-tuning controller combines the advantages of a low-order generalized FIR filter with the attractive properties of FeLMS in MIMO systems.*

1.5.4 Combined feedback and feedforward control

To minimize noise amplification, mixed-sensor feedback control strategies are developed that use different types of sensors dedicated to different frequency ranges. A linear mixed-sensor control strategy for soft mounts is presented in [10] where position feedback at low frequencies is combined with acceleration feedback at high frequencies. In [116, 25], sliding mode control strategies are proposed that improve performance by mixing relative position feedback and absolute velocity (or acceleration) feedback. Amongst others, [39] studied the combination of velocity and force feedback, using a geophone and a load cell, to reduce the structural power transmission from a vibrating source to a receiving plate. For hard mounts, an example of mixed-sensor feedback control is pre-

sented in [90], where a combination of acceleration feedback and force feedback is used. This improves the disturbance rejection properties at lower frequencies while at the same time adding damping to internal modes at higher frequencies. However, this requires high-bandwidth feedback controllers, which can lead to stability problems and inherently induce noise amplification.

DFC can be used in addition to feedback control to further improve vibration isolation. Compared to feedback control, disturbance feedforward control has the advantage of a better SNR, and disturbance suppression at high frequencies is not restricted by bandwidth limitations. However, DFC can only reject indirect disturbances. Therefore, a combination of feedback control and DFC is promising to maximize the suppression of both direct and indirect disturbances. However, it is not straightforward to find a combination that gives a well-designed tradeoff between disturbance rejection, noise amplification, and robust stability. Amongst others, [20, 68] designed H_∞ feedback controllers which improve disturbance rejection and appear robust to plant variation, but do not take into account noise amplification and rejection of floor vibrations. [115, 11] present an H_∞ control design method that takes into account disturbances, noise, and plant uncertainty to track a modeled skyhook target. However, once designed, it is hard to retune H_∞ controllers in practice. Therefore, a physics-driven control design approach is desired to find the requested tradeoff and which is easy to (re-)tune in an industrial environment.

Contribution IV: *A novel physics-based control design method to combine DFC with relative position feedback and inertial feedback control in active vibration isolation systems. Feedback control is used to reject direct disturbances, whereas feedforward control is used to reject indirect disturbances.*

1.6 Outline of the thesis

An overview of the thesis is provided in Figure 1.5, showing that each of the chapters is related to one of the specific contributions. The main chapters (2–6) of this thesis are submitted for publication. Therefore, they are self-contained and can be read independently of each other.

Chapter 1	Introduction
Chapter 2	Contribution I: A novel Bode integral constraint to describe the fundamental performance limitations of DFC in MIMO systems, together with an H_2 control design method to optimize performance under these limitations by adequate modeling of disturbances both in terms of magnitudes and directions.
Chapter 3	Contribution II: A novel systematic approach to identify the transmissibility matrix for heavy-weight MIMO vibration isolation systems using a combination of environmental vibrations and shaker excitations by appropriate input design.
Chapter 4 Chapter 5	Contribution III: A novel self-tuning controller for MIMO vibration isolation systems. The self-tuning controller combines the advantages of a low-order generalized FIR filter with the attractive properties of FeLMS in MIMO systems.
Chapter 6	Contribution IV: A novel physics-based control design method to combine DFC with relative position feedback and inertial feedback control in active vibration isolation systems. Feedback control is used to reject direct disturbances, whereas feedforward control is used to reject indirect disturbances.
Chapter 7	Conclusions

Figure 1.5. Overview of the thesis.

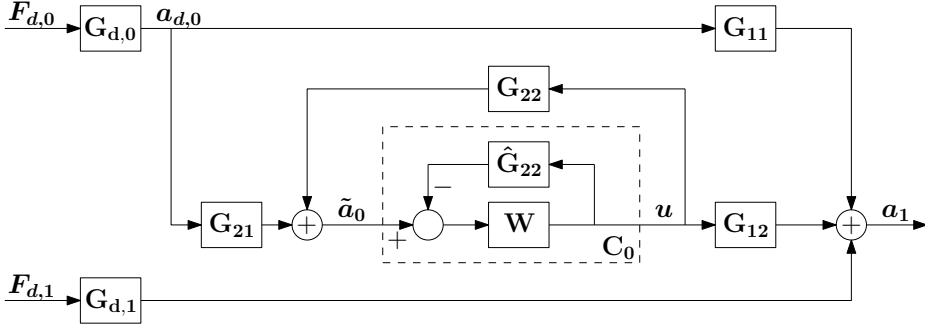


Figure 1.6. Block diagram for Approach 1, using internal model compensation.

1.7 Appendix: Flexible base frames

This appendix discusses the effects of flexibilities in the base frame on model-based disturbance feedforward control. Flexibility in the base frame has two consequences: (i) reaction forces of the actuators on the base frame can induce additional base frame motion, hence an intrinsic feedback loop is created; (ii) structural modes of the base frame could appear as part of the plant dynamics, and therefore affect the optimal controller design.

Two model-based feedforward control design approaches are presented to deal with these aspects. The first approach is a two-step design method that uses internal model compensation (IMC), whereas the second approach is a one-step design method where stability is guaranteed by the generalized Nyquist criterion. Both approaches are illustrated by means of an example at the end of the appendix.

1.7.1 Approach 1: two-step design method

Using the disturbance factorization $(F_{d,0}, F_{d,1})$ given by Assumption 1.1, the block diagram for disturbance feedforward control can be drawn as in Figure 1.6. In this figure, $a_{d,0}$ represents all base frame accelerations caused by $F_{d,0}$ via $G_{d,0}$, whereas $G_{d,1}$ describes the effect of $F_{d,0}$ on the metrology frame vibrations a_1 . The other plant transfer functions in Figure 1.6 are represented by G_{ji} , with $j, i = 1, 2$. From a physical point of view, these transfer functions can be obtained or measured as follows. G_{21} represents (potential) sensor dynamics or sensor filtering which can be determined from a priori measurements in a test bench, or obtained from data sheets provided by the sensor manufacturer. G_{11} represents the transmissibility matrix from base frame vibrations to metrology frame vibrations, which can be identified by setting $u = 0$ and measuring $\tilde{a}_0$, a_1 under the assumptions that G_{21} is known and $a_{d,0}$ is sufficiently exciting. G_{12} and G_{22} are transfer functions from the actuator signal u to $\tilde{a}_0$ and a_1 ,

respectively, which can be obtained from FRF measurements using $\mathbf{u}$ as an input for system identification.

For the control configuration shown in Figure 1.6, a model-based two-step design method has been proposed in [32, 94]. In the first step, internal model compensation (IMC, see, e.g., [84]) is used to recast the control configuration from Figure 1.6 into a disturbance feedforward control configuration with an internal feedforward controller $\mathbf{W}$, see Figure 1.6, whereas the total controller equals

$$\mathbf{C}_0(z) = \mathbf{W}(z)(\mathbf{I} + \hat{\mathbf{G}}_{22}(z)\mathbf{W}(z))^{-1}. \quad (1.8)$$

Assuming $\hat{\mathbf{G}}_{22} = \mathbf{G}_{22}$, it is known from the theory of IMC that the closed-loop system is stable given that $\mathbf{C}_0$ and $\mathbf{G}_{22}$ are stable [84]. The second step is then to design the internal feedforward controller $\mathbf{W}$, which is ideally designed as the so-called Wiener filter [32, 94],

$$\mathbf{W}(z) = -\mathbf{G}_{12}^{-1}(z)\mathbf{G}_{11}(z)\mathbf{G}_{21}^{-1}(z), \quad (1.9)$$

where only square systems are considered, and leading to perfect cancellation of $\mathbf{F}_{d,0}$. This expression is similar to the Wiener filter as proposed in (1.3), and the reader is referred back to Chapter 1 for more details regarding the feasibility of this controller.

As can be seen from Figure 1.6, IMC effectively cancels the intrinsic feedback path $\mathbf{G}_{22}$. However, this requires a perfect model $\hat{\mathbf{G}}_{22}$ which is hard to obtain in practice when the base frame has a large stiffness. In that case, it is difficult to sufficiently excite $\tilde{\mathbf{a}}_0$ via $\mathbf{u}$, resulting in a poor FRF measurement of $\mathbf{G}_{22}$. Therefore, it is preferred to omit IMC in such cases.

1.7.2 Approach 2: one-step design method

For this approach, consider the corresponding block diagram in Figure 1.7 which is, without loss of generality, redrawn from the block diagram of Figure 1.6. The ideal feedforward controller is now given by

$$\mathbf{C}_0(z) = -\mathbf{P}_2^{-1}(z)\mathbf{P}_1(z)\mathbf{P}_3^{-1}(z). \quad (1.10)$$

Note that, in contrast to (1.8), the optimal controller in (1.10) has become independent of the intrinsic feedback loop created by $\mathbf{P}_0$. As such, the feedback loop no longer has to be considered in the design of $\mathbf{C}_0$, as long as $\mathbf{C}_0$ does not destabilize the intrinsic feedback loop, which can be easily checked using the generalized Nyquist criterion.

Theorem 1.3 (Closed-loop stability). *Define the intrinsic feedback loop with loop transfer function $\mathbf{L}_0 = \mathbf{C}_0\mathbf{P}_3\mathbf{P}_0$. Under the assumption that $\mathbf{C}_0$, $\mathbf{P}_3$, and $\mathbf{P}_0$ are stable, i.e. have no poles outside the unit disk, the closed-loop system is stable if and only if the Nyquist plot of $\det(\mathbf{I} - \mathbf{L}_0(j\omega))$ has no anti-clockwise encirclements of the origin, and does not pass the origin.*

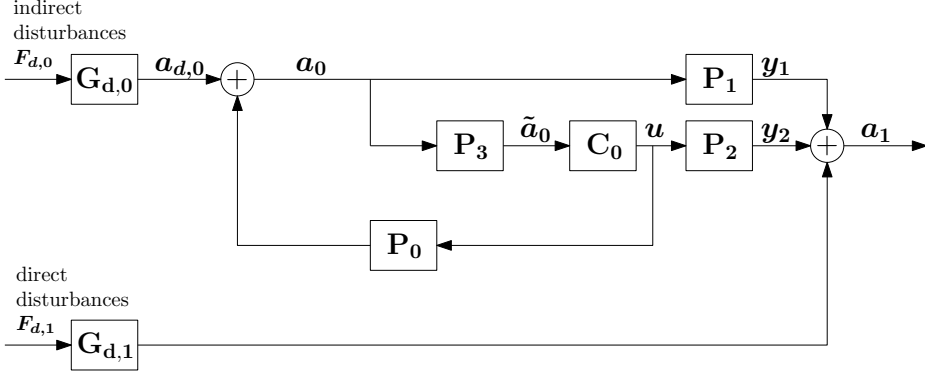


Figure 1.7. Block diagram for the physics-based AVIS control approach.

Proof. See [84, Sec. 4.9.2]. □

The plant transfer functions in Figure 1.7 are related to the ones in Figure 1.6 via

$$\mathbf{P}_1 = \mathbf{G}_{11}, \quad \mathbf{P}_2 = \mathbf{G}_{12} - \mathbf{G}_{11}\mathbf{G}_{21}^{-1}\mathbf{G}_{22}, \quad \mathbf{P}_3 = \mathbf{G}_{21}, \quad \mathbf{P}_0 = \mathbf{G}_{21}^{-1}\mathbf{G}_{22}. \quad (1.11)$$

Compared to Approach 1, (1.11) shows that the transmissibility matrix $\mathbf{P}_1 = \mathbf{G}_{11}$ and the sensor dynamics matrix $\mathbf{P}_3 = \mathbf{G}_{21}$ are identical for both approaches. The reaction path $\mathbf{P}_0$ differs only slightly from the reaction path $\mathbf{G}_{22}$ in Approach 1 in the sense that it now acts on the actual acceleration $\mathbf{a}_0$ instead of the measured acceleration $\tilde{\mathbf{a}}_0$. The main difference, however, is found in the actuator path. Compared to the actuator path $\mathbf{G}_{12}$ in Approach 1, the new actuator path $\mathbf{P}_2$ in (1.11) has a correction term $\mathbf{G}_{11}\mathbf{G}_{21}^{-1}\mathbf{G}_{22}$. In fact, this correction term removes the effects of base frame dynamics in $\mathbf{P}_2$, such that $\mathbf{P}_2$ can be regarded as the actuator path corresponding to the case the base frame would have been infinitely stiff ($\mathbf{P}_0 = 0$).

Under the following assumption, the controller gets an attractive physical interpretation.

Assumption 1.4. *The base frame sensors measuring $\mathbf{a}_0$ are placed exactly at the points where the isolators are connected to the base frame.*

Under this assumption (see also Section 3.2.2 for further details), $\mathbf{P}_1$ becomes independent of base frame dynamics, because the base frame is no longer part of the transmission path from the sensor measuring $\mathbf{a}_0$ to the sensor measuring $\mathbf{a}_1$. Moreover, it was already shown that $\mathbf{P}_2$ was independent of base frame dynamics, and it is reasonable that the sensor dynamics in $\mathbf{P}_3$ are independent of base frame dynamics. As such, $\mathbf{C}_0 = -\mathbf{P}_2^{-1}\mathbf{P}_1\mathbf{P}_3^{-1}$ becomes independent of base

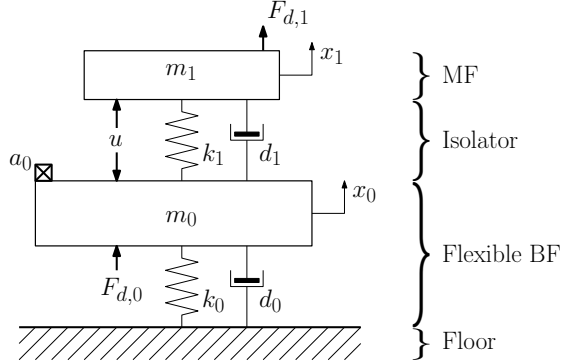


Figure 1.8. Example of a simplified model for a vibration isolation system.

frame dynamics, which is intuitive because in fact $\mathbf{C}_0$ only has to counteract the isolator dynamics. This interpretation is attractive because it enables modeling of base frame motion as an external disturbance input which is not influenced by the actuator forces, which greatly simplifies controller design. If one afterwards wants to account for a finite mass and stiffness of the base frame, it is sufficient to check closed-loop stability of the intrinsic feedback loop by measuring the FRF $\mathbf{P}_0(j\omega)$ and applying the Nyquist criterion given in Theorem 1.3.

Remark 1.5. A possible problem with this one-step design approach is that the new secondary path $\mathbf{P}_2$ cannot be measured directly, but has to be calculated from measured FRF data of $\mathbf{G}_{12} - \mathbf{G}_{11}\mathbf{G}_{21}^{-1}\mathbf{G}_{22}$, see (1.11). However, if the base frame is much stiffer and heavier than the isolator and the metrology frame, $\mathbf{G}_{22}$ is relatively small such that $\mathbf{P}_2 \approx \mathbf{G}_{12}$, hence directly measurable.

1.7.3 Illustrative example

Consider the mechanical model in Figure 1.8, which shows a great simplification of a high-precision system with an isolated metrology frame and a flexible base frame. Mass m_0 represents the base frame, and mass m_1 represents the metrology frame. The base frame is connected to the floor via spring k_0 and damper d_0 . The metrology frame and base frame are connected via an isolator represented by spring k_1 and damper d_1 . A perfect sensor is assumed to be available in the model, measuring $\tilde{\mathbf{a}}_0 = \mathbf{a}_0 = \ddot{x}_0$. As such, $\mathbf{G}_{21} = \mathbf{P}_3 = 1$. The system has two incoming disturbances, i.e. $\mathbf{F}_{d,0}$ and $\mathbf{F}_{d,1}$, and contains a force actuator $\mathbf{u}$. The equations of motion for this system are given by

$$m_1\ddot{x}_1 = F_{d,1} + u - k_1(x_1 - x_0) - d_1(\dot{x}_1 - \dot{x}_0), \quad (1.12)$$

$$m_0\ddot{x}_0 = F_{d,0} - u + k_1(x_1 - x_0) + d_1(\dot{x}_1 - \dot{x}_0) - k_0x_0 - d_0\dot{x}_0. \quad (1.13)$$

The equations of motion are used below to derive the feedforward controllers using either of the two approaches.

Approach 1: internal model compensation

From these equations of motion, it is derived that

$$p_1(s)F_d(s) + (p_1(s) - p_2(s))F_a(s) + p_2(s)F_0(s) = (p_2(s)p_3(s) - p_1^2(s))X_0(s), \quad (1.14)$$

$$p_3(s)F_d(s) + (p_3(s) - p_1(s))F_a(s) + p_1(s)F_0(s) = (p_2(s)p_3(s) - p_1^2(s))X_1(s), \quad (1.15)$$

with $p_1(s) = d_1s + k_1$, $p_2(s) = m_1s^2 + d_1s + k_1$, and $p_3(s) = m_0s^2 + (d_1 + d_0)s + (k_1 + k_0)$. Using (1.14), the following plant transfer functions are derived:

$$\begin{aligned} G_{11}(s) &= \frac{p_1(s)}{p_2(s)}, & G_{12}(s) &= \frac{s^2(p_3(s) - p_1(s))}{p_2(s)p_3(s) - p_1^2(s)}, \\ G_{21}(s) &= 1, & G_{22}(s) &= \frac{s^2(p_1(s) - p_2(s))}{p_2(s)p_3(s) - p_1^2(s)}, \\ G_{d,0}(s) &= \frac{p_2(s)}{p_2(s)p_3(s) - p_1^2(s)}, & G_{d,1}(s) &= \frac{s^2p_3(s)}{p_2(s)p_3(s) - p_1^2(s)}. \end{aligned} \quad (1.16)$$

Substitution of these transfer functions in (1.8) gives

$$C_0 = -\frac{1}{s^2}(d_1s + k_1), \quad (1.17)$$

As predicted, this controller only needs information about the isolator to obtain perfect cancellation. As such, the two-step approach to design $\mathbf{C}_0$ according to (1.8) leads to several pole-zero cancellations and unnecessarily complex modeling.

Design 2: one-step method

In this approach, base frame dynamics are neglected during controller design. Therefore, one only needs to consider the equation of motion for m_1 given in (1.12). From this equation, it follows that

$$P_1(s) := \frac{X_1(s)}{X_0(s)} = \frac{d_1s + k_1}{m_1s^2 + d_1s + k_1}, \quad P_2(s) = \frac{s^2}{m_1s^2 + d_1s + k_1}. \quad (1.18)$$

Substitution of these transfer functions in (1.10), and using $P_3(s) = 1$, gives (again)

$$C_0 = -\frac{1}{s^2}(d_1s + k_1), \quad (1.19)$$

which is the same result as obtained in (1.17), but with much less modeling effort.

1.7.4 Conclusions

It is shown that the two-step design approach is not useful in practice when the base frame is relatively stiff, because it needs model knowledge to implement IMC which is hard to obtain in such cases. As an alternative, the one-step design approach is presented for which model-based design is greatly simplified and closed-loop stability could be evaluated via measured frequency response data. The latter has proven to be of great value in other fields of mechatronics such as high-performance motion control.

Chapter 2

Design and performance limitations

This chapter presents design and performance tradeoffs in disturbance feedforward control (DFC) for Multi-Input Multi-Output (MIMO) systems. These tradeoffs, which are the result of causality aspects, imply that performance improvements at certain frequencies and directions generally lead to performance deterioration at other frequencies and directions. These tradeoffs are expressed in terms of the feedforward sensitivity matrix. Given these tradeoffs, the cost of disturbance rejection can be minimized by only reducing the principal gains of the feedforward sensitivity matrix that correspond to specific disturbance directions, and only at frequencies that are relevant for disturbance rejection. This generally results in a centralized feedforward controller being the optimal solution if the incoming disturbances are correlated, even in cases where the plant dynamics themselves are fully decoupled, which is demonstrated by a simulation example of a 2-DOF vibration isolation system.¹

¹This chapter is based on:
M.A. Beijen, B. Cong, and M.F. Heertjes, *Fundamental limitations and performance tradeoffs in MIMO disturbance feedforward control*, in preparation for journal publication.

2.1 Introduction

Disturbance feedforward control (DFC) can be used to reject external disturbances that are a priori unknown. An example of DFC is found in active vibration control, where sensors are used to measure external vibrations (e.g. floor vibrations), and a controller is used to counteract the induced vibrations of a dynamically coupled structure using actuator forces [95, 58]. A second example from the field of advanced motion control is found in the synchronization of positioning stages of wafer scanners [46, 8, 79]. In [46], the tracking error of a low-bandwidth wafer stage is forwarded as a setpoint to a high-bandwidth reticle stage, with the aim to minimize the synchronization error between the two stages. A third example is found in active noise cancellation [29, 7, 13], in which a microphone picks up sound waves from the environment. A controller and a loudspeaker are then used to generate a counteracting sound wave that ideally cancels the incoming sound wave, the latter can be seen as the disturbance.

DFC essentially differs from setpoint feedforward control (SFC) used in motion control systems, see e.g. [44, 98]. In SFC the future reference is assumed to be a priori known from a setpoint generator, while in DFC this reference must be measured prior to being used for cancellation. Due to the causality aspect introduced by DFC, i.e. the reference signals needs to be measured first before computing and applying the appropriate feedforward controller output, performance is bounded by limitations similar to the waterbed effect, which for feedback systems is expressed by the Bode sensitivity integral [36]. In [41], it is shown that a similar performance limitation exists for filtering, prediction, and smoothing problems.

For SISO systems, the waterbed effect only refers to performance tradeoffs along the frequency axis. For Multi-Input Multi-Output (MIMO) systems, however, the directions of the disturbances also play an important role in performance optimization. Amongst others, [32, 94] give expressions of the expected residual mean-square-error for a MIMO system with DFC that is optimal in the sense of an L_2 -norm, but give no general expression for fundamental performance limitations in the frequency domain. In [16], design and performance tradeoffs based on the MIMO sensitivity matrix are extensively discussed for feedback control systems, but not within the context of feedforward design. In [81], fundamental limitations are derived for the MIMO filtering sensitivity matrix, which is very similar to the feedforward sensitivity matrix, but with the difference of being formulated in terms of a Poisson integral constraint and not in terms of a Bode integral constraint. The latter, however, is preferred in exploiting the waterbed effect for controller design, which will be shown as part of the discussion in this chapter. More specifically, fundamental limitations for MIMO systems with DFC are presented in terms of the MIMO sensitivity matrix, while considering tradeoffs in both frequency and disturbance direction.

This chapter has two main contributions. First, the Bode integral constraint

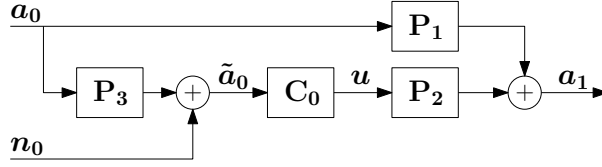


Figure 2.1. General configuration of a DFC system.

for SISO systems is extended to describe the fundamental limitations for feedforward sensitivity matrices in MIMO systems. More specifically, it is shown that it is possible to exchange frequency-wise tradeoffs over principal gains, but not over directions. Second, it is shown how to deal with these frequency-wise tradeoffs in the feedforward control design using H_2 optimization. It turns out that a centralized feedforward controller is the optimal solution if the incoming disturbances are correlated, even when the plant dynamics themselves are fully decoupled.

This chapter is organized as follows. Section 2.2 presents the problem formulation for a general DFC system. Arguments underlying the performance limitations in MIMO feedforward systems are derived in Section 2.3, whereas directionality aspects and design interpretations of these limitations are presented in Section 2.4. Design interpretations regarding these limitations are given in Section 2.5. An illustrative example, which considers a 2-DOF vibration isolation system, is presented in Section 2.6. The main conclusions of this chapter are given in Section 2.7.

2.2 Problem formulation

Consider the block diagram in Figure 2.1, where all systems are considered continuous-time and linear-time-invariant (LTI) $n \times n$ transfer function matrices, loosely called transfer matrices. The primary path $\mathbf{P}_1(s)$, with s the Laplace variable, transmits the input disturbance $\mathbf{a}_0(t) \in \mathbb{R}^n$, whereas the secondary path $\mathbf{P}_2(s)$ transmits the control signal $\mathbf{u}(t) \in \mathbb{R}^n$. The control signal is calculated from a disturbance feedforward controller $\mathbf{C}_0$ having input $\tilde{\mathbf{a}}_0(t) \in \mathbb{R}^n$ that represents a measurement of $\mathbf{a}_0$ filtered by $\mathbf{P}_3(s)$ and affected by measurement noise $\mathbf{n}_0(t) \in \mathbb{R}^n$. The filter $\mathbf{P}_3$ represents (potential) sensor dynamics and (anti-aliasing) sensor filtering.

The residual error (with $\mathbf{A}_1$, $\mathbf{A}_0$, $\mathbf{N}_0$ the Laplace transform of $\mathbf{a}_1$, $\mathbf{a}_0$, $\mathbf{n}_0$, respectively) reads

$$\mathbf{A}_1(s) = \mathcal{T}(s)\mathbf{A}_0(s) + \mathcal{S}_0\mathbf{N}_0(s), \quad (2.1)$$

with the transmissibility matrix $\mathcal{T}$ and the noise sensitivity matrix $\mathcal{S}_0$ given by

$$\mathcal{T}(s) = \mathbf{S}_{\mathbf{FF}}(s)\mathbf{P}_1(s), \quad (2.2)$$

$$\mathcal{S}_0(s) = -\mathbf{S}_{\mathbf{C},\mathbf{FF}}(s)\mathbf{P}_1(s)\mathbf{P}_3^{-1}(s), \quad (2.3)$$

where

$$\mathbf{S}_{\mathbf{FF}}(s) = (\mathbf{P}_1(s) + \mathbf{P}_2(s)\mathbf{C}_0(s)\mathbf{P}_3(s))\mathbf{P}_1^{-1}(s), \quad (2.4)$$

$$\mathbf{S}_{\mathbf{C},\mathbf{FF}}(s) = -\mathbf{P}_2(s)\mathbf{C}_0(s)\mathbf{P}_3(s)\mathbf{P}_1^{-1}(s). \quad (2.5)$$

The term $\mathbf{S}_{\mathbf{FF}}$ indicates the so-called (output) feedforward sensitivity matrix. Considering $s = j\omega$, with angular frequency $\omega \in \mathbb{R}$, leads to the frequency response function (FRF) matrix $\mathbf{S}_{\mathbf{FF}}(j\omega)$. This FRF matrix describes the relative improvement of $\mathcal{T}$, i.e. the amount of disturbance rejection that is obtained by adding DFC. In SISO systems, disturbance rejection is improved at frequencies where $|S_{FF}(j\omega)| < 1$, while disturbance rejection is deteriorated at frequencies where $|S_{FF}(j\omega)| > 1$. For MIMO systems, this concept is generalized by considering the principal gains (singular values per frequency) of $\mathbf{S}_{\mathbf{FF}}$. Without control ($\mathbf{C}_0 = \mathbf{0}$), $\mathbf{S}_{\mathbf{FF}}$ becomes an identity matrix such that $\mathcal{T}$ reduces to $\mathbf{P}_1$. The function $\mathbf{S}_{\mathbf{C},\mathbf{FF}}$ represents the complementary feedforward sensitivity matrix, which serves as a measure for disturbances induced by noise amplification.

The main problem considered in this chapter is to design a feedforward controller $\mathbf{C}_0$ such that an error signal $\mathbf{a}_1(t) \in \mathbb{R}^{n_y}$, with time $t \geq 0$, is minimized in the sense of the L_2 -norm, or, using $\mathbf{A}_1^H \mathbf{A}_1 = \text{tr}(\mathbf{A}_1 \mathbf{A}_1^H)$,

$$\begin{aligned} \mathbf{C}_{0,\text{opt}} &= \arg \min_{\mathbf{C}_0} (\mathbf{A}_1^H \mathbf{A}_1) \\ &= \arg \min_{\mathbf{C}_0} \text{tr} \left(\mathbf{S}_{\mathbf{FF}} \mathbf{P}_1 \mathbf{S}_{\mathbf{a}_0 \mathbf{a}_0} \mathbf{P}_1^H \mathbf{S}_{\mathbf{FF}}^H + \mathbf{S}_{\mathbf{C},\mathbf{FF}} \mathbf{P}_1 \mathbf{P}_3^{-1} \mathbf{S}_{\mathbf{n}_0 \mathbf{n}_0} \mathbf{P}_3^{-H} \mathbf{P}_1^H \mathbf{S}_{\mathbf{C},\mathbf{FF}}^H \right), \end{aligned} \quad (2.6)$$

with $\mathbf{S}_{\mathbf{a}_0 \mathbf{a}_0}$ and $\mathbf{S}_{\mathbf{n}_0 \mathbf{n}_0}$ representing the power spectra of $\mathbf{a}_0$ and $\mathbf{n}_0$, respectively. It will be shown in Sections 2.3 and 2.4 of this chapter that fundamental limitations and design tradeoffs exist, and that they can be related directly to $\mathbf{S}_{\mathbf{FF}}$ which is a function of $\mathbf{C}_0$, see (2.4). This makes it generally impossible to make all entries of $\mathbf{S}_{\mathbf{FF}}$ arbitrarily low, hence perfect cancellation of the effect of $\mathbf{a}_0$ at $\mathbf{a}_1$ is not possible. To solve the minimization problem in (2.6), it will be shown that proper modeling of $\mathbf{S}_{\mathbf{a}_0 \mathbf{a}_0}$, including cross-correlation between the channels in $\mathbf{a}_0$, is essential. This will be further discussed in Sections 2.5 and 2.6.

Throughout this chapter, the following assumptions are made.

Assumption 2.1. *The transfer matrices satisfy the following conditions:*

- (1) $\mathbf{P}_1(s)$, $\mathbf{P}_2(s)$, $\mathbf{P}_3(s)$, and $\mathbf{C}_0(s)$ are Hurwitz matrices,
- (2) All entries of $\mathbf{P}_2(s)\mathbf{P}_3(s)\mathbf{P}_1^{-1}(s)$ and $\mathbf{C}_0(s)$ represent proper transfer functions,

- (3) for z being a right-half-plane (RHP) transmission zero of $\mathbf{P}_2(s)$, i.e. $\mathbf{P}_2(s)$ loses rank at $s = z$, $\mathbf{P}_1(z)$ is invertible.

Condition (1) requires the stability of all subsystems in the feedforward control loop, thus sufficiently guaranteeing the stability of the system as a whole. As a result, $\mathbf{u}$ and $\mathbf{a}_1$ are bounded for any bounded inputs $\mathbf{a}_0, \mathbf{n}_0$. Assuming $\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3$ to be stable transfer matrices makes sense since many passive mechanical systems and sensors are stable by themselves. Condition (2) assures that $\mathbf{S}_{\mathbf{FF}}$ is proper, which is a necessary condition in the formulation of the Bode integral constraints. $\mathbf{C}_0$ being proper is needed to realize a causal feedforward controller in a discrete-time implementation. Condition (3) prevents $\mathbf{S}_{\mathbf{FF}}$ from being singular at each of the RHP transmission zeros of $\mathbf{P}_2$.

Remark 2.2. *It is worth noting that $\mathbf{S}_{\mathbf{FF}}$ can have RHP poles even under Assumption 2.1. These RHP poles result from the term $\mathbf{P}_1^{-1}$ in (2.4) in case $\mathbf{P}_1$ has RHP transmission zeros that are not shared by the matrix transfer $\mathbf{P}_2 \mathbf{C}_0 \mathbf{P}_3$. However, such RHP poles only exist in the definition of $\mathbf{S}_{\mathbf{FF}}$ and do not appear in the controlled system as shown in Figure 2.1, hence they will not influence the system's stability properties.*

2.3 Fundamental limitations

For SISO systems, the matrix expressions $\mathbf{S}_{\mathbf{FF}}$ and $\mathbf{S}_{\mathbf{C},\mathbf{FF}}$ in respectively (2.4) and (2.5) reduce to the (complementary) feedforward sensitivity functions denoted by S_{FF} and $S_{C,FF}$, respectively. Fundamental limitations for these functions are derived in [81]. More specifically, S_{FF} and $S_{C,FF}$ satisfy the so-called complementarity constraint, interpolation constraints, and integral constraints. In view of these constraints, it is not possible to control $\mathbf{A}_1(j\omega) \rightarrow \mathbf{0}$ for all ω . Moreover, in [81] Poisson integrals are derived for describing fundamental limitations of $\mathbf{S}_{\mathbf{FF}}$ and $\mathbf{S}_{\mathbf{C},\mathbf{FF}}$ in MIMO systems. In addition to [81], this section proposes an extension of the Bode integral constraint to MIMO systems. For clarity of presentation, first the integral constraint for SISO systems is revisited in Section 2.3.1. Hereafter, the proposed extension to MIMO systems is posed in Section 2.3.2.

2.3.1 SISO constraints

Performance constraints for SISO systems are given by the Bode integral constraint which is adopted from [81] and presented in the following theorem.

Theorem 2.3 (Bode integral constraint). *Suppose Assumption 2.1 holds such that S_{FF} is a proper transfer function. Assuming $S_{FF}(j\infty) \neq 0$, the Bode*

integral for S_{FF} is given by

$$\int_0^\infty \log \left| \frac{S_{FF}(j\omega)}{S_{FF}(j\infty)} \right| d\omega = \frac{\pi}{2} \lim_{s \rightarrow \infty} \frac{s [S_{FF}(s) - S_{FF}(\infty)]}{S_{FF}(\infty)} + \pi \sum_{i=1}^{n_q} q_i - \pi \sum_{l=1}^{n_p} p_l, \quad (2.7)$$

in which n_q is the number of open right-half-plane (RHP) zeros q_i of S_{FF} , and n_p is the number of open RHP poles of S_{FF} , i.e. the open RHP zeros p_i of P_1 that are not open RHP zeros of $P_2(s)C_0(s)P_3(s)$.

Proof. For the proof, the reader is referred to [81]. $\square$

Note that, although q_i and p_l can be complex-valued, the right-hand side of (2.7) is always real-valued since complex zeros and poles must appear in conjugated pairs. Due to the integral constraint in (2.7), perfect cancellation is not encountered in practice, evidencing the existence of performance tradeoffs similar to the waterbed effect in feedback systems. Although it follows from (2.7) that DFC systems have the performance properties of a feedback control system, they have the stability properties of a setpoint feedforward control system. RHP zeros in S_{FF} are limiting performance similar to the role of RHP poles in the Bode sensitivity integral for feedback systems. At the same time, RHP poles in S_{FF} can improve performance since those correspond with RHP zeros in P_1 which, according to [81], can be thought of as ‘mild’ time delays in the disturbance path. This means that their effect on performance is similar to the effect of a physical time delay in the disturbance path, which provides more time to compute the compensating control force. As a final remark, note that the limit term in the right-hand-side of (2.7) is zero if the relative degree of $S_{C,FF}$ is at least two.

2.3.2 MIMO Integral constraints

The following theorem and corollary, which are inspired by the work on feedback systems in [37], introduce the Bode sensitivity integral for MIMO DFC systems.

Theorem 2.4. *Under Assumption 2.1, let the feedforward sensitivity matrix $\mathbf{S}_{FF}$ be defined as in (2.4). If $\mathbf{S}_{FF}$ has RHP transmission zeros $q_i, i \in \{1, \dots, n_q\}$, RHP poles $p_l, l \in \{1, \dots, n_p\}$, and all entries of $\mathbf{S}_{C,FF}$ as defined in (2.5) have at least two more poles than zeros, then*

$$\int_0^\infty \log |\det(\mathbf{S}_{FF}(j\omega))| d\omega = \pi \sum_{i=1}^{n_q} \Re(q_i) - \pi \sum_{l=1}^{n_p} \Re(p_l). \quad (2.8)$$

Proof. Using the complementarity constraint $\mathbf{S}_{FF} + \mathbf{S}_{C,FF} = \mathbf{I}$ [81], and applying cofactor expansion, the determinant of $\mathbf{S}_{FF}$ can be expressed by

$$\det(\mathbf{S}_{FF}(s)) = \det(\mathbf{I} - \mathbf{S}_{C,FF}(s)) = 1 + \sum_k \tilde{S}_{C,FF}^k(s), \quad (2.9)$$

where all k functions $\tilde{S}_{C,FF}^k(s)$ contain at least one factor corresponding to an entry of $\mathbf{S}_{C,FF}(s)$ and therefore have at least two more poles than zeros. From the latter observation, it follows that the term $\sum_k \tilde{S}_{C,FF}^k$ approaches zero at infinity, and therefore $\det(\mathbf{S}_{FF}(j\infty)) \neq 0$. Moreover, since $\det(\mathbf{S}_{FF})$ is proper, Theorem 2.3 can be applied to $\det(\mathbf{S}_{FF}(s))$, resulting in

$$\int_0^\infty \log \left| \frac{\det(\mathbf{S}_{FF}(j\omega))}{\det(\mathbf{S}_{FF}(j\infty))} \right| d\omega = \frac{\pi}{2} \lim_{s \rightarrow \infty} \frac{s [\det(\mathbf{S}_{FF}(s)) - \det(\mathbf{S}_{FF}(\infty))]}{\det(\mathbf{S}_{FF}(\infty))} + \pi \sum_{i=1}^{n_q} \Re(q_i) - \pi \sum_{l=1}^{n_p} \Re(p_l). \quad (2.10)$$

Notice that, by means of definition, the RHP poles p_l and RHP transmission zeros q_i for $\det(\mathbf{S}_{FF}(s))$ are identical to those of $\mathbf{S}_{FF}(s)$. Hence, the last two terms at the right-hand side of (2.10) are related to the RHP poles and RHP transmission zeros of $\mathbf{S}_{FF}(s)$. In addition, because $\tilde{S}_{C,FF}^k(s)$ has at least two more poles than zeros, it follows straightforwardly that $\lim_{s \rightarrow \infty} s \sum_k \tilde{S}_{C,FF}^k(s) = 0$ and $\det(\mathbf{S}_{FF}(j\infty)) = \det(\mathbf{S}_{FF}(\infty)) = 1$, which simplifies (2.10) to (2.8) and which completes the proof. $\square$

Corollary 2.5. *Let $\sigma_k(\mathbf{S}_{FF}(\omega))$ be the k^{th} principal gain (k^{th} singular value at frequency ω) of the feedforward sensitivity matrix $\mathbf{S}_{FF}$ as defined in Theorem 2.4. If $\mathbf{S}_{FF}$ has no RHP transmission zeros and no RHP poles, Theorem 2.4 implies that*

$$\int_0^\infty \log \sigma_k(\mathbf{S}_{FF}(j\omega)) d\omega = F_k, \quad \text{and} \quad \sum_{k=1}^{n_y} F_k = 0. \quad (2.11)$$

Proof. If $\mathbf{S}_{FF}$ has no RHP transmission zeros and no RHP poles, the summation terms at the right-hand side of (2.8) become zero. Then, using the fact that $|\det(A)| = \prod_{k=1}^{n_y} \sigma_k(A)$ completes the proof. $\square$

Similar to feedback systems [16], it follows from Corollary 2.5 that there exists a tradeoff for the *sum* of the logarithmic magnitudes of the principal gains. The individual terms F_k can be non-zero, hence it is possible to exchange frequency-wise tradeoffs between principal gains. However, Corollary 2.5 does not give any information about the possibilities and limitations to exchange frequency-wise tradeoffs between directions. This will be further discussed in the next section.

2.4 Directionality aspects

The goal of this section is to extend the formulation of fundamental limitations to the directions as well. To this end, first consider the following definition.

Definition 2.6 (Directional sensitivity function). *Let vector $\mathbf{v}_1 \in \mathbb{C}^n$ be the normalized output direction and vector $\mathbf{v}_2 \in \mathbb{C}^n$ be the normalized input direction of the feedforward sensitivity matrix $\mathbf{S}_{\mathbf{FF}}(s) \in \mathbb{C}^{n \times n}$. For $\mathbf{v}_1^T \mathbf{v}_2 \neq 0$, the scalar sensitivity function $S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}(s) \in \mathbb{C}$ is defined as*

$$S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}(s) = \mathbf{v}_1^T \mathbf{S}_{\mathbf{FF}}(s) \mathbf{v}_2. \quad (2.12)$$

Definition 2.6 enables the possibility to analyze limitations in a specific input and output direction of $\mathbf{S}_{\mathbf{FF}}$. Based on this definition, the following directional Bode integral constraint for MIMO feedforward control systems is proposed.

Theorem 2.7 (Directional Bode Integral Constraint). *Under Assumption 2.1, let the scalar sensitivity function $S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}$ be given by Definition 2.6 with RHP zeros $q_i, i \in \{1, \dots, n_q\}$ and RHP poles $p_l, l \in \{1, \dots, n_p\}$. If it is further assumed that each element of $\mathbf{S}_{\mathbf{C}, \mathbf{FF}}$ has at least two more poles than zeros, then*

$$\int_0^\infty (\log |S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}(j\omega)| - \log |\mathbf{v}_1^T \mathbf{v}_2|) d\omega = \pi \sum_{i=1}^{n_q} q_i - \pi \sum_{l=1}^{n_p} p_l. \quad (2.13)$$

Proof. Since $S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}$ is a proper transfer function, and under the assumption that $S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}(j\infty) \neq 0$, which is verified later on, applying (2.7) to $S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}$ gives

$$\begin{aligned} \int_0^\infty \log \left| \frac{S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}(j\omega)}{S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}(j\infty)} \right| d\omega = \\ \frac{\pi}{2} \lim_{s \rightarrow \infty} s \frac{[S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}(s) - S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}(\infty)]}{S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}(\infty)} + \pi \sum_{i=1}^{n_q} q_i - \pi \sum_{l=1}^{n_p} p_l. \end{aligned} \quad (2.14)$$

If each element of $\mathbf{S}_{\mathbf{C}, \mathbf{FF}}$ has at least two more poles than zeros, then it holds that $\lim_{s \rightarrow \infty} \mathbf{S}_{\mathbf{C}, \mathbf{FF}}(s) = \mathbf{0}$ and $s \cdot \lim_{s \rightarrow \infty} \mathbf{S}_{\mathbf{C}, \mathbf{FF}}(s) = \mathbf{0}$. Using the complementarity constraint $\mathbf{S}_{\mathbf{FF}} + \mathbf{S}_{\mathbf{C}, \mathbf{FF}} = \mathbf{I}$ [81], it holds that $\lim_{s \rightarrow \infty} \mathbf{S}_{\mathbf{FF}}(s) = \mathbf{I} - \lim_{s \rightarrow \infty} \mathbf{S}_{\mathbf{C}, \mathbf{FF}}(s) = \mathbf{I}$. Therefore,

$$\begin{aligned} \lim_{s \rightarrow \infty} \frac{s[S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}(s) - S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}(\infty)]}{S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}(\infty)} \\ = \lim_{s \rightarrow \infty} \frac{\mathbf{v}_1^T s[\mathbf{S}_{\mathbf{FF}}(s) - \mathbf{S}_{\mathbf{FF}}(\infty)] \mathbf{v}_2}{\mathbf{v}_1^T \mathbf{S}_{\mathbf{FF}}(\infty) \mathbf{v}_2} \\ = \lim_{s \rightarrow \infty} \frac{\mathbf{v}_1^T s[\mathbf{S}_{\mathbf{FF}}(s) - \mathbf{I}] \mathbf{v}_2}{\mathbf{v}_1^T \mathbf{v}_2} \\ = \lim_{s \rightarrow \infty} \frac{\mathbf{v}_1^T s \mathbf{S}_{\mathbf{C}, \mathbf{FF}}(s) \mathbf{v}_2}{\mathbf{v}_1^T \mathbf{v}_2} = 0. \end{aligned} \quad (2.15)$$

Similar to (2.15), it can be shown that $S_{\mathbf{FF}}^{\mathbf{v}_1, \mathbf{v}_2}(j\infty) = \mathbf{v}_1^T \mathbf{v}_2$ which is non-zero by the grace of Definition 2.6, recall the earlier assumption, such that

$\log \left| \frac{S_{FF}^{v_1, v_2}(j\omega)}{S_{FF}^{v_1, v_2}(j\infty)} \right| = \log |S_{FF}^{v_1, v_2}(j\omega)| - \log |\mathbf{v}_1^T \mathbf{v}_2|$. Substitution of this result together with (2.15) in (2.14) completes the proof. $\square$

Theorem 2.7 shows a generalization of Theorem 2.3, and includes information about the input and output directions by including the vectors $\mathbf{v}_1, \mathbf{v}_2$. Next, two corollaries are derived describing two extremes: (1) the input and output directions are given by two unit vectors pointing in the same direction, and (2) the input and output directions are given by two orthonormal unit vectors.

Corollary 2.8 (Same input and output directions). *Suppose a system satisfies the conditions of Theorem 2.7, and let $\mathbf{v}_1 = \mathbf{v}_2 = \mathbf{u}$ be given by a unit vector $\mathbf{u}$ such that $\mathbf{u}^T \mathbf{u} = 1$. Then, application of Theorem 2.7 gives*

$$\int_0^\infty \log |S_{FF}^{\mathbf{u}, \mathbf{u}}(j\omega)| d\omega = \pi \sum_{i=1}^{n_q} q_i - \pi \sum_{l=1}^{n_p} p_l, \quad (2.16)$$

where $q_i, i \in \{1, \dots, n_q\}$ are the RHP zeros of $S_{FF}^{\mathbf{u}, \mathbf{u}}$, and $p_l, l \in \{1, \dots, n_p\}$ are the RHP poles of $S_{FF}^{\mathbf{u}, \mathbf{u}}$.

Corollary 2.9 (Orthonormal input and output directions). *Suppose a system satisfies the conditions of Theorem 2.7, and let $\mathbf{v}_1$ and $\mathbf{v}_2$ approach to two orthonormal vectors such that $\log |\mathbf{v}_1^T \mathbf{v}_2| \rightarrow -\infty$. Then, Theorem 2.7 implies that no Bode integral constraint is imposed on the magnitude of $S_{FF}^{v_1, v_2}$.*

It follows from Corollary 2.8 that the waterbed effect exists if the considered input direction $\mathbf{v}_1$ and output direction $\mathbf{v}_2$ coincide. This implies that it is impossible to manipulate the waterbed effect from one direction to another. Contrarily, Corollary 2.9 implies that there is no relation between the Bode integral constraints of two orthonormal directions. Interpretations of these main results are discussed in the remainder of this section by considering sensor directions and unitary transformations.

2.4.1 Fundamental limitations in sensor directions

Consider an orthonormal set of unit vectors $\mathbf{E} = \text{span}\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ with $\mathbf{e}_k = \underbrace{[0, \dots, 0, 1, 0, \dots]}_{k-1}^T$. Selection of $\mathbf{v}_1$ and $\mathbf{v}_2$ from this set enables the possibility

to evaluate the existence of a waterbed effect for each entry of $\mathbf{S}_{FF}$. Choosing $\mathbf{v}_1 = \mathbf{v}_2 = \mathbf{e}_k$ and applying Corollary 2.8, shows that any k^{th} entry on the main diagonal of $\mathbf{S}_{FF}$ is subject to the Bode integral constraint. In other words, it is not possible to manipulate the waterbed effect from one direction of $\mathbf{a}_1$ to another. It also implies that performance improvement at each sensor location is subject to a waterbed effect. Contrarily, Corollary 2.9 implies that the off-diagonal entries of $\mathbf{S}_{FF}$ are not subject to a Bode integral constraint, but these entries are anyway constrained under the assumption that $\mathbf{S}_{FF} \rightarrow \mathbf{I}$, see Theorem 2.7. As such, the off-diagonal terms must approach zero if $s \rightarrow \infty$.

2.4.2 Fundamental limitations under unitary transformations

Another interpretation of the fundamental limitations is found by considering unitary transformations.

Corollary 2.10 (Unitary transformation). *Suppose a system satisfies the conditions of Theorem 2.7. Assume vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ are selected from a given $\mathbf{U} = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ with $\mathbf{u}_k \in \mathbb{R}^n$, $\mathbf{u}_k^T \mathbf{u}_k = 1$, and $\mathbf{u}_k^T \mathbf{u}_h = 0$ if $k \neq h$. It is concluded that if $\mathbf{v}_1 = \mathbf{v}_2$, then*

$$\int_0^\infty \log |S_{FF}^{\mathbf{v}_1, \mathbf{v}_2}(j\omega)| d\omega = \pi \sum_{i=1}^{n_q} q_i - \pi \sum_{l=1}^{n_p} p_l, \quad (2.17)$$

where $q_i, i \in \{1, \dots, n_q\}$, are the RHP zeros, and $p_l, l \in \{1, \dots, n_p\}$, are the RHP poles of $S_{FF}^{\mathbf{v}_1, \mathbf{v}_2}$, respectively. Moreover, $S_{FF}^{\mathbf{v}_1, \mathbf{v}_2}$ with $\mathbf{v}_1 \neq \mathbf{v}_2$ is not subject to the Bode integral constraint.

Proof. The proof follows directly from replacing $\mathbf{v}_1$ and $\mathbf{v}_2$ by unitary vectors $\mathbf{u}_k$ and $\mathbf{u}_h$, respectively, in the proofs of Corollaries 2.8 and 2.9. $\square$

Corollary 2.10 shows that fundamental limitations on performance also apply after coordinate transformation. Using a unitary transformation matrix $\mathbf{U}$, one can find a transformed sensitivity function matrix $\mathbf{S}_{FF}^d$ via

$$\begin{aligned} \mathbf{S}_{FF}^d &= \mathbf{U}^T \mathbf{S}_{FF} \mathbf{U} = \begin{bmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix} \mathbf{S}_{FF} [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n] \\ &= \begin{bmatrix} S_{FF}^{\mathbf{u}_1, \mathbf{u}_1} & S_{FF}^{\mathbf{u}_1, \mathbf{u}_2} & \dots & S_{FF}^{\mathbf{u}_1, \mathbf{u}_n} \\ S_{FF}^{\mathbf{u}_2, \mathbf{u}_1} & S_{FF}^{\mathbf{u}_2, \mathbf{u}_2} & \dots & S_{FF}^{\mathbf{u}_2, \mathbf{u}_n} \\ \vdots & \vdots & \ddots & \vdots \\ S_{FF}^{\mathbf{u}_n, \mathbf{u}_1} & S_{FF}^{\mathbf{u}_n, \mathbf{u}_2} & \dots & S_{FF}^{\mathbf{u}_n, \mathbf{u}_n} \end{bmatrix}, \end{aligned} \quad (2.18)$$

where the argument (s) is omitted in view of space considerations. Each scalar entry of $\mathbf{S}_{FF}^d$ satisfies Corollary 2.10. In other words, each main diagonal entry of $\mathbf{S}_{FF}^d$ is restricted by the Bode integral constraint, while the off-diagonal entries of $\mathbf{S}_{FF}^d$ are not subject to a Bode integral constraint. Hence, the fundamental limitations found on the main diagonal for a feedforward sensitivity matrix remain valid also after unitary transformations.

2.5 Design interpretations

Corollary 2.5 showed that the reduction of each principal gain comes at the cost of one or more increased principal gains at other frequencies. Moreover,

Corollary 2.8 showed that it is impossible to exchange performance tradeoffs between directions in MIMO disturbance feedforward control (DFC). Therefore, to minimize the cost of disturbance rejection, vibration isolation provided by DFC should be restricted to the relevant disturbance directions and only at frequencies that are relevant for disturbance rejection. To obtain this level of system knowledge, one needs (a) identification methods to obtain an accurate model for the disturbances to distinguish a minimum number of independent disturbance sources, and (b) control design methods to optimize performance by exploiting the modeled disturbances and their directions. Since the goal of this chapter is mainly to discuss the inherent design limitations, these topics are not further pursued. Instead, a short literature overview is given on the subject.

Several identification methods can be used to provide disturbance models. Principal component analysis (PCA) can be used to find the principal directions in a disturbance spectrum. However, the principal directions generally change over frequency and do not give information about the independent sources. A technique that is able to do this is independent component analysis (ICA). In case of Gaussian disturbances with second-order statistics, one can use the second-order blind identification (SOBI) method instead of using general ICA. For an extensive discussion and comparison between these methods, the reader is referred to [16].

Having obtained a disturbance model, several control design methods can be used to account for correlation between disturbances. For feedback systems, disturbance decoupling is proposed in [16]. The latter means that the disturbances are transformed to a set of independent disturbances. Then, given that the plant dynamics are also decoupled, the MIMO design problem reduces to a multi-loop SISO problem which can be solved by loop shaping of the individual SISO loops. For feedforward control, however, such an approach is not suitable because the disturbances are filtered with the primary path $\mathbf{P}_1$ introducing additional dynamic coupling, and also because the concept of loop shaping does not translate well to feedforward control, see Remark 2.11. As an alternative, norm-based design methods are available such as H_2 and H_∞ optimization. The following section shows how H_2 control, which is commonly used in vibration isolation problems [32, 94], can be used to find an optimal MIMO controller.

Remark 2.11 (Loop shaping). *In feedback control, loop shaping is a widely used strategy for controller design. This strategy exploits the inverse proportional relation between the magnitudes of the open-loop system and the closed-loop sensitivity. In DFC, such an explicit relation between the open-loop magnitude and the closed-loop sensitivity does not exist, which impedes controller design via loop shaping.*

2.6 Numerical example: 2-DOF system

This section presents a numerical example to show how the cost of disturbance rejection in MIMO DFC could be minimized by proper modeling of the disturbances. This enables the possibility to minimize the principal gains of the sensitivity matrix only in the disturbance directions and at frequencies that are relevant for disturbance rejection. Two controller design strategies are presented. The first strategy neglects correlation between the disturbance channels in $\mathbf{a}_0$, while the second strategy exploits proper modeling of the correlated disturbances to increase performance.

2.6.1 System description

The system under study is the two-degree-of-freedom (2-DOF) vibration isolation system shown in Figure 2.2. The model consists of two spring-damper pairs installed underneath an isolated platform with mass m and inertia J . The platform can translate vertically in z -direction and rotate in the θ -direction. To enable disturbance rejection of base vibrations on the platform, actuator forces F_1, F_2 can be applied at the interconnection of the respective spring-damper pairs. Since the effect of cross-correlation between the disturbance channels in $\mathbf{a}_0$ is of key interest in this section, to an extension that much exceeds the effect of coupling in the system, the system parameters are chosen such that the model is symmetric and the plant dynamics become fully decoupled with respect to the (z, θ) coordinates. The following transfer matrices are used to describe the dynamics of the system in Figure 2.2:

$$\mathbf{P}_1(s) = \begin{bmatrix} \frac{33.3s+333.3}{s^2+33.3s+333.3} & 0 \\ 0 & \frac{25s+250}{s^2+25s+250} \end{bmatrix}, \quad (2.19)$$

$$\mathbf{P}_2(s) = \begin{bmatrix} \frac{0.00333s^2}{s^2+33.3s+333.3} & 0 \\ 0 & \frac{0.01s^2}{s^2+25s+250} \end{bmatrix}. \quad (2.20)$$

The input accelerations a_0 are assumed to be measured by a sensor including a second-order low-pass filter at $\omega_f = 2\pi \cdot 35$ rad/s such that

$$\mathbf{P}_3(s) = \frac{\omega_f^2}{s^2 + 1.4\omega_f s + \omega_f^2} \mathbf{I}_2. \quad (2.21)$$

2.6.2 Input disturbances

In the example, it is assumed that the base is vibrating due to two independent disturbance sources D_{s1}, D_{s2} . Simulated PSDs of D_{s1}, D_{s2} are shown in Figure 2.3. The source locations are linked to the simulated measurements of the

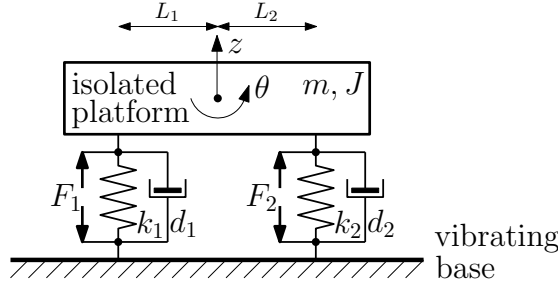


Figure 2.2. Simple 2-DOF model of a platform being isolated from base vibrations using DFC; in the example, $k_1 = k_2 = 50 \cdot 10^3$ N/m, $d_1 = d_2 = 5000$ Ns/m, $m_1 = 300$ kg, $J_1 = 100$ k·m², and $L_1 = L_2 = 0.5$ m.

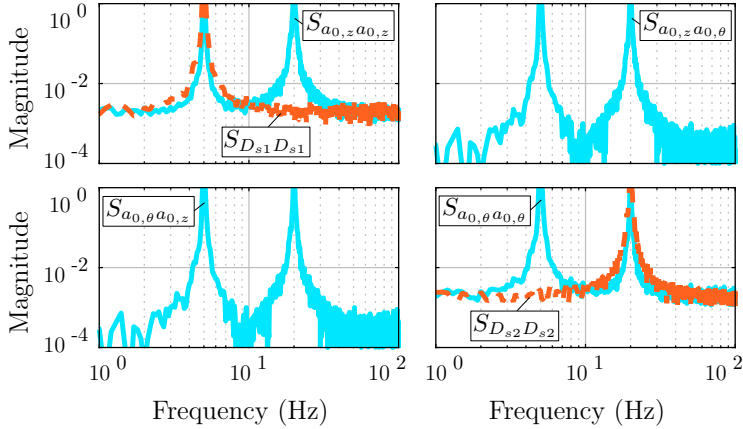


Figure 2.3. Auto- and cross-power spectral density plots of the base disturbances ($A_{0,x}, A_{0,\theta}$) in sensor coordinates (solid light blue), and the decorrelated disturbances (D_{s1}, D_{s2}) after coordinate transformation (dashed red).

base vibrations $a_0 = [a_{0,z}, a_{0,\theta}]^T$ in sensor locations via the unitary transformation matrix $\mathbf{U}$,

$$\begin{bmatrix} a_{0,z} \\ a_{0,\theta} \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}}_{\mathbf{U}} \begin{bmatrix} D_{s1} \\ D_{s2} \end{bmatrix}. \quad (2.22)$$

Due to the transformation, the sensor spectra $a_{0,z}, a_{0,\theta}$ become a mix of the spectra of D_{s1} and D_{s2} , see Figure 2.3. Moreover, the sensor signals possess clear cross-correlation, which can be seen by the non-zero off-diagonal entries in Figure 2.3.

2.6.3 H_2 Control design

The objective is to find a disturbance feedforward controller $\mathbf{C}_0$ that minimizes the L_2 -norm of $\mathbf{a}_1$, recall Figure 2.1. To find this controller, H_2 -optimization will be used that minimizes

$$\|\mathbf{F}(s)\|_2 = \sqrt{\frac{1}{2\pi} \int_{-\infty}^{\infty} \text{tr} [\mathbf{F}(j\omega)\mathbf{F}(j\omega)^H] d\omega}, \quad (2.23)$$

where $\mathbf{F}$ represents the closed-loop transfer function from the generalized disturbance vector $\mathbf{w} = [\mathbf{w}_1, \mathbf{w}_2]^T$ to the objective vector $\mathbf{z} = \mathbf{a}_1$, and which is given by

$$\mathbf{z} = \underbrace{(\mathbf{G}_{11} + \mathbf{G}_{12}\mathbf{C}_0(\mathbf{I} - \mathbf{G}_{22}\mathbf{C}_0)^{-1}\mathbf{G}_{21})}_{\mathbf{F}} \mathbf{w},$$

with $\mathbf{G}_{ij}$ representing the entries of the augmented plant, or

$$\begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_0 \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{P}_1\mathbf{V}_1 & \mathbf{0} & \mathbf{P}_2 \\ \mathbf{P}_3\mathbf{V}_1 & \mathbf{V}_2 & \mathbf{0} \end{bmatrix}}_{\begin{bmatrix} \mathbf{G}_{11}(s) & \mathbf{G}_{12}(s) \\ \mathbf{G}_{21}(s) & \mathbf{G}_{22}(s) \end{bmatrix}} \cdot \begin{bmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \mathbf{u} \end{bmatrix}. \quad (2.24)$$

Minimization of the H_2 -norm in (2.23) is equivalent to minimization of the L_2 -norm of $\mathbf{a}_1$ in case the disturbances in $\mathbf{w}_1, \mathbf{w}_2$ represent a set of independent white Gaussian noise disturbances with unit variance [84]. However, in the context of vibration isolation the disturbances should be defined according to the measured base and noise spectra. Therefore, weighting filters $\mathbf{V}_1, \mathbf{V}_2$ are included in (2.24) to specify the disturbance spectra correctly.

In the following, two different control strategies based on H_2 -optimization are compared. Input $\mathbf{w}_1$ represents a set of independent white Gaussian noise disturbances with unit variance. These inputs become colored and correlated by weighting function $\mathbf{V}_1$, which contains the information obtained from disturbance modeling. As such, $\mathbf{a}_0 = \mathbf{V}_1\mathbf{w}_1$ represents the actual, possibly correlated, base disturbances as measured by the input sensors. Weight $\mathbf{V}_1$ depends on the control strategy, as will be discussed below.

Input $\mathbf{w}_2$ represents another set of independent white Gaussian noise disturbances colored by weighting function $\mathbf{V}_2$ such that $\mathbf{n}_0 = \mathbf{V}_2\mathbf{w}_2$ represents additive sensor noise. $\mathbf{V}_2$ is the same for the two strategies, and corresponds to a typical non-quadratic noise spectrum of an accelerometer suffering from low-frequency noise, or

$$\mathbf{V}_2(s) = 0.03 \frac{(s + 2\pi \cdot 5)^2}{(s + 2\pi \cdot 0.1)^2} \mathbf{I}_2. \quad (2.25)$$

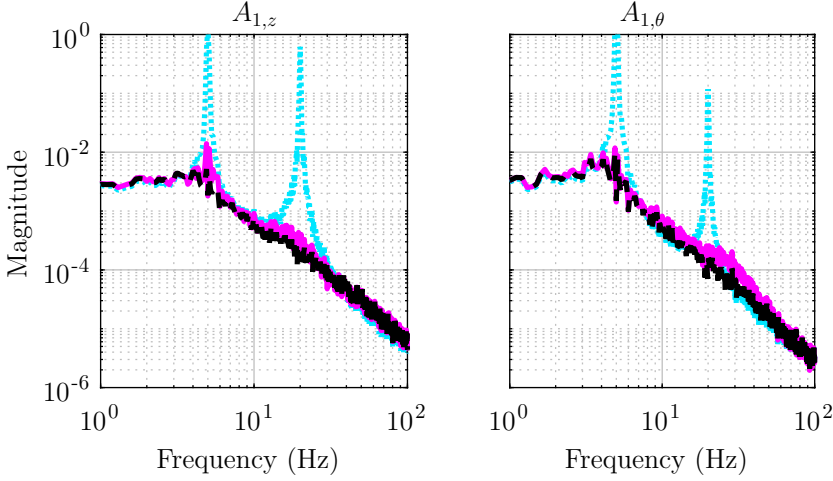


Figure 2.4. Power spectral density plots of the simulated output channels in the objective vector $\mathbf{A}_1(j\omega)$ for the system without control (dotted light blue), the system with control in sensor coordinates (solid magenta), and the system with control in decorrelated disturbance coordinates (dashed black).

Although the noise spectrum of accelerometers generally tends to infinity for $\omega \rightarrow 0$, this filter is designed bi-proper to satisfy the assumptions that are typically made in H_2 -control [84]. For the choice of $\mathbf{V}_1$, two control strategies are adopted.

Control strategy using sensor coordinates

In this strategy, $\mathbf{V}_1$ is a diagonal matrix in which the main entries correspond to the diagonal entries of the disturbance spectra in sensor coordinates ($A_{0,x}$, $A_{0,\theta}$), see Figure 2.3. Since cross-correlation between the disturbances is neglected in this strategy, the off-diagonal terms in $\mathbf{V}_1$ are set to zero, leading to

$$\mathbf{V}_1(s) = 0.045 \cdot \text{diag}\left(\frac{(s^2 + 8.2s + 1039)(s^2 + 94s + 15010)}{(s^2 + 0.31s + 987)(s^2 + 1.3s + 15790)}, \frac{(s^2 + 25s + 1039)(s^2 + 31s + 15010)}{(s^2 + 0.31s + 987)(s^2 + 1.3s + 15790)} \right). \quad (2.26)$$

Control strategy using decorrelated disturbances

In this strategy, $\mathbf{V}_1$ is constructed based on the decorrelated disturbance spectra in Figure 2.3 and transformation matrix $\mathbf{U}$,

$$\mathbf{V}_1(s) = \mathbf{U} \cdot 0.045 \cdot \text{diag} \left(\frac{(s^2 + 1.4s + 44)}{(s^2 + 0.31s + 987)}, \frac{(s^2 + 5.6s + 706)}{(s^2 + 1.3s + 15790)} \right). \quad (2.27)$$

Notice that in (2.27) only a single notch filter per disturbance channel is required, while (2.26) requires two notch filters per sensor channel. Thanks to this proper modeling of cross-correlation between the channels in $\mathbf{a}_0$, (2.27) has less strict requirements on performance improvement, and therefore less deterioration of performance due to the waterbed effect is expected. In contrast to the diagonal matrix designed in (2.26), the design in (2.27) results in a full transfer matrix by means of the matrix $\mathbf{U}$.

2.6.4 Results

Figure 2.4 presents PSDs of the residual vibrations in the sensor directions. The figure shows that both strategies succeed in suppressing the resonance peaks. However, strategy 1 neglects cross-correlation and therefore the residual noise spectra are higher, see for example the plot of $A_{1,\theta}$ at frequencies beyond 20 Hz. This increased spectrum is due to incomplete knowledge about the disturbances that increase the waterbed effect. To explain this, consider the singular values of $\mathbf{S}_{\mathbf{F}\mathbf{F}}$ which are shown in Figure 2.5. This figure shows that for strategy 1 both singular values are largely dropped at each of the two notch filters at 5 and 20 Hz. Contrarily, for strategy 2 merely one of both singular values drops, which corresponds to a principal gain in the disturbance direction at that specific frequency. Since strategy 2 only needs one singular value drop per notch, it can be understood from Corollary 2.5 that the corresponding areas with deteriorated performance also remain smaller, hence a decreased cost expressed by the waterbed effect.

Figure 2.6 shows Bode plots of the sensitivity function matrix for both control strategies and expressed in the sensor coordinates (z, θ) . For strategy 1 that neglects correlation, a decentralized $\mathbf{S}_{\mathbf{F}\mathbf{F}}$ is obtained. This is expected, because (a) $\mathbf{P}_1$ is diagonal and (b) the power spectrum of the modeled $\mathbf{a}_0$ is diagonal, hence no information about cross-correlation between the signals in $\mathbf{a}_0$ is included in (2.26). The figure clearly shows that the main diagonal entries of $\mathbf{S}_{\mathbf{F}\mathbf{F}}$ suffer from a waterbed effect because the magnitude exceeds zero dB for multiple frequency intervals, which is in line with Corollary 2.8. The off-diagonal terms in strategy 1 are zero, i.e. unconstrained, which is in line with Corollary 2.9. For strategy 2, that includes correlation, a centralized sensitivity matrix is obtained (in sensor coordinates) despite the fact that the plant dynamics were fully decoupled in sensor coordinates. The off-diagonal entries are largely raised because correlation is taken into account by the unitary transformation matrix $\mathbf{U}$.

In the field of active vibration isolation, it is convenient to evaluate performance in terms of the transmissibility matrix $\mathcal{T}$, recall (2.2), rather than $\mathbf{S}_{\mathbf{FF}}$. Therefore, Figure 2.7 shows the singular values of $\mathcal{T}$ which give a remarkable result. Although the first impression is that $\mathcal{T}$ has slightly larger singular values using strategy 2, the overall performance is better because the singular values are only decreased in the disturbance directions and at relevant frequencies. In addition, Figure 2.8 shows Bode magnitude plots of the elements in $\mathcal{T}$. Notice the paradox that in the second strategy $\mathcal{T}$ has *increased* off-diagonal entries as a means to *decrease* the overall performance, i.e. residual vibrations in $\mathbf{A}_1$. Hence, this coupling induced by DFC is a necessity for improved performance.

2.7 Conclusions

Performance tradeoffs are shown to be present in causal feedforward control systems like DFC. The sensitivity matrix describes performance improvement and deterioration obtained for different frequencies and directions, and is constrained by fundamental limitations. More specifically, it is shown that it is possible to exchange frequency-wise tradeoffs between the principal gains, but not between the directions. In view of this, the H_2 control design framework is demonstrated to include both an accurate model of the system and the correlated disturbances to optimally deal with the fundamental limitations. Application to a numerical example of a 2-DOF vibration isolation system shows how performance is improved by proper modeling of the disturbances. It also shows the main message of this chapter, namely that a centralized controller is the optimal solution if the disturbances are correlated, even though the plant dynamics are fully decoupled.

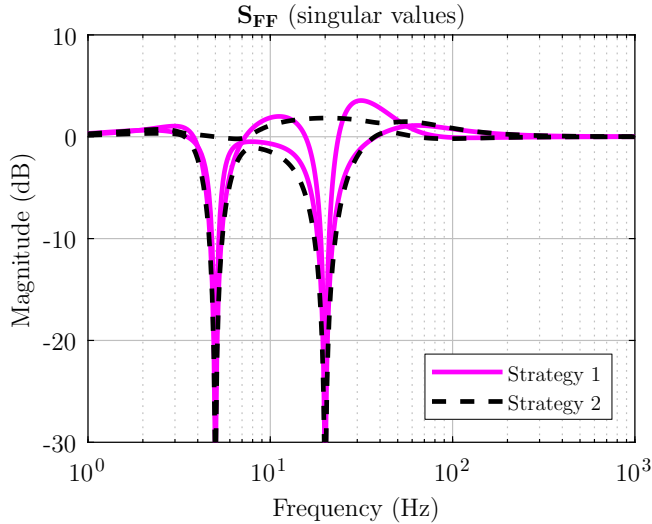


Figure 2.5. Singular value plots of $\mathbf{S}_{\mathbf{FF}}$ for the two control strategies; observe that strategy 2 has lower areas of performance deterioration because it needs less singular value drops around the notches, hence better overall performance.

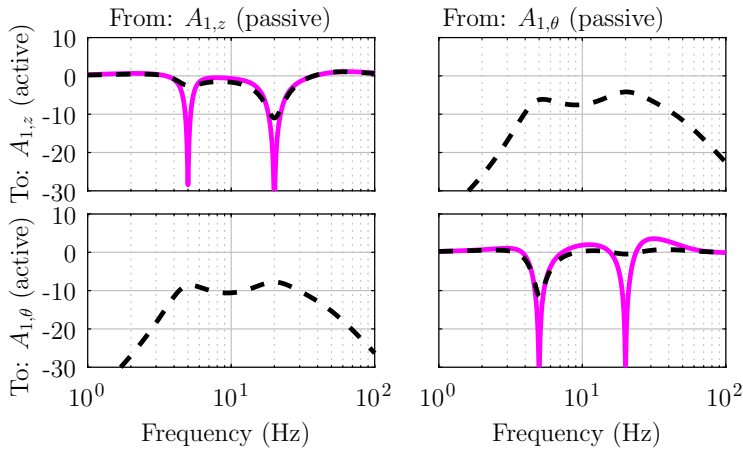


Figure 2.6. Bode magnitude plots of the elements in $\mathbf{S}_{\mathbf{FF}}$; Strategy 1 (magenta, solid) neglects correlation and results in a decentralized controller with two notches per entry, whereas Strategy 2 accounts for correlation and only needs one notch per element, hence less deterioration at higher frequencies; moreover, Strategy 2 increases the off-diagonal entries to define the proper principal directions for disturbance suppression.

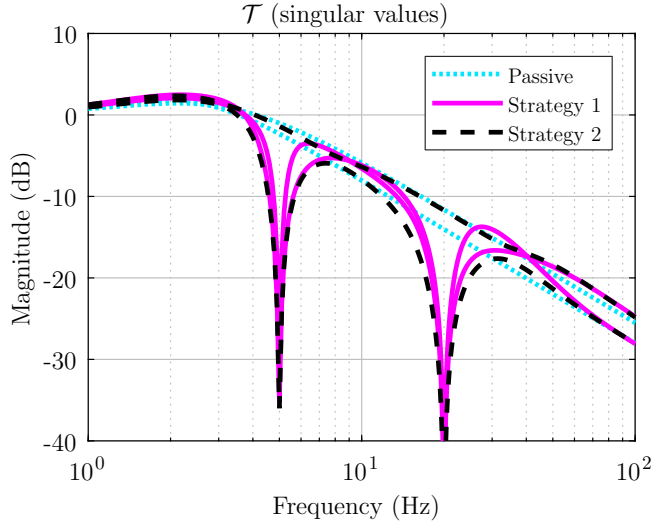


Figure 2.7. Singular value plots of $\mathcal{T}$. Observe that the second strategy has lower areas of performance deterioration because it needs less singular value drops around the notches, hence better overall performance.

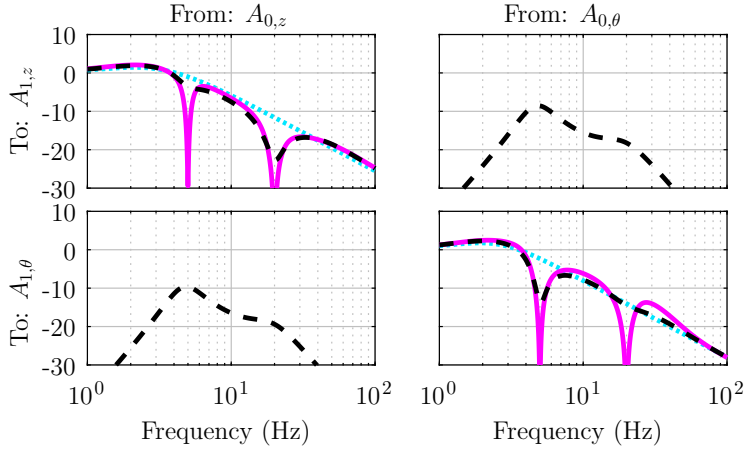


Figure 2.8. Bode magnitude plots of the elements in $\mathcal{T}$; Note that Strategy 1 (magenta, solid) preserves the property of a diagonal $\mathcal{T}$ matrix; whereas Strategy 2 (black, dashed) increases the off-diagonal terms but with the desired result of a lower residual output power, recall Figure 2.4.

Chapter 3

Experimental estimation of transmissibility matrices

Vibration isolation is essential for industrial high-precision systems to suppress external disturbances. The aim of this chapter is to develop a general identification approach to estimate the frequency response function (FRF) of the transmissibility matrix, which is a key performance indicator for vibration isolation systems. Since estimation of a transmissibility matrix requires sufficient excitation in terms of base frame accelerations, the major challenge lies in obtaining a good signal-to-noise ratio in view of a large system weight. A non-parametric system identification method is proposed that combines floor and shaker excitations. Furthermore, a method is presented to analyze the input power spectrum of the floor excitations, both in terms of magnitude and direction. In turn, the input design of the shaker excitation signals is investigated to obtain sufficient excitation power in all directions with minimum experiment cost. The proposed methods are shown to provide an accurate FRF of the transmissibility matrix in three relevant directions on an industrial active vibration isolation system over a large frequency range. This demonstrates that, despite their heavy weight, industrial vibration isolation systems can be accurately identified using this approach.¹

¹This chapter is based on:

M.A. Beijen, R. Voorhoeve, M.F. Heertjes, and T. Oomen, *Experimental estimation of transmissibility matrices for industrial multi-axis vibration isolation systems*, accepted for Mechanical Systems and Signal Processing, 2018.

3.1 Introduction

Vibration isolators are widely used in high-precision systems, such as wafer scanners [45], scanning tunneling microscopes [53, 52], and measurement systems in general [91, 61]. These isolators are used to isolate machinery from floor and base frame vibrations. A key performance indicator is the transmissibility function [61, 74]. For single-axis systems, the transmissibility function describes the transfer function from base frame vibrations to payload vibrations. For multi-axis systems, the transmissibility function is extended to a transmissibility matrix, where its performance metric is often posed in terms of scalar norms [74, 43]. Amongst vibration isolation, the concept of transmissibility functions has a key interest in a wide range of applications, for example operational modal analysis [76, 104], and operational transfer path analysis [38, 96].

The frequency response function (FRF) of the transmissibility matrix can be estimated from experimental data by placing accelerometers or geophones on both the base frame and the isolated payload of the machine [6]. In, e.g., [74, 62], several methods for transmissibility matrix measurements are developed where external shaker constructions are used to ensure that the base frame is sufficiently excited. This enables an accurate estimation of the FRF, both unbiased and with a small variance [69]. However, these methods are often not applicable because industrial machines are often too heavy for commercially available shakers to support and excite the system. Moreover, in industry it is important to provide sufficiently accurate models on the one hand with the least-costly identification experiment on the other hand [4]. Therefore, estimation methods where the machine runs under operating conditions are highly preferred. Such methods rely on environmental base frame excitations, e.g., floor vibrations [82]. These methods can result in a reasonable estimation, as long as the environmental vibrations provide sufficient spectral power in all relevant directions to obtain a sufficient signal-to-noise ratio (SNR). Available methods are restricted to those that can deal with random excitations, because the environmental vibrations are not known beforehand and are typically non-periodic. In this regard, spectral analysis is often used [70, 59], which assumes that either the base frame or the payload excitation is measured free of noise. However, sensor noise on both the base frame and payload measurement leads to an error-in-variables identification problem [85, 114], which may subsequently lead to biased estimates [2, 71].

Several methods are proposed in literature to improve the estimation results for systems that can only be partly identified due to an insufficient SNR. A first example is output-only or operational modal analysis [76, 104] which can provide eigenfrequencies and mode shapes, but does not give an FRF which represents an input-output relation. A second example is found in methods that augment FRF measurements with finite-element modeling, e.g., [26], but these methods need intervention regarding meshing, model order selection, and so on. A third example is frequency-based substructuring [100, 65] in which separate

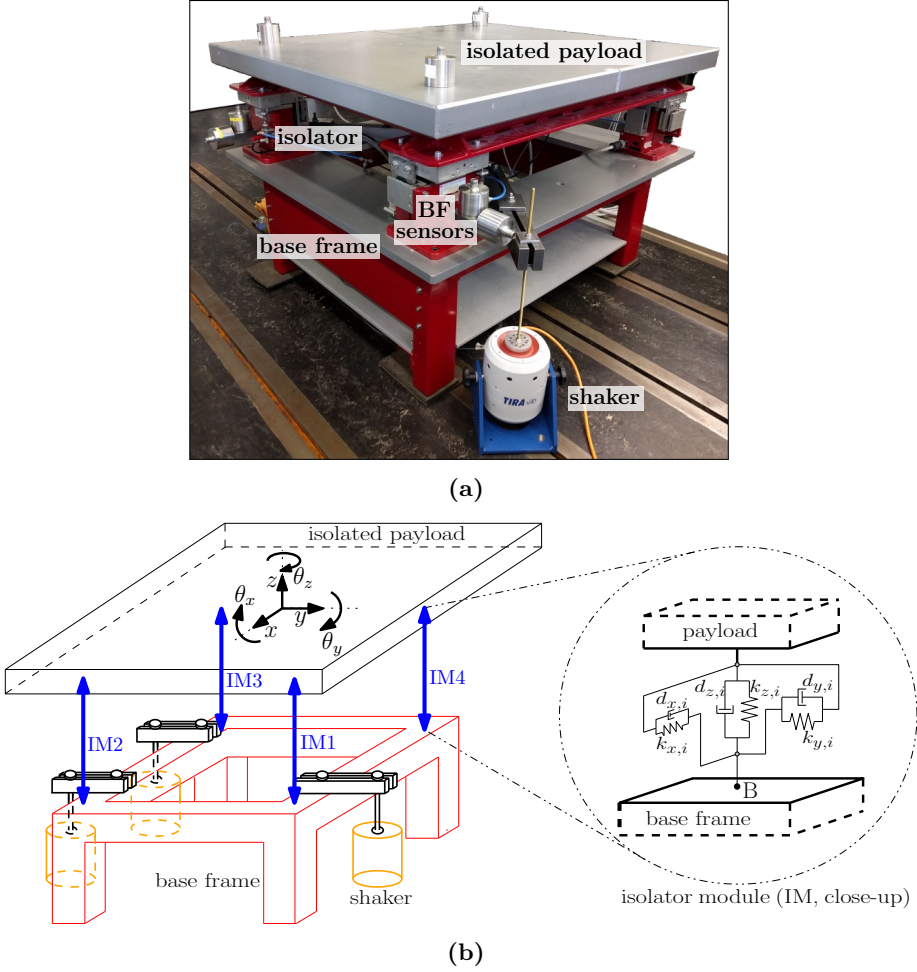


Figure 3.1. AVIS used for validation of the non-parametric identification method: (a) photograph, (b) schematic representation showing the base frame, payload, four isolator module locations, and three shaker locations.

parts of the system are identified independently and merged afterwards, but this method is sensitive for large uncertainty propagation errors [100]. Moreover, none of these methods provides a validation regarding the actual performance of the assembled system in closed-loop. The latter is important, since control loops might deteriorate performance due to changing system dynamics and noise amplification [115, 90]. Therefore, none of these methods is further pursued in this chapter.

Although vibration isolation is essential in high-tech equipment, there is a lack of a systematic approach to estimate the FRF of the transmissibility matrix for heavy-weight systems. The first contribution of this chapter is to show a non-parametric identification method to estimate the FRF for such systems in multiple directions, where a combination of floor and shaker excitations is used to maximize the base frame excitation power. The second contribution is a method to evaluate the spectral power and directions of the base frame excitations, and how its result can be used for input design of the shaker excitation signals. In this respect, the proposed method can be seen as optimal input design [4]. A third contribution is given by the application of the identification and input design methods to a heavy-weight active vibration isolation system. This leads to an accurately estimated FRF of the transmissibility matrix in the frequency range of interest, i.e. between 1–100 Hz, and in three relevant directions. These results show that the presented methods, which are applicable for vibration isolation systems in general, are particularly suited for systems that are too heavy to be sufficiently excited by shakers during operating conditions.

The chapter is organized as follows. The experimental setup and the transmissibility matrix are described in Section 3.2. The main problem considered in this chapter is defined in Section 3.3. The non-parametric identification method is presented in Section 3.4, and the input excitation analysis and design method is presented in Section 3.5. The experimental results and validation are presented in Section 3.6, and the major conclusions are given in Section 3.7.

3.2 Active vibration isolation system

3.2.1 System description

The system used throughout this chapter for validation purposes is the Active Vibration Isolation System (AVIS) shown in Figure 3.1. The AVIS consists of two main parts: (i) an isolated payload of 289 kg, and (ii) a base frame (BF) that is supported by the floor. The payload and BF are connected by four isolator modules (IM). These isolator modules provide a low stiffness and damping through pneumatically controlled air mounts to obtain vibration isolation in both horizontal and vertical directions. At three of the four modules, horizontal and vertical accelerometers are placed to measure the absolute BF accelerations denoted by $\mathbf{u}_1(t) \in \mathbb{R}^6$. Moreover, three out of four modules contain horizontal and vertical accelerometers on the payload to measure the absolute payload accelerations denoted by $\mathbf{y}_1(t) \in \mathbb{R}^6$. Measurements of $\mathbf{u}_1$ and $\mathbf{y}_1$ are filtered with analogue band-pass filters from 0.1 Hz to 450 Hz to reduce aliasing, sensor noise, cross-talk, and drift. For ease of presentation, $\mathbf{u}_1$ and $\mathbf{y}_1$ are transformed to the same coordinate frame, i.e. the set of Cartesian coordinates in the center of gravity (CoG) of the payload, see Figure 3.1b. To this end, both the BF and payload are considered as a rigid body, such that the BF motion is de-

defined by $\mathbf{x}_b(t) = [x_b, y_b, z_b, \theta_{x,b}, \theta_{y,b}, \theta_{z,b}]^T$ and the payload motion is defined by $\mathbf{x}_p(t) = [x_p, y_p, z_p, \theta_{x,p}, \theta_{y,p}, \theta_{z,p}]^T$. In the absence of sensor noise, and only considering small displacements, the Cartesian coordinates are related to the measured signals through

$$\ddot{\mathbf{x}}_b(t) = \mathbf{T}_u \mathbf{u}_1(t), \quad \ddot{\mathbf{x}}_p(t) = \mathbf{T}_y \mathbf{y}_1(t), \quad (3.1)$$

with transformation matrices $\mathbf{T}_u, \mathbf{T}_y \in \mathbb{R}^{6 \times 6}$.

Three shakers providing vertical shaker forces $\mathbf{r}(t) \in \mathbb{R}^3$ are installed between the floor and the BF to increase the excitation level of the BF. The shaker locations are shown in Fig. 3.1b. Data acquisition is performed by a 14-bit real-time digital signal processor (DSP) running at $f_s = 2$ kHz.

3.2.2 The transmissibility matrix

The FRF of the $n_y \times n_u$ transmissibility matrix $\mathcal{T}(j\omega)$, with frequency $\omega \in \mathbb{R}$, defines the relation between a set of BF coordinates $\mathbf{x}_b \in \mathbb{R}^{n_u}$ and a set of payload coordinates $\mathbf{x}_p \in \mathbb{R}^{n_y}$ in the frequency domain. In general, a transmissibility matrix is not uniquely defined by $\mathbf{x}_b$ and $\mathbf{x}_p$ only, but also depends on the locations of the floor and shaker excitation sources $\mathbf{f}$ causing the responses measured at $\mathbf{x}_b$ and $\mathbf{x}_p$. This property is extensively investigated in, e.g., [38] for linear time-invariant (LTI) systems. In order to make $\mathcal{T}$ uniquely defined and independent of $\mathbf{f}$ in case of vibration isolation systems, the number of considered BF coordinates n_u must equal the number of transmission paths between the BF and the payload. Moreover, the BF sensors must be co-located with the attachment points of the transmission paths, for example, the vertical BF acceleration of each air mount must be measured at the point B in the IM close-up in Fig. 3.1b. By doing so, structural dynamics of the BF will not be involved in the transmission path from $\mathbf{x}_b$ to $\mathbf{x}_p$. Under these conditions, $\mathcal{T}$ becomes independent of BF dynamics and the locations of the excitation sources, such that $\mathcal{T}$ can be regarded as an input-output relation from $\mathbf{x}_b$ to $\mathbf{x}_p$, or

$$\mathbf{X}_p(\omega) = \mathcal{T}(j\omega) \mathbf{X}_b(\omega), \quad (3.2)$$

with $\mathbf{X}_p(\omega), \mathbf{X}_b(\omega)$ representing the payload and BF motion expressed in the frequency domain.

An expression for $\mathcal{T}(j\omega)$ could be either obtained from parametric first-principles modeling or from measured frequency response data. A first-principles model for the transmissibility matrix is derived in Appendix B in Section 3.9. In this model, the payload is considered as a rigid body while the air mounts are described using the three-spring-damper model as shown in Figure 3.1b. As such, there are twelve transmission paths, i.e. four air mounts each containing three spring-damper connections between the BF and payload. This can be reduced to six transmission paths by considering the BF as a rigid body, such that

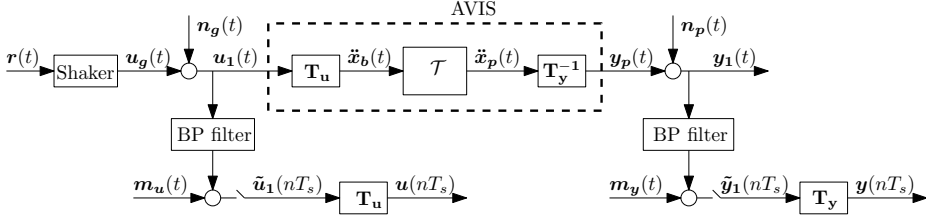


Figure 3.2. Block diagram for non-parametric system identification; base frame motion is measured by $\mathbf{u}_1$, payload motion by $\mathbf{y}_1$; band-pass (BP) filters are applied in the signal conditioners to prevent aliasing and to suppress sensor noise; the transformed signals $\mathbf{u}, \mathbf{y}$ represent base frame and payload motion, respectively, expressed in the same Cartesian coordinate frame, see (3.1).

all twelve attachment points of the transmission paths only depend on the six rigid-body degrees of freedom of the BF. Due to the modeling assumptions this model is only valid at low frequencies where no structural dynamics or air mount dynamics occur. The model is therefore not sufficient to provide an accurately estimated FRF of the transmissibility matrix in the full frequency range of interest between 1–100 Hz. As such, the model will only be used for input design of the shaker signals, and to validate FRF measurements of the transmissibility matrix obtained from experiments at low frequencies.

3.3 Problem definition

The aim of this chapter is to develop a general identification approach to estimate the transmissibility matrix $\mathcal{T}$ for industrial systems running under operating conditions, such as the system in Figure 3.1. More specifically, the key interest is to measure the FRF $\mathcal{T}(j\omega)$ in a broad frequency of interest, which is typically 1–100 Hz for industrial high-precision machines, and in multiple directions. The main challenge in the identification of $\mathcal{T}$ lies in the fact that excitation of $\mathbf{X}_b$ is limited due to physical constraints. Environmental vibrations always excite the BF, but these excitations are uncertain in spectral power and directions. Nevertheless, these vibrations are exploited as useful excitation signal. In addition, these excitations can be augmented with shaker excitations to sufficiently excite the BF in the remaining directions and frequencies that are not sufficiently excited by the environment. These shaker excitations are designed carefully to obtain sufficient excitation power with minimum experiment cost.

3.4 Non-parametric system identification

In this section, the non-parametric identification method to identify the FRF of $\mathcal{T}$ is presented. Spectral analysis is used to simultaneously benefit from environmental floor and shaker excitations. Bias analysis and 95% confidence regions are presented to determine the quality of the measured FRF.

Figure 3.2 shows the block diagram used for non-parametric system identification. The goal is to identify $\mathcal{T}$. Signals $\mathbf{u}_1$ and $\mathbf{y}_1$ represent the BF and payload motions in sensor coordinates as described in Section 3.2. Signal $\mathbf{u}_1$ is the result of two contributions, i.e. $\mathbf{u}_g$ containing the shaker excitations, and $\mathbf{n}_g$ containing the environmental floor vibrations. The shaker block represents the transfer function from reference $\mathbf{r}$ to $\mathbf{u}_g$, with $\mathbf{r}$ being the output of a pseudo-random noise signal generator. The output signal $\mathbf{y}_1$ consists of $\mathbf{y}_p$ representing payload motion caused by BF vibrations, and process noise $\mathbf{n}_p$ representing payload motion due to direct payload disturbances (e.g., acoustic waves). Measurements of $\mathbf{u}_1$ and $\mathbf{y}_1$ are filtered with band-pass filters, which are assumed to behave as an ideal band-pass filter, see Assumption 3.1. Moreover, the measured signals contain additive measurement noise $\mathbf{m}_u$ and $\mathbf{m}_y$, respectively. After sampling, the BF and payload measurements are converted to Cartesian coordinates using the blocks $\mathbf{T}_u$ and $\mathbf{T}_y$, respectively, which results in the signals $\mathbf{u}$ and $\mathbf{y}$. These signals represent sampled-data signals $\mathbf{u}(nT_s), \mathbf{y}(nT_s)$ having sampling time T_s and $n \in \{0, 1, \dots, N-1\}$.

Assumption 3.1 (Ideal band-pass filter). *An ideal band-pass filter with a frequency band ranging from $\omega_{BP,min}$ to $\omega_{BP,max}$ produces a band-limited output signal $u_{BP}(t)$ of input signal $u(t)$, i.e. $U_{BP}(\omega) = U(\omega)$ for $\omega_{BP,min} < |\omega| < \omega_{BP,max}$, and $U_{BP}(\omega) = 0$ for all other frequencies ω .*

3.4.1 Spectral analysis

To identify MIMO systems using random excitations and spectral analysis, a full record of data $\mathbf{y}, \mathbf{u}$ is measured for a sufficiently long time. Subsequently, the full record is split into M possibly overlapping subrecords of equal length denoted $\mathbf{y}^{[l]}, \mathbf{u}^{[l]}$, with $l \in \{1, \dots, M\}$. These subrecords are multiplied with a Hanning window $w(nT_s)$ to reduce the effects of leakage [69, Section 2.6]. Then, the discrete Fourier transforms (DFT) $\mathbf{Y}^{[l]}(\omega)$ and $\mathbf{U}^{[l]}(\omega)$ of the windowed signals are defined as

$$\mathbf{U}^{[l]}(\omega_k) = \frac{1}{S_2} \sum_{n=0}^{N-1} w(nT_s) \mathbf{u}^{[l]}(nT_s) e^{-j\omega_k nT_s}, \quad (3.3)$$

$$\mathbf{Y}^{[l]}(\omega_k) = \frac{1}{S_2} \sum_{n=0}^{N-1} w(nT_s) \mathbf{y}^{[l]}(nT_s) e^{-j\omega_k nT_s}, \quad (3.4)$$

where only the N frequencies $\omega_k = (2\pi k)/(NT_s)$, $k \in \{0, \dots, N-1\}$ are considered. In view of Assumption 3.1, it is assumed that $\omega_{\text{BP},\min} < \omega_k < \omega_{\text{BP},\max}$ for all k . In the remainder of this chapter, the abuse of notation $\omega \equiv \omega_k$ is used with the intention to improve readability. The scaling factor

$$S_2 = \sqrt{\sum_{n=0}^{N-1} |w(nT_s)|^2}, \quad (3.5)$$

is chosen such that the transformation preserves the root mean square (RMS) value of the signal [69, Section 7.2].

Having calculated all $\mathbf{Y}^{[l]}(\omega)$ and $\mathbf{U}^{[l]}(\omega)$, estimates for the cross power spectral density (PSD) matrix $\hat{\mathbf{S}}_{\mathbf{YU}}(\omega)$ and auto PSD matrix $\hat{\mathbf{S}}_{\mathbf{UU}}(\omega)$ are obtained by averaging over the M blocks, i.e.

$$\hat{\mathbf{S}}_{\mathbf{YU}}(\omega) = \left(\frac{2}{f_s}\right) \frac{1}{M} \sum_{l=1}^M \mathbf{Y}^{[l]}(\omega) (\mathbf{U}^{[l]}(\omega))^H, \quad (3.6)$$

$$\hat{\mathbf{S}}_{\mathbf{UU}}(\omega) = \left(\frac{2}{f_s}\right) \frac{1}{M} \sum_{l=1}^M \mathbf{U}^{[l]}(\omega) (\mathbf{U}^{[l]}(\omega))^H. \quad (3.7)$$

In (3.6), (3.7), a scaling factor $2/f_s$ is included to obtain a single-sided PSD matrix with the physical unit $(\text{m/s}^2)^2/\text{Hz}$ [12]. Neglecting the effects of leakage and windowing, and under the assumption of perfect anti-aliasing provided by the band-pass filters, an estimate for $\mathcal{T}$ is obtained using the so-called H_1 estimator, see, e.g. [69, 105],

$$\hat{\mathcal{T}}(j\omega) = \hat{\mathbf{S}}_{\mathbf{YU}}(\omega) \hat{\mathbf{S}}_{\mathbf{UU}}^{-1}(\omega). \quad (3.8)$$

The H_1 estimator is a suitable choice when the input SNR of $\mathbf{u}$ is much better than the output SNR of $\mathbf{y}$. Contrarily, when the output SNR is much better than the input SNR, the H_2 estimator as defined in [59] is preferred. When the input and output SNR are similar, and there is a priori knowledge about the noise spectra, more advanced estimators can be defined that result in reduced bias, see, e.g., [87, 106]. However, in the context of vibration isolation, the SNR of $\mathbf{u}$ is typically much higher than the SNR of $\mathbf{y}$, at least beyond the suspension frequency where isolation occurs. Therefore, the H_1 estimator typically gives the best estimates.

3.4.2 Bias, variance, and the 95% confidence regions

This section presents expressions for bias and variance of the H_1 estimator in (3.8), where the variance can be used to construct the 95% confidence region

which is defined below. To this end, the inputs and outputs are written in the frequency domain as

$$\mathbf{U}(\omega) = \mathbf{U}_0(\omega) + \mathbf{N}_U(\omega), \quad \mathbf{Y}(\omega) = \mathbf{Y}_0(\omega) + \mathbf{N}_Y(\omega), \quad (3.9)$$

with corresponding noise-free input and output

$$\mathbf{U}_0(\omega) = \mathbf{T}_u(\mathbf{U}_G(\omega) + \mathbf{N}_G(\omega)), \quad \mathbf{Y}_0(\omega) = \mathcal{T}(j\omega)\mathbf{U}_0(\omega), \quad (3.10)$$

and input and output noise

$$\mathbf{N}_U(\omega) = \mathbf{T}_u \mathbf{M}_U(\omega), \quad \mathbf{N}_Y(\omega) = \mathbf{T}_y(\mathbf{N}_P(\omega) + \mathbf{M}_Y(\omega)). \quad (3.11)$$

In these equations, $\mathbf{U}_G, \mathbf{N}_G, \mathbf{N}_P, \mathbf{M}_U, \mathbf{M}_Y$ represent frequency-domain representations of the time-domain signals $\mathbf{u}_g, \mathbf{n}_g, \mathbf{n}_p, \mathbf{m}_u, \mathbf{m}_y$ indicated in Figure 3.2.

The following assumptions are introduced regarding the noise, and are consistent with [69, Section 16.16]:

Assumption 3.2 (Noise disturbance). *The DFTs $\mathbf{N}_U(\omega), \mathbf{N}_Y(\omega)$ of the noise signals in (3.11) are circular complex normally distributed. This leads to the following properties:*

$$\begin{aligned} \mathbb{E}[\mathbf{N}_U(\omega)] &= \mathbf{0}, \quad \mathbb{E}[\mathbf{N}_Y(\omega)] = \mathbf{0}, \\ \mathbb{E}[\mathbf{N}_U(\omega)\mathbf{N}_U^T(\omega)] &= \mathbf{0}, \quad \mathbb{E}[\mathbf{N}_Y(\omega)\mathbf{N}_Y^T(\omega)] = \mathbf{0}, \\ \mathbb{E}[\mathbf{N}_Y(\omega)\mathbf{N}_U^T(\omega)] &= \mathbf{0}, \\ \mathbb{E}[\mathbf{N}_U(\omega)\mathbf{N}_U^H(\omega)] &= \mathbf{C}_U(\omega), \quad \mathbb{E}[\mathbf{N}_Y(\omega)\mathbf{N}_Y^H(\omega)] = \mathbf{C}_Y(\omega), \\ \mathbb{E}[\mathbf{N}_Y(\omega)\mathbf{N}_U^H(\omega)] &= \mathbf{C}_{YU}(\omega). \end{aligned} \quad (3.12)$$

Assumption 3.3 (Noise disturbance - continued). *The noise signals $\mathbf{N}_U, \mathbf{N}_Y$ in (3.11) are independent of the signals $\mathbf{U}_0, \mathbf{Y}_0$ in (3.10).*

Under Assumptions 3.2 and 3.3, and under the assumption that there is no input noise, i.e. $\mathbf{C}_U = \mathbf{C}_{YU} = \mathbf{0}$, it is shown in [69] that the H_1 estimator (3.8) is asymptotically unbiased. The latter assumption holds if $\mathbf{U}$ is measured with a sufficiently large SNR, which is discussed further in Section 3.5. Under the same assumptions, $\hat{\mathcal{T}}$ is asymptotically circular complex normally distributed where an estimation for the covariance matrix of $\hat{\mathcal{T}}$ is given by [69, Sec. 7.2.3]

$$\text{Cov}(\text{vec}(\hat{\mathcal{T}}(j\omega))) = \frac{1}{M}(\hat{\mathbf{S}}_{UU}^{-1}(\omega))^T \otimes \hat{\mathbf{C}}_Y(\omega), \quad (3.13)$$

where the output noise covariance is estimated as

$$\hat{\mathbf{C}}_Y(\omega) = \frac{M}{q} \left(\hat{\mathbf{S}}_{YY}(\omega) - \hat{\mathbf{S}}_{YU}(\omega)\hat{\mathbf{S}}_{UU}^{-1}(\omega)\hat{\mathbf{S}}_{YU}^H(\omega) \right), \quad (3.14)$$

with $q = M - n_u$ the number of degrees of freedom in the residual [69]. Eq. (3.13) results in an $(n_y n_u) \times (n_y n_u)$ matrix. The entries on the main diagonal of this matrix represent estimations of the squared standard deviations $\hat{\sigma}^2(\hat{\mathcal{T}}_{i,j})$ for all entries $[i, j]$ of $\hat{\mathcal{T}}$, with $i \in \{1, \dots, n_y\}; j \in \{1, \dots, n_u\}$. Subsequently, the most compact 95% confidence region for each entry $[i, j]$ of $\hat{\mathcal{T}}$ is defined as a circle with radius $\sqrt{3}\hat{\sigma}(\hat{\mathcal{T}}_{i,j})$ [69, Section 7.2.4]. This region is used to indicate the uncertainty due to output noise in the measurements presented in Section 3.6.

Note that the previous conditions regarding bias and the 95% confidence region assume that $\mathbf{C}_U = \mathbf{C}_{YU} = \mathbf{0}$. Moreover, the H_1 estimator assumes that $\hat{\mathbf{S}}_{UU}$ is invertible, i.e. there is sufficient uncorrelated spectral power in all input directions. However, these assumptions are violated when one or more input directions are strongly correlated or have a poor SNR due to insufficient spectral power. As a consequence, the estimated FRF and the 95% confidence regions might become inaccurate. To assess the quality of $\hat{\mathbf{S}}_{UU}$, an input excitation analysis and design method is presented in Section 3.5.

Remark 3.4. In (3.8), the direct estimation method has been applied [69], i.e. the estimation of $\mathcal{T}$ is based on measurement data of $\mathbf{u}$ and $\mathbf{y}$ only. An alternative is to use the indirect method [69], which also uses data of the user-defined shaker forces $\mathbf{r}$ and therefore can provide unbiased estimates for $\mathcal{T}$ even if there is a significant contribution of (input) noise. However, the drawback of indirect methods is that $\mathbf{n}_g$, which includes useful environmental floor excitations, cannot be used for identification. Therefore, indirect methods are not useful in view of the considered problem in this chapter.

3.5 Input excitation analysis and design

In order to accurately estimate $\mathcal{T}$, the input excitations of the base frame must satisfy two requirements. First, $\hat{\mathbf{S}}_{UU}$ must be invertible, and second, the 95% confidence regions must be sufficiently small. These requirements are generally not satisfied in experiments with only environmental vibrations. In that case, BF vibrations can be augmented with shaker excitations. Proper design of the shaker input signals is essential to achieve sufficient BF excitation while minimizing the experiment cost. This design problem is known in literature as optimal input design [4]. This section presents two steps towards such an input design method. In the first step, requirements for the spectral input power are derived to accurately estimate $\mathcal{T}$. In the second step, Cartesian input directions are detected that need shaker excitations to fulfill the requirements with minimum experiment cost.

Remark 3.5. Many examples can be found in literature to optimize the input signal such that a parametric model can be fitted with a sufficiently low variance, see, e.g., [4] and the references therein. In this chapter, however, the main objective is to find an input signal to minimize the bias and variance of an estimated

non-parametric model (FRF). Since the frequency data points are independent, input design will be performed by optimizing the SNR at all frequency points of interest. Afterwards, optimal time-domain signals for shaker excitations can be generated, see, e.g., [4, 78].

3.5.1 Requirements on spectral input power

Requirements on spectral input power should be such that $\hat{\mathbf{S}}_{\mathbf{U}\mathbf{U}}$ is invertible and the 95% confidence regions are sufficiently small, see Section 3.4.2. In the following, it will be shown that these two requirements can be assessed via principal component analysis [87, 15], which evaluates the uncorrelated input excitations. For this analysis method, the original sensor signals $\tilde{\mathbf{U}}_1$ are used rather than the transformed Cartesian coordinates $\mathbf{U}$ for two reasons. First, it ensures that all sensor units are the same, i.e. m/s^2 , to prevent unit scaling problems. Second, it preserves a specific structure of the noise covariance matrix, which will be further explained in Section 3.5.1. This structure would otherwise have been lost by multiplication of the signals with transformation matrix $\mathbf{T}_u$. First, principal component analysis will be presented, and, second, conditions are derived to satisfy the requirements.

Principal component analysis

The principal components are calculated by the eigenvalue decomposition (EVD) of $\hat{\mathbf{S}}_{\tilde{\mathbf{U}}_1\tilde{\mathbf{U}}_1}$, which is the PSD matrix of $\tilde{\mathbf{U}}_1$ and which is defined similar to (3.7). By definition, $\hat{\mathbf{S}}_{\tilde{\mathbf{U}}_1\tilde{\mathbf{U}}_1}$ is a Hermitian matrix ($\mathbf{A} = \mathbf{A}^H$) which has an EVD

$$\hat{\mathbf{S}}_{\tilde{\mathbf{U}}_1\tilde{\mathbf{U}}_1}(\omega) = \tilde{\mathbf{T}}_{\mathbf{Q}}(\omega)\tilde{\mathbf{\Lambda}}(\omega)\tilde{\mathbf{T}}_{\mathbf{Q}}^H(\omega). \quad (3.15)$$

In (3.15), $\tilde{\mathbf{\Lambda}}(\omega) = \text{diag}(\tilde{\lambda}_1(\omega), \dots, \tilde{\lambda}_n(\omega))$ contains the eigenvalues of $\hat{\mathbf{S}}_{\tilde{\mathbf{U}}_1\tilde{\mathbf{U}}_1}(\omega)$ that represent spectral powers of the principal components, and the columns of the $n_u \times n_u$ unitary matrix $\tilde{\mathbf{T}}_{\mathbf{Q}}(\omega)$ contain the normalized eigenvectors that indicate the directions of the principal components. Matrix $\tilde{\mathbf{T}}_{\mathbf{Q}}$ can be used to transform the noisy sensor signals $\tilde{\mathbf{U}}_1$ to the noisy principal coordinates $\tilde{\mathbf{Q}}$ via

$$\begin{aligned} \tilde{\mathbf{Q}}(\omega) &= \tilde{\mathbf{T}}_{\mathbf{Q}}^H(\omega)\tilde{\mathbf{U}}_1(\omega) \\ &= \tilde{\mathbf{T}}_{\mathbf{Q}}^H(\omega)\mathbf{U}_1(\omega) + \tilde{\mathbf{T}}_{\mathbf{Q}}^H(\omega)\mathbf{M}_U(\omega), \end{aligned} \quad (3.16)$$

with noise-free excitation signal $\mathbf{U}_1$ and noise $\mathbf{M}_U$ as defined in Figure 3.2. The noise term $\tilde{\mathbf{T}}_{\mathbf{Q}}^H\mathbf{M}_U$ contributes to $\tilde{\mathbf{Q}}$, which can lead to bias in the estimated power and directions of the principal components. To deal with this problem, the following assumption is introduced regarding the noise.

Assumption 3.6. *The noise signals in $\mathbf{M}_U(\omega)$ are uncorrelated over the channels, have zero mean, and variance $\sigma_{acc}^2(\omega)$ such that $\mathbf{U}_1(\omega)$ has a noise covariance matrix $\mathbf{C}_{\mathbf{U}_1\mathbf{U}_1}(\omega) = \sigma_{acc}^2(\omega)\mathbf{I}_{n_u}$.*

Assumption 3.6, which has also been used in [15], can be motivated from the fact that accelerometers of the same type are used to measure $\mathbf{U}_1$. Therefore, it is reasonable to assume that all accelerometers have uncorrelated measurement noise with similar standard deviation. Under Assumptions 3.3 and 3.6, and for the number of averages $M \rightarrow \infty$, the eigenvectors of the principal components derived from the noisy measurements $\tilde{\mathbf{U}}_1$ asymptotically converge to the eigenvectors of the principal components that would be obtained from a noise-free measurement of $\mathbf{U}_1$. Under the same conditions, each principal component is measured with signal-to-noise ratio

$$(SNR)_i(\omega) = \frac{\tilde{\lambda}_i(\omega)}{\sigma_{acc,u}^2(\omega)}, \quad i \in \{1, \dots, n_u\}. \quad (3.17)$$

(see Appendix A in Section 3.8) with $\tilde{\lambda}_i$ the diagonal elements from $\tilde{\mathbf{\Lambda}}$ in (3.15).

Conditions for input SNR

The result in (3.17) can be used to check if the input excitations in $\mathbf{u}_1$ are sufficiently powerful and uncorrelated such that the input spectrum is invertible. To ensure this, all principal components should have sufficient SNR, or

$$\tilde{\lambda}_i(\hat{\mathbf{S}}_{\tilde{\mathbf{U}}_1} \tilde{\mathbf{U}}_1(\omega)) > (SNR_{\min}) \sigma_{acc,u}^2(\omega), \quad \forall i, \quad (3.18)$$

with $(SNR_{\min})$ the user-defined minimum SNR.

Given that the input spectrum is invertible, identification of $\mathcal{T}(j\omega)$ with a low variance and a small 95% confidence region is desired. To this end, conditions regarding the variance are translated to conditions on the spectral input power in Appendix A in Section 3.8. The resulting condition is given by

$$\tilde{\lambda}_i(\hat{\mathbf{S}}_{\tilde{\mathbf{U}}_1} \tilde{\mathbf{U}}_1(\omega)) > \frac{1}{M} \frac{\sigma_{acc,y}^2(\omega)}{(\text{VAR}_{\max}) \varsigma^2(\omega)}, \quad \forall i. \quad (3.19)$$

with M the number of averaged windows, and $\sigma_{acc,y}^2$ the sensor noise level of the payload accelerometers. $(\text{VAR}_{\max}) \in \mathbb{R}$ is user-defined and gives the allowed variance relative to ς which is the lowest singular value of $\mathcal{T}_l = \mathbf{T}_y^{-1} \mathcal{T} \mathbf{T}_u$ representing the transmissibility matrix in local sensor coordinates, i.e. from $\mathbf{U}_1$ to $\mathbf{Y}_1$.

Conditions (3.18) and (3.19) can be replaced by the single condition

$$\tilde{\lambda}_i(\hat{\mathbf{S}}_{\tilde{\mathbf{U}}_1} \tilde{\mathbf{U}}_1(\omega)) > \max \left((SNR_{\min}) \sigma_{acc,u}^2(\omega), \frac{1}{M} \frac{\sigma_{acc,y}^2(\omega)}{(\text{VAR}_{\max}) \varsigma^2(\omega)} \right), \quad \forall i. \quad (3.20)$$

If (3.20) is not satisfied for all i , the BF excitation power needs to be increased using shakers. In view of minimizing experiment cost, only the power of insufficiently excited directions should be increased. This requires knowledge about the weakest excited directions. This information can be obtained from a method that calculates the so-called projection similarity, which is explained in Section 3.5.2.

3.5.2 Projection similarity

This section proposes the projection similarity, which is a performance measure that can be used to discriminate between the input directions with sufficient and insufficient excitation power. This information can subsequently be used for input design of the shaker signals. To determine the projection similarity, first the set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_{n_u}\}$ representing the Cartesian components expressed in sensor coordinates is defined:

$$\mathbf{T}_u^{-1} = [\mathbf{x}_1(\omega) \ \mathbf{x}_2(\omega) \ \dots \ \mathbf{x}_{n_u}(\omega)]. \quad (3.21)$$

When the vectors with Cartesian components are normalized, one obtains

$$\bar{\mathbf{x}}_i(\omega) = \frac{1}{\|\mathbf{x}_i\|_2} \mathbf{x}_i, \quad i \in \{1, \dots, n_u\}, \quad (3.22)$$

with $\|\dots\|_2$ representing the 2-norm. A second set of eigenvectors $\{\mathbf{q}_1, \dots, \mathbf{q}_{n_u}\}$ corresponding to the principal components expressed in sensor coordinates is defined as

$$\mathbf{T}_Q(\omega) = [\mathbf{q}_1(\omega) \ \mathbf{q}_2(\omega) \ \dots \ \mathbf{q}_{n_u}(\omega)]. \quad (3.23)$$

Next, the normalized Cartesian components $\bar{\mathbf{x}}_i$ are written as a linear combination of the principal components $\mathbf{q}_i$:

$$\begin{aligned} \bar{\mathbf{x}}_i(\omega) &= c_1(\omega)\mathbf{q}_1(\omega) + \dots + c_{n_u}(\omega)\mathbf{q}_{n_u}(\omega), \\ \text{with } \begin{bmatrix} c_1(\omega) \\ \vdots \\ c_{n_u}(\omega) \end{bmatrix} &= \mathbf{T}_Q^{-1}(\omega)\bar{\mathbf{x}}_i(\omega), \quad i \in \{1, \dots, n_u\}. \end{aligned} \quad (3.24)$$

Now suppose that only the first m principal components have sufficient spectral power. Then, (3.24) can be written as

$$\begin{aligned} \bar{\mathbf{x}}_i(\omega) &= \underbrace{c_1(\omega)\mathbf{q}_1(\omega) + \dots + c_m(\omega)\mathbf{q}_m(\omega)}_{\hat{\bar{\mathbf{x}}}_i(\omega)} \\ &\quad + \underbrace{c_{m+1}(\omega)\mathbf{q}_{m+1}(\omega) + \dots + c_{n_u}(\omega)\mathbf{q}_{n_u}(\omega)}_{\bar{\mathbf{x}}_i^\perp(\omega)}, \quad i \in \{1, \dots, n_u\}. \end{aligned} \quad (3.25)$$

In (3.25), $\hat{\bar{\mathbf{x}}}_i$ represents the orthonormal projection of $\bar{\mathbf{x}}_i$ in the subspace spanned by the sufficiently excited principal components $\{\mathbf{q}_1, \dots, \mathbf{q}_m\}$. This is because the vectors $\mathbf{q}_i$ form an orthonormal basis for $\{\mathbf{q}_1, \dots, \mathbf{q}_{n_u}\}$ since $\mathbf{T}_Q$ is a unitary matrix by definition of (3.15). Define $(PE)_i \in [0, 1]$ as the projection error of the i^{th} Cartesian component, i.e.,

$$(PE)_i(\omega) = \|\bar{\mathbf{x}}_i(\omega) - \hat{\bar{\mathbf{x}}}_i(\omega)\|_2, \quad i \in \{1, \dots, n_u\}. \quad (3.26)$$

The maximum projection error equals 1, because the length of $\bar{\mathbf{x}}_i$ is 1. From $(PE)_i$, the projection similarity in the i^{th} Cartesian component $(PS)_i \in [0, 1]$ is defined as

$$(PS)_i(\omega) = 1 - (PE)_i(\omega), \quad i \in \{1, \dots, n_u\}. \quad (3.27)$$

The projection similarity (PS) gives information about the ability to express a Cartesian component as a linear combination of the sufficiently excited principal components. On the one hand, a high PS means a large spectral power in that Cartesian component of the BF excitation. On the other hand, a low PS means that there will be little spectral power in that Cartesian component, and shakers should be used to increase the power. As such, the PS provides a tool for input design of the shaker signals, which is demonstrated during the experimental validation in the next section.

3.6 Experimental results

The results of two identification experiments are presented in this section to show the viability of the proposed approach to deal with heavy-weight systems. In the first experiment, the shakers are switched off such that the BF is only excited by environmental floor vibrations. The results of this experiment are used to gain insight in the spectral input power and directions of the environmental excitations. Based on these results, the shaker excitation signals are designed such that the BF excitation power is sufficiently increased at relevant frequencies and in the directions (z, θ_x, θ_y) . For clarity of presentation, only the directions (z, θ_x, θ_y) are considered, but the results directly apply to the directions (x, y, θ_z) .

The shaker experiment design step is described in Section 3.6.1. The resulting FRF estimates for the transmissibility matrix $\mathcal{T}$ from both experiments are presented and validated with the parametric model in Section 3.6.2. For both experiments, a 10-minute long measurement sequence of $\mathbf{u}$ and $\mathbf{y}$ is conducted, which is split up into 60 subrecords of 10 seconds each. Each sub-record is filtered using a Hanning window. To compensate for data loss due to windowing, a 50% overlap factor is used. Using the Fourier transform, this results in $M = 119$ subrecords with a linear frequency grid having a step size of 0.1 Hz.

3.6.1 Experiment design: shaker signals

As explained in Section 3.5, the input spectral power should be such that $\hat{\mathbf{S}}_{\mathbf{U}\mathbf{U}}$ is invertible and that the 95% confidence region is sufficiently small. These requirements are validated by considering the SNR condition in (3.20). To this end, consider the plot in Figure 3.3 representing the spectral powers of the principal components of $\hat{\mathbf{S}}_{\tilde{\mathbf{U}}_1\tilde{\mathbf{U}}_1}$. For the experiment without shaker excitation it is observed that only one principal power is above the line of minimum power in

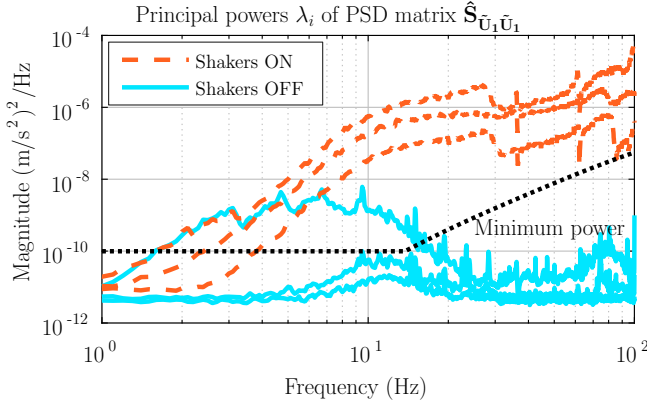


Figure 3.3. Principal powers of the auto-power spectrum matrix $\hat{\mathbf{S}}_{\hat{\mathbf{u}}_1\hat{\mathbf{u}}_1}$ for both experiments; the black dotted line represents the minimum spectral power defined by (3.20) with $\text{SNR}_{\min} = 30$ and $\text{VAR}_{\max} = 0.1$; it follows from (3.20) that the minimum power line at low frequencies is flat and determined by the required input SNR, whereas at high frequencies the line has an increasing magnitude to get a sufficient low variance; this plot shows that the experiment without shaker excitations only has sufficient power in one principal direction, and only between 2 and 15 Hz.

the frequency region from 2 to 15 Hz. The minimum power is defined according to (3.20) with $(\text{SNR}_{\min}) = 30$, such that the SNR is sufficiently above the ADC noise floor of $3.3 \cdot 10^{-12} (\text{m/s}^2)^2/\text{Hz}$, and the relative variance $(\text{VAR}_{\max}) = 0.1$ such that the 95% confidence region is sufficiently small. The local transmissibility matrix is estimated as $\mathcal{T}_l = \mathbf{T}_y^{-1} \mathcal{T} \mathbf{T}_u$ using the parametric model for $\mathcal{T}$ in (3.41). When analyzing the projection similarity (PS) plot in Fig. 3.4, it is observed that for frequencies between 2 and 15 Hz the floor is mainly exciting the BF in the z -direction, while there is almost no excitation in the θ_x - and θ_y -directions. Outside this frequency region, the excitation level is clearly insufficient in all directions.

For the second experiment, shakers are used to increase the spectral input power. Since Fig. 3.4 shows that the floor is not providing sufficient power in any of the three components beyond 15 Hz, three shakers are enabled simultaneously to increase the spectral power in all three directions. The input signals for the shakers consist of three uncorrelated white noise signals, which are filtered such that all principal powers are exceeding the minimum required power at most frequencies. Using the shakers, Figure 3.3 shows that the spectral power of all principal components is greatly increased. For frequencies below 10 Hz it is difficult to largely increase the BF excitation level due to the mechanical limits of the shakers, but the figure shows that for frequencies beyond 4 Hz (almost)

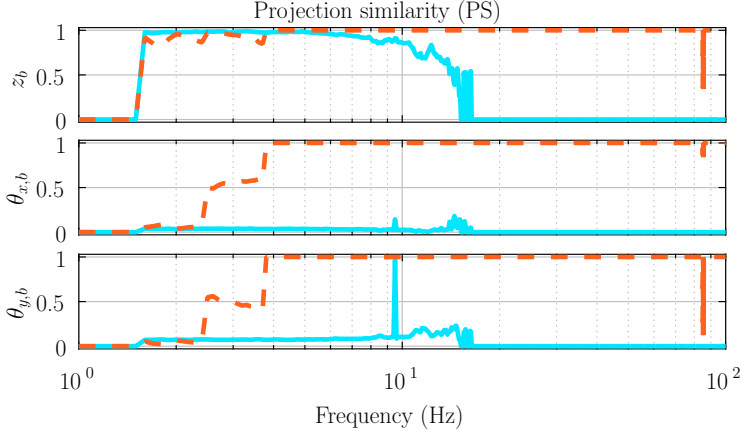


Figure 3.4. Projection similarity (PS) plot, see (3.27); the legend for this figure is identical to the legend in Figure 3.3; the experiment without shaker excitation shows insufficient power in multiple frequency regions and directions; by enabling the shakers, sufficient power is obtained for almost all frequencies.

all directions now have a sufficient SNR. The figure also shows the benefit of combining the spectral power from the floor and the shakers. The spectral power from the floor drops for frequencies beyond 10 Hz, while the shakers are not able to dominate all principal components of the floor for frequencies below 7 Hz. Considering Figure 3.4, it is observed that all three Cartesian coordinates now have a PS of 1, hence sufficient excitation, for frequencies beyond 4 Hz. At low frequencies there is still insufficient excitation, in particular in the θ_x - and θ_y directions, to provide a reliable estimate. For these frequencies and directions, the parametric model can be used to estimate $\mathcal{T}$.

3.6.2 Transmissibility matrix measurements

Bode plots are presented for both experiments to show $\hat{\mathcal{T}}$ as defined in (3.8), and the most compact 95% confidence region as defined in Section 3.4.2. The measurement results are compared with the parametric model presented in Section 3.2.2 to validate the measurements.

Experiment with only floor excitations

Figure 3.5 shows measurement results for $\hat{\mathcal{T}}$ in the first experiment where the shakers are disabled. Below 20 Hz, only the subplots having z_b as input are trusted because the floor is mainly exciting the BF in this direction, recall Figure 3.4. Only in these parts of Figure 4 the confidence regions are sufficiently small, and the measured FRF generally coincides with the parametric model.

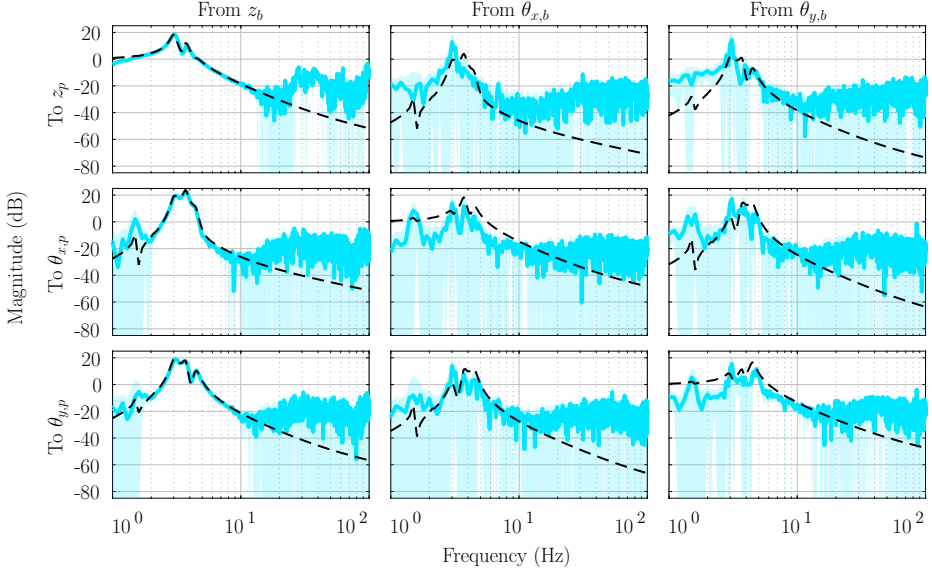


Figure 3.5. Bode magnitude plot showing the transmissibility matrix $\hat{\mathcal{T}}$ for the first experiment with only floor excitations. The solid blue lines represent $\hat{\mathcal{T}}$, while the shaded areas represent the 95% confidence regions as defined in Section 3.4.2. The dashed black lines correspond to the parametric model of the transmissibility matrix, see (3.41) in Appendix B in Section 3.9. In this experiment, $\hat{\mathcal{T}}$ is only accurate in the z_b input directions between 2–10 Hz because only this region has sufficient excitation power, see Figure 3.4.

The latter indicates that the parametric model is sufficiently accurate at the lower frequencies. The measured FRFs also show the roll-off as predicted by the model up to 20 Hz, clearly showing the benefit of vibration isolation. Moreover, all three subplots have a desired small confidence region between 2 and 10 Hz, indicating sufficient output SNR. Beyond 15 Hz, the roll-off in $\hat{\mathcal{T}}$ stops, but here the requirements regarding the input are not satisfied (see Figure 3.4), which leads to a large uncertainty (see Figure 3.5), hence an unreliable FRF estimate.

In general, the FRF measurements for this experiment should be taken with care because $\hat{\mathbf{S}}_{\mathbf{U}\mathbf{U}}$ is ill-conditioned. Although the projection similarity for z_b is close to 1 for frequencies 2–15 Hz, the FRF could be heavily biased due to cross-coupling in the transmissibility matrix. In that case, a small perturbation of a discarded principal component in (3.25) might still give a large difference in the output responses, leading to inaccurate FRF estimates.

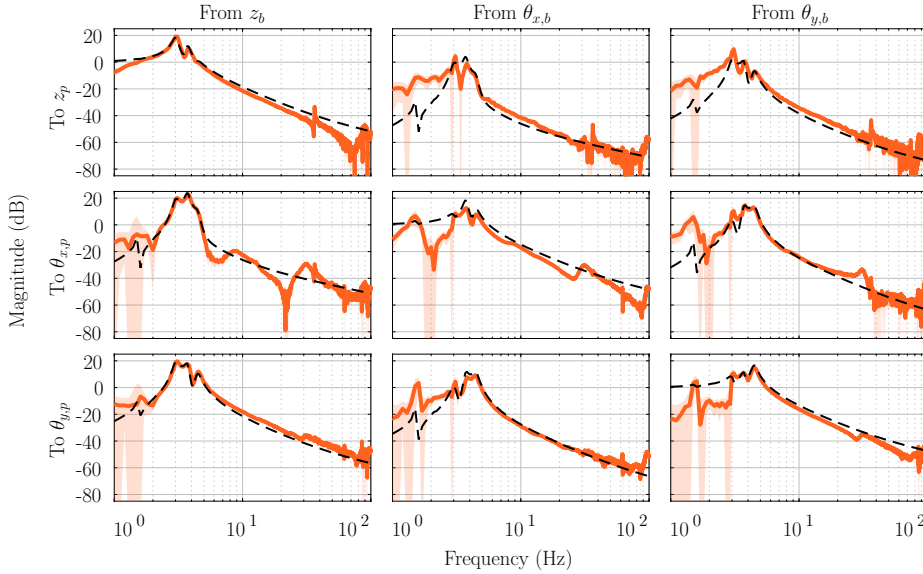


Figure 3.6. Bode magnitude plot showing the transmissibility matrix for the second experiment with both floor and shaker excitations. The solid red lines represent $\hat{T}$, while the shaded areas represent the 95% confidence regions as defined in Section 3.4.2. The dashed black lines correspond to the parametric model of the transmissibility matrix, see (3.41) in Appendix B in Section 3.9. Compared to Figure 3.5, $\hat{T}$ is now accurately estimated for most frequencies and input directions due to increased excitation power, see Figure 3.4.

Experiment with floor and shaker excitations

Figure 3.6 shows measurement results for $\hat{T}$ in the second experiment with enabled shaker excitations. Along the plots, three vertical suspension modes appear around 3–5 Hz. At 1.5 Hz, a horizontal suspension mode appears due to cross-coupling in $\mathcal{T}$ and non-perfect alignment of the setup. Below 4 Hz, significant discrepancies are visible between the model and the measurement. In this region, the model should be correct because below the suspension modes the main diagonal of the transmissibility matrix should have 0 dB magnitude while the off-diagonal terms tend to zero for lower frequencies, see Remark 3.7 in 3.9. The incorrect measurement at low frequencies is mainly due to an insufficient input SNR, see Figure 3.4. For the mid- and higher frequencies, Figure 3.6 shows a good agreement between the measurement and the parametric model. Compared to Figure 3.5 of the first experiment, Figure 3.6 shows better roll-off beyond 15 Hz in all directions, as predicted from the parametric model. The 95% confidence regions are very small, indicating a sufficient output SNR and a reliable FRF estimate. At the higher frequencies, some (anti-)resonance peaks

appear due to flexibilities in the BF. In fact, beyond the first BF mode around 27 Hz. Then, the BF can no longer be considered as a rigid body, such that four accelerometers at the BF are necessary to uniquely measure $\mathcal{T}$. Note that the accelerometers are not perfectly aligned with the air mounts which can induce the visibility of structural dynamics of the BF, recall Section 3.2.2. In addition, the first resonance frequency of the payload occurs at 121 Hz, which is also responsible for the occurrence of anti-resonances at the lower frequencies.

Another potential cause of additional anti-(resonances) and bias showing up is found in the fact that horizontal BF excitations are neglected. However, the shakers also induce parasitic horizontal motion at the isolator locations because the shakers are not perfectly aligned with the isolators. Since the validation was only concerned with estimations in the directions (z, θ_x, θ_y) , vertical payload motion due to horizontal BF excitation and mechanical cross-coupling in the transmissibility matrix shows up as output noise $\mathbf{n}_p$, which is correlated with the vertical BF excitations. As such, Assumption 3.2 for uncorrelated output noise no longer holds, which leads to bias in $\hat{\mathcal{T}}$. However, since both the parasitic horizontal excitation and the horizontal-to-vertical cross-coupling are expected to be sufficiently small with respect to the vertical excitation and the vertical-to-vertical coupling in the transmissibility matrix, it is generally considered fair to neglect this parasitic effect.

3.7 Conclusions

A non-parametric identification approach based on spectral analysis is presented to estimate the transmissibility matrix of an industrial active vibration isolation system. Measurements without shaker excitation can only provide a good FRF estimate for the outputs related to the z_b input direction, because floor vibrations mainly occur in this input direction. This follows from the developed analysis method based on principal component analysis. Measurements with enabled shaker excitation can provide reliable measurements for frequencies from 4 to 100 Hz in three directions. This result is obtained by proper input design of the shaker signals as described in this chapter. For lower frequencies, the shakers are unable to sufficiently excite the BF. The measurements are validated using a parametric model for the transmissibility matrix. This validation generally shows a good fit of the measurement with the parametric model, and completes the transmissibility matrix estimation as a whole.

3.8 Appendix A: Mathematical derivations

3.8.1 Proof of Eq. (3.19)

Using the definition of $\tilde{\mathbf{Q}}$ from (3.16), its power spectrum can be written as

$$\begin{aligned} \hat{\mathbf{S}}_{\tilde{\mathbf{Q}}\tilde{\mathbf{Q}}}(\omega) = & \tilde{\mathbf{T}}_{\mathbf{Q}}^H(\omega) \hat{\mathbf{S}}_{\mathbf{U}_1\mathbf{U}_1}(\omega) \tilde{\mathbf{T}}_{\mathbf{Q}}(\omega) + \tilde{\mathbf{T}}_{\mathbf{Q}}^H(\omega) \hat{\mathbf{S}}_{\mathbf{U}_1\mathbf{M}_U}(\omega) \tilde{\mathbf{T}}_{\mathbf{Q}}(\omega) \\ & + \tilde{\mathbf{T}}_{\mathbf{Q}}^H(\omega) \hat{\mathbf{S}}_{\mathbf{M}_U\mathbf{U}_1}(\omega) \tilde{\mathbf{T}}_{\mathbf{Q}}(\omega) + \tilde{\mathbf{T}}_{\mathbf{Q}}^H(\omega) \hat{\mathbf{C}}_{\mathbf{U}_1\mathbf{U}_1}(\omega) \tilde{\mathbf{T}}_{\mathbf{Q}}(\omega). \end{aligned} \quad (3.28)$$

By definition of the EVD in (3.15), $\tilde{\mathbf{Q}}$ has a diagonal power spectrum $\hat{\mathbf{S}}_{\tilde{\mathbf{Q}}\tilde{\mathbf{Q}}} = \mathbf{\Lambda}$. Under Assumption 3.3 such that $\mathbf{U}_1$ and $\mathbf{M}_U$ are uncorrelated, it follows that $\hat{\mathbf{S}}_{\mathbf{U}_1\mathbf{M}_U}$ and $\hat{\mathbf{S}}_{\mathbf{M}_U\mathbf{U}_1}$ asymptotically converge almost sure to zero for $M \rightarrow \infty$. Under Assumption 3.6, it holds that $\mathbf{C}_{\mathbf{U}_1\mathbf{U}_1} \rightarrow \sigma_{\text{acc}}^2 \mathbf{I}_{n_u}$. Moreover, $\tilde{\mathbf{T}}_{\mathbf{Q}} \tilde{\mathbf{T}}_{\mathbf{Q}}^H = \mathbf{I}_{n_u}$ because $\tilde{\mathbf{T}}_{\mathbf{Q}}$ is a unitary matrix, see (3.15). Then, (3.28) reduces to

$$\tilde{\mathbf{\Lambda}}(\omega) = \mathbf{\Lambda}(\omega) + \sigma_{\text{acc}}^2(\omega) \mathbf{I}_{n_u}, \quad \text{with } \mathbf{\Lambda}(\omega) = \tilde{\mathbf{T}}_{\mathbf{Q}}^H(\omega) \hat{\mathbf{S}}_{\mathbf{U}_1\mathbf{U}_1}(\omega) \tilde{\mathbf{T}}_{\mathbf{Q}}(\omega). \quad (3.29)$$

Since $\tilde{\mathbf{\Lambda}}$ is a diagonal matrix by the grace of (3.15), and $\sigma_{\text{acc}}^2 \mathbf{I}_{n_u}$ is a diagonal matrix, it follows from (3.29) that $\mathbf{\Lambda}$ must be a diagonal matrix. In other words, the noise-free power spectrum $\hat{\mathbf{S}}_{\mathbf{U}_1\mathbf{U}_1}(\omega)$ must have an EVD that equals $\hat{\mathbf{S}}_{\mathbf{U}_1\mathbf{U}_1} = \tilde{\mathbf{T}}_{\mathbf{Q}} \mathbf{\Lambda} \tilde{\mathbf{T}}_{\mathbf{Q}}^H$, hence the eigenvectors of the noisy and noise-free principal components coincide. Moreover, since (3.29) represents a set of decoupled equations, it follows that for $M \rightarrow \infty$ the spectral power of the i^{th} principal component is given by

$$\mathbb{E}[\tilde{Q}_i(\omega) \tilde{Q}_i^H(\omega)] = \tilde{\lambda}_i(\omega) = \lambda_i(\omega) + \sigma_{\text{acc}}^2(\omega), \quad (3.30)$$

with noise-free spectral power λ_i . Then, the SNR is given by the total power $\tilde{\lambda}_i$ divided by the noise power σ_{acc}^2 , which results in (3.17) and which completes the proof.

3.8.2 Derivation of required input power for low variance

This section derives requirements for the spectral input power to obtain a sufficiently low variance of the FRF. Recall the expression for the covariance matrix from (3.13), and note that, when using local sensor coordinates, this variance is given by

$$\text{Cov}(\text{vec}(\hat{\mathcal{T}}_l(j\omega))) = \frac{1}{M} (\hat{\mathbf{S}}_{\tilde{\mathbf{U}}_1\tilde{\mathbf{U}}_1}^{-1}(\omega))^T \otimes \hat{\mathbf{C}}_{\mathbf{Y}_1}(\omega). \quad (3.31)$$

Under the assumption that $n_y = n_u$, the input is measured free of noise, and the output covariance matrix is given by $\mathbf{C}_{\mathbf{Y}_1\mathbf{Y}_1}(\omega) = \sigma_{\text{acc},y}^2(\omega) \mathbf{I}_{n_u}$ (similar to

Assumption 3.6), it follows that the covariance matrix has n_u unique singular values given by

$$\sigma_i \left(\text{Cov}(\text{vec}(\hat{\mathcal{T}}_l(j\omega))) \right) = \sigma_i \left(\frac{\sigma_{acc,y}^2(\omega)}{M} (\hat{\mathbf{S}}_{\tilde{\mathbf{U}}_1 \tilde{\mathbf{U}}_1}^{-1}(\omega))^T \right), \quad i \in \{1, \dots, n_u\}. \quad (3.32)$$

This can be rewritten to

$$\sigma_i \left(\text{Cov}(\text{vec}(\hat{\mathcal{T}}_l(j\omega))) \right) = \frac{\sigma_{acc,y}^2(\omega)}{M} \frac{1}{\sigma_i \left(\hat{\mathbf{S}}_{\tilde{\mathbf{U}}_1 \tilde{\mathbf{U}}_1}(\omega) \right)}, \quad i \in \{1, \dots, n_u\}. \quad (3.33)$$

Next, an upper bound on the variance is defined as $\sigma_i \left(\text{Cov}(\text{vec}(\hat{\mathcal{T}}_l(j\omega))) \right) \leq (\text{VAR}_{\max}) \varsigma^2(\omega)$ for all $i \in \{1, \dots, n_u\}$. Here, $(\text{VAR}_{\max}) \in \mathbb{R}$ is user-defined and gives the allowed variance relative to ς which is the lowest singular value of $\mathcal{T}_l = \mathbf{T}_y^{-1} \mathcal{T} \mathbf{T}_u$ representing the transmissibility matrix in local sensor coordinates, i.e. from $\mathbf{U}_1$ to $\mathbf{Y}_1$. Substitution of this upper bound in (3.33), the following condition regarding the input spectral power is obtained:

$$\sigma_i \left(\hat{\mathbf{S}}_{\tilde{\mathbf{U}}_1 \tilde{\mathbf{U}}_1}(\omega) \right) \geq \frac{\sigma_{acc,y}^2(\omega)}{M} \frac{1}{(\text{VAR}_{\max}) \varsigma^2(\omega)}, \quad \forall i. \quad (3.34)$$

Since $\hat{\mathbf{S}}_{\tilde{\mathbf{U}}_1 \tilde{\mathbf{U}}_1}$ is a hermitian matrix ($\mathbf{A} = \mathbf{A}^H$), the singular values are identical to the eigenvalues $\tilde{\lambda}_i$ that would be obtained from an eigenvalue decomposition. Therefore, (3.34) can also be written as

$$\tilde{\lambda}_i \left(\hat{\mathbf{S}}_{\tilde{\mathbf{U}}_1 \tilde{\mathbf{U}}_1}(\omega) \right) \geq \frac{1}{M} \frac{\sigma_{acc,y}^2(\omega)}{(\text{VAR}_{\max}) \varsigma^2(\omega)}, \quad \forall i. \quad (3.35)$$

with $\tilde{\lambda}_i$ the eigenvalues of $\hat{\mathbf{S}}_{\tilde{\mathbf{U}}_1 \tilde{\mathbf{U}}_1}$ which correspond with the spectral powers of the principal components.

3.9 Appendix B: Parametric model of the AVIS

3.9.1 Derivation of the parametric model

A parametric model of the transmissibility matrix can be derived indirectly from the so-called compliance matrix $\mathcal{C}$. The latter represents the transfer function matrix from payload forces $\mathbf{f}_l$ to payload motion $\mathbf{x}_p$, see Figure 3.7. Since the AVIS considered in Section 3.2 is a so-called *active* vibration isolation system, each isolator contains a horizontal and a vertical Lorentz actuator intended for active vibration isolation and denoted by $\mathbf{f}_l(t) \in \mathbb{R}^8$. This appendix shows that, under some modeling assumptions, these actuators can also be used to indirectly identify $\mathcal{T}$ using three steps. First, the FRF of $\mathcal{C}$ is measured, see, e.g., [99].

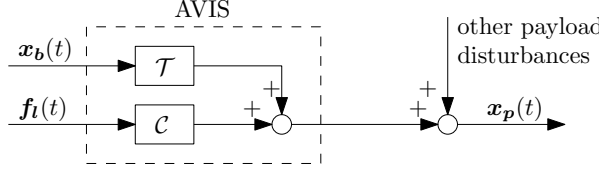


Figure 3.7. Block diagram for the AVIS. The input vectors represent BF displacements $\mathbf{x}_b$ and Lorentz actuator forces $\mathbf{f}_l$. The output vector $\mathbf{x}_p$ represents the payload displacements. The transfer functions $\mathcal{T}$ and $\mathcal{C}$ are called the transmissibility matrix and compliance matrix, respectively. Since the AVIS is placed on a stiff floor, and actuator forces are small, it is assumed that $\mathbf{x}_b$ and $\mathbf{f}_l$ can be considered as independent input signals.

Second, a parametric model $\mathcal{C}(s)$ is fitted on the FRF to provide numerical values for the system's mass matrix $\mathbf{M}$, damping matrix $\mathbf{D}$, and stiffness matrix $\mathbf{K}$. Third, the system matrices are used to derive a parametric model describing the FRF of the transmissibility matrix $\mathcal{T}$.

The first step is to measure the FRF of $\mathcal{C}$. To this end, the actuator forces $\mathbf{f}_l$ are transformed to the Cartesian coordinate frame, such that they represent three forces acting on the payload in the directions x_p, y_p, z_p , and three moments acting on the payload in the directions $\theta_{x,p}, \theta_{y,p}, \theta_{z,p}$. The transformed forces are denoted by $\mathbf{f}_l(t) \in \mathbb{R}^6$ and, considering again small displacements, are given by

$$\mathbf{f}_l(t) = \mathbf{B}\mathbf{f}_l(t), \quad \mathbf{B} \in \mathbb{R}^{6 \times 8}. \quad (3.36)$$

Figure 3.8 shows the measured FRF of $\mathcal{C}$. It is shown in [99] that this FRF can be measured accurately with a low bias and variance by using periodic excitations provided by $\mathbf{f}_l$.

The second step is to fit the measured FRF of $\mathcal{C}$ with a parametric model. In this model, the BF and payload are considered as rigid bodies. Define $\mathbf{x}_p$ and $\mathbf{x}_b$ as the vector with payload and BF displacements locally at each air mount, and $\mathbf{x}_p$ and $\mathbf{x}_b$ as the payload and BF displacements described in Cartesian coordinates from the CoG, see (3.1). Furthermore, each air mount is modeled as a combination of three springs $k_{x,i}, k_{y,i}, k_{z,i}$ and three dampers $(d_{x,i}, d_{y,i}, d_{z,i})$, with $i \in \{1, 2, 3, 4\}$, see Figure 3.1b. Each air mount generates, in all three directions $q \in \{x, y, z\}$, a reaction force $-k_{q,i}(q_{p,i} - q_{b,i}) - d_{q,i}(\dot{q}_{p,i} - \dot{q}_{b,i})$, with $q_{p,i}, q_{b,i}$ representing the local payload and BF displacements at the location of air mount i in direction q . Then, the equations of motion are derived from Newton's second law applied to all six rigid-body coordinates of the payload's CoG:

$$\mathbf{M}\ddot{\mathbf{x}}_p(t) + \mathbf{D}\dot{\mathbf{x}}_p(t) + \mathbf{K}\mathbf{x}_p(t) = \mathbf{D}\dot{\mathbf{x}}_b(t) + \mathbf{K}\mathbf{x}_b(t) + \mathbf{f}_l(t), \quad (3.37)$$

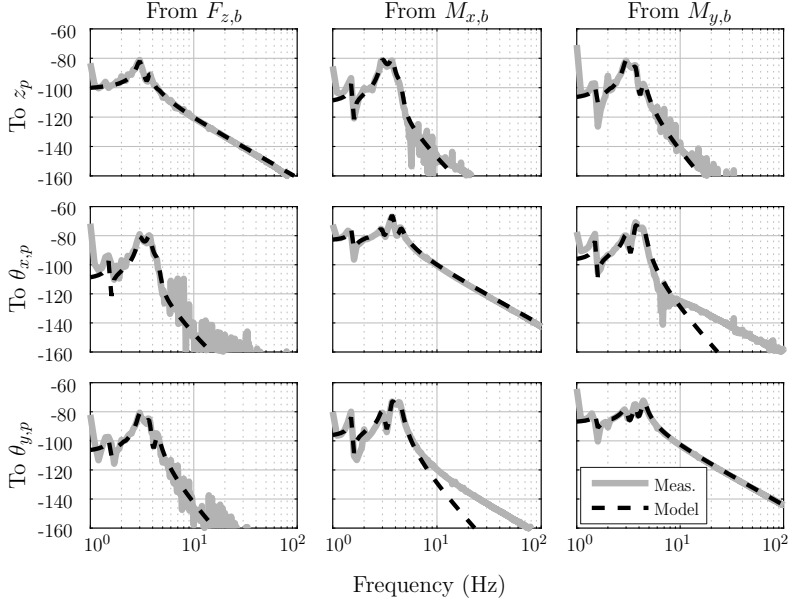


Figure 3.8. Bode magnitude plot of the compliance matrix $\mathcal{C}$. solid gray: measured FRF, dashed black: fitted parametric model.

with global mass $\mathbf{M}$, local damping matrix $\mathbf{D}$, and local stiffness matrix $\mathbf{K}$ as given in 3.9.2. Under the assumption that the air mounts are perfectly aligned, there exists a transformation matrix $\mathbf{T}_b$, see 3.9.2, such that $\underline{\mathbf{x}}_b = \mathbf{T}_b \mathbf{x}_b$ and $\underline{\mathbf{x}}_p = \mathbf{T}_b \mathbf{x}_p$. Then, (3.37) is transformed to

$$\mathbf{M} \ddot{\mathbf{x}}_p(t) + \mathbf{D} \dot{\mathbf{x}}_p(t) + \mathbf{K} \mathbf{x}_p(t) = \mathbf{D} \dot{\mathbf{x}}_b(t) + \mathbf{K} \mathbf{x}_b(t) + \mathbf{f}_l(t), \quad (3.38)$$

with global damping and stiffness matrix

$$\mathbf{D} = \mathbf{D} \mathbf{T}_b, \quad \mathbf{K} = \mathbf{K} \mathbf{T}_b. \quad (3.39)$$

Taking the Laplace transform of (3.38),

$$(\mathbf{M}s^2 + \mathbf{D}s + \mathbf{K}) \mathbf{X}_p(s) = (\mathbf{D}s + \mathbf{K}) \mathbf{X}_b(s) + \mathbf{F}_l(s), \quad (3.40)$$

with Laplace variable s , $\mathcal{L}\{\mathbf{x}_b(t)\} = \mathbf{X}_b(s)$, $\mathcal{L}\{\mathbf{x}_p(t)\} = \mathbf{X}_p(s)$, $\mathcal{L}\{\mathbf{f}_l(t)\} = \mathbf{F}_l(s)$. Recalling the definitions for $\mathcal{T}$ and $\mathcal{C}$ from Figure 3.7, parametric expressions are derived from (3.40) as

$$\mathcal{T}(s) = (\mathbf{M}s^2 + \mathbf{D}s + \mathbf{K})^{-1}(\mathbf{D}s + \mathbf{K}), \quad (3.41)$$

$$\mathcal{C}(s) = (\mathbf{M}s^2 + \mathbf{D}s + \mathbf{K})^{-1}. \quad (3.42)$$

Table 3.1. Model parameters of the six-axis vibration isolator

k_{1x}	8455	N/m	k_{1y}	7824	N/m	k_{1z}	39574	N/m
k_{2x}	6476	N/m	k_{2y}	7824	N/m	k_{2z}	30376	N/m
k_{3x}	6476	N/m	k_{3y}	6757	N/m	k_{3z}	33131	N/m
k_{4x}	8455	N/m	k_{4y}	6757	N/m	k_{4z}	15401	N/m
d_{1x}	33	Ns/m	d_{1y}	48	Ns/m	d_{1z}	173	Ns/m
d_{2x}	12	Ns/m	d_{2y}	48	Ns/m	d_{2z}	79	Ns/m
d_{3x}	12	Ns/m	d_{3y}	14	Ns/m	d_{3z}	70	Ns/m
d_{4x}	33	Ns/m	d_{4y}	14	Ns/m	d_{4z}	110	Ns/m
J_{xx}	30	kg m ²	h_x	0.475	m			
J_{yy}	41	kg m ²	h_y	0.375	m			
J_{zz}	60	kg m ²	h_z	0.185	m			
m	289	kg						

Figure 3.8 shows a comparison between a measured FRF for $\mathcal{C}$ and a fit of the parametric model given in (3.42) for the directions (z, θ_x, θ_y) . Note that, for clarity of presentation, the directions x, y, θ_z are omitted. The figure shows that the parametric model fits well at the lower frequencies where the model assumptions are valid. In the figure, three vertical suspension modes of the AVIS are visible in the frequency range 3–5 Hz. The system shows good decoupling for frequencies beyond the suspension frequencies (>5 Hz). At 1.5 Hz, a horizontal suspension mode is visible due to a small cross-coupling effect in the AVIS between the horizontal force inputs and the vertical payload motion.

The third step is to calculate the modeled FRF $\mathcal{T}(j\omega)$ which can be obtained for every frequency ω by substitution of $s = j\omega$ in (3.41). Note that this lumped-element model does not include higher-order dynamics due to limited BF and payload stiffness. Moreover, the model does not include higher-order dynamics due to non-perfect air mounts. Therefore, the modeled FRF is expected to be valid only at low frequencies, and definitely not sufficient for the main goal of Chapter 3 as described in Section 3.3.

Remark 3.7. By using the same coordinate system for $\mathbf{x}_b$ and $\mathbf{x}_p$, the transmissibility matrix reduces to an identity matrix at very low frequencies, i.e. $\mathcal{T} \rightarrow \mathbf{I}_6$. Indeed, at low frequencies the complete system behaves as a single rigid-body such that $\mathbf{x}_b = \mathbf{x}_p$. This result follows immediately from (3.41) by substitution of $s = j\omega$, and taking the limit for $\omega \rightarrow 0$.

3.9.2 Matrices and vectors for the parametric model

Local coordinate vectors:

$$\begin{aligned}\underline{\mathbf{x}}_m &= \begin{bmatrix} x_{m,1} & y_{m,1} & z_{m,1} & x_{m,2} & y_{m,2} & z_{m,2} & x_{m,3} & y_{m,3} & z_{m,3} & x_{m,4} & y_{m,4} & z_{m,4} \end{bmatrix}, \\ \underline{\mathbf{x}}_b &= \begin{bmatrix} x_{b,1} & y_{b,1} & z_{b,1} & x_{b,2} & y_{b,2} & z_{b,2} & x_{b,3} & y_{b,3} & z_{b,3} & x_{b,4} & y_{b,4} & z_{b,4} \end{bmatrix}.\end{aligned}$$

Global coordinate vectors:

$$\begin{aligned}\mathbf{x}_m &= \begin{bmatrix} x_m & y_m & z_m & \theta_{m,x} & \theta_{m,y} & \theta_{m,z} \end{bmatrix}, \\ \mathbf{x}_b &= \begin{bmatrix} x_b & y_b & z_b & \theta_{b,x} & \theta_{b,y} & \theta_{b,z} \end{bmatrix}.\end{aligned}$$

Global mass matrix:

$$\mathbf{M} = \begin{bmatrix} m & 0 & 0 & 0 & 0 & 0 \\ 0 & m & 0 & 0 & 0 & 0 \\ 0 & 0 & m & 0 & 0 & 0 \\ 0 & 0 & 0 & J_{xx} & 0 & 0 \\ 0 & 0 & 0 & 0 & J_{yy} & 0 \\ 0 & 0 & 0 & 0 & 0 & J_{zz} \end{bmatrix}.$$

Local damping matrix:

$$\underline{\mathbf{D}} = \begin{bmatrix} \underline{\mathbf{D}}_1 & \underline{\mathbf{D}}_2 \end{bmatrix},$$

with

$$\begin{aligned}\underline{\mathbf{D}}_1 &= \begin{bmatrix} d_{1x} & 0 & 0 & d_{2x} & 0 & 0 \\ 0 & d_{1y} & 0 & 0 & d_{2y} & 0 \\ 0 & 0 & d_{1z} & 0 & 0 & d_{2z} \\ 0 & h_z d_{1y} & h_y d_{1z} & 0 & h_z d_{2y} & -h_y d_{2z} \\ -h_z d_{1x} & 0 & -h_x d_{1z} & -h_z d_{2x} & 0 & -h_x d_{2z} \\ -h_y d_{1x} & h_x d_{1y} & 0 & h_y d_{2x} & h_x d_{2y} & 0 \end{bmatrix}, \\ \underline{\mathbf{D}}_2 &= \begin{bmatrix} d_{3x} & 0 & 0 & d_{4x} & 0 & 0 \\ 0 & d_{3y} & 0 & 0 & d_{4y} & 0 \\ 0 & 0 & d_{3z} & 0 & 0 & d_{4z} \\ 0 & h_z d_{3y} & -h_y d_{3z} & 0 & h_z d_{4y} & h_y d_{4z} \\ -h_z d_{3x} & 0 & h_x d_{3z} & -h_z d_{4x} & 0 & h_x d_{4z} \\ h_y d_{3x} & -h_x d_{3y} & 0 & -h_y d_{4x} & -h_x d_{4y} & 0 \end{bmatrix}.\end{aligned}$$

Local stiffness matrix:

$$\underline{\mathbf{K}} = \begin{bmatrix} \underline{\mathbf{K}}_1 & \underline{\mathbf{K}}_2 \end{bmatrix},$$

with

$$\underline{\mathbf{K}}_1 = \begin{bmatrix} k_{1x} & 0 & 0 & k_{2x} & 0 & 0 \\ 0 & k_{1y} & 0 & 0 & k_{2y} & 0 \\ 0 & 0 & k_{1z} & 0 & 0 & k_{2z} \\ 0 & h_z k_{1y} & h_y k_{1z} & 0 & h_z k_{2y} & -h_y k_{2z} \\ -h_z k_{1x} & 0 & -h_x k_{1z} & -h_z k_{2x} & 0 & -h_x k_{2z} \\ -h_y k_{1x} & h_x k_{1y} & 0 & h_y k_{2x} & h_x k_{2y} & 0 \end{bmatrix},$$

$$\underline{\mathbf{K}}_2 = \begin{bmatrix} k_{3x} & 0 & 0 & k_{4x} & 0 & 0 \\ 0 & k_{3y} & 0 & 0 & k_{4y} & 0 \\ 0 & 0 & k_{3z} & 0 & 0 & k_{4z} \\ 0 & h_z k_{3y} & -h_y k_{3z} & 0 & h_z k_{4y} & h_y k_{4z} \\ -h_z k_{3x} & 0 & h_x k_{3z} & -h_z k_{4x} & 0 & h_x k_{4z} \\ h_y k_{3x} & -h_x k_{3y} & 0 & -h_y k_{4x} & -h_x k_{4y} & 0 \end{bmatrix}.$$

Transformation matrix

$$\mathbf{T}_b = \begin{bmatrix} 1 & 0 & 0 & 0 & -h_z & -h_y \\ 0 & 1 & 0 & +h_z & 0 & +h_x \\ 0 & 0 & 1 & +h_y & -h_x & 0 \\ 1 & 0 & 0 & 0 & -h_z & +h_y \\ 0 & 1 & 0 & +h_z & 0 & +h_x \\ 0 & 0 & 1 & -h_y & -h_x & 0 \\ 1 & 0 & 0 & 0 & -h_z & +h_y \\ 0 & 1 & 0 & +h_z & 0 & -h_x \\ 0 & 0 & 1 & -h_y & +h_x & 0 \\ 1 & 0 & 0 & 0 & -h_z & -h_y \\ 0 & 1 & 0 & +h_z & 0 & -h_x \\ 0 & 0 & 1 & +h_y & +h_x & 0 \end{bmatrix}.$$

Chapter 4

Self-tuning control for spring-damper compensation

This chapter proposes a multi-input multi-output (MIMO) disturbance feedforward controller to improve the rejection of floor vibrations in active vibration isolation systems for high-precision machinery. To minimize loss of performance due to model uncertainties, the feedforward controller is implemented as a self-tuning generalized FIR filter. This filter contains a priori knowledge of the poles, such that relatively few parameters have to be estimated which makes the algorithm computationally efficient. The zeros of the filter are estimated using the filtered-error least mean squares (FeLMS) algorithm. Residual noise shaping is used to reduce bias. Conditions on convergence speed, stability, bias, and the effects of sensor noise on the self-tuning algorithm are discussed in detail. The combined control strategy is validated on a 6-DOF Stewart platform, which serves as a multi-axis and hard-mounted vibration isolation system, and shows significant improvement in the rejection of floor vibrations.¹

¹This chapter is based on:

M.A. Beijen, M.F. Heertjes, J. Van Dijk, and W.B.J. Hakvoort, *Self-tuning MIMO Disturbance Feedforward Control for Active Hard-mounted Vibration Isolators*, accepted for Control Engineering Practice, 2018.

4.1 Introduction

Active vibration isolators are widely used to isolate high-precision machines from floor vibrations [45, 53, 52]. A classic example is the addition of skyhook damping via absolute velocity feedback to damp the suspension modes [55]. To further increase vibration isolation performance in a broad frequency range, developments are made towards high-gain feedback control [90, 73, 88] and disturbance feedforward control [13, 29, 1]. In contrast to feedback control, where measured machine vibrations are used as controller input, feedforward control exploits measurements from external disturbances such as floor vibrations. Compared to feedback control, the advantages of feedforward control are two-fold. First, it often leads to better signal-to-noise ratios because the measured disturbance signals are not (mechanically) low-pass filtered by the passive vibration isolator itself [93]. Second, feedforward control does not affect the closed-loop stability properties, whereas high-gain feedback control requires a high bandwidth that can lead to stability problems. However, disturbance feedforward control requires an accurate dynamic model of the transfer function from floor vibrations to machine vibrations, which is often difficult to obtain from modeling and identification experiments.

To suppress the problem of obtaining an accurate model in the application of disturbance feedforward control, self-tuning filters can be used to accurately estimate the controller parameters online. Often used filters in this regard are of the finite-impulse-response (FIR) type, see for example [29, 14, 56, 107]. A disadvantage of FIR filters, however, is that usually many parameters are necessary to accurately describe the behavior of lightly damped resonances and low-frequency poles. This is because all FIR filter poles are located at the origin. Alternatively, one can use infinite-impulse-response (IIR) filters in which poles can be placed at locations other than the origin [83]. As a result, a more computationally efficient algorithm can be obtained in which the system can be accurately described with generally less parameters. Examples are found in [34, 58], but these works use IIR filters with self-tuning poles which can lead to instability because the poles are not bounded to the open unit disk. An alternative is found in using IIR filters with fixed poles [110, 66], also called generalized FIR filters [23]. These filters lead to a linear formulation of the estimation problem for which a unique closed-form solution exists, while the poles are not restricted to the origin.

For self-tuning filters several recursive update algorithms exist. Typical examples are least mean squares (LMS) and recursive least squares (RLS) algorithms. LMS-type algorithms are attractive because they perform relatively simple computations compared to RLS algorithms [80]. This is important for practical implementation. There are two main classes of LMS algorithms that can deal with plant dynamics in the compensation path (or secondary path) from actuator forces to payload accelerations. Namely, filtered-reference LMS (FxLMS) and filtered-error LMS (FeLMS), which involve filtering of the reference

and the error signal with the (modeled) plant dynamics, respectively. Compared to FxLMS, FeLMS algorithms require less computational power in case of multi-input multi-output (MIMO) systems [101], but with the disadvantage of having additional requirements on the estimated plant dynamics. Hybrid methods using FeLMS [24] or Filtered-reference Filtered-error LMS (Fx-Fe-LMS) [63, 86] are proposed in an attempt to combine the advantages of several algorithms. However, none of the above works apply self-tuning of generalized FIR filters.

This chapter has three main contributions. The first contribution is a model-based MIMO feedforward controller for multi-axis vibration isolation systems. It is shown that, despite the presence of flexible modes in the payload, the MIMO feedforward controller only depends on the damping and stiffness properties of the suspension system. Hence, an accurate estimation of these properties is necessary for the feedforward control design in achieving improved performance. The second contribution is a MIMO self-tuning algorithm that combines the advantages of a low-order generalized FIR filter with the fast convergence and MIMO computational efficiency of FeLMS. This contribution also includes a detailed discussion regarding stability, convergence, and bias. Self-tuning is used to minimize uncertainty in the estimated damping and stiffness properties, hence providing the accuracy needed to maximize vibration isolation performance. The third contribution is given by the application of the self-tuning control approach to a hard-mounted vibration isolation system.

The remainder of this chapter is organized as follows. The vibration isolation system, its modeling, and a feedback controller to damp the suspension modes are given in Section 4.2. The main problem considered in this chapter is formulated in Section 4.3. The model-based feedforward controller is discussed in Section 4.4, while the self-tuning feedforward controller is described in Section 4.5. An experimental validation is presented in Section 4.6. The conclusions and main findings of this chapter are given in Section 4.7.

4.2 Hard-mounted vibration isolation system

The control strategy in this chapter is intended for high-precision machinery with an active vibration isolation system. Figure 4.1a shows an experimental setup having suspension frequencies that are representative for a hard-mounted isolator [89]. Compared to soft-mounted isolators, these isolators have a relatively high stiffness and suspension frequency. This setup will be used throughout this chapter. The setup consists of a Stewart platform which basically comprises a mass of 5.4 kg connected to six voice coil motors (VCMs, type Geeplus VM4032-250) via wire springs. The VCMs provide the actuator forces $\mathbf{u}(t) \in \mathbb{R}^6$ and are guided by circular leaf springs, the latter associate with six suspension modes of the system with resonance frequencies varying between 15 and 45 Hz. On top of the Stewart platform a payload with a mass of 3.9 kg is mounted. This is done with leaf springs that give rise to additional eigenmodes in the setup

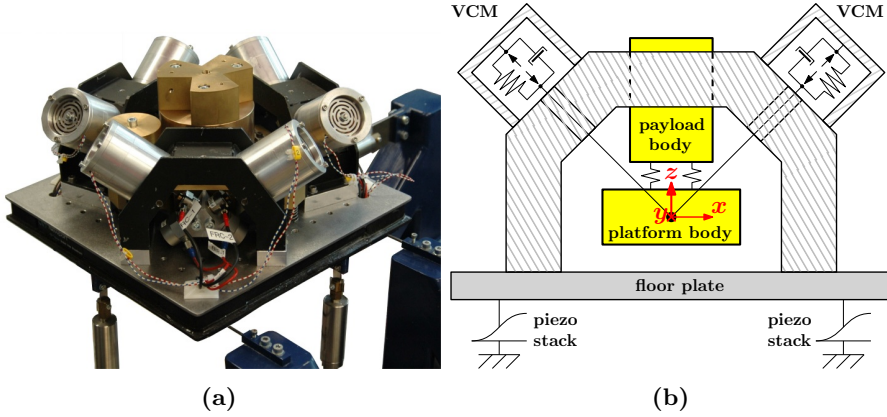


Figure 4.1. (a) Photo of the test setup [89]; and (b) schematic picture including the coordinate frame of the elastic center (e.c.); the VCMs provide actuator forces $\mathbf{u}$ and are suspended in circular leaf springs providing passive stiffness and damping; both the floor plate and the platform body are equipped with six accelerometers measuring $\ddot{\mathbf{q}}_0$ and $\ddot{\mathbf{q}}_1$, respectively; the payload body is connected to the platform using leaf springs.

at frequencies beyond 90 Hz. The Stewart platform as a whole is mounted on a rectangular floor plate. The floor plate can be excited by three piezo stacks (type PiezoMechanik PSt 150/5/40 PE) providing vibrations in z -, R_x - and R_y -direction. The floor plate is equipped with six accelerometers (type Endevco 7703A-1000), measuring the floor plate accelerations denoted by sensor coordinates $\ddot{\mathbf{q}}_0(t) \in \mathbb{R}^6$. Moreover, six accelerometers (type Endevco 7703A-1000) are attached to the Stewart platform, measuring the platform accelerations denoted by sensor coordinates $\ddot{\mathbf{q}}_1(t) \in \mathbb{R}^6$. The sensors are connected to signal conditioners containing a second-order high-pass filter at 0.1 Hz and a second-order low-pass filter at 3000 Hz to filter sensor noise. The controllers are implemented on a dSPACE digital signal processor running at a sampling frequency of $f_s = 6400$ Hz.

4.2.1 Modeling

The vibration isolation system in Figure 4.1 is modeled in SPACAR, which is a non-linear finite element software package for flexible multi-body dynamics [54]. For both the setup and the SPACAR model the reader is referred to [89]. The linearized equations of motion for the rigid-body model, in which the payload is assumed to be rigidly connected to the platform, are given by

$$\mathbf{M}_r \ddot{\mathbf{x}}_1(t) + \mathbf{D}_r \dot{\mathbf{x}}_1(t) + \mathbf{K}_r \mathbf{x}_1(t) = \mathbf{D}_0 \dot{\mathbf{x}}_0(t) + \mathbf{K}_0 \mathbf{x}_0(t) + \mathbf{B} \mathbf{u}(t). \quad (4.1)$$

In this rigid-body model, $\mathbf{M}_r, \mathbf{D}_r, \mathbf{K}_r \in \mathbb{R}^{6 \times 6}$ are the mass-, damping-, and stiffness matrices, and $\mathbf{D}_0, \mathbf{K}_0 \in \mathbb{R}^{6 \times 6}$ are the damping- and stiffness matrices.

ces related to the input vector $\mathbf{x}_0 = [x_0, y_0, z_0, R_{x0}, R_{y0}, R_{z0}]^T$, which describe the floor plate displacements. In this model the hard mounts are regarded as massless elements which only induce linear damping and stiffness forces. Matrix $\mathbf{B} \in \mathbb{R}^{6 \times 6}$ describes the relation between actuator forces $\mathbf{u}$ and platform displacements $\mathbf{x}_1 = [x_1, y_1, z_1, R_{x1}, R_{y1}, R_{z1}]^T$. Note that both $\mathbf{x}_0$ and $\mathbf{x}_1$ are expressed in the coordinate frame of the elastic center (e.c.) shown in Figure 4.1b, with R_x, R_y, R_z denoting rotations about the x, y, z -axis of the e.c., respectively. The coordinates $\mathbf{x}_0, \mathbf{x}_1$ are related to the sensor coordinates $\mathbf{q}_0, \mathbf{q}_1$ of the experimental setup by transformation matrices $\mathbf{R}_0, \mathbf{R}_1 \in \mathbb{R}^{6 \times 6}$, such that

$$\mathbf{x}_0(t) = \mathbf{R}_0 \mathbf{q}_0(t), \quad \mathbf{x}_1(t) = \mathbf{R}_1 \mathbf{q}_1(t). \quad (4.2)$$

To include the non-rigid-body part of the model, the platform mass and payload mass are split in two separate rigid bodies that are interconnected by leaf springs. By doing so, the equations of motion in (4.1) are augmented with six additional equations and dynamic DOFs $\Delta \mathbf{q}_f(t) \in \mathbb{R}^6$. These DOFs are related to the deformations of the leaf springs between the Stewart platform and the payload. As a result, the matrices $\mathbf{M}_r, \mathbf{D}_r, \mathbf{K}_r$ at the left-hand side in (4.1) are augmented to become $\mathbf{M}_f, \mathbf{D}_f, \mathbf{K}_f \in \mathbb{R}^{12 \times 12}$. Note that the right-hand side of (4.1) is augmented with zeros, since none of the inputs act directly on the payload. The augmented equations of motion are given by

$$\begin{aligned} \mathbf{M}_f \begin{bmatrix} \ddot{\mathbf{x}}_1(t) \\ \Delta \ddot{\mathbf{q}}_f(t) \end{bmatrix} + \mathbf{D}_f \begin{bmatrix} \dot{\mathbf{x}}_1(t) \\ \Delta \dot{\mathbf{q}}_f(t) \end{bmatrix} + \mathbf{K}_f \begin{bmatrix} \mathbf{x}_1(t) \\ \Delta \mathbf{q}_f(t) \end{bmatrix} = \\ \begin{bmatrix} \mathbf{D}_0 \\ \mathbf{0} \end{bmatrix} \dot{\mathbf{x}}_0(t) + \begin{bmatrix} \mathbf{K}_0 \\ \mathbf{0} \end{bmatrix} \mathbf{x}_0(t) + \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \mathbf{u}(t). \end{aligned} \quad (4.3)$$

Using the Laplace transform, it can be derived that

$$[\mathbf{M}_f s^2 + \mathbf{D}_f s + \mathbf{K}_f] \begin{bmatrix} \mathbf{X}_1(s) \\ \Delta \mathbf{Q}_f(s) \end{bmatrix} = \begin{bmatrix} \mathbf{D}_0 s + \mathbf{K}_0 \\ \mathbf{0} \end{bmatrix} \mathbf{X}_0(s) + \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \mathbf{U}(s), \quad (4.4)$$

with Laplace variable s , $\mathcal{L}\{\mathbf{x}_0(t)\} = \mathbf{X}_0(s)$, $\mathcal{L}\{\mathbf{x}_1(t)\} = \mathbf{X}_1(s)$, $\mathcal{L}\{\Delta \mathbf{q}_f(t)\} = \Delta \mathbf{Q}_f(s)$, and $\mathcal{L}\{\mathbf{u}(t)\} = \mathbf{U}(s)$. Numerical values for the model in (4.4) are given in Appendix B in Section 4.9.

4.2.2 Feedback control

Although the main focus of this chapter is on feedforward control, a feedback controller is used to add skyhook damping [55] to the suspension and internal modes. Without feedback, these modes would heavily dominate the payload response and drive the system to its mechanical limits. Since the setup in Figure 4.1 is designed such that the 6×6 transfer function matrix from actuator forces $\mathbf{u}$ to sensor coordinates $\ddot{\mathbf{q}}_1$ is diagonally dominant, a diagonal feedback

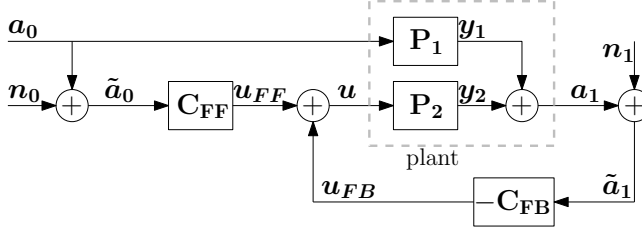


Figure 4.2. Combined feedforward and feedback control block diagram for active vibration isolation.

controller is considered to be the appropriate choice. The controller is post-multiplied with $\mathbf{R}_1^{-1}$ because the transformed coordinates $\tilde{\mathbf{a}}_1$ are used (instead of the sensor coordinates $\tilde{\mathbf{q}}_1$) as feedback controller inputs. The feedback controller $\mathbf{C}_{FB}$ is thus given by

$$\mathbf{C}_{FB}(s) = \left(\frac{\omega_i}{s + \omega_i} k_v \mathbf{I}_6 \right) \mathbf{R}_1^{-1}. \quad (4.5)$$

This controller integrates the inputs (accelerations) using weak integrators with a cut-off frequency of $\omega_i = 1$ rad/s and thus provides output (feedback forces) proportional to the platform velocities needed for adding skyhook damping. Tuning of the feedback gain k_v should be such that the suspension modes of the closed-loop system obtain sufficient relative damping, typically 0.7. To obtain sufficient damping via feedback control that in turn guarantee sufficiently good robustness margins in practice, tuning of k_v is based on the measured frequency response data of the system which is common practice for the considered industry. For the modeled system, $k_v = 2000$ Ns/m is used. Note that, for the aim of this chapter, the actual feedback tuning is not that relevant, as long as the feedback loop provides sufficient robust stability margins, which is guaranteed by the earlier mentioned data-based feedback design, and which keeps the system within its mechanical limits. For further information regarding feedback tuning, the reader is referred to [90].

4.3 Problem formulation

In this chapter, active vibration isolation of the Stewart platform will be obtained through the combination of feedforward and feedback control. The main focus in this chapter is on the feedforward controller design such that the vibration isolation performance is greatly improved in a broad frequency range. As discussed in Section 4.2.2, feedback control is only used to damp the suspension modes and its design is not further considered in this chapter.

A representative block diagram is shown in Figure 4.2. The input disturbance signals representing floor vibrations are denoted by $\mathbf{a}_0(t) \in \mathbb{R}^6$ with time $t \in \mathbb{R}_{\geq 0}$. The signals representing platform vibrations are denoted by $\mathbf{a}_1(t) \in \mathbb{R}^6$. The latter are decomposed in two parts because there are two transmission paths: signals $\mathbf{y}_1(t) \in \mathbb{R}^6$ are caused by disturbances $\mathbf{a}_0$ via the so-called primary path $\mathbf{P}_1$, and signals $\mathbf{y}_2(t) \in \mathbb{R}^6$ are caused by the control actions $\mathbf{u}(t) \in \mathbb{R}^6$ via the secondary path $\mathbf{P}_2$. The control actions $\mathbf{u}$ are obtained by summation of the feedback controller output $\mathbf{u}_{\mathbf{FB}}(t) \in \mathbb{R}^6$ and the feedforward controller output $\mathbf{u}_{\mathbf{FF}}(t) \in \mathbb{R}^6$ induced by the controllers $\mathbf{C}_{\mathbf{FB}}$ and $\mathbf{C}_{\mathbf{FF}}$, which react on the measured platform vibrations and the floor vibrations respectively. The signals $\tilde{\mathbf{a}}_0(t), \tilde{\mathbf{a}}_1(t) \in \mathbb{R}^6$ represent measurements of $\mathbf{a}_0, \mathbf{a}_1$ with additive sensor noise signals $\mathbf{n}_0(t), \mathbf{n}_1(t) \in \mathbb{R}^6$, respectively.

Transfer functions for $\mathbf{P}_1$ and $\mathbf{P}_2$ are obtained from the model in Section 4.2. Since the setup is equipped with accelerometers, one can define $s^2 \mathbf{X}_0(s) = \mathbf{A}_0(s)$, $s^2 \mathbf{X}_1(s) = \mathbf{A}_1(s)$. Then, using (4.4) and Figure 4.2 the input-output equations are obtained as

$$\mathbf{A}_1(s) = \mathbf{P}_1(s)\mathbf{A}_0(s) + \mathbf{P}_2(s)\mathbf{U}(s), \quad (4.6)$$

with

$$\mathbf{P}_1(s) = \mathbf{G}(s) [\mathbf{D}_0 s + \mathbf{K}_0], \quad \mathbf{P}_2(s) = s^2 \mathbf{G}(s) \mathbf{B}, \quad (4.7)$$

where $\mathbf{G}(s)$ represents the common terms in $\mathbf{P}_1$ and $\mathbf{P}_2$ which are given by

$$\mathbf{G}(s) = [\mathbf{I}_6 \quad \mathbf{0}] [\mathbf{M}_f s^2 + \mathbf{D}_f s + \mathbf{K}_f]^{-1} \begin{bmatrix} \mathbf{I}_6 \\ \mathbf{0} \end{bmatrix}. \quad (4.8)$$

Since $\mathbf{U}(s) = \mathbf{C}_{\mathbf{FF}}(s)\tilde{\mathbf{A}}_0(s) - \mathbf{C}_{\mathbf{FB}}(s)\tilde{\mathbf{A}}_1(s)$ it follows that (4.6) can be rewritten:

$$\mathbf{A}_1(s) = \mathcal{T}(s)\mathbf{A}_0(s) + \mathcal{S}_0(s)\mathbf{N}_0(s) + \mathcal{S}_1(s)\mathbf{N}_1(s), \quad (4.9)$$

with $\mathcal{L}\{\mathbf{a}_1(t)\} = \mathbf{A}_1(s)$, $\mathcal{L}\{\mathbf{a}_0(t)\} = \mathbf{A}_0(s)$, $\mathcal{L}\{\mathbf{n}_0(t)\} = \mathbf{N}_0(s)$, $\mathcal{L}\{\mathbf{a}_1(t)\} = \mathbf{A}_1(s)$, and

$$\begin{aligned} \mathcal{T}(s) &= \mathbf{S}_{\mathbf{FB}}(s)(\mathbf{P}_1(s) + \mathbf{P}_2(s)\mathbf{C}_{\mathbf{FF}}(s)), \\ \mathcal{S}_0(s) &= \mathbf{S}_{\mathbf{FB}}(s)\mathbf{P}_2(s)\mathbf{C}_{\mathbf{FF}}(s), \\ \mathcal{S}_1(s) &= -\mathbf{S}_{\mathbf{FB}}(s)\mathbf{P}_2(s)\mathbf{C}_{\mathbf{FB}}(s), \end{aligned} \quad (4.10)$$

where the feedback sensitivity function $\mathbf{S}_{\mathbf{FB}}$ is defined as

$$\mathbf{S}_{\mathbf{FB}}(s) = (\mathbf{I}_6 + \mathbf{P}_2(s)\mathbf{C}_{\mathbf{FB}}(s))^{-1}. \quad (4.11)$$

In (4.10), $\mathcal{T}(s)$ represents the transmissibility function matrix that describes the input-output relation between $\mathbf{a}_0$ and $\mathbf{a}_1$, while $\mathcal{S}_0(s), \mathcal{S}_1(s)$ represent the noise sensitivity functions with respect to $\mathbf{n}_0$ and $\mathbf{n}_1$, respectively.

In general, the control objective in active vibration isolation is to minimize the platform vibrations $\mathbf{a}_1$, which implies a trade-off between minimizing $\mathcal{T}\mathbf{A}_0$, $\mathcal{S}_0\mathbf{N}_0$, and $\mathcal{S}_1\mathbf{N}_1$ in (4.9). In the absence of sensor noise, i.e. $\mathbf{N}_0 = \mathbf{N}_1 = \mathbf{0}$, minimization of $\mathbf{A}_1$ in (4.9) reduces to minimization of $\mathcal{T}$. In that case, it is derived from (4.10) that the ideal (model-based) feedforward controller is given by

$$\mathbf{C}_{\mathbf{FF}}(s) = -\mathbf{P}_2^{-1}(s)\mathbf{P}_1(s). \quad (4.12)$$

Hence, substitution of (4.12) in (4.10) results in $\mathcal{T} = \mathbf{0}$ such that $\mathbf{A}_1 = \mathbf{0}$ in (4.9) in the case that $\mathbf{N}_0 = \mathbf{N}_1 = \mathbf{0}$. The controller in (4.12) is known as the Wiener controller [34], and is used in Section 4.4 to derive a model-based controller. To minimize performance loss due to model uncertainty, and to account for sensor noise, the model-based controller will be turned into a self-tuning controller in Section 4.5. Note that without control, i.e. $\mathbf{C}_{\mathbf{FB}} = \mathbf{C}_{\mathbf{FF}} = \mathbf{0}$, $\mathcal{T}$ in (4.10) equals $\mathbf{P}_1$.

4.4 Model-based feedforward control

In this section, a model-based feedforward controller is derived for the model in Section 4.2.1 such that the transmissibility is lowered for frequencies beyond 2 Hz. First, the model-based controller design is discussed. Second, practical aspects are discussed that limit performance and impede a straightforward implementation of the model-based feedforward controller.

4.4.1 Model-based controller design

Recall from (4.12) that perfect cancellation of floor vibrations is obtained when the feedforward controller is designed as $\mathbf{C}_{\mathbf{FF}}(s) = \mathbf{P}_2^{-1}(s)\mathbf{P}_1(s)$. When the expressions for $\mathbf{P}_1$ and $\mathbf{P}_2$ from (4.7) are substituted in (4.12) it is obtained that

$$\mathbf{C}_{\mathbf{FF},\text{opt}}(s) = -\frac{1}{s^2}(\bar{\mathbf{D}}s + \bar{\mathbf{K}}), \quad (4.13)$$

with

$$\bar{\mathbf{D}} = \mathbf{B}^{-1}\mathbf{D}_0, \quad \bar{\mathbf{K}} = \mathbf{B}^{-1}\mathbf{K}_0. \quad (4.14)$$

Note that in (4.14) it is assumed that $\mathbf{B}$ is invertible, which is reasonable since the actuators of the system in Figure 4.1 are placed such that the Stewart platform can be actuated in all directions. Moreover, observe from (4.13) that $\mathbf{C}_{\mathbf{FF},\text{opt}}$ only requires information from the rigid-body model to obtain perfect cancellation of floor vibrations despite the presence of internal non-rigid-body modes. In other words, the ideal feedforward controller is independent of the internal

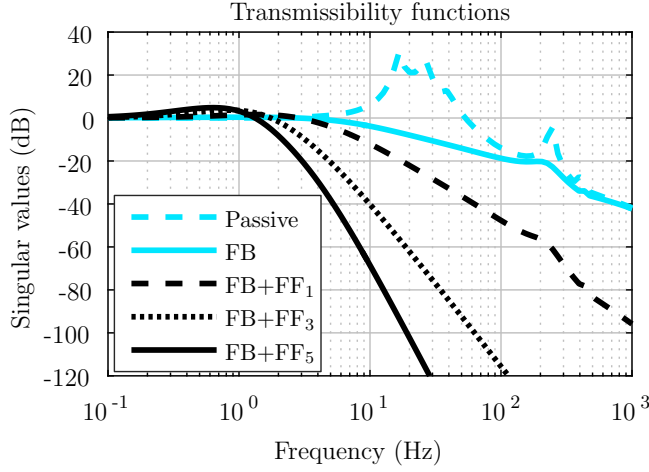


Figure 4.3. Maximum singular values of $\mathcal{T}$ for the six-axis vibration isolation system, in case of the passive system (dashed blue), the feedback (FB) controlled system using (4.5) (solid blue), and the combined feedback plus model-based feedforward (FF) controlled system using (4.5) and (4.15) with $\alpha = 2 \times 2\pi$ rad/s and $n = 1, 3, 5$ (black).

modes of the payload. From a physical point of view, the ideal feedforward controller perfectly compensates the disturbance forces transmitted through the hard mounts, such that the net forces on the platform become zero.

In practice, the controller in (4.13) is not feasible because it induces drift and actuator saturation as a result of using pure integrators. To cope with this problem, the pure integrators are replaced by n^{th} -order weak integrators. That is, the integrating actions are cut off below a certain frequency α . By doing so, an approximate Wiener controller is given by

$$\mathbf{C}_{\mathbf{FF},n}(s) = -H_{(\alpha,n)}^2(s) (\bar{\mathbf{D}}s + \bar{\mathbf{K}}), \quad (4.15)$$

with

$$H_{(\alpha,n)}(s) = \frac{1 - L_{(\alpha,n)}(s)}{s}, \quad (4.16)$$

that describes an n^{th} -order weak integrator with relative degree one, and

$$L_{(\alpha,n)}(s) = \left(\frac{\alpha}{s + \alpha} \right)^n, \quad (4.17)$$

representing an n^{th} -order low-pass filter with cut-off frequency α . The feedfor-

ward controller in (4.15) can be rewritten as

$$\begin{aligned}\mathbf{C}_{\mathbf{FF},n}(s) &= - \left(\frac{1 - L_{(\alpha,n)}(s)}{s} \right)^2 (\bar{\mathbf{D}}s + \bar{\mathbf{K}}) \\ &= - (1 - L_{(\alpha,n)}(s))^2 \frac{1}{s^2} (\bar{\mathbf{D}}s + \bar{\mathbf{K}}) \\ &= - (1 - L_{(\alpha,n)}(s))^2 \mathbf{P}_2^{-1}(s) \mathbf{P}_1(s).\end{aligned}\quad (4.18)$$

Substitution of (4.18) in (4.10) and after some algebra gives

$$\mathcal{T}(s) = (2 - L_{(\alpha,n)}(s)) L_{(\alpha,n)}(s) \mathbf{S}_{\mathbf{FB}}(s) \mathbf{P}_1(s), \quad (4.19)$$

with $\mathbf{S}_{\mathbf{FB}}(s) \mathbf{P}_1(s)$ representing the primary path in case the feedback loop is closed, see (4.7) and (4.11). For $s = j\omega$ and $\omega > \alpha$, (4.19) reduces to

$$\mathcal{T}(j\omega) \rightarrow 2L_{(\alpha,n)}(j\omega) \mathbf{S}_{\mathbf{FB}}(j\omega) \mathbf{P}_1(j\omega), \quad (4.20)$$

because $L_{(\alpha,n)}$ dominates over $L_{(\alpha,n)}^2$ for $\omega > \alpha$. Eq. (4.20) shows that, potentially, a transmissibility function with arbitrary roll-off can be created for frequencies $\omega > \alpha$ by increasing the amount of roll-off in $L_{(\alpha,n)}$. By means of example, Figure 4.3 shows various plots of $\mathcal{T}$ through singular values. Three cases of feedforward control with $n = 1, 3, 5$ and $\alpha = 2 \times 2\pi$ rad/s (2 Hz) are depicted. Beyond the frequency α , the figure clearly shows a reduction of $\mathcal{T}$ with a decay of $n \times 20$ dB/decade. Note that for frequencies smaller than α , performance is slightly deteriorated with respect to the passive system. This performance deterioration could be reduced by replacing $H_{(\alpha,n)}^2$ in (4.15) by $H_{(\alpha_1,n)} H_{(\alpha_2,n)}$ with $\alpha_1 > \alpha_2$, i.e. using different values of α . However, this solution direction is not pursued in order to keep clarity of presentation.

To illustrate the behavior of the controllers containing n^{th} -order weak integrators in terms of Bode plots, Figure 4.4 shows a comparison between the (1,1)-components of $\mathbf{C}_{\mathbf{FF},1}$, $\mathbf{C}_{\mathbf{FF},3}$, and $\mathbf{C}_{\mathbf{FF},5}$ and the corresponding component of the Wiener solution $\mathbf{C}_{\mathbf{FF},\text{opt}}$ from (4.13). From Figure 4.4 it can be observed that increasing n leads to less phase shifts in $\mathbf{C}_{\mathbf{FF},n}$ with respect to $\mathbf{C}_{\mathbf{FF},\text{opt}}$ beyond the cut-off frequency α (2 Hz) and, therefore, improved roll-off in $\mathcal{T}$ for $\omega > \alpha$ such as shown in Figure 4.3. However, increasing n comes at the expense of a larger controller gain for the lower frequency interval,

$$\mathbf{C}_{\mathbf{FF},n}(0) = - \left(\frac{n}{\alpha} \right)^2 \bar{\mathbf{K}}, \quad (4.21)$$

which is undesired because of drift and actuator saturation. Note that (4.21) can be derived from the static gain of the weak integrator, or

$$\lim_{s \rightarrow 0} H_{(\alpha,n)}(s) = \lim_{s \rightarrow 0} \frac{1 - L_{(\alpha,n)}(s)}{s} = \frac{n}{\alpha}. \quad (4.22)$$

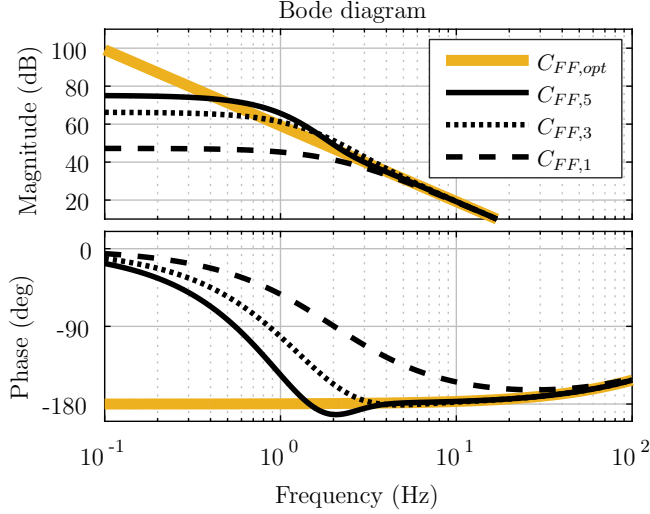


Figure 4.4. Bode plots of the (1,1)-components, i.e. the controller transfer function from input 1 to output 1, in $\mathbf{C}_{FF,1}$, $\mathbf{C}_{FF,3}$, and $\mathbf{C}_{FF,5}$ compared to the (1,1)-component in $\mathbf{C}_{FF,opt}$. Increasing n results in a trade-off: (a) a better approximation of the phase with respect to $\mathbf{C}_{FF,opt}$, but (b) increased controller gain for low frequencies.

4.4.2 Practical aspects

The theoretical performance improvements shown in Figure 4.3 will not easily be obtained in practice. Important performance limiters in practice are actuator-floor interaction, sensor filtering, and parameter uncertainty, which are discussed below.

Actuator-floor interaction

In this chapter, it is assumed that the reaction forces of the actuator exerted on the floor plate do not influence $\ddot{\mathbf{q}}_0$. For the given setup, this is a valid assumption since both the control forces and guidance stiffnesses of the VCMs are small compared to the piezo shaker forces and piezo and floor plate stiffnesses. In general, however, the floor acceleration might be significantly influenced by the reaction forces of the actuators. This can result in a possibly unstable feedback loop from the reaction forces to the floor plate accelerometers. An extensive discussion about this topic is found in, e.g., [58]. However, for the setup considered in this chapter, no stability problems occurred during the measurements. Therefore, this stability aspect is not further addressed.

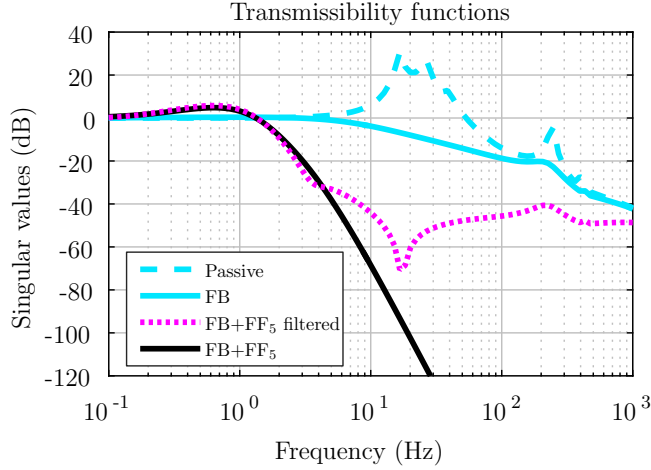


Figure 4.5. Maximum singular values of $\mathcal{T}$ for the six-axis vibration isolation system. The blue and black lines are identical to the lines in Figure 4.3. The magenta dotted line shows that disturbance suppression in terms of $\mathcal{T}$ is limited when including sensor filtering, see Section 4.4.2.

Sensor filtering

As discussed in Section 4.2, the signal conditioners for the sensor signals contain a 0.1 Hz high-pass filter and a 3000 Hz low-pass filter to suppress sensor noise, internal sensor dynamics, and the effects of aliasing. To analyze performance including the effects of sensor filtering, $\mathbf{P}_2$ in (4.7) is extended to

$$\mathbf{P}_{2,\text{new}} = \left(\frac{s^2}{s^2 + 2\zeta_1\omega_1 s + \omega_1^2} \right) \left(\frac{\omega_2^2}{s^2 + 2\zeta_2\omega_2 s + \omega_2^2} \right) \mathbf{P}_2,$$

with $\omega_1 = 0.1 \times 2\pi$ rad/s, $\omega_2 = 3000 \times 2\pi$ rad/s representing the filter frequencies, and $\zeta_1 = \zeta_2 = 0.7$ representing the relative damping coefficients of the filters. The corresponding transmissibility is plotted in Figure 4.5. The figure clearly shows that sensor filtering deteriorates performance for frequencies beyond 4 Hz, even though the high cut-off frequency ω_2 in the sensor filter is more than 700 times higher. Below 4 Hz, the effect of sensor filtering on performance deterioration is hardly visible. This is because the lower cut-off frequency ω_1 in the sensor filter is much lower than the cut-off frequency α (2 Hz) in the weak integrators which already limit performance improvements at low frequencies.

Parameter uncertainty

Performance as discussed in Figure 4.3 is obtained in case the feedforward controller parameters exactly match the plant parameters. However, in practice it

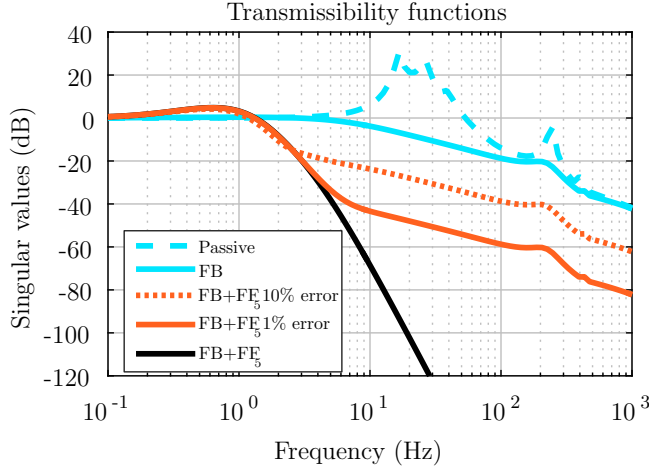


Figure 4.6. Maximum singular values of $\mathcal{T}$ for the six-axis vibration isolation system. The blue and black lines are identical to the lines in Figure 4.3. The red lines show that disturbance suppression in terms of $\mathcal{T}$ is limited when having parameter estimation errors, see Section 4.4.2.

is often difficult to accurately identify $\mathbf{P}_1$, hence $\bar{\mathbf{D}}, \bar{\mathbf{K}}$ in (4.15), because it is often not possible to excite the floor sufficiently. As a result, the controller parameters will not match the plant parameters exactly. For example, Figure 4.6 shows the maximum singular values of the transmissibility function in case of a 1% and 10% parameter estimation error, i.e. $\bar{\mathbf{D}} \rightarrow 0.99\bar{\mathbf{D}}, \bar{\mathbf{K}} \rightarrow 0.99\bar{\mathbf{K}}$ and $\bar{\mathbf{D}} \rightarrow 0.9\bar{\mathbf{D}}, \bar{\mathbf{K}} \rightarrow 0.9\bar{\mathbf{K}}$, respectively. It follows that the maximum achievable performance is proportional with the parameter estimation error. For example, when there is an estimation error of 1%, the reduction in transmissibility is also limited to 1% (or $0.01 = -40$ dB) with respect to performance without feedforward control. It thus makes sense to minimize parameter uncertainty, which is the aim in the next section by applying self-tuning feedforward control.

4.5 Self-tuning feedforward control

The motivation for residing to self-tuning methods is to minimize performance limitations due to parameter uncertainty, see Section 4.4.2. The first step in self-tuning controller design is to rewrite the model-based feedforward controller from Section 4.4 to a generalized FIR filter structure, which is presented in Section 4.5.1. The second step is to determine the update law for the controller parameters, which is derived in Section 4.5.2. A convergence analysis and simulation studies for the self-tuning controller are presented in Section 4.5.3 and Section 4.5.4, respectively.

4.5.1 Generalized FIR filter structure

In this chapter, the generalized FIR filter structure is used for self-tuning feedforward control. In this structure, an explicit separation between the zeros and the poles of the controller is made. The poles are fixed a priori in so-called basis functions. The zeros are subsequently determined online by a self-tuning algorithm. This section shows how the generalized FIR filter structure can be derived from the model-based controller in (4.15). Defining $\tilde{\mathbf{A}}_0(s) \in \mathbb{C}^6$ as the Laplace transform of $\tilde{\mathbf{a}}_0$ in Figure 4.2, the output of a feedforward controller similar to (4.15) can be written as

$$\mathbf{u}_{FF}(s) = \begin{bmatrix} -\bar{\mathbf{D}} & -\bar{\mathbf{K}} \end{bmatrix} \begin{bmatrix} H_{(\alpha,n)}(s) \tilde{\mathbf{A}}_0(s) \\ H_{(\alpha,n)}^2(s) \tilde{\mathbf{A}}_0(s) \end{bmatrix}. \quad (4.23)$$

Since the self-tuning algorithm operates in discrete-time at sampling frequency f_s , a discrete-time representation of (4.23) is used which reads

$$\mathbf{u}_{FF}(k) = \begin{bmatrix} -\bar{\mathbf{D}}_d & -\bar{\mathbf{K}}_d \end{bmatrix} \begin{bmatrix} H_{(\alpha,n)}(q) \tilde{\mathbf{a}}_0(k) \\ H_{(\alpha,n)}^2(q) \tilde{\mathbf{a}}_0(k) \end{bmatrix}, \quad (4.24)$$

with $\bar{\mathbf{D}}_d, \bar{\mathbf{K}}_d \in \mathbb{R}^{6 \times 6}$ containing the discrete-time controller parameters, q representing the backward shift operator, and $k \in \mathbb{N}$ referring to time samples $t_k = kT_s$ with sampling time $T_s = 1/f_s$. To improve the convergence properties of the update algorithm, (4.24) is rewritten as

$$\mathbf{u}_{FF}(k) = H_{(\alpha,n)}(q) \underbrace{\begin{bmatrix} -\bar{\mathbf{D}}_d & -\frac{1}{\beta} \bar{\mathbf{K}}_d \end{bmatrix}}_{\mathbf{W}^*} \underbrace{\begin{bmatrix} \tilde{\mathbf{a}}_0(k) \\ \beta H_{(\alpha,n)}(q) \tilde{\mathbf{a}}_0(k) \end{bmatrix}}_{\tilde{\boldsymbol{\psi}}(k)}. \quad (4.25)$$

In (4.25), $\beta \in \mathbb{R}$ is a scaling constant (Section 4.5.3), the matrix $\mathbf{W}^* \in \mathbb{R}^{6 \times 12}$ contains the model-based controller parameters, and $\tilde{\boldsymbol{\psi}}(k) \in \mathbb{R}^{12}$ represents the so-called regression vector. To find (online) the parameters in $\mathbf{W}^*$, which are assumed to be a priori unknown, (4.25) is rewritten such that the parameters from $\mathbf{W}^*$ are stored in a single column vector $\mathbf{w}^*$, or

$$\mathbf{u}_{FF}(k) = H_{(\alpha,n)}(q) \cdot \underbrace{\begin{bmatrix} \tilde{\boldsymbol{\psi}}^T(k) & \dots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \dots & \tilde{\boldsymbol{\psi}}^T(k) \end{bmatrix}}_{\tilde{\boldsymbol{\Psi}}(k)} \underbrace{\begin{bmatrix} (\mathbf{W}_{(1,:)}^*)^T \\ \vdots \\ (\mathbf{W}_{(6,:)}^*)^T \end{bmatrix}}_{\mathbf{w}^*}, \quad (4.26)$$

with regression matrix $\tilde{\boldsymbol{\Psi}}(k) \in \mathbb{R}^{6 \times 72}$, parameter vector $\mathbf{w}^* \in \mathbb{R}^{72}$ containing the model-based controller parameters, and $(\mathbf{W}_{(i,:)})^T$ denoting the transposed

i^{th} row of $\mathbf{W}^*$. Eq. (4.26) turns into a self-tuning controller if the model-based parameter vector $\mathbf{w}^*$ is replaced by the time-varying vector $\mathbf{w}(k)$ that is estimated online:

$$\mathbf{u}_{\mathbf{F}\mathbf{F}}(k) = H_{(\alpha,n)}(q)\tilde{\Psi}(k)\mathbf{w}(k). \quad (4.27)$$

Since (4.27) is affine in the parameters, minimization of a quadratic cost criterion to estimate $\mathbf{w}(k)$ renders a convex optimization problem that (under certain conditions) has a unique closed-form solution. The poles of the self-tuning filter are fixed in $\tilde{\Psi}$ such that adaptation of $\mathbf{w}(k)$ only involves estimation of its zeros, which makes that this filter has the structure of a generalized FIR filter [23].

4.5.2 Update law using the FeLMS algorithm

The update law for $\mathbf{w}(k)$ will be based on the filtered-error least mean squares (FeLMS) algorithm [24] combined with residual noise shaping [57, 7]. As mentioned in the introduction, FeLMS is attractive because it combines relatively low computational power with good convergence properties. Residual noise shaping is used to add frequency weighting to the error signal that is minimized.

The optimization algorithm [108] uses the steepest descent method [80] with corresponding update law

$$\mathbf{w}(k+1) = \mathbf{w}(k) - \frac{\mu(k)}{2} \left(\frac{\partial J(k)}{\partial \mathbf{w}} \right)^T, \quad (4.28)$$

with adaptation rate $\mu(k)$, which will be determined in Section 4.5.3, and gradient $\frac{\partial J(k)}{\partial \mathbf{w}}$ of a scalar-valued cost function J to be minimized. The parameters are updated at every time sample t_k . The cost function J is chosen as the squared filtered-error at time sample t_k , or

$$J(k) = \mathbf{e}^T(k)\mathbf{e}(k), \quad (4.29)$$

such that

$$\left(\frac{\partial J(k)}{\partial \mathbf{w}} \right)^T = \left(\frac{\partial J(k)}{\partial \mathbf{e}} \frac{\partial \mathbf{e}(k)}{\partial \mathbf{w}} \right)^T = 2 \left(\frac{\partial \mathbf{e}(k)}{\partial \mathbf{w}} \right)^T \mathbf{e}(k), \quad (4.30)$$

with filtered error $\mathbf{e}(k) \in \mathbb{R}^6$ obtained from the measured output signal $\tilde{\mathbf{a}}_1$ (see Figure 4.2) by

$$\mathbf{e}(k) = H_{(\alpha,n)}^{-1}(q)\mathbf{N}(q)\hat{\mathbf{P}}_2^{-1}(q)\tilde{\mathbf{a}}_1(k) \quad (4.31)$$

$$= H_{(\alpha,n)}^{-1}(q)\mathbf{N}(q)\hat{\mathbf{P}}_2^{-1}(q)(\mathbf{y}_1(k) + \mathbf{y}_2(k) + \mathbf{n}_1(k)). \quad (4.32)$$

make the product $H_{(\alpha,n)}^{-1} \mathbf{N} \hat{\mathbf{P}}_2^{-1}$ in Figure 4.7 proper. Also, $\hat{\mathbf{P}}_2$ may contain transmission zeros outside the unit disk, that render $\hat{\mathbf{P}}_2^{-1}$ unstable. There are two possible solutions for this problem. First, the sensors and actuators can be relocated. Second, modified versions of the FeLMS can be used. An example is the Filtered-reference Filtered-error LMS (Fx-Fe-LMS) method [86, 63], in which the error signals are only filtered by the invertible (and minimum-phase) part of $\hat{\mathbf{P}}_2^{-1}$, while the reference signals $\tilde{\mathbf{a}}_0$ are filtered by the non-minimum phase part of $\hat{\mathbf{P}}_2$. Another alternative is using hybrid Filtered-error LMS [24], which combines inversion-based FeLMS [51, 108] with adjoint FeLMS. Using this combination, the filter can be constructed by taking the inverse of the minimum-phase part of $\hat{\mathbf{P}}_2$ and the adjoint of the maximum-phase part of $\hat{\mathbf{P}}_2$.

4.5.3 Convergence analysis

This section addresses the convergence properties of the self-tuning algorithm subject to both modeling errors in $\hat{\mathbf{P}}_2$ and the presence of sensor noise $\mathbf{n}_0$ and $\mathbf{n}_1$. These properties are used to design the residual noise shaping filter $\mathbf{N}$ and to find an expression for μ in (4.34).

To compare the result of the self-tuning controller with the model-based controller from Section 4.4, the weight estimation error will be defined as $\Delta \mathbf{w}(k) = \mathbf{w}(k) - \mathbf{w}^*$ with $\mathbf{w}^*$ from (4.26). Subtracting $\mathbf{w}^*$ from both sides of (4.34) it is found that

$$\Delta \mathbf{w}(k+1) = \Delta \mathbf{w}(k) - \mu(k) \left[\mathbf{N}(q) \tilde{\Psi}(k) \right]^T \mathbf{e}(k). \quad (4.35)$$

Eq. (4.35) describes the development of $\Delta \mathbf{w}(k)$ over time, i.e. convergence and bias of the self-tuning filter with respect to the model-based controller. To analyze the asymptotic behavior of $\Delta \mathbf{w}(k)$ with respect to the disturbances, (4.35) must be rewritten because $\mathbf{e}(k)$ is still a function of $\Delta \mathbf{w}(k)$. Reconsider $\mathbf{e}(k)$ in (4.32), which contains $\mathbf{y}_1(k)$ and note that without loss of generality $\mathbf{y}_1$ can be written as

$$\mathbf{y}_1(k) = \underbrace{-H_{(\alpha,n)}(q) \mathbf{P}_2(q) \Psi(k) \mathbf{w}^*}_{\mathbf{y}_2^*(k)} + \boldsymbol{\epsilon}_0(k), \quad (4.36)$$

with $\mathbf{y}_2^*(k)$ representing the (noise-free) secondary path output in case $\mathbf{w}(k) := \mathbf{w}^*$, and $\boldsymbol{\epsilon}_0(k) \in \mathbb{R}^6$ describing the residual error for which $\mathbf{y}_2^*$ cannot compensate. In (4.36), the noise-free regression matrix Ψ is used rather than $\tilde{\Psi}$ since $\mathbf{y}_1$ is obtained from the noise-free input signal $\mathbf{a}_0$, see Figure 4.7. Similar to the definition of $\tilde{\Psi}$ in (4.26), Ψ is defined as

$$\Psi(k) = \begin{bmatrix} \boldsymbol{\psi}^T(k) & \dots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \dots & \boldsymbol{\psi}^T(k) \end{bmatrix}, \quad \boldsymbol{\psi}(k) = \begin{bmatrix} \mathbf{a}_0(k) \\ \beta H_{(\alpha,n)}(q) \mathbf{a}_0(k) \end{bmatrix}. \quad (4.37)$$

Substitution of (4.36) in (4.32) and after some rewriting it is found that

$$\begin{aligned} e(k) = & \left[\mathbf{N}(q) \hat{\mathbf{P}}_2^{-1}(q) \mathbf{P}_2(q) \tilde{\Psi}(k) \right] \Delta \mathbf{w}(k) + \mathbf{N}(q) \hat{\mathbf{P}}_2^{-1}(q) \mathbf{P}_2(q) \left(\tilde{\Psi}(k) - \Psi(k) \right) \mathbf{w}^* \\ & + H_{(\alpha, n)}^{-1}(q) \mathbf{N}(q) \hat{\mathbf{P}}_2^{-1}(q) (\epsilon_0(k) + \mathbf{n}_1(k)). \end{aligned} \quad (4.38)$$

Moreover, when (4.38) is substituted in (4.35) it is obtained that

$$\Delta \mathbf{w}(k+1) = [\mathbf{I}_6 - \mu(k) \mathbf{G}_1(k)] \Delta \mathbf{w}(k) + \mu(k) \mathbf{G}_2(k) + \mu(k) \mathbf{G}_3(k), \quad (4.39)$$

with

$$\begin{aligned} \mathbf{G}_1(k) &= \left[\mathbf{N}(q) \tilde{\Psi}(k) \right]^T \left[\mathbf{N}(q) \hat{\mathbf{P}}_2^{-1}(q) \mathbf{P}_2(q) \tilde{\Psi}(k) \right], \\ \mathbf{G}_2(k) &= \left[\mathbf{N}(q) \tilde{\Psi}(k) \right]^T \cdot \left[\mathbf{N}(q) \hat{\mathbf{P}}_2^{-1}(q) \mathbf{P}_2(q) (\Psi(k) - \tilde{\Psi}(k)) \right] \mathbf{w}^*, \\ \mathbf{G}_3(k) &= - \left[\mathbf{N}(q) \tilde{\Psi}(k) \right]^T \cdot \left[H_{(\alpha, n)}^{-1}(q) \mathbf{N}(q) \hat{\mathbf{P}}_2^{-1}(q) (\epsilon_0(k) + \mathbf{n}_1(k)) \right]. \end{aligned} \quad (4.40)$$

Eq. (4.39) describes the development of $\Delta \mathbf{w}(k)$ over time as a function of the disturbances $\mathbf{a}_0$, $\mathbf{n}_0$ (via Ψ , $\tilde{\Psi}$), $\mathbf{n}_1$ and ϵ_0 . Convergence of $\Delta \mathbf{w}(k)$ in terms of mean values should be evaluated given the stochastic nature of these disturbances. In so doing, the average updating direction can be obtained by computing the expected value $\mathbb{E}[\Delta \mathbf{w}(k+1)]$. Using again the assumption that $\mathbf{w}(k)$ is adapted slowly with respect to $\tilde{\Psi}(k)$, see also Appendix A in Section 4.8, it is obtained that $\mathbb{E}[\mathbf{G}_1(k) \Delta \mathbf{w}(k)] = \mathbb{E}[\mathbf{G}_1(k)] \mathbb{E}[\Delta \mathbf{w}(k)]$. Under this assumption, it is found that

$$\begin{aligned} \mathbb{E}[\Delta \mathbf{w}(k+1)] &= \mathbb{E}[\mathbf{I}_6 - \mu(k) \mathbf{G}_1(k)] \mathbb{E}[\Delta \mathbf{w}(k)] \\ &\quad + \mu(k) \mathbb{E}[\mathbf{G}_2(k)] + \mu(k) \mathbb{E}[\mathbf{G}_3(k)]. \end{aligned} \quad (4.41)$$

Note that (4.41) describes the development of the mean weight estimation error $\mathbb{E}[\Delta \mathbf{w}(k)]$ in terms of expected values. Consequently, this equation provides information 1) regarding stability of the discrete-time recursive algorithm (Section 4.5.3), 2) the speed of convergence (Section 4.5.3), and 3) the steady-state mean estimation error $\mathbb{E}[\Delta \mathbf{w}(\infty)]$ (Section 4.5.3).

Stability analysis

Basically (4.41) shows that $\Delta \mathbf{w}(k)$ converges in terms of mean values if and only if all eigenvalues λ_i of the term $\mathbb{E}[\mathbf{I}_6 - \mu(k) \mathbf{G}_1(k)]$ lie within the unit circle and under the assumption that $\mathbb{E}[\mathbf{G}_2(k)]$ and $\mathbb{E}[\mathbf{G}_3(k)]$ are uniformly bounded. Using (4.40) and assuming that $\mathbf{N}(q) = n(q) \mathbf{I}_6$ with single-input single-output transfer function $n(q)$ such that the order in which $\mathbf{N}$ and $\hat{\mathbf{P}}_2^{-1}(q) \mathbf{P}_2(q)$ appear can be exchanged, it follows that $\mathbb{E}[\mathbf{G}_1(k)]$ can be written as

$$\mathbb{E}[\mathbf{G}_1(k)] = \mathbb{E} \left[\left[\mathbf{N}(q) \tilde{\Psi}(k) \right]^T \cdot \left[\hat{\mathbf{P}}_2^{-1}(q) \mathbf{P}_2(q) \mathbf{N}(q) \tilde{\Psi}(k) \right] \right]. \quad (4.42)$$

In the case of perfect estimation $\hat{\mathbf{P}}_2 := \mathbf{P}_2$, (4.42) reduces to the autocorrelation matrix of $\mathbf{N}(q)\tilde{\Psi}(k)$ which is per definition positive definite (assuming that $\tilde{\Psi}$ is persistently exciting). As such, there exists $\mu(k) > 0$ for which all eigenvalues λ_i of $\mathbb{E}[\mathbf{I}_6 - \mu(k)\mathbf{G}_1(k)]$ lie within the unit circle, hence stability is guaranteed. When the model $\hat{\mathbf{P}}_2$ is not perfect, a sufficient condition for stability is given by the strictly positive real (SPR) condition [102], or

$$\left(\hat{\mathbf{P}}_2^{-1}(e^{j\omega})\mathbf{P}_2(e^{j\omega})\right)^H + \hat{\mathbf{P}}_2^{-1}(e^{j\omega})\mathbf{P}_2(e^{j\omega}) \succ 0. \quad (4.43)$$

The derivation of this condition, which is based on the work of [102], is given in Appendix A in Section 4.8.

Remark 4.2. Eq. (4.43) requires an accurate model for $\mathbf{P}_2$ which is the transfer function from actuator forces to platform accelerations. This transfer function is generally easy to obtain from frequency-domain system identification methods as described in, e.g., [69]. Therefore, it might be expected that $\hat{\mathbf{P}}_2$ is accurate over a broad frequency range. Nevertheless, it is very difficult to perfectly estimate $\mathbf{P}_2$ by $\hat{\mathbf{P}}_2$ for all frequencies such that (4.43) will not likely be satisfied in practice. However, this does not imply instability because (4.43) only gives a sufficient condition. Furthermore, the noise shaping filter can be used to suppress the destabilizing effect of frequencies for which (4.43) is not satisfied. Moreover, [13] explains that robust convergence can be improved by adding damping to resonance peaks using feedback control. Another method to stabilize convergence is to add plant regularization [13, 33] which means that the plant is augmented with virtual plant dynamics. This method improves convergence but generally leads to biased estimates of the parameters.

Provided that the eigenvalues of $[\mathbf{I}_6 - \mu(k)\mathbf{G}_1(k)]$ lie within the unit circle, and assuming $\hat{\mathbf{P}}_2 := \mathbf{P}_2$ such that the eigenvalues of $\mathbf{G}_1(k)$ are positive, lower and upper bounds for $\mu(k)$ are defined as

$$0 < \mu(k) < \frac{2}{\|\mathbf{N}(q)\tilde{\Psi}(k)\|_2^2}, \quad (4.44)$$

explicitly using the fact that the maximum eigenvalue $\lambda_{\max}$ of $\mathbf{G}_1$ is always smaller or equal to the maximum singular value of $\mathbf{G}_1$ which is given by the 2-norm of $\mathbf{G}_1$. To satisfy (4.44) the adaptation rate is normalized, or

$$\mu(k) = \frac{\bar{\mu}}{\epsilon + \|\mathbf{N}(q)\tilde{\Psi}(k)\|_2^2}, \quad (4.45)$$

with $\epsilon > 0$ a small positive regularization constant to provide an upper bound on $\mu(k)$, and $\bar{\mu} \in (0, 2)$ the normalized adaptation rate. Using (4.45) the self-tuning algorithm becomes a special form of the normalized LMS algorithm [80] in which $\mu(k)$ is dependent on $\tilde{\Psi}(k)$ which by itself is a function of $\tilde{\mathbf{a}}_0$. As

such, stability of the algorithm becomes independent of the power of $\tilde{\mathbf{a}}_0$. On the one hand, $\bar{\mu}$ must be chosen small enough to prevent instability by violating the slow-adaptation assumption; the largest stabilizing step size may be very small if $\hat{\mathbf{P}}_2 \neq \mathbf{P}_2$. Moreover, the steady-state variance on the parameters depends on the choice for $\bar{\mu}$ [80, 94]. To ensure that the estimation errors in the parameters are sufficiently small to obtain the desired transmissibility function, see Section 4.4.2, $\bar{\mu}$ must be sufficiently small. On the other hand, too slow adaptation rates will induce unnecessarily long convergence times, which is generally undesired.

Convergence speed

In Section 4.5.3 conditions on the eigenvalues λ_i of $(\mathbf{I}_6 - \mu(k)\mathbf{G}_1(k))$ were given to achieve convergence of the self-tuning algorithm. In addition, to get uniform convergence speeds for all parameters in $\mathbf{w}(k)$, all eigenvalues λ_i should be identical [29]. In this section, the convergence speed of the parameters is analyzed for the case $\hat{\mathbf{P}}_2 = \mathbf{P}_2$. Furthermore, for simplicity, neither input sensor noise nor residual noise shaping will be considered, i.e. $\mathbf{n}_0 = \mathbf{0}$ and $\mathbf{N} = \mathbf{I}_6$, and $\mathbf{a}_0$ is assumed to be a ZMWN process, see Definition 4.3. Under these conditions, uniform convergence is obtained if

$$\beta = \frac{1}{\|H_{(\alpha,n)}(q)\|_2}, \quad (4.46)$$

with $\|H_{(\alpha,n)}(q)\|_2 \in \mathbb{R}$. The full derivation of (4.46) can be found in Appendix A in Section 4.8.

Definition 4.3. *Two uncorrelated zero-mean Gaussian white noise (ZMWN) processes $\mathbf{u}_1$ and $\mathbf{u}_2$ have at every time sample $k \in \mathbb{Z}$ the properties that*

$$\begin{aligned} \mathbb{E} [\mathbf{u}_j(k)\mathbf{u}_j^T(k)] &= \sigma_{u_j}^2 \mathbf{I}_6, \quad j \in \{1, 2\}, \\ \mathbb{E} [\mathbf{u}_j(k)\mathbf{u}_j^T(k-i)] &= \mathbf{0} \quad \forall i \in \mathbb{Z} \setminus \{0\}, \quad j \in \{1, 2\}, \\ \mathbb{E} [\mathbf{u}_1(i)\mathbf{u}_2^T(k)] &= \mathbf{0} \quad \forall i \in \mathbb{Z}, \end{aligned}$$

with $\sigma_{u_j}^2$ representing the variance of the signals in $\mathbf{u}_j$.

Steady-state mean weight estimation error

Having discussed the conditions for stability and convergence of the self-tuning algorithm, explicit expressions for the steady-state mean weight estimation error $\mathbb{E} [\Delta \mathbf{w}(\infty)]$, hence the bias in the estimates, can be derived. Bias will be related to the fact that the self-tuning filter aims at minimizing $\mathbf{e}^T(k)\mathbf{e}(k)$, while the model-based controller design is based on minimization of $\mathcal{T}$. The latter is posed in the frequency domain and obviously neglecting the contributions of sensor noise and unmodeled dynamics to $\mathbf{e}(k)$. As a result, bias will appear if

$\mathbb{E}[\mathbf{G}_2(k)]$ and $\mathbb{E}[\mathbf{G}_3(k)]$ in (4.41) are non-zero. In Appendix A in Section 4.8, the expression for bias is derived as

$$\mathbb{E}[\Delta \mathbf{w}(\infty)] = -\frac{1}{\sigma_{a_0}^2 + \sigma_{n_0}^2} \mathbb{E} \left[\left[\tilde{\Psi}(k) \right]^T \left[H_{(\alpha, n)}^{-1}(q) \hat{\mathbf{P}}_2^{-1}(q) \boldsymbol{\epsilon}_0(k) \right] \right] - \frac{\sigma_{n_0}^2}{\sigma_{a_0}^2 + \sigma_{n_0}^2} \mathbf{w}^*. \quad (4.47)$$

The first term in (4.47) describes the contribution of process noise $\boldsymbol{\epsilon}_0$. This term accounts for the effects of all dynamics the feedforward controller cannot compensate for, see (4.36). In practice, $\boldsymbol{\epsilon}_0 \neq \mathbf{0}$ because the pure integrators of the Wiener solution have been replaced by weak integrators in (4.23). As such, $\boldsymbol{\epsilon}_0$ will possess low-frequency contents if $\mathbf{a}_0$ contains frequencies $\omega < \alpha$. Furthermore, the feedforward control law $\tilde{\Psi}(q)\mathbf{w}^*$ is based on the assumption that the hard mounts are massless and can be fully described by a stiffness and a damping matrix. This excludes the effect of structural dynamics of the suspension itself that may occur at higher frequencies. Therefore, it is quite natural to expect that $\boldsymbol{\epsilon}_0$ has high-frequency contents too. Hence, the residual noise shaping filter $\mathbf{N}$ should be designed in view of filtering both low- and high-frequency contents from $\mathbf{e}$, such that the effect of $\boldsymbol{\epsilon}_0$ in terms of bias remains sufficiently small.

The second term in (4.47) describes the contribution of input sensor noise $\mathbf{n}_0$. If $\mathbf{n}_0$ and $\mathbf{a}_0$ are colored signals, this bias may be decreased (but not avoided) by appropriate design of $\mathbf{N}$ (note that $\mathbf{N} = \mathbf{I}_6$ in the foregoing analysis). Since $\tilde{\mathbf{a}}_0 = \mathbf{a}_0 + \mathbf{n}_0$ is filtered by $\mathbf{N}$, it follows that $\mathbf{N}$ must be designed such that the frequencies where $\mathbf{n}_0$ is dominant over $\mathbf{a}_0$ induce appropriate filtering, while the frequencies where $\mathbf{a}_0$ is dominant over $\mathbf{n}_0$ remain unaffected.

Remark 4.4. *The described convergence analysis only discusses convergence in terms of mean values, i.e. $\mathbb{E}[\Delta \mathbf{w}(k)]$. Convergence in terms of mean square values, i.e. the variance $\mathbb{E}[(\Delta \mathbf{w}(k))(\Delta \mathbf{w}(k))^T]$, is not discussed. It can be stated that $\mathbf{n}_0 \neq \mathbf{0}$, $\mathbf{n}_1 \neq \mathbf{0}$, $\boldsymbol{\epsilon}_0 \neq \mathbf{0}$ and $\hat{\mathbf{P}}_2 \neq \mathbf{P}_2$ all contribute to a larger mean-square-error (MSE); also a larger value for $\bar{\mu}$ increases the MSE. For a detailed discussion on calculating the mean-square performance in case $\mathbf{n}_1 \neq \mathbf{0}$ the reader is referred to [63].*

4.5.4 Simulation results

Three simulation studies without sensor noise ($\mathbf{n}_0 = \mathbf{n}_1 = \mathbf{0}$) have been performed to show convergence and to provide guidelines on how to choose the self-tuning controller parameters. Herein, different designs are used for $H_{(\alpha, n)}$ in (4.16), namely $n = 1, 3, 5$, to show the effect of increasing n . The cut-off frequency $\alpha = 2 \times 2\pi$ rad/s is chosen to be close to the lowest measurable frequency (1 Hz) of the sensors on the experimental setup to maximize performance. The

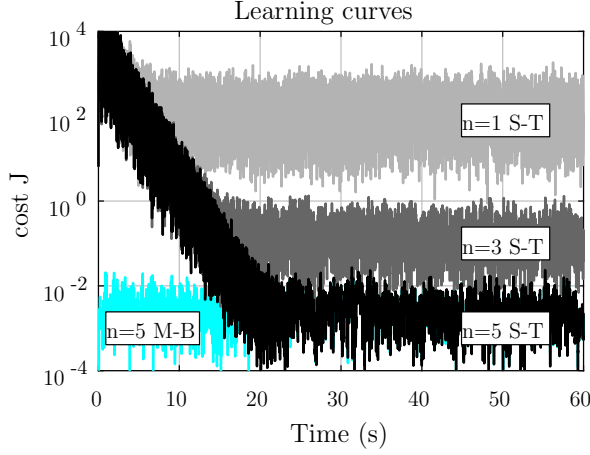


Figure 4.8. Learning Curves, showing convergence of the cost function defined in (4.29) when using self-tuning (S-T) controllers with n^{th} -order weak integrators and $n \in \{1, 3, 5\}$. For $n = 5$, the plot shows that the cost of a self-tuning controller converges to the cost obtained in a simulation with model-based (M-B) control.

primary and secondary path are obtained from the non-rigid-body model in (4.7). The feedback controller is given by (4.5). The inputs in $\mathbf{a}_0$ satisfy Definition 4.3 with variance $\sigma_{a_0}^2 = 1$. The adaptation rate is set to $\bar{\mu} = 10^{-3}$ which represents a tradeoff between stability, convergence time, and a sufficiently low variance on the steady-state estimates of the parameters. The constant $\epsilon = 10^{-3}$ is 0.1% of the average input power to prevent division by zero. The sampling frequency $f_s = 6400$ Hz is set identical to that of the experimental setup. The parameter vector $\mathbf{w}(k)$ is initialized with zeros and is updated at every time sample t_k . In all simulation studies it is assumed that $\hat{\mathbf{P}}_2 = \mathbf{P}_2$. The noise shaping filter is designed as

$$\mathbf{N}(s) = \left(\frac{s}{s + 5\alpha} \right)^5 \left(\frac{1000}{s + 1000} \right)^2 \mathbf{I}_6. \quad (4.48)$$

In view of the design rules derived at the end of Section 4.5.3, (4.48) contains a high-pass filter with a cut-off frequency of 5α rad/s to suppress low-frequency noise and a low-pass filter with two poles at 1000 rad/s to suppress the effects of unmodeled dynamics. Furthermore, the low-pass filter poles in $\mathbf{N}$ render $H_{(\alpha,n)}^{-1} \mathbf{N} \hat{\mathbf{P}}_2^{-1}$ strictly proper to ensure a causal implementation in discrete-time.

Figure 4.8 shows the learning curves in terms of cost J for the self-tuning controller. From the figure, it is clear that the controller parameters converge to minimum costs. It can also be observed that increasing n in (4.16) leads to

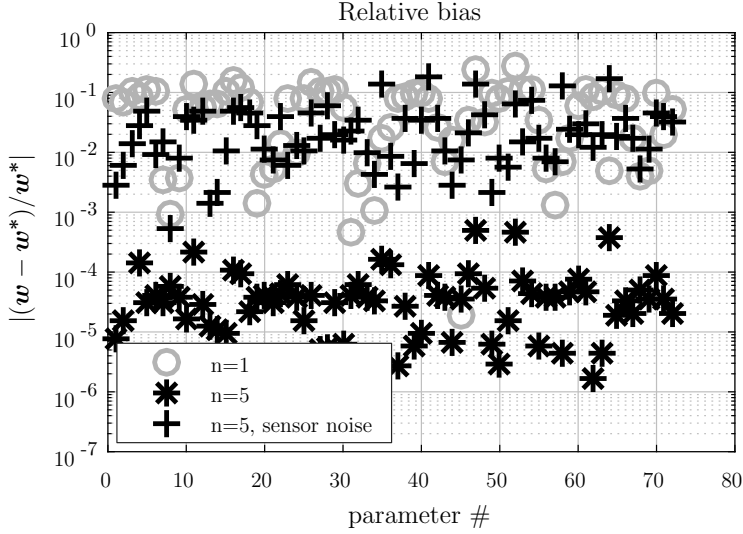


Figure 4.9. Average relative bias on the 72 parameters in $\mathbf{w}$, i.e. the parameters after convergence for $n = 1$, $n = 5$ and $n = 5$ with input sensor noise $\mathbf{n}_0$, with respect to the model-based parameters $\mathbf{w}^*$ according to (4.26); the figure shows that increasing n leads to less bias.

smaller residual costs, which is expected because $\mathbf{a}_0$ is better rejected. Increasing n leads to a longer convergence time to reach the accordingly smaller steady-state cost level, but even for $n = 5$ the algorithm only needs 20 seconds to converge. Figure 4.9 shows the average relative bias of the parameters after convergence. It is observed that increasing n leads to less bias, because the residual error in ϵ_0 becomes smaller. For $n = 5$, the average relative bias is less than 0.01%, which implies that the parameters of the self-tuning controller converge to the actual damping and stiffness parameters of the system. The figure also shows the result of a fourth study with $n = 5$ and additional sensor noise $\mathbf{n}_0$ consisting of six uncorrelated ZMWN signals having variance $\sigma_{n_0}^2 = 0.01$. As expected from Section 4.5.3, the steady-state bias level increases to approximately $\frac{\sigma_{n_0}^2}{\sigma_{a_0}^2 + \sigma_{n_0}^2} \approx 10^{-2}$.

Two important benefits of self-tuning feedforward control with respect to model-based control are derived from the bias plot in Figure 4.9. First, it is shown that the self-tuning parameters converge to the actual system parameters with sufficient accuracy. This accuracy is said to be sufficient because for parameter uncertainties less than 1% the effect of sensor filtering is the dominant restriction on performance at most frequencies, see Figures 4.5 and 4.6. It is important to note that, using self-tuning control, this convergence is reached

without the need of doing an accurate (and costly) system identification procedure which would be necessary when using model-based control. Second, it is shown that the bias increases in case of sensor noise, which is the result from minimization of the cost function in (4.29). In other words, the model-based controller is not the optimal solution for the noisy case, and this aspect is automatically compensated for in the self-tuning controller design.

4.6 Experimental results

The self-tuning feedforward controller from Section 4.5 is implemented on the experimental setup as discussed in Section 4.2 to validate the control strategy. Performance is evaluated in terms of the transmissibility matrix $\mathcal{T}$ in z -, R_x - and R_y -direction, since these are the only three directions that can be excited using the vertical floor plate shakers, see Figure 4.1. Three uncorrelated Pseudo-Random-Binary-Sequence (PRBS) signals are provided to the floor plate shakers. A parametric model $\hat{\mathbf{P}}_2$, needed for self-tuning, is adopted from [89]. The self-tuning feedforward controller from Section 4.5 is implemented using $n = 1, 3, 5$ and $\alpha = 2$ Hz, together with the feedback controller from (4.5). After convergence of the parameters, the adaptation rate is set to $\bar{\mu} = 0$ to make the system time-invariant such that the transmissibilities can be measured in terms of frequency response functions.

4.6.1 Single-input single-output measurements

Figure 4.10 shows the transmissibilities $\mathcal{T}$ in z -direction for the passive, the feedback-controlled, and the feedback- plus feedforward-controlled system. It is observed that adding feedforward control leads to large performance improvements, because the transmissibility is decreased up to 40 dB with respect to the feedback-only controlled system for frequencies in the interval between 2 and 300 Hz. Figure 4.10 also shows that increasing n leads to more roll-off after the frequency α (2 Hz) as predicted in Figure 4.3, but only up to the saturation limit of -40 dB. This saturation limit is similar to the limits predicted (and discussed) by sensor filtering in Figure 4.5. Therefore it is reasonable to assume that this saturation limit is dominantly caused by the effect of sensor filtering. This implies that the problems regarding parameter uncertainty in model-based control, see Figure 4.6, are sufficiently solved by applying self-tuning control. Furthermore, performance is limited due to the causal nature of the input disturbance measurement, see Chapter 2, resulting in the fact that active vibration isolation deteriorates performance at high frequencies due to a waterbed effect. For frequencies below 2 Hz, performance is slightly deteriorated due to application of the weak integrators, as discussed in Section 4.4. For frequencies beyond 400 Hz and at the 90 Hz anti-resonance frequency, sensor noise dominates the error response.

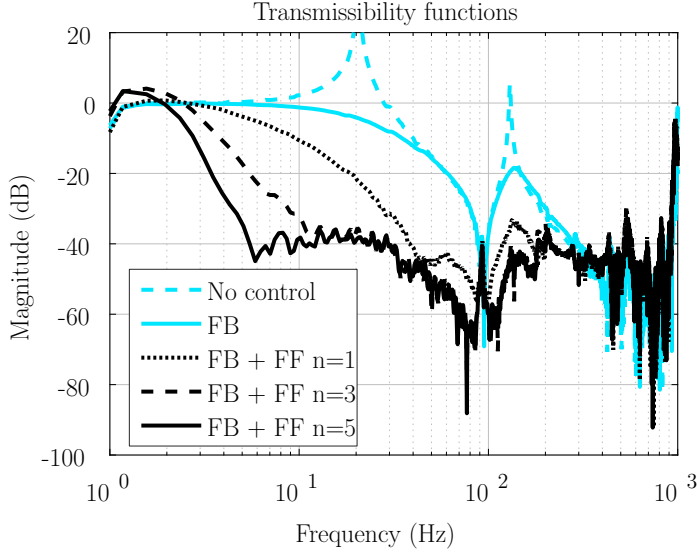


Figure 4.10. Measured transmissibility functions in the z -direction. The passive and feedback controlled system response are compared with the system response including feedforward control. Notice that higher-order weak integrators in the feedforward controller result in a stronger roll-off for frequencies beyond $\alpha = 2$ Hz.

4.6.2 Multi-input multi-output measurements

Figure 4.11 shows the measured MIMO transmissibility matrix $\mathcal{T}$ in z , R_x , and R_y directions for the passive system, the system with feedback control, and the system with feedback plus feedforward control using $n = 5$. The transmissibility matrix is obtained by doing three sequential experiments with different input signals for the floor plate shakers. The figure shows that the main diagonals of the transmissibility matrix associate with 40 dB reduction in all directions, similar to the performance of the measurements in Figure 4.10. The off-diagonal terms in the transmissibility matrix have different lower bounds because not all inputs and outputs have the same units (meters vs. radians).

4.7 Conclusions

This chapter presents a feedforward control strategy for active vibration isolation of precision machinery. A model-based feedforward controller to achieve perfect cancellation of floor vibrations is derived that only depends on the stiffness and damping properties of the suspension. To prevent problems with drift and actuator saturation, the feedforward controller gain is limited for frequencies

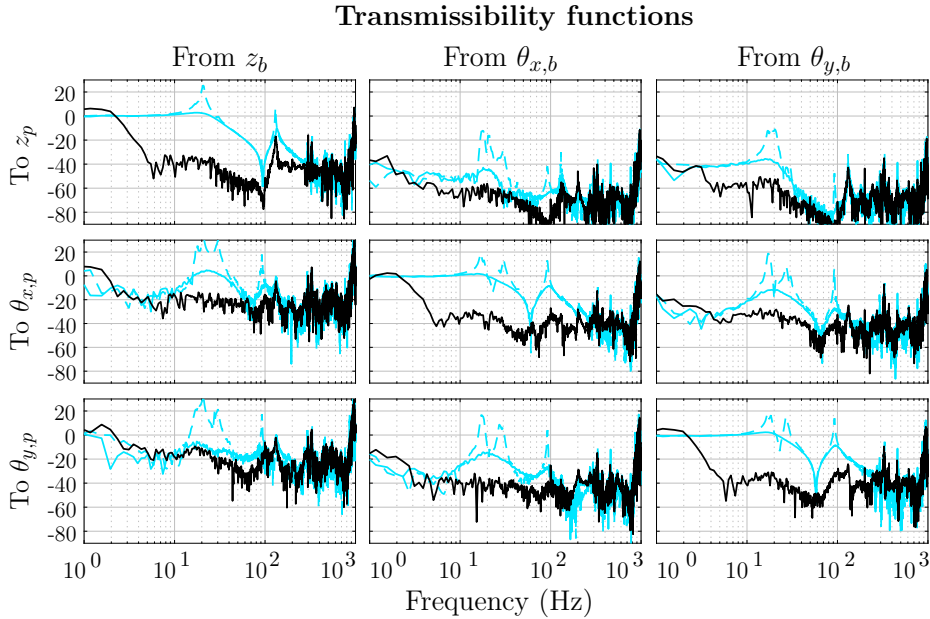


Figure 4.11. Measured transmissibility functions for the MIMO system, for the cases passive (dashed blue), feedback-controlled (solid blue) and feedback-plus feedforward-controlled (black, using $n = 5$ and $\alpha = 2 \times 2\pi$ rad/s). Observe that all three plots on the main diagonal show up to 40 dB performance improvement, hence a natural extension of the results for single-axis control shown in Figure 4.10.

below 2 Hz using higher-order weak integrators. Since pure model-based feedforward control is shown to be very sensitive for parameter estimation errors, the feedforward controller parameters are obtained by applying a self-tuning algorithm. Stability and minimization of bias in the parameter estimates are obtained by using a residual noise shaping filter. Simulations show that the parameters of the self-tuning controller converge to the actual system parameters with a predictable bias and in the presence of input sensor noise. An experimental validation of the self-tuning feedforward control strategy on an active hard-mounted vibration isolation system confirms the simulation results. Moreover, the validation shows that suppression of floor vibrations is improved up to 40 dB for frequencies between 2 and 300 Hz in the measured 3×3 MIMO transmissibility matrix.

4.8 Appendix A: Mathematical derivations

4.8.1 Derivation of the gradient

This Appendix derives a further expression for $e(k)$ and the error gradient signal in (4.30). To this end, note from Figure 4.2 that $\tilde{\mathbf{a}}_1$ can be defined as

$$\tilde{\mathbf{a}}_1(k) = \mathbf{y}_1(k) + \mathbf{P}_2(q)\mathbf{u}(k) + \mathbf{n}_1(k). \quad (4.49)$$

Substitution of (4.49) in (4.31) and neglecting feedback control ($\mathbf{u}_{FB} = \mathbf{0}$) such that $\mathbf{u} = \mathbf{u}_{FF}$ (see (4.27)) leads to

$$\begin{aligned} e(k) = & H_{(\alpha,n)}^{-1}(q)\mathbf{N}(q)\hat{\mathbf{P}}_2^{-1}(q)\mathbf{y}_1(k) + \mathbf{N}(q)\hat{\mathbf{P}}_2^{-1}(q)\mathbf{P}_2(q) \left[\tilde{\Psi}(k)\mathbf{w}(k) \right] \\ & + H_{(\alpha,n)}^{-1}(q)\mathbf{N}(q)\hat{\mathbf{P}}_2^{-1}(q)\mathbf{n}_1(k). \end{aligned} \quad (4.50)$$

Eq. (4.50) is not an explicit function of $\mathbf{w}(k)$, since $\mathbf{w}(k)$ is influenced by the time-shifting filters $\mathbf{N}(q)$, $\hat{\mathbf{P}}_2^{-1}(q)$, and $\mathbf{P}_2(q)$. Hence, no direct (explicit) expression for the gradient $\frac{\partial e(k)}{\partial \mathbf{w}}$ results. However, if $\mathbf{w}(k)$ is assumed to vary slowly with respect to $\tilde{\Psi}(k)$, then $\tilde{\Psi}(k)$ and $\mathbf{w}(k)$ can be considered independent [102] such that

$$\begin{aligned} e(k) \approx & H_{(\alpha,n)}^{-1}(q)\mathbf{N}(q)\hat{\mathbf{P}}_2^{-1}(q)\mathbf{y}_1(k) + \left[\mathbf{N}(q)\hat{\mathbf{P}}_2^{-1}(q)\mathbf{P}_2(q)\tilde{\Psi}(k) \right] \mathbf{w}(k) \\ & + H_{(\alpha,n)}^{-1}(q)\mathbf{N}(q)\hat{\mathbf{P}}_2^{-1}(q)\mathbf{n}_1(k). \end{aligned} \quad (4.51)$$

As a result, the derivative of (4.51) with respect to $\mathbf{w}$ becomes

$$\begin{aligned} \frac{\partial e(k)}{\partial \mathbf{w}} & \approx \mathbf{N}(q)\hat{\mathbf{P}}_2^{-1}(q)\mathbf{P}_2(q)\tilde{\Psi}(k) \\ & \approx \mathbf{N}(q)\tilde{\Psi}(k) \quad \text{if } \hat{\mathbf{P}}_2(q) = \mathbf{P}_2(q). \end{aligned} \quad (4.52)$$

Using (4.52), the gradient in (4.30) can be estimated by

$$\left(\frac{\partial J(k)}{\partial \mathbf{w}} \right)^T \approx 2 \left[\mathbf{N}(q)\tilde{\Psi}(k) \right]^T e(k) \quad \text{if } \hat{\mathbf{P}}_2(q) = \mathbf{P}_2(q). \quad (4.53)$$

4.8.2 Derivation of uncertainty bound for convergence

In this appendix a sufficient condition is derived for convergence of the self-tuning algorithm presented in Section 4.5. The derivation is inspired by the work of Wang and Ren [102].

The derivation starts with (4.41), which shows that convergence in terms of mean values is obtained if the eigenvalues λ_i of $\mathbb{E}[\mathbf{I}_6 - \mu(k)\mathbf{G}_1(k)]$ lie within the unit circle. Provided that all eigenvalues of $\mathbf{A} = \mathbb{E}[\mathbf{G}_1(k)]$ have positive real

parts, it follows that a positive step-size $\mu(k)$ can be found such that all $|\lambda_i| < 1$. According to (4.42) this matrix is given by

$$\mathbf{A} = \mathbb{E} \left[[\Psi_{\mathbf{N}}(k)]^T \left[\hat{\mathbf{P}}_2^{-1}(q) \mathbf{P}_2(q) \Psi_{\mathbf{N}}(k) \right] \right]. \quad (4.54)$$

with $\Psi_{\mathbf{N}}(k) = \mathbf{N}(q) \tilde{\Psi}(k)$. Next, define a random vector $\mathbf{L} \in \mathbb{R}^{72}$ and compute

$$\begin{aligned} \mathbf{L}^T (\mathbf{A} + \mathbf{A}^T) \mathbf{L} &= \mathbb{E} \left[[\Xi(k)]^T \left[\hat{\mathbf{P}}_2^{-1}(q) \mathbf{P}_2(q) \Xi(k) \right] \right] \\ &\quad + \mathbb{E} \left[\left[\hat{\mathbf{P}}_2^{-1}(q) \mathbf{P}_2(q) \Xi(k) \right]^T [\Xi(k)] \right], \end{aligned} \quad (4.55)$$

with

$$\Xi(k) = \Psi_{\mathbf{N}}(k) \mathbf{L}. \quad (4.56)$$

The output of (4.55) is scalar valued. Using the property $\mathbf{u}_1^T \mathbf{u}_2 = \text{tr} \{ \mathbf{u}_1 \mathbf{u}_2^T \}$, it follows that

$$\begin{aligned} \mathbf{L}^T (\mathbf{A} + \mathbf{A}^T) \mathbf{L} &= \text{tr} \left\{ \mathbb{E} \left[\Xi(k) \left[\hat{\mathbf{P}}_2^{-1}(q) \mathbf{P}_2(q) \Xi(k) \right]^T \right] \right. \\ &\quad \left. + \mathbb{E} \left[\left[\hat{\mathbf{P}}_2^{-1}(q) \mathbf{P}_2(q) \Xi(k) \right] [\Xi(k)]^T \right] \right\}. \end{aligned} \quad (4.57)$$

Next, Parseval's theorem is used to convert (4.57) to the frequency domain, or

$$\begin{aligned} \mathbf{L}^T (\mathbf{A} + \mathbf{A}^T) \mathbf{L} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \text{tr} \{ \mathbf{S}_{\Xi}(e^{j\omega}) \\ &\quad \left[\left(\hat{\mathbf{P}}_2^{-1}(e^{j\omega}) \mathbf{P}_2(e^{j\omega}) \right)^H + \hat{\mathbf{P}}_2^{-1}(e^{j\omega}) \mathbf{P}_2(e^{j\omega}) \right] \} d\omega, \end{aligned} \quad (4.58)$$

with $\mathbf{S}_{\Xi}(e^{j\omega})$ representing the power spectrum matrix of $\Xi(k)$. In (4.58) the cyclic property for the trace operator, or $\text{tr} \{ ABC \} = \text{tr} \{ BCA \}$, is used to isolate $\mathbf{S}_{\Xi}$. Observe that the integrand of (4.58) is the trace of two matrices. The matrix $\mathbf{S}_{\Xi}$ represents a power spectrum matrix which is per definition positive definite if the input is sufficiently exciting. If now the strictly positive real (SPR) condition [102] is used, or

$$\left(\hat{\mathbf{P}}_2^{-1}(e^{j\omega}) \mathbf{P}_2(e^{j\omega}) \right)^H + \hat{\mathbf{P}}_2^{-1}(e^{j\omega}) \mathbf{P}_2(e^{j\omega}) \succ 0, \quad (4.59)$$

(where $\mathbf{M} \succ 0$ means that the Hermitian matrix $\mathbf{M}$ is positive definite, i.e., all of its eigenvalues are greater than zero) then the trace contains the product of two symmetric positive definite matrices (under the assumption that the input is sufficiently exciting such that $\mathbf{S}_{\Xi} \succ 0$). Using (4.59), it follows that

$$\mathbf{L}^T (\mathbf{A} + \mathbf{A}^T) \mathbf{L} > 0 \quad \text{for } \mathbf{L} \neq \mathbf{0}. \quad (4.60)$$

Now, reconsider the mean weight update equation in (4.41). Define $\mathbb{E}[\Delta \mathbf{w}(k)] = \mathbf{x}(k)$ and assume there is no sensor noise and process noise such that $\mathbf{G}_2(k) = \mathbf{G}_3(k) = \mathbf{0}$. Then, (4.41) reduces to

$$\mathbf{x}(k+1) = [\mathbf{I} - \mu \mathbf{A}] \mathbf{x}(k). \quad (4.61)$$

Next, a Lyapunov function $V(k) = \mathbf{x}^T(k) \mathbf{x}(k) \geq 0$ is defined, with

$$\begin{aligned} \Delta V(k) &= V(k+1) - V(k) \\ &= \mu^2 \mathbf{x}^T \mathbf{A}^T \mathbf{A} \mathbf{x} - \mu \mathbf{x}^T (\mathbf{A} + \mathbf{A}^T) \mathbf{x}. \end{aligned} \quad (4.62)$$

Since $\mathbf{x}^T \mathbf{A}^T \mathbf{A} \mathbf{x} > 0$ because of its quadratic form, and $\mathbf{x}^T (\mathbf{A} + \mathbf{A}^T) \mathbf{x} > 0$ because of (4.60), it holds that there always exists a sufficiently small μ such that $\mu^2 \mathbf{x}^T \mathbf{A}^T \mathbf{A} \mathbf{x} < \mu \mathbf{x}^T (\mathbf{A} + \mathbf{A}^T) \mathbf{x}$. Then, it follows from combining (4.62) and (4.60), and for sufficiently small μ , that $\Delta V(k) < 0$ which implies that the self-tuning algorithm converges.

4.8.3 Derivation of expected values and optimal convergence speed

This appendix shows that for the conditions $\hat{\mathbf{P}}_2 = \mathbf{P}_2$, $\mathbf{N} = \mathbf{I}_6$, $\beta = \frac{1}{\|H_{(\alpha,n)}(q)\|_2}$ with $\|H_{(\alpha,n)}(q)\|_2 \in \mathbb{R}$, and $\mathbf{a}_0, \mathbf{n}_0, \mathbf{n}_1$ are ZMWN processes satisfying the properties from Definition 4.3, the following two properties hold: (1) $\mathbb{E}[\mathbf{G}_1(k)] = (\sigma_{a_0}^2 + \sigma_{n_0}^2) \mathbf{I}_{72}$, and (2) uniform convergence speed of the parameters $\mathbf{w}(k)$ is obtained in terms of mean values.

Under the given assumptions, $\mathbb{E}[\mathbf{G}_1(k)]$ in (4.41) reduces to

$$\begin{aligned} \mathbb{E}[\mathbf{G}_1(k)] &= \mathbb{E}[\tilde{\Psi}^T(k) \tilde{\Psi}(k)] \\ &= \mathbb{E} \begin{bmatrix} \tilde{\psi}(k) \tilde{\psi}^T(k) & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & \tilde{\psi}(k) \tilde{\psi}^T(k) \end{bmatrix}. \end{aligned} \quad (4.63)$$

Because $\mathbb{E}[\tilde{\Psi}^T(k) \tilde{\Psi}(k)]$ is block-diagonal, the eigenvalues of $\mathbb{E}[\tilde{\Psi}^T(k) \tilde{\Psi}(k)]$ are determined by the eigenvalues of $\mathbb{E}[\tilde{\psi}(k) \tilde{\psi}^T(k)]$. From the definition of $\tilde{\psi}(k)$ in (4.37), the following expression is derived:

$$\begin{aligned} \mathbb{E}[\tilde{\psi}(k) \tilde{\psi}^T(k)] &= \\ \mathbb{E} \left[\begin{array}{c|c} \tilde{\mathbf{a}}_0(k) \tilde{\mathbf{a}}_0^T(k) & \tilde{\mathbf{a}}_0(k) [\beta H_{(\alpha,n)}(q) \tilde{\mathbf{a}}_0(k)]^T \\ \hline [\beta H_{(\alpha,n)}(q) \tilde{\mathbf{a}}_0(k)] \tilde{\mathbf{a}}_0^T(k) & [\beta H_{(\alpha,n)}(q) \tilde{\mathbf{a}}_0(k)] [\beta H_{(\alpha,n)}(q) \tilde{\mathbf{a}}_0(k)]^T \end{array} \right]. \end{aligned} \quad (4.64)$$

Assuming that $\tilde{\mathbf{a}}_0 = \mathbf{a}_0 + \mathbf{n}_0$ consists of uncorrelated ZMWN signals $\mathbf{a}_0, \mathbf{n}_0$, it follows that the upper left part of (4.64) may be written as

$$\mathbb{E} [\tilde{\mathbf{a}}_0(k) \tilde{\mathbf{a}}_0^T(k)] = (\sigma_{a_0}^2 + \sigma_{n_0}^2) \mathbf{I}_6. \quad (4.65)$$

If the discrete-time weak integrator $H_{(\alpha,n)}(q)$ is derived by discretization of (4.16) using the zero-order-hold method, the relative degree of $H_{(\alpha,n)}(q)$ becomes one, which implies that there is no direct feedthrough term in the output equation of the filter. As a result, $H_{(\alpha,n)}(q)\mathbf{a}_0(k)$ can be expressed as a function of $\tilde{\mathbf{a}}_0(k-1), \dots, \tilde{\mathbf{a}}_0(k-n)$. Given Definition 4.3, it follows that $\mathbb{E} [\tilde{\mathbf{a}}_0(k) \tilde{\mathbf{a}}_0^T(k-i)] = \mathbf{0}$ for $i = 1, 2, \dots$, and therefore

$$\mathbb{E} [\tilde{\mathbf{a}}_0(k) [H_{(\alpha,n)}(q)\tilde{\mathbf{a}}_0(k)]^T] = \mathbf{0}, \quad (4.66)$$

such that the upper right and lower left parts of (4.64) are zero too. Finally, an expression for the lower right part of the matrix in (4.64) can be obtained using Parseval's theorem, or

$$\begin{aligned} & \mathbb{E} [\beta H_{(\alpha,n)}(q) \tilde{\mathbf{a}}_0(k) [\beta H_{(\alpha,n)}(q) \tilde{\mathbf{a}}_0(k)]^T] \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \beta^2 (H_{(\alpha,n)}(e^{j\omega})(\sigma_{a_0}^2 + \sigma_{n_0}^2) \mathbf{I}_6 H_{(\alpha,n)}^*(e^{j\omega})) d\omega \\ &= \frac{\beta^2 (\sigma_{a_0}^2 + \sigma_{n_0}^2)}{2\pi} \mathbf{I}_6 \int_{-\pi}^{\pi} (H_{(\alpha,n)}(e^{j\omega}) H_{(\alpha,n)}^*(e^{j\omega})) d\omega \\ &= \beta^2 (\sigma_{a_0}^2 + \sigma_{n_0}^2) \mathbf{I}_6 \|H_{(\alpha,n)}(q)\|_2^2. \end{aligned}$$

With $\beta = \frac{1}{\|H_{(\alpha,n)}(q)\|_2}$ it follows that

$$\mathbb{E} [\beta H_{(\alpha,n)}(q) \tilde{\mathbf{a}}_0(k) [\beta H_{(\alpha,n)}(q) \tilde{\mathbf{a}}_0(k)]^T] = (\sigma_{a_0}^2 + \sigma_{n_0}^2) \mathbf{I}_6, \quad (4.67)$$

Combining the results of (4.65), (4.66) and (4.67) it is found that

$$\mathbb{E} [\tilde{\boldsymbol{\psi}}(k) \tilde{\boldsymbol{\psi}}^T(k)] = (\sigma_{a_0}^2 + \sigma_{n_0}^2) \mathbf{I}_{12}, \quad (4.68)$$

hence all eigenvalues of $\mathbb{E} [\tilde{\boldsymbol{\psi}}(k) \tilde{\boldsymbol{\psi}}^T(k)]$ equal $(\sigma_{a_0}^2 + \sigma_{n_0}^2)$, which implies uniform convergence speed for all parameters. Furthermore, if the result of (4.68) is used in (4.63), it follows that $\mathbb{E} [\mathbf{G}_1(k)] = (\sigma_{a_0}^2 + \sigma_{n_0}^2) \mathbf{I}_{72}$.

4.8.4 Derivation of the bias

An expression for the bias is obtained as follows. Assuming the self-tuning algorithm is stable (Section 4.5.3), it holds that $\mathbb{E} [\Delta \mathbf{w}(k+1)] \rightarrow \mathbb{E} [\Delta \mathbf{w}(k)]$ as $k \rightarrow \infty$. Then, from (4.41) it can be seen that

$$\mathbb{E} [\Delta \mathbf{w}(\infty)] = \mathbb{E} [\mathbf{G}_1(\infty)]^{-1} (\mathbb{E} [\mathbf{G}_2(\infty)] + \mathbb{E} [\mathbf{G}_3(\infty)]). \quad (4.69)$$

For the simplified case $\mathbf{N} = \mathbf{I}_6$, $\hat{\mathbf{P}}_2 = \mathbf{P}_2$, setting $\beta = \frac{1}{\|H_{(\alpha,n)}(q)\|_2}$ as proposed in Section 4.5.3, and assuming that $\mathbf{a}_0$, $\mathbf{n}_0$, and $\mathbf{n}_1$ are uncorrelated ZMWN processes satisfying the properties from Definition 4.3, the terms in (4.69) are given by

$$\mathbb{E}[\mathbf{G}_1(\infty)] = \mathbb{E}\left[\tilde{\Psi}^T(k)\tilde{\Psi}(k)\right] = (\sigma_{a_0}^2 + \sigma_{n_0}^2)\mathbf{I}_{72}, \quad (4.70)$$

$$\begin{aligned} \mathbb{E}[\mathbf{G}_2(\infty)] &= \mathbb{E}\left[\tilde{\Psi}^T(k)(\Psi(k) - \tilde{\Psi}(k))\right]\mathbf{w}^* \\ &= -\sigma_{n_0}^2\mathbf{w}^*, \end{aligned} \quad (4.71)$$

$$\mathbb{E}[\mathbf{G}_3(\infty)] = -\mathbb{E}\left[\left[\tilde{\Psi}(k)\right]^T \left[H_{(\alpha,n)}^{-1}(q)\hat{\mathbf{P}}_2^{-1}(q)\epsilon_0(k)\right]\right]. \quad (4.72)$$

The derivations regarding (4.70) and (4.71) are given in 4.8.3. Eq. (4.72) is obtained from (4.40) as follows. Given $\tilde{\mathbf{a}}_0 = \mathbf{a}_0 + \mathbf{n}_0$ it follows that $\tilde{\Psi}(k)$ is affine in $\mathbf{n}_0, \mathbf{a}_0$. Moreover, the term $\left[H_{(\alpha,n)}^{-1}(q)\mathbf{N}(q)\hat{\mathbf{P}}_2^{-1}(q)\mathbf{n}_1(k)\right]$ is affine in $\mathbf{n}_1$. Since $\mathbf{n}_1$ is assumed to be uncorrelated with $\mathbf{a}_0, \mathbf{n}_0$, the term $\mathbb{E}[\mathbf{G}_3(k)]$ no longer depends on $\mathbf{n}_1$, thus $\mathbf{n}_1$ will not contribute to bias.

Substitution of (4.70), (4.71), and (4.72) in (4.69) gives an expression for the bias in the parameter estimates, or

$$\begin{aligned} \mathbb{E}[\Delta\mathbf{w}(\infty)] &= -\frac{\sigma_{n_0}^2}{\sigma_{a_0}^2 + \sigma_{n_0}^2}\mathbf{w}^* - \frac{1}{\sigma_{a_0}^2 + \sigma_{n_0}^2} \cdot \\ &\quad \mathbb{E}\left[\left[\tilde{\Psi}(k)\right]^T \left[H_{(\alpha,n)}^{-1}(q)\hat{\mathbf{P}}_2^{-1}(q)\epsilon_0(k)\right]\right]. \end{aligned} \quad (4.73)$$

4.9 Appendix B: Numerical values for the model

Global mass matrix:

$$\mathbf{M}_f = \begin{bmatrix} \mathbf{M}_{f,1} & \mathbf{M}_{f,2} \end{bmatrix}$$

with

$$\mathbf{M}_{f,1} = \begin{bmatrix} 9.05 & 0 & 0 & 0 & 0.49 & 0 \\ 0 & 9.05 & 0 & -0.49 & 0 & 0 \\ 0 & 0 & 9.05 & 0 & 0 & 0 \\ 0 & -0.49 & 0 & 0.17 & 0 & 0 \\ 0.49 & 0 & 0 & 0 & 0.17 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.15 \\ -2.82 & 2.82 & 0 & -0.4 & -0.4 & 0 \\ 12.1 & 12.1 & 0 & -2.14 & 2.14 & 0 \\ -10.07 & -10.07 & 0 & 1.43 & -1.43 & 0.62 \\ -7.25 & -7.25 & 0 & 1.03 & -1.03 & 0.62 \\ 12.1 & -12.1 & 14.24 & 2.14 & 2.14 & 0 \\ 12.1 & -12.1 & 10.25 & 2.14 & 2.14 & 0 \end{bmatrix}$$

$$\mathbf{M}_{f,2} = \begin{bmatrix} -2.82 & 12.1 & -10.07 & -7.25 & 12.1 & 12.1 \\ 2.82 & 12.1 & -10.07 & -7.25 & -12.1 & -12.1 \\ 0 & 0 & 0 & 0 & 14.24 & 10.25 \\ -0.4 & -2.14 & 1.43 & 1.03 & 2.14 & 2.14 \\ -0.4 & 2.14 & -1.43 & -1.03 & 2.14 & 2.14 \\ 0 & 0 & 0.62 & 0.62 & 0 & 0 \\ 3.99 & 0 & 0 & 0 & -17.11 & -17.11 \\ 0 & 95.06 & -61.11 & -44 & 0 & 0 \\ 0 & -61.11 & 73.15 & 58.91 & 0 & 0 \\ 0 & -44 & 58.91 & 48.66 & 0 & 0 \\ -17.11 & 0 & 0 & 0 & 145.92 & 131.68 \\ -17.11 & 0 & 0 & 0 & 131.68 & 121.43 \end{bmatrix}$$

Global damping matrix:

$$\mathbf{D}_f = \begin{bmatrix} \mathbf{D}_{f,1} & \mathbf{D}_{f,2} \end{bmatrix}$$

$$\mathbf{K}_{f,2} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 4.78 \cdot 10^8 & 0 & 0 & 0 & 0 & 0 \\ 0 & 7.07 \cdot 10^7 & 0 & 0 & 0 & 0 \\ 0 & 0 & 8.33 \cdot 10^9 & 6.62 \cdot 10^9 & 0 & 0 \\ 0 & 0 & 6.62 \cdot 10^9 & 5.40 \cdot 10^9 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2.58 \cdot 10^8 & 2.06 \cdot 10^8 \\ 0 & 0 & 0 & 0 & 2.06 \cdot 10^8 & 1.68 \cdot 10^8 \end{bmatrix}$$

Input damping matrix:

$$\mathbf{D}_0 = \begin{bmatrix} 104.0 & 0 & 0 & 0 & -0.85 & 0 \\ 0 & 104.0 & 0 & 0.85 & 0 & 0 \\ 0 & 0 & 105.0 & 0 & 0 & 0 \\ 0 & 0.85 & 0 & 2.1 & 0 & 0 \\ -0.85 & 0 & 0 & 0 & 2.1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 8.32 \end{bmatrix}$$

Input stiffness matrix:

$$\mathbf{K}_0 = \begin{bmatrix} 117800 & 0 & 0 & 0 & 0 & 0 \\ 0 & 117800 & 0 & 0 & 0 & 0 \\ 0 & 0 & 169794 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3279 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3279 & 0 \\ 0 & 0 & 0 & 0 & 0 & 8685 \end{bmatrix}$$

Input force matrix:

$$\mathbf{B} = \begin{bmatrix} 0.577 & -0.577 & -0.789 & 0.789 & 0.211 & -0.211 \\ 0.577 & -0.577 & 0.211 & -0.211 & -0.789 & 0.789 \\ -0.577 & -0.577 & -0.577 & -0.577 & -0.577 & -0.577 \\ 0.086 & 0.077 & -0.110 & -0.113 & 0.023 & 0.036 \\ 0.077 & 0.086 & 0.036 & 0.023 & -0.113 & -0.110 \\ 0.163 & -0.163 & 0.163 & -0.163 & 0.163 & -0.163 \end{bmatrix}$$

Transformation matrix:

$$\mathbf{R}_1 = \begin{bmatrix} 0.305 & -0.272 & -0.388 & 0.400 & 0.083 & -0.129 \\ 0.272 & -0.305 & 0.129 & -0.083 & -0.400 & 0.388 \\ -0.289 & -0.289 & -0.289 & -0.289 & -0.289 & -0.289 \\ 2.041 & 2.041 & -2.788 & -2.788 & 0.747 & 0.747 \\ 2.041 & 2.041 & 0.747 & 0.747 & -2.788 & -2.788 \\ 1.021 & -1.021 & 1.021 & -1.021 & 1.021 & -1.021 \end{bmatrix}$$

Chapter 5

Compensation for internal isolator dynamics

This chapter presents a disturbance feedforward control strategy for active vibration isolation systems that have internal isolator dynamics. The structure of the feedforward controller follows from modeling, but to achieve robust performance the tuning of the controller parameters will be based on measurement data. More specifically, the controller poles, which correspond to internal isolator dynamics, are obtained offline from system identification experiments, while the controller zeros are obtained from online self-tuning using measured sampled-data systems. Self-tuning is used to reduce parameter uncertainty when having too limited identification possibilities, and to account automatically for the effect of noise amplification. Measurement results obtained from an industrial vibration isolation system shows that residual vibrations are largely reduced up to the level of output noise limitations.¹

¹This chapter is based on:

M.A. Beijen, M.F. Heertjes, H. Butler, and M. Steinbuch, *Disturbance feedforward control for active vibration isolation systems with internal isolator dynamics*, submitted to the Journal of Sound and Vibration, 2018.

5.1 Introduction

Vibration isolation systems are widely used in high-precision machines to achieve the often extreme demands on accuracy [19, 53, 92, 3]. The basic idea is to create a low suspension stiffness between an isolated payload and a corresponding vibrating base. As a result, passive vibration isolation is provided beyond the low suspension frequency. However, systems with low suspension frequencies generally have internal isolator dynamics at relatively low frequencies that may compromise isolation performance. An example is given by an air mount system [60, 75, 5], which achieves a low stiffness using large air tank volumes. Systems with large air tanks generally lead to low internal resonance frequencies within the air tank as a result of air acoustics, as will be shown later on in this chapter. Another example is found in isolators using metal springs such as coil springs [103] or leaf springs [90]. For such springs, the internal resonance frequency corresponding to the first structural eigenmode is generally proportional to the suspension frequency [42, 111].

Several methods have been proposed to improve performance of vibration isolators with internal isolator dynamics. In [27], dynamic vibration absorbers (DVAs) are proposed to counteract internal resonances, but these absorbers require hardware modifications to the isolator. Alternatively, feedback control strategies for active vibration isolation are found in [73, 45]. Feedback control, however, may associate with a poor signal-to-noise ratio (SNR), because the sensor is placed on the payload which by itself is passively isolated. Moreover, a high bandwidth is required to control internal isolator dynamics, which can result in stability problems and large amplification of sensor noise. A feasible solution is then given by disturbance feedforward control that uses measurements of the base vibrations as controller input instead and therefore has a much better SNR. However, to provide an effective feedforward control force an accurate model of the vibration isolator is needed. In Chapter 4, feedforward controllers are presented which rely on a spring-damper model for the isolator. This model may be suitable for isolators with a relatively high suspension frequency and as such having internal isolator dynamics occurring well beyond the frequency range of interest. For isolators with a low suspension frequency, however, internal isolator dynamics should be included in the feedforward controller model, because these usually impact performance within the frequency range of interest.

To obtain accurate models for mechanical systems [71] such as vibration isolators, one generally resides to system identification tools. For heavy-weight industrial vibration isolation systems, however, system identification may give unsatisfactory results, because it is difficult (if not impossible) to sufficiently excite the base of the system, see Chapter 3. To avoid these problems, self-tuning feedforward controllers then provide an efficient solution to estimate the controller parameters online while keeping the model-based controller structure intact. In Chapter 4, a self-tuning algorithm was presented to estimate the

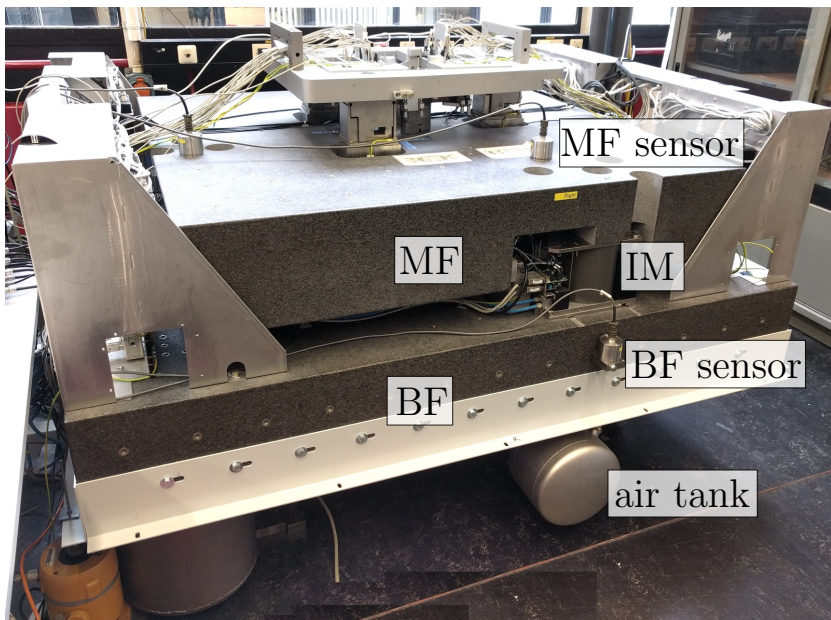


Figure 5.1. industrial vibration isolation system consisting of an isolated metrology frame (MF), base frame (BF), and three isolation modules (IMs) with external air tanks. The MF dimensions are $1.50 \times 1.40 \times 0.26$ m, and has a mass of 1300 kg.

stiffness and damping properties induced by air mounts, but this approach did not include the internal isolator dynamics. Differently, [56, 29, 95] have considered adaptive feedforward control using FIR filters, with the disadvantage of requiring many adaptive parameters to accurately estimate low-frequency poles and lightly damped resonances. In [58] an adaptive IIR filter is proposed that generally needs much less parameters compared to FIR filters, but the IIR filter can become unstable if the filter poles are shifted outside the unit circle during adaptation. The solution that is pursued in this chapter considers an IIR filter with fixed poles [66], also called a generalized FIR filter [23].

This chapter has two main contributions. The first contribution is an extension of the Multi-Input Multi-Output (MIMO) feedforward controller design from Chapter 4. The extension involves the inclusion of an arbitrary number of pre-selected poles from FRF measurements, while the zeros are obtained from data-based self-tuning. The second contribution is the application of the MIMO feedforward control strategy to an industrial multi-axis vibration isolation system that contains air mounts having performance-limiting internal isolator dynamics. By measurements obtained from a real industrial system, significant improvements in vibration isolation performance are obtained with respect to

systems that only use feedback control and feedforward control consisting of spring-damper compensation. These improvements are demonstrated in the frequency range of interest, i.e. in the interval 1–100 Hz.

The remainder of this chapter is organized as follows. The industrial vibration isolation system along with the disturbance feedforward control setting are discussed in Section 5.2. The control problem is formulated in Section 5.3, and the self-tuning algorithm is presented in Section 5.4. An experimental validation using measurement results obtained from the industrial vibration isolation system is given in Section 5.5, and the main conclusions are summarized in Section 5.6.

5.2 Vibration isolation system

An example of an industrial vibration isolation system is shown in Figure 5.1. This system basically consists of: (a) 6 degree-of-freedom movable metrology frame (MF), (b) base frame (BF), and (c) three isolation modules (IMs) that are installed between the base frame and the metrology frame to isolate the latter from base frame vibrations. Each isolation module contains a pneumatically controlled dual-chamber air mount system that provides gravity compensation, damping, and is characterized by a low suspension stiffness. To enable active vibration isolation, each isolation module is equipped with two Lorentz motors: one to apply horizontal forces and one to apply vertical forces. Furthermore, both the base frame and the metrology frame are equipped with accelerometers to measure the vertical accelerations at the isolation module locations. The accelerometers contain analog band-pass filters with an operating range from 0.1 to 450 Hz. The considered frequency range of interest is from 1 to 100 Hz, which falls inside the range of the band-pass filters and which is representative for high-precision applications such as the wafer scanning industry.

5.2.1 Internal isolator dynamics

A simplified schematic representation of the considered vibration isolation system is depicted in Figure 5.2. Each isolation module is considered as a dual-chamber air mount system that acts as a single-axis vibration isolator in vertical direction. The lower chamber V_1 corresponds to an air tank as shown in Figure 5.1, while the upper chamber V_2 corresponds to the air volume inside an isolation module. V_2 is sealed with a freely moving piston that supports the metrology frame whose movement associates with the residual acceleration denoted by a_1 . The lower and upper chamber are connected via a small air neck. Actuator force u can be used for active vibration isolation. Define $A_0(s)$, $A_1(s)$, and $U(s)$ as the Laplace transforms of a_0 , a_1 , and u , respectively. Using A_0 , U

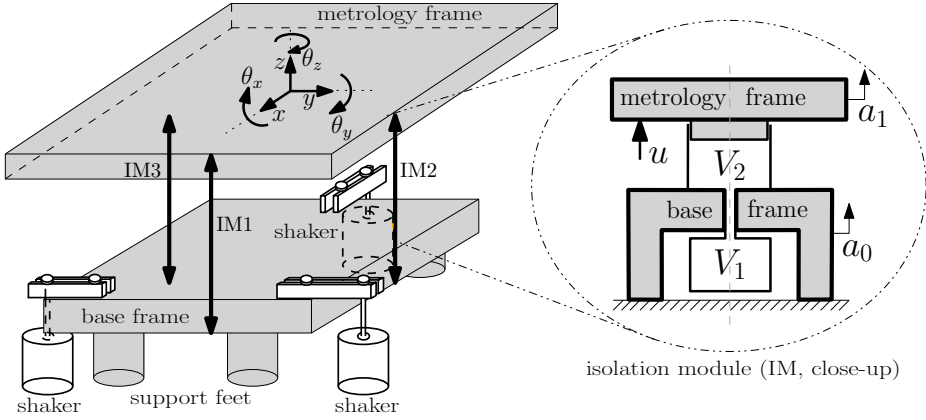


Figure 5.2. simplified model; a_0 and a_1 represent base frame (BF) and metrology frame (MF) accelerations, respectively; each isolation module (IM) is regarded as a dual-chamber air mount with air tanks V_1 and V_2 ; the actuator force u acts directly on the MF.

as inputs and A_1 as output, the input-output relations are given by

$$\begin{aligned}
 P_1(s) &:= \frac{A_1(s)}{A_0(s)} \approx \underbrace{\frac{ds + k}{ms^2 + ds + k}}_{\text{suspension dynamics}} \underbrace{\frac{s^2 + \zeta_{ar}\omega_{ar}s + \omega_{ar}^2}{s^2 + \zeta_r\omega_r s + \omega_r^2} \frac{\omega_{ar}^2}{\omega_r^2}}_{\text{internal isolator dynamics}}, \\
 P_2(s) &:= \frac{A_1(s)}{U(s)} \approx \frac{s^2}{ms^2 + ds + k},
 \end{aligned} \tag{5.1}$$

see also Appendix A in Section 5.7. Bode plots for P_1, P_2 are shown in Figure 5.3. This figure clearly shows the suspension mode at 1 Hz in both P_1 and P_2 . Moreover, the plot of P_1 shows the presence of internal isolator dynamics, i.e. the effect of air tank acoustics due to a Helmholtz resonator effect. This follows from the anti-resonance at 44 Hz and the resonance at 99 Hz. In P_2 , the effect of air acoustics is not visible. Physically this makes sense, because the actuator force is applied in parallel to the effect induced by air acoustics in the tanks. And since the metrology frame mass is much larger than the total (resonating) air mass inside the tanks, the impact of the air mass on the metrology frame is negligible compared to u . Different from input u , input a_0 is transmitted to the metrology frame via the air tanks, hence a series connection. As such, the effect of air acoustics is clearly visible in P_1 .

5.2.2 Active vibration isolation

The goal of active vibration isolation is to improve the rejection of base frame disturbances by control. In this chapter, only feedforward control is considered,

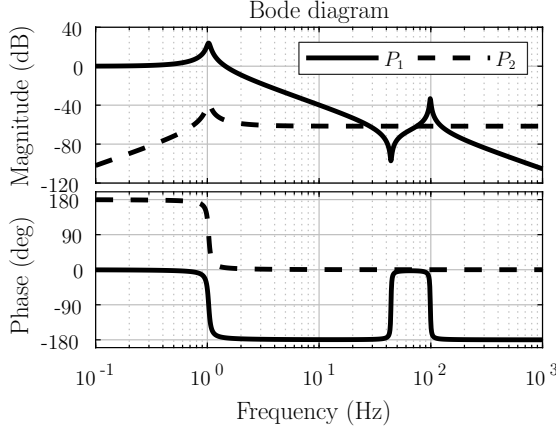


Figure 5.3. Bode plots for $P_1(j\omega) = A_1(j\omega)/A_0(j\omega)$ (solid), and $P_2(s) = A_1(j\omega)/U(j\omega)$ (dashed) according to (5.1) with $m = 1200$ kg, $d = 500$ Ns/m, $k = 50 \cdot 10^3$ N/m, $\omega_{ar} = 2\pi \cdot 44$ rad/s, $\zeta_{ar} = 0.01$, $\omega_r = 2\pi \cdot 99$ rad/s, $\zeta_r = 0.01$.

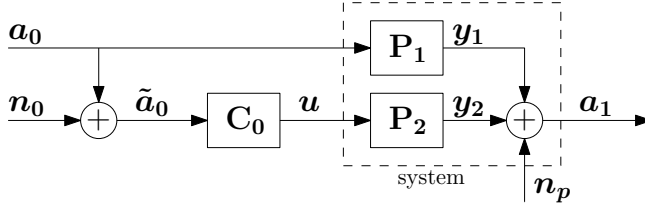


Figure 5.4. block diagram of a vibration isolation system with disturbance feedforward control.

for which the general MIMO block diagram is shown in Figure 5.4. The base frame acceleration $\mathbf{a}_0(t) \in \mathbb{R}^{n_r}$ is considered as input disturbance while the metrology frame acceleration $\mathbf{a}_1 = \mathbf{y}_1 + \mathbf{y}_2 + \mathbf{n}_p$ is considered as the signal to be minimized in some norm-based sense. The latter is formed by three signals: $\mathbf{y}_1(t) \in \mathbb{R}^{n_y}$ is caused by $\mathbf{a}_0$ via the primary path $\mathbf{P}_1$, $\mathbf{y}_2(t) \in \mathbb{R}^{n_y}$ represents the impact of the control signal $\mathbf{u}(t) \in \mathbb{R}^{n_u}$ transferred via the secondary path $\mathbf{P}_2$, and $\mathbf{n}_p(t) \in \mathbb{R}^{n_y}$ represents metrology frame accelerations due to the process noise caused by direct disturbance forces, actuator noise, et cetera. The feed-forward controller $\mathbf{C}_0$ provides disturbance feedforward control based on the measured signal $\tilde{\mathbf{a}}_0 = \mathbf{a}_0 + \mathbf{n}_0$, representing the base frame acceleration with additive measurement noise represented by $\mathbf{n}_0(t) \in \mathbb{R}^{n_r}$. In the Laplace domain, the signal $\mathbf{a}_1$ can be written as

$$\mathbf{A}_1(s) = \mathcal{T}(s)\mathbf{A}_0(s) + \mathcal{S}_0(s)\mathbf{N}_0(s) + \mathbf{N}_p(s), \quad (5.2)$$

with closed-loop transmissibility matrix $\mathcal{T}$ and noise sensitivity matrix $\mathcal{S}_0$ defined as

$$\mathcal{T}(s) = \mathbf{P}_1(s) + \mathbf{P}_2(s)\mathbf{C}_0(s), \quad \mathcal{S}_0(s) = \mathbf{P}_2(s)\mathbf{C}_0(s). \quad (5.3)$$

The transmissibility matrix $\mathcal{T}$ is a measure for vibration isolation performance, whereas the noise sensitivity matrix $\mathcal{S}_0$ is a measure for noise amplification.

5.3 Problem formulation

The control problem can now be formulated as follows. Given a set of (possibly correlated and colored) disturbances $\mathbf{A}_0(j\omega)$, find a feedforward controller $\mathbf{C}_0$ such that the power of $\mathbf{A}_1(j\omega)$ is minimized in the frequency range of interest, i.e. in the frequency interval between 1–100 Hz. Minimization will be done for a vibration isolation system that has internal isolator dynamics. Under the assumptions that $n_y = n_u$ such that $\mathbf{P}_2$ is a square system, $\mathbf{a}_0$ and $\mathbf{n}_p$ are uncorrelated, and in the absence of $\mathbf{n}_0$, perfect cancellation of base frame vibrations is obtained if $\mathcal{T} = \mathbf{0}$ in (5.3). A controller that achieves perfect cancellation is given by

$$\mathbf{C}_0(s) = -\mathbf{P}_2^{-1}(s)\mathbf{P}_1(s), \quad (5.4)$$

which follows from substitution of (5.4) in (5.3) and is known as the Wiener solution [94]. However, a discrete-time implementation of the design in (5.4) generally results in a non-causal and unstable feedforward controller due to time delays and non-minimum phase zeros in $\mathbf{P}_2$. In the context of disturbance feedforward control such unfavorable properties should be avoided. In practice, the aim is therefore to obtain the best causal and stable approximation of (5.4) [34]. Such an approximation is generally unable to improve performance at all frequencies, see Chapter 2. Therefore, an additional frequency weighting Φ is considered to optimize for a specific frequency range of interest. The approximated solution is then given by [94]

$$\mathbf{C}_0(z) = -\mathbf{P}_{2,o}^{-1}(z) \{ \mathbf{P}_{2,i}^*(z)\mathbf{P}_1(z)\Phi(z)\Phi_{ci}^*(z) \}_+ \Phi_{co}^{-1}(z), \quad (5.5)$$

with $z \in \mathbb{C}$. In (5.5), inner-outer factorization is applied such that $\mathbf{P}_2(z) = \mathbf{P}_{2,i}(z)\mathbf{P}_{2,o}(z)$. The outer factor $\mathbf{P}_{2,o}$ contains all the dynamics of $\mathbf{P}_2$ that can be inverted in a stable and causal manner. All remaining dynamics are included in the inner factor $\mathbf{P}_{2,i}$. Note that $\mathbf{P}_{2,i}^*(z) = \mathbf{P}_{2,i}^T(z^{-1})$ represents the adjoint which is the transposed and time-reversed system [32]. Similarly, co-inner-outer factorization is applied to Φ such that $\Phi(z) = \Phi_{co}(z)\Phi_{ci}(z)$ with co-outer-factor Φ_{co} and co-inner-factor Φ_{ci} . The causality operator $\{ \cdot \}_+$ is used in (5.5) to find the best causal and stable approximation of the non-invertible part. For more information regarding (co-)inner-outer factorization and the causality operator, the reader is referred to [32].

Although (5.5) shows a model-based solution to the feedforward control problem, two practical aspects limit a straightforward implementation of (5.5). First, (5.5) needs parametric models for $\mathbf{P}_1$ and $\mathbf{P}_2$. Model estimates of $\mathbf{P}_1$ are only available with limited accuracy because in identification experiments it is often difficult (if not impossible) to sufficiently excite the base frame, see Chapter 3. This impedes an accurate model-based design of $\mathbf{C}_0$. Second, (5.5) does not consider the noise contribution $\mathbf{S}_0\mathbf{N}_0$, while minimization of $\mathbf{A}_1$ in (5.2) implies finding a balanced tradeoff between minimizing $\mathcal{T}\mathbf{A}_0$ and $\mathbf{S}_0\mathbf{N}_0$. To deal with these two practical aspects, self-tuning is proposed as a solution. Namely, through self-tuning the necessity of requiring an accurate model for $\mathbf{P}_1$ is avoided, whereas the tradeoff between $\mathcal{T}\mathbf{A}_0$ and $\mathbf{S}_0\mathbf{N}_0$ is automatically considered depending on the significance of each contribution in the data. Also, a stable and causal $\mathbf{C}_0$ is guaranteed by fixing the controller poles a priori in so-called basis functions. The controller zeros are obtained online via self-tuning, which will be discussed in Section 4.

5.3.1 Example: simplified SISO air mount control

Reconsider the simplified model for the single-axis air mount system given in (5.1). A perfect feedforward controller for such an air mount system is given by substitution of (5.1) in (5.4), giving

$$C_0(s) = \underbrace{\frac{ds + k}{s^2}}_{\text{spring-damper compensation}} \cdot \underbrace{\frac{s^2 + \zeta_{ar}\omega_{ar}s + \omega_{ar}^2}{s^2 + \zeta_r\omega_r s + \omega_r^2} \frac{\omega_r^2}{\omega_{ar}^2}}_{\text{internal isolator dynamics compensation}}. \quad (5.6)$$

The first rational transfer function in (5.6) describes the compensation of the fundamental stiffness and damping properties of the air mount system, similar to the spring-damper compensation as proposed in Chapter 4. The second rational transfer function in (5.6) can be seen as the compensation part that accounts for internal isolator dynamics.

Remark 5.1. As shown in (5.6), the disturbance feedforward controller requires two poles at $s = 0$, hence it has two pure integrators. However, since the input to $\mathbf{C}_0$ is a measured acceleration signal, pure integrators are generally undesired, because they may induce the amplification of low-frequency sensor noise and signal drift. To circumvent this problem, the pure integrators will be replaced by n^{th} -order weak integrators $H_{(\alpha,n)}$ as proposed in Chapter 4, or

$$H_{(\alpha,n)} = \frac{1 - \left(\frac{\alpha}{s+\alpha}\right)^n}{s}, \quad (5.7)$$

with cut-off frequency $\alpha \in \mathbb{R}$ and order $n \geq 1$. Large n implies large roll-off obtained in $\mathcal{T}$ for frequencies beyond α , but more amplification of low-frequency sensor noise. In this chapter, $n = 3$ and $\alpha = 2\pi \cdot 0.5 \text{ rad/s}$ (0.5 Hz).

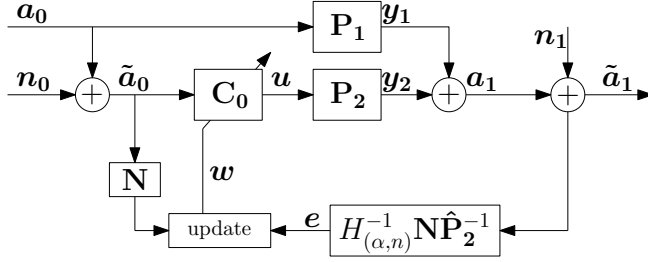


Figure 5.5. layout of the self-tuning controller using the FeLMS algorithm with residual noise shaping, see also Chapter 4.

5.4 Self-tuning disturbance feedforward control

In this section, the self-tuning feedforward controller C_0 is presented. The key idea is that the controller poles are obtained from a priori measurements and will be embedded in basis functions, while the zeros are obtained during a self-tuning process. For a simple spring-damper compensation, a similar MIMO self-tuning control approach is presented in Chapter 4, see the block diagram in Figure 5.5, but without addressing internal isolator dynamics. To address these dynamics in the self-tuning controller structure, an extension to the approach in Chapter 4 is required that will be presented in Section 5.4.1. The update law for the self-tuning parameters is shortly summarized in Section 5.4.2. A method to construct the basis functions is presented in Section 5.4.3, and design guidelines for the noise shaping filter N are presented in Section 5.4.4.

5.4.1 Self-tuning controller structure

Consider Figure 5.6 that shows the proposed structure in the form of a generalized FIR filter for the j^{th} controller signal u_j , with $j \in \{1, \dots, n_u\}$. The input consists of the measured base frame accelerations $\tilde{\mathbf{a}}_0(k) \in \mathbb{R}^{n_r}$, with $k \in \mathbb{N}$ referring to time samples $t_k = kT_s$ with sampling time T_s . The poles are stored in basis functions $\mathcal{B}^{[i]}$, with $i \in \{1, \dots, n_p + 2\}$. The self-tuning parameter vectors are given by $\mathbf{w}_j^{[i]}(k) \in \mathbb{R}^{n_r}$. Figure 5.6 basically consists of two blocks. The left block is used for compensating internal isolator dynamics and represents the extension with respect to Chapter 4. The right block represents the straightforward spring-damper compensation as can be found in Chapter 4. Without compensation for internal isolator dynamics ($n_p = 0$), the structure reduces to a spring-damper compensation scheme where $\mathcal{B}^{[n_p+1]} = 1$, $\mathcal{B}^{[n_p+2]} = H_{(\alpha,n)}$ acting as a weak integrator, and $\mathbf{w}_j^{[n_p+1]}$ and $\mathbf{w}_j^{[n_p+2]}$ correspond to the estimated damping and stiffness parameters.

The control signals are calculated as follows. First, the measured disturbances $\tilde{\mathbf{a}}_0$ are filtered making use of the basis functions $\mathcal{B}^{[i]}(q)$, with q represent-

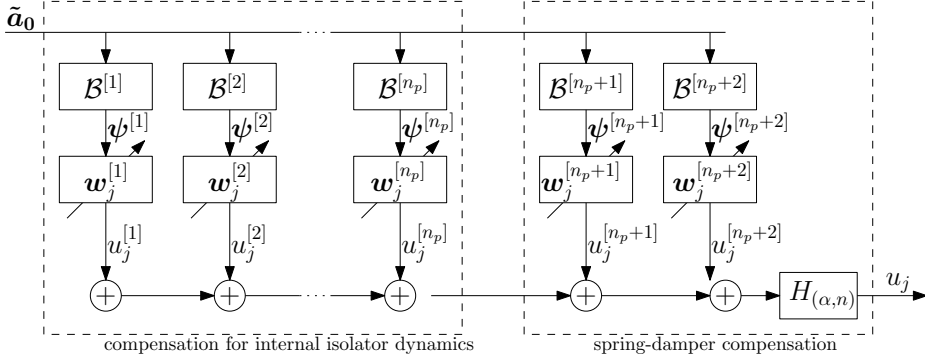


Figure 5.6. general structure of a self-tuning controller, computing the j^{th} controller output $u_j, j \in \{1, \dots, n_u\}$; the controller poles are fixed in the basis functions $\mathcal{B}^{[i]}, i \in \{1, \dots, n_p + 2\}$, while the controller zeros are determined by the self-tuning parameter vectors $w_j^{[i]}$.

ing the backward time-shift operator. Hence, the filtered disturbance vectors are given by $\tilde{\psi}^{[i]}(k) = \mathcal{B}^{[i]}(q)\tilde{a}_0(k)$. Second, the i^{th} contribution to the j^{th} control signal follows from the inner product $u_j^{[i]}(k) = (w_j^{[i]}(k))^T \tilde{\psi}^{[i]}(k)$. Third, all contributions for $i = 1, \dots, n_p + 2$ are summed and integrated by an n^{th} -order weak integrator $H_{(\alpha,n)}(q)$, resulting in the j^{th} control output $u_j(k) \in \mathbb{R}$, or

$$u_j(k) = H_{(\alpha,n)}(q) \sum_{i=1}^{N_p+2} (w_j^{[i]}(k))^T \tilde{\psi}^{[i]}(k). \quad (5.8)$$

Fourth, each u_j given by (5.8) is stacked in the control vector which is denoted by $\mathbf{u}(k) = [u_1(k), \dots, u_{n_u}(k)]^T$, leading to

$$\mathbf{u}(k) = H_{(\alpha,n)}(q) \underbrace{[\mathbf{W}^{[1]}(k) \ \dots \ \mathbf{W}^{[n_p+2]}(k)]}_{\mathbf{W}(k)} \underbrace{\begin{bmatrix} \tilde{\psi}^{[1]}(k) \\ \vdots \\ \tilde{\psi}^{[n_p+2]}(k) \end{bmatrix}}_{\tilde{\psi}(k)}, \quad (5.9)$$

$$\text{with } \mathbf{W}^{[i]}(k) = \begin{bmatrix} (w_1^{[i]}(k))^T \\ \vdots \\ (w_{n_u}^{[i]}(k))^T \end{bmatrix}.$$

In (5.9), $\tilde{\psi}(k) \in \mathbb{R}^{(n_p+2)n_u}$ represents the so-called regression vector, and matrix $\mathbf{W}(k) \in \mathbb{R}^{n_u \times (n_p+2)n_r}$ contains all the parameters that are required from self-tuning. To find the parameters in $\mathbf{W}(k)$, (5.9) is rewritten such that the

parameters are stored in a single column vector $\mathbf{w}(k) \in \mathbb{R}^{n_u(n_p+2)n_r}$, or

$$\mathbf{u}(k) = H_{(\alpha,n)}(q) \underbrace{\begin{bmatrix} \tilde{\boldsymbol{\psi}}^T(k) & \dots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \dots & \tilde{\boldsymbol{\psi}}^T(k) \end{bmatrix}}_{\tilde{\boldsymbol{\Psi}}(k)} \underbrace{\begin{bmatrix} (\mathbf{W}_{(1,:)}^T)^T \\ \vdots \\ (\mathbf{W}_{(N_u,:)}^T)^T \end{bmatrix}}_{\mathbf{w}}, \quad (5.10)$$

with regression matrix $\tilde{\boldsymbol{\Psi}}(k) \in \mathbb{R}^{n_u \times n_u(n_p+2)n_r}$, and $(\mathbf{W}_{(j,:)}^T)^T$ denoting the transposed j^{th} row of $\mathbf{W}$. Since (5.10) is affine in the parameters, minimization of a quadratic cost criterion to estimate $\mathbf{w}(k)$ renders a convex optimization problem. Moreover, since the poles in the self-tuning filter have become pre-determined (fixed) parameters in $\tilde{\boldsymbol{\psi}}$, the adaptation of $\mathbf{w}(k)$ only involves estimation of its zeros, which renders the structure of a generalized FIR filter [23].

5.4.2 Update law

Given input $\tilde{\mathbf{a}}_0$, the goal is to find the optimal parameter vector $\mathbf{w}(k) = \mathbf{w}^*$ such that $\mathbf{a}_1$ is minimized in a frequency range of interest. This is done using the Filtered-error Least Mean Squares (FeLMS) algorithm [108] with residual noise shaping [57]. This section only summarizes the update law briefly. For a detailed description, the reader is referred to Chapter 4. In particular, the instantaneous squared filtered-error $\mathbf{e}$ is minimized, where $\mathbf{e}$ is given by

$$\mathbf{e}(k) = H_{(\alpha,n)}^{-1}(q) \mathbf{N}(q) \hat{\mathbf{P}}_2^{-1}(q) \tilde{\mathbf{a}}_1(k). \quad (5.11)$$

This expression for $\mathbf{e}(k)$ can be explained as follows. From FeLMS [56], it is known that $\tilde{\mathbf{a}}_1$ must be filtered by the model inverse² of the secondary path denoted by $\hat{\mathbf{P}}_2^{-1}$, and by $H_{(\alpha,n)}^{-1}$ to compensate for the weak integrators used as pre-multiplication filter in (5.10). Residual noise shaping [57] requires an additional filter $\mathbf{N}$ to add frequency weighting, which should be designed such that minimization of $\mathbf{e}$ implies minimization of $\mathbf{a}_1$ in the frequency range of interest and under a fast and robust convergence of $\mathbf{w}$. Design guidelines for $\mathbf{N}$ are further discussed in Section 5.4.4, and a parametric model $\hat{\mathbf{P}}_2$ is given in Appendix B in Section 5.8. Note that, although $\hat{\mathbf{P}}_2^{-1}$ might become non-causal, it is sufficient to render the product $H_{(\alpha,n)}^{-1} \mathbf{N} \hat{\mathbf{P}}_2^{-1}$ causal by designing $\mathbf{N}$ with sufficiently large relative degree.

Given the minimization problem, it is shown in Chapter 4 that the parameters can be computed online using the update law

$$\mathbf{w}(k+1) = \mathbf{w}(k) + \mu(k) \left[\mathbf{N}(q) \tilde{\boldsymbol{\Psi}}(k) \right]^T \mathbf{e}(k), \quad (5.12)$$

²Remarks on a causal and stable implementation of $\hat{\mathbf{P}}_2^{-1}$ are given in Chapter 4.

with time-varying adaptation rate $\mu(k)$ that is defined by

$$\mu(k) = \frac{\bar{\mu}}{\epsilon + \|\mathbf{N}(q)\tilde{\Psi}(k)\|_2^2}, \quad (5.13)$$

see [56], with $\epsilon > 0$ a small positive regularization constant to prevent division by zero, and the normalized adaptation rate $0 < \bar{\mu} < 2$. In this chapter, $\bar{\mu} = 0.0001$, which is chosen to provide a good tradeoff between convergence speed on the one hand and a sufficient low steady-state variance in the parameters on the other hand, and ϵ is 0.1% of the RMS power of the signals in $\mathbf{N}(q)\tilde{\Psi}(k)$.

5.4.3 Basis functions

Two important aspects play a role when designing basis functions. First, the linear combination of basis functions must include all desired controller poles. This requirement can be fulfilled by using rational basis functions [49]. Second, the basis functions must be designed such that all parameters in $\mathbf{w}(k)$ have a fast and uniform convergence speed when updated according to (5.12). To obtain this, it is shown in Chapter 4 that all eigenvalues of $\mathbb{E} \left[\tilde{\Psi}^T(k) \mathbf{N}^T(q) \mathbf{N}(q) \tilde{\Psi}(k) \right]$ must be identical. This requirement implies careful design of both the filter $\mathbf{N}$ and the basis functions (which determine $\tilde{\Psi}$). In general, it is not straightforward to design both $\mathbf{N}$ and the basis functions in such a way that all eigenvalues become identical. However, a solution is found under the following assumption.

Assumption 5.2. *The system satisfies the following conditions:*

- (I) *The noise shaping filter is multi-loop SISO, i.e. $\mathbf{N}(q) = n(q)\mathbf{I}_{n_r}$, with SISO filter $n(q)$ and $n_r \times n_r$ identity matrix $\mathbf{I}_{n_r}$;*
- (II) *The set of filtered input disturbance signals $n(q)\tilde{\mathbf{a}}_0(k)$ has a power spectral density matrix corresponding to a set of (possibly correlated) Gaussian disturbances with a flat frequency spectrum.*

Condition (I) in Assumption 5.2 is easily satisfied since $\mathbf{N}$ is user-defined. Condition (II) in Assumption 5.2 is often reasonable, because the controller aims at broadband disturbance rejection. This means that the spectral power of base frame vibrations should be sufficiently distributed over a broad frequency range. Note that a power spectrum that not exactly satisfies Condition (II) still leads to convergence, although some of the parameters might suffer from reduced convergence speeds. For more information regarding implementation aspects of basis functions for non-Gaussian disturbances, the reader is referred to [17].

Under Assumption 5.2, it is derived in Appendix C in Section 5.9 that fast and uniform convergence of the parameters is obtained if orthonormal basis functions are designed [66, 49], satisfying

$$\langle \mathcal{B}^{[m]}, \mathcal{B}^{[n]} \rangle = 0, \quad \langle \mathcal{B}^{[m]}, \mathcal{B}^{[m]} \rangle = \langle \mathcal{B}^{[n]}, \mathcal{B}^{[n]} \rangle = 1, \quad (5.14)$$

with basis functions $\mathcal{B}^{[m]}, \mathcal{B}^{[n]}$, where $m, n \in \{1, \dots, n_p + 2\}$, $m \neq n$, and the inner product defined as

$$\langle \mathcal{B}^{[m]}, \mathcal{B}^{[n]} \rangle := \frac{1}{2\pi} \int_{-\pi}^{\pi} \mathcal{B}^{[m]}(e^{j\omega}) \mathcal{B}^{[n]H}(e^{j\omega}) d\omega. \quad (5.15)$$

Internal isolator dynamics

For the left block in Figure 5.6, which compensates internal isolator dynamics, a set of orthonormal basis functions is derived based on Takenaka-Malmquist functions [66, 50]:

$$\mathcal{B}^{[i]}(z) = \frac{\sqrt{(1 - |\xi_i|^2)}}{z - \xi_i} \prod_{k=1}^{i-1} \left(\frac{1 - \xi_k^* z}{z - \xi_k} \right), \quad i \in \{1, \dots, n_p\}, \quad (5.16)$$

with $\xi_i \in \mathbb{C}$ representing the desired controller poles. When these poles appear in complex pairs, an implementation according to (5.16) results in transfer functions with complex coefficients. To circumvent this problem, these basis functions will be subsequently implemented using Kautz functions [50], which are defined as

$$\begin{aligned} \mathcal{B}^{[2i-1]}(z) &= \frac{\sqrt{1 - c_i^2}(z - b_i)}{z^2 + b_i(c_i - 1)z - c_i} \prod_{k=1}^{i-1} \left(\frac{-c_k z^2 + b_k(c_k - 1)z + 1}{z^2 + b_k(c_k - 1)z - c_k} \right), \\ \mathcal{B}^{[2i]}(z) &= \frac{\sqrt{1 - c_i^2} \sqrt{1 - b_i^2}}{z^2 + b_i(c_i - 1)z - c_i} \prod_{k=1}^{i-1} \left(\frac{-c_k z^2 + b_k(c_k - 1)z + 1}{z^2 + b_k(c_k - 1)z - c_k} \right), \end{aligned} \quad (5.17)$$

with $i \in \{1, \dots, \frac{n_p}{2}\}$.

Spring-damper compensation

The basis functions for the spring-damper compensation block in Figure 5.6 are chosen similar to Chapter 4,

$$\mathcal{B}^{[n_p+1]} = 1, \quad \mathcal{B}^{[n_p+2]}(z) = \prod_{k=1}^{n_p} \left(\frac{-c_k z^2 + b_k(c_k - 1)z + 1}{z^2 + b_k(c_k - 1)z - c_k} \right) \beta H_{(\alpha, n)}(z). \quad (5.18)$$

Orthonormality of the basis functions in (5.18) can be proven by showing the relevant orthogonality and normalization properties. Orthogonality is shown in three steps. First, let the weak integrator in $\mathcal{B}^{[n_p+2]}$ be obtained from ZOH discretization of (5.7) such that $\mathcal{B}^{[n_p+2]}$ has a relative degree of one, rendering it orthogonal to $\mathcal{B}^{[n_p+1]}$ as can be verified from (5.15). Second, following the same line of reasoning, it is concluded that $\mathcal{B}^{[n_p+1]}$ is orthogonal to the Kautz functions in (5.17). Third, the all-pass filter included in $\mathcal{B}^{[n_p+2]}$ renders the function

orthogonal to the Kautz functions in (5.17), see Appendix C in Section 5.9. To prove orthonormality, observe that $\langle \mathcal{B}^{[n_p+1]}, \mathcal{B}^{[n_p+1]} \rangle = 1$, and let the normalization constant $\beta \in \mathbb{R}$ be determined such that $\langle \mathcal{B}^{[n_p+2]}, \mathcal{B}^{[n_p+2]} \rangle = 1$.

5.4.4 Guidelines for the design of filter $\mathbf{N}$

As explained in Section 5.3, it is generally not possible to achieve perfect cancellation such that $\mathcal{T}(\omega) = 0$ in (5.3) for all ω . Therefore, it is important to specify the frequency range of interest where vibration isolation is desired. This is the purpose of the noise shaping filter $\mathbf{N}$. In line with Assumption 5.2, a multi-loop SISO design for $\mathbf{N}$ is proposed:

$$\mathbf{N}(q) = n_{\text{flat}}(q)n_{\text{perf}}(q)\mathbf{I}_{N_u}. \quad (5.19)$$

The flattening filter n_{flat} is used to make the frequency spectrum of $H_{(\alpha,n)}^{-1} \mathbf{N} \mathbf{P}_2^{-1}$ as flat as possible, at least in the frequency range of interest. By doing so, frequency characteristics of the original error signal $\mathbf{a}_1$ are preserved in the filtered error signal $\mathbf{e}$, hence minimization of $\mathbf{e}$ implies minimization of $\mathbf{a}_1$. In this chapter, n_{flat} contains a low-pass filter with a pole at $s = \alpha$ to create a -1 slope that compensates the +1 slope induced by $H_{(\alpha,n)}^{-1}$. The performance filter n_{perf} is used to specify the frequency range where vibration isolation is required. According to the frequency range of interest in this chapter, n_{perf} is designed as a fifth-order band-pass filter from 1 to 100 Hz. A side effect of the large roll-off in n_{perf} is that high-frequency content is strongly filtered, which prevents convergence problems of $\mathbf{w}(k)$ under the presence of non-modeled plant dynamics at high frequencies, see Chapter 4.

Remark 5.3. *The design of $\mathbf{N}$ can also be used to improve the spectral flatness of $\mathbf{S}_{\mathbf{a}_0, \mathbf{N} \mathbf{a}_0, \mathbf{N}}(e^{j\omega})$ which increases convergence speed, see Section 5.4.3. However, this generally conflicts with the design rules given above, and therefore is often not preferred.*

5.5 Experimental validation

This section presents an experimental validation of the self-tuning control strategy. First, the controller implementation is described and, second, the performance measurement results are presented.

5.5.1 Controller implementation

Two controllers are implemented on the industrial vibration isolation system shown in Figure 5.1, namely a controller of (a) reduced-order (RO), and (b) full-order (FO). For all three input directions, the RO controller only provides spring-damper compensation using the two basis functions in (5.18), and serves

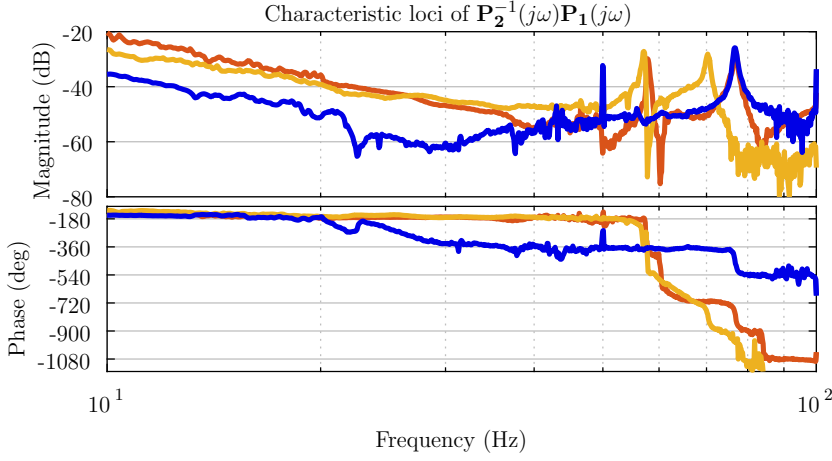


Figure 5.7. Measured characteristic loci plots of $\mathbf{P}_2^{-1}(j\omega)\mathbf{P}_1(j\omega)$; the peak at 50 Hz is not caused by a pole, but related to the electrical power grid.

as a performance reference. The FO controller also includes 25 pairs of Kautz functions, see (5.17), to compensate for internal isolator dynamics. The poles for these Kautz functions depend on the poles of $\mathbf{P}_1$ and the transmission zeros of $\mathbf{P}_2$, see (5.4). Therefore, frequency response function (FRF) measurements of $\mathbf{P}_1(j\omega)$ and $\mathbf{P}_2(j\omega)$ are performed, see Appendix B in Section 5.8. Based on these measurements, an estimated FRF matrix of $\mathbf{P}_2^{-1}(j\omega)\mathbf{P}_1(j\omega)$ is obtained for which Figure 5.7 shows the characteristic loci, i.e. the eigenvalues of the FRF matrix along the frequency axis [84], which are used to estimate the required poles. An advantage of using characteristic loci is that closely spaced poles can be separated from each other. A set of poles is determined using peak picking, the result of which is listed in Table 5.1. Note that all poles have an estimated relative damping ratio of 0.01, although it is unlikely that this value matches the true damping ratios. Namely, it is not straightforward to accurately estimate the damping ratio, since the peak value of a resonance is generally underestimated in spectral analysis due to averaging and a finite frequency grid [40]. To enable better estimations of the true controller poles, five different pairs of basis functions are implemented for every complex pair of poles, each using the same frequency, but with different damping ratios $\zeta_i \in \{0.1, 0.03, 0.01, 0.003, 0.001\}$. The characteristic loci are not plotted for frequencies below 10 Hz, because internal isolator dynamics are not expected to occur below 10 Hz, see also Figure 5.12 in Appendix B in Section 5.8. Note that at low frequencies, the characteristic loci have a -2 slope caused by two poles at $s = 0$. These correspond to the poles for spring-damper compensation, similar to the example in (5.6).

Table 5.1. Frequencies and damping ratios of the poles estimated from Figure 5.7.

Char. locus	Frequency (Hz)	Damping ratio
yellow	57.1	0.01
yellow	70.7	0.01
red	57.8	0.01
red	77.0	0.01
blue	76.7	0.01

5.5.2 Results

The feedforward controllers are self-tuned to optimize performance for two different cases of base frame excitation. For both cases, Figure 5.8 shows the power spectral density plots of the measured base frame accelerations that served as input disturbances. In the first case, the environmental base frame excitation power under normal operating conditions is used. In the second case, base frame shakers are used to increase the excitation power in the frequency range from 60–90 Hz. The second case is used to demonstrate the benefit of FO control with respect to RO control in suppressing resonances due to internal isolator dynamics.

The self-tuning algorithm for both the RO and FO controller is turned on during 10 minutes. Afterwards, it is turned off ($\bar{\mu} = 0$) such that the parameter vector $\mathbf{w}$ becomes fixed with the result of the feedforward controller becoming linear time-invariant (LTI). Using the LTI controllers, a second measurement of 10 minutes is conducted to serve as an assessment of performance. Each measurement is split into 60 subrecords of 10 seconds, and each subrecord is filtered with a Hanning window to reduce the effects of leakage. To compensate for data loss due to windowing, additional subrecords are created by using an overlap factor of 50%. Then, all subrecords are averaged which leads to the averaged (cross) power spectral density (PSD) matrices $\hat{\mathbf{S}}_{\mathbf{a}_0, \mathbf{a}_0}$, $\hat{\mathbf{S}}_{\mathbf{a}_1, \mathbf{a}_1}$, and $\hat{\mathbf{S}}_{\mathbf{a}_1, \mathbf{a}_0}$. Additionally, the output noise covariance matrix $\hat{\mathbf{C}}_{\mathbf{a}_1}$ is estimated similar to (3.14) in Chapter 3. Under the assumption that the signals in $\mathbf{n}_p$ and $\mathbf{n}_1$ are uncorrelated with $\mathbf{a}_0$, recall Figure 5.4, $\hat{\mathbf{C}}_{\mathbf{a}_1}$ gives a lower bound for $\hat{\mathbf{S}}_{\mathbf{a}_1, \mathbf{a}_1}$, i.e. the best performance that can be obtained using disturbance feedforward control.

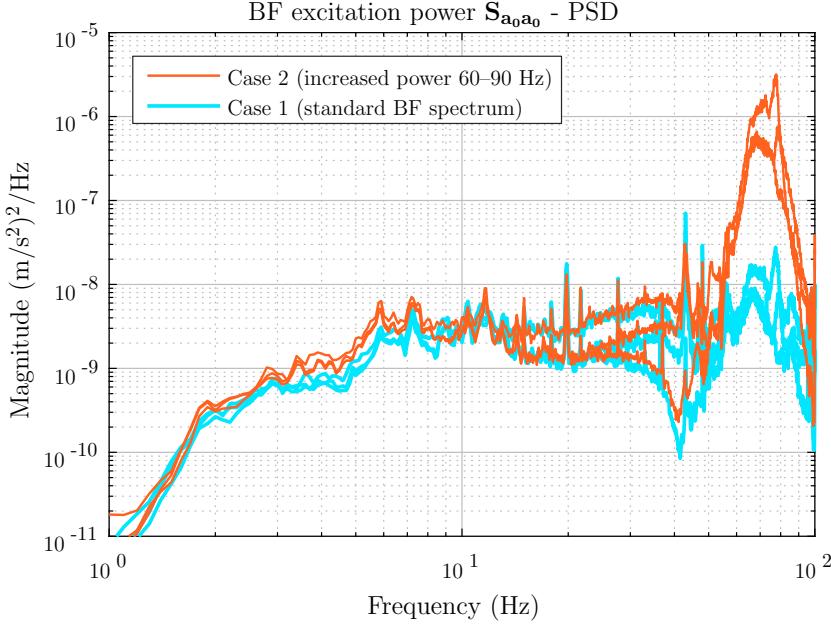


Figure 5.8. power spectral density (PSD) plots of the measured base frame accelerations at all three sensor locations; in blue the accelerations for case 1, and in red the accelerations for case 2; observe that for case 2 the excitation is largely increased for frequencies between 60 and 90 Hz.

CASE 1: Base frame excitation under operating conditions

In this case, the base frame excitation roughly has a flat frequency spectrum with a magnitude of $10^{-9} \text{ (m/s}^2\text{)}^2/\text{Hz}$ for frequencies between 2 and 100 Hz in the three directions (z_{b1} , z_{b2} , z_{b3}), see Figure 5.8. Vibration isolation performance in terms of output power is shown in Figure 5.9. In the middle frequency interval, Figure 5.9a shows performance improvements up to a factor of 22 (at 10.0 Hz) using FO control, and a factor of 11 (at 11.5 Hz) using RO control. At high frequencies, where acoustic resonances occur at 70 and 77 Hz, both controllers are unable to improve performance, because the output power is dominated by output noise. This can be explained from the fact that there is a continuous air flow through the air mounts, needed to realize the zero-stiffness in horizontal direction. This induces a form of actuator noise uncorrelated with $\mathbf{a}_0$ and therefore cannot be compensated by feedforward control.

Figure 5.9b shows the cumulative PSD plots. It is observed that the cumulative power is decreased with 15% in z -direction, and with 23% in θ_y -direction. In θ_x -direction, no significant performance improvement is obtained with FF control. The latter is attributed to the fact that the contribution of base frame

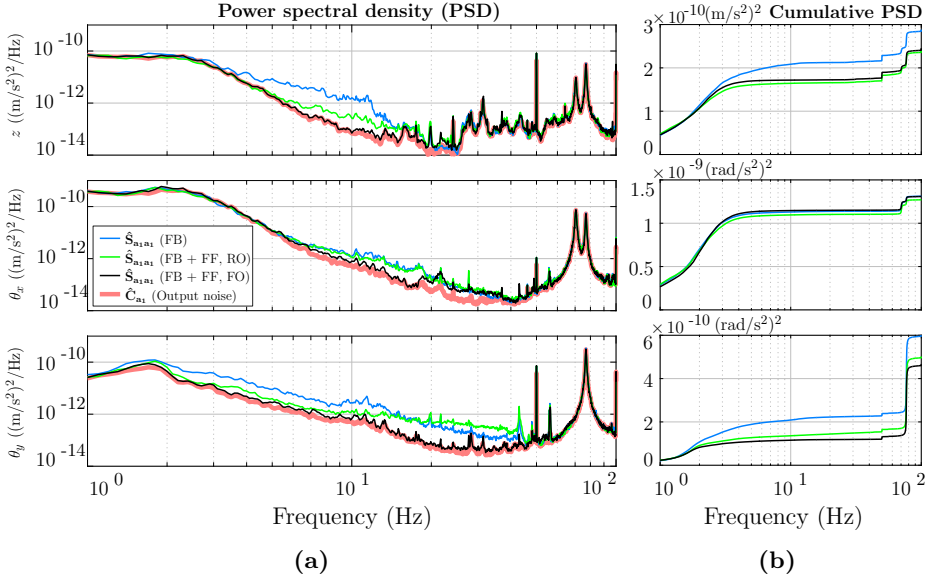


Figure 5.9. measurement results for case 1 (operating conditions), (a) power spectral density, and (b) cumulative power spectral density; the benefit of applying either reduced-order (RO) or full-order (FO) disturbance feedforward control is clearly visible in the z - and θ_y -directions; in θ_x -direction no net benefit is observed because output noise dominates the PSD.

excitations to the output power is not significantly higher than the contribution of the output noise itself. For all three directions, performance for FO and RO feedforward control are comparable. In other words, it seems not very useful to add basis functions to the feedforward controller when the high-frequency metrology frame accelerations are dominated by output noise.

CASE 2: Base frame excitation with increased high-frequency power

In this second case, base frame shakers are used to increase the excitation power for frequencies between 60 and 90 Hz, see Figure 5.8. By doing so, metrology frame vibrations are also increased at these frequencies. The corresponding vibration isolation performance is shown in Figure 5.10. The effect of increasing the signal-to-noise ratio at high frequencies is clearly visible in Figure 5.10a for feedback and RO feedforward control. However, when using FO feedforward control, the output power is again reduced to (almost) the noise power, indicating large performance improvements obtained with FO feedforward control.

Figure 5.10b shows cumulative PSD plots. From these plots, it follows that FO feedforward control outperforms the other controllers. Compared to feedback

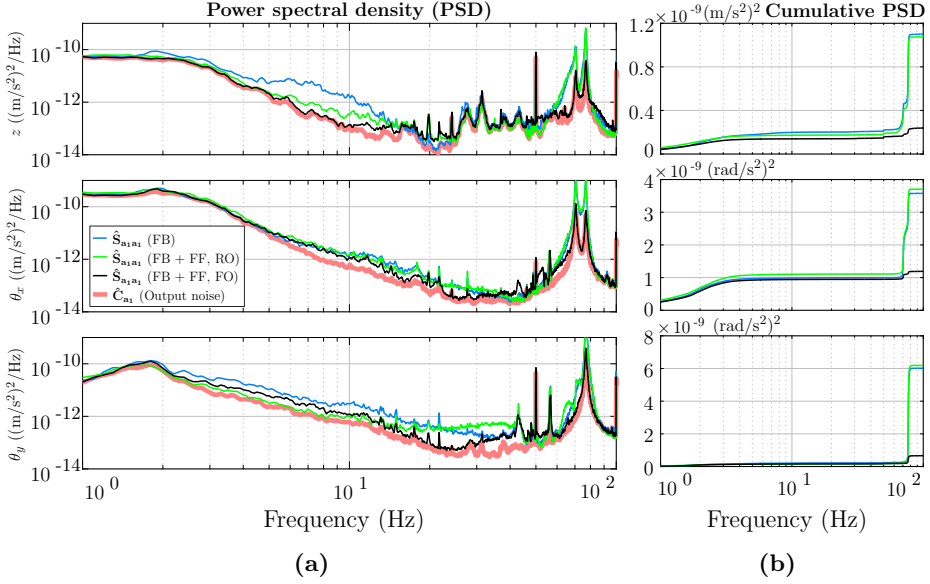


Figure 5.10. Measurement results for case 2 (operating conditions + increased content at higher frequencies), (a) power spectral density, and (b) cumulative power spectral density; Using full-order (FO) control, the cumulative PSD is largely reduced; in fact, it reduces to the same level as obtained in case 1, despite the strongly increased excitation levels, demonstrating the benefit of FO control.

control, FO control reduces the cumulative power with approximately 79% in Z -direction, 67% in θ_x -direction, and 89% in θ_y -direction. This clearly shows that there is a benefit when using FO control under the condition of having sufficiently good signal-to-noise ratios.

Remark 5.4. *Reconsidering the model-based causal Wiener solution in (5.5), it follows that it includes a parametric description of the base frame spectrum given by $\Phi(z)$. In that sense, it may be surprising that the self-tuning approach is able to reduce the output error back to sensor noise level while no additional poles for describing the base frame spectrum are included in the basis functions. However, each pole listed in Table 5.1 corresponds to five pairs of basis functions with different damping ratios, recall Section 5.5.1. As such, many basis functions related to the higher frequency range between 50 and 100 Hz are available, and it is therefore reasonable to assume that there is sufficient design freedom for the self-tuning algorithm to adapt the parameters such that the base frame spectrum is accurately described.*

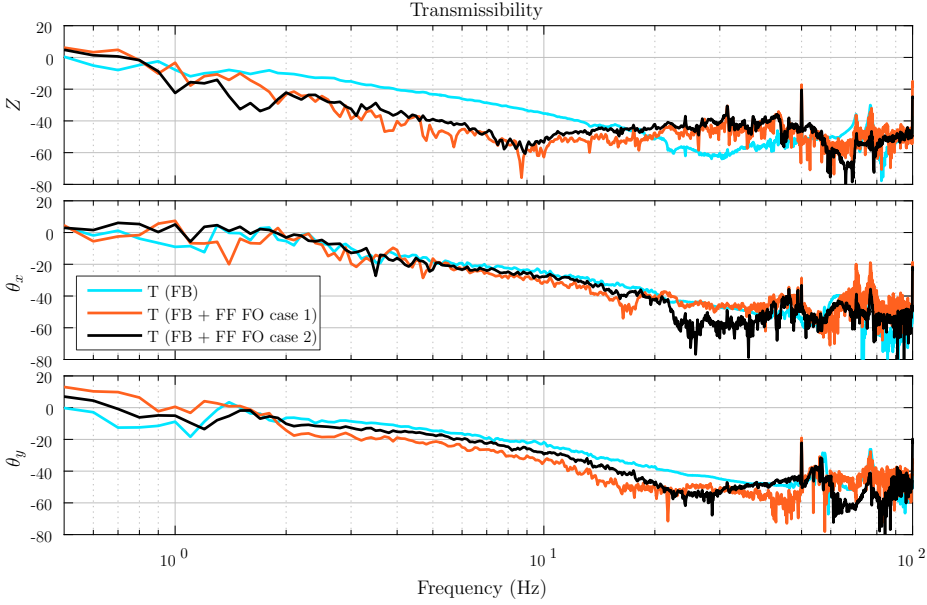


Figure 5.11. Bode magnitude plots of the transmissibility functions $\mathcal{T}$ for the systems with feedback and feedforward control; compared to the system with only feedback control, self-tuning of a full-order (FO) feedforward controller leads in both cases to a significant reduction in $\mathcal{T}$ at the lower frequencies; at the higher frequencies, the actual shape of $\mathcal{T}$ after self-tuning depends on the provided disturbance spectrum; for case 2, where strong excitations appear at frequencies between 60 and 90 Hz, the self-tuning algorithm reduces $\mathcal{T}$ at these frequencies at the cost of increasing $\mathcal{T}$ at other frequencies due to a waterbed effect.

Transmissibility functions

Figure 5.11 shows Bode magnitude plots of the transmissibility functions for the systems with feedback and feedforward control. Below 2 Hz, the measurements have a significant uncertainty due to an insufficient signal-to-noise ratio. In the z -direction, application of the self-tuning FO controllers leads in both cases to a significant reduction of the transmissibilities at frequencies below 15 Hz. However, this reduction comes at the cost of performance deterioration at the mid-frequencies (between 15 and 50 Hz) due to a waterbed effect, see Chapter 2. In the θ_x - and θ_y -directions, this shift in performance is less visible because the signal-to-noise ratio at low frequencies is much smaller, recall Figures 5.9a and 5.10a. For frequencies beyond 50 Hz, the reduction in transmissibility clearly depends on the provided input spectrum. It follows that for case 2, where the base frame disturbances induce increased excitation power, the self-tuning

algorithm renders the transmissibility to be lower in this frequency range such that the resulting output power is again minimized, recall Figures 5.9a and 5.10a. However, this comes again at the cost of performance deterioration at other frequencies. This is clearly visible in the z - and θ_y -directions, where, for the second case, the transmissibility is increased at mid-frequencies compared to the first case.

5.6 Conclusions

This chapter presented a disturbance feedforward control strategy for active vibration isolation systems with internal air mount dynamics. The controller is implemented as a self-tuning controller for which the poles are fixed in rational basis functions and the zeros are obtained online via a self-tuning approach. The poles are obtained from measured characteristic loci plots of the system using spectral analysis. Implementation of the self-tuning controller shows a reduction up to 23% of residual output vibrations under normal operating conditions, and a reduction up to 89% under additional shaker-induced base frame vibrations. For the latter case, it is shown that the proposed full-order self-tuning controller may clearly outperform a feedforward controller which is only based on spring-damper compensation.

5.7 Appendix A: Parametric model for a 1-DOF air mount

The air mount model considered in Fig. 5.2 basically consists of two interconnected air chambers. Let V_1, V_2 represent the volumes of the two chambers, and p_1, p_2 , represent the pressures in the lower and upper chamber, respectively. The upper chamber is sealed with a freely moving piston with cross-sectional area A_p that supports the payload. The upper and lower chamber are connected via a small air neck. The air in the neck is assumed to behave as an incompressible volume with mass $m_n = \rho L_n A_n$, where ρ is the density of air and L_n, A_n are the air neck length and cross-sectional area, respectively. The variables x_b, x_n , and x_p denote the displacement of the base, the air neck mass, and the payload, respectively.

To obtain the equations of motion for the parametric air mount model, the air in the chambers is regarded as an ideal gas, and it is assumed that there is no heat exchange with the environment. Then, the adiabatic ideal gas law can be used to describe the relation between instant pressure and volume of the air in a tank sealed with a freely moving piston [47],

$$pV^\gamma = p_0 V_0^\gamma = \text{constant}, \quad (5.20)$$

with p the isolator pressure, V the isolator volume, p_0 the static isolator pressure, V_0 the corresponding volume, and γ the heat capacity ratio. The static isolator pressure equals

$$p_0 = p_{atm} + \frac{m_p g}{A_p}, \quad (5.21)$$

with p_{atm} the atmospheric pressure, m_p the payload mass supported by the isolator, and g the gravitational constant. From (5.20), it can be derived that

$$dp = -\frac{\gamma p_0 V_0^\gamma}{V^{\gamma+1}} dV \approx -\frac{\gamma p_0}{V_0} dV, \quad (5.22)$$

which yields a linear relation between the pressure change and volume change, under the assumption that the chamber volume is large compared to the changes in the volume. The volume changes in the lower and upper chamber are

$$\begin{aligned} dV_1 &= A_n(x_n - x_b), \\ dV_2 &= A_p x_p - A_n x_n - (A_p - A_n)x_b. \end{aligned} \quad (5.23)$$

Combining (5.22) and (5.23), the relations between pressure changes and displacements are

$$\begin{aligned} dp_1 &= -\frac{\gamma p_0}{V_{1,0}} A_n(x_n - x_b), \\ dp_2 &= -\frac{\gamma p_0}{V_{2,0}} [A_p x_p - A_n x_n - (A_p - A_n)x_b], \end{aligned} \quad (5.24)$$

where $V_{1,0}$, $V_{2,0}$ represent the static volumes for the lower and upper chamber, respectively. To obtain the equations of motion, Newton's second law is applied to both the air neck mass and the supporting payload mass, which gives

$$dp_1 A_n - dp_2 A_n = m_n \ddot{x}_n, \quad F_p + dp_2 A_p = m_p \ddot{x}_p, \quad (5.25)$$

with F_p including the static weight $m_p g$ and actuator forces u . When the results from (5.24) and (5.25) are combined, the equations of motion for the undamped model read

$$\mathbf{M} \ddot{\mathbf{x}} + \mathbf{K} \mathbf{x} = \mathbf{K}_0 x_b + \mathbf{B}_0 F_p, \quad (5.26)$$

with

$$\begin{aligned} \mathbf{x} &= \begin{pmatrix} x_n \\ x_p \end{pmatrix}, \quad \mathbf{M} = \begin{bmatrix} m_n & 0 \\ 0 & m_p \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} (\frac{\gamma p_0}{V_{1,0}} + \frac{\gamma p_0}{V_{2,0}}) A_n^2 & -\frac{\gamma p_0}{V_{2,0}} A_p A_n \\ -\frac{\gamma p_0}{V_{2,0}} A_p A_n & \frac{\gamma p_0}{V_{2,0}} A_n^2 \end{bmatrix}, \\ \mathbf{K}_0 &= \begin{bmatrix} (\frac{\gamma p_0}{V_{1,0}} + \frac{\gamma p_0}{V_{2,0}}) A_n^2 - \frac{\gamma p_0}{V_{2,0}} A_p A_n \\ \frac{\gamma p_0}{V_{2,0}} A_p (A_p - A_n) \end{bmatrix}, \quad \mathbf{B}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \end{aligned} \quad (5.27)$$

Damping is added by assuming modal damping, giving

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{D}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{D}_0\dot{x}_b + \mathbf{K}_0x_b + \mathbf{B}_0F_p, \quad (5.28)$$

with

$$\mathbf{D} = \mathbf{M}\mathbf{T} \begin{bmatrix} 2\beta_1\omega_1 & 0 \\ 0 & 2\beta_2\omega_2 \end{bmatrix} \mathbf{T}^{-1}, \quad \mathbf{D}_0 = \mathbf{M}\mathbf{T} \begin{bmatrix} 2\beta_1\omega_1 \\ 2\beta_2\omega_2 \end{bmatrix}, \quad (5.29)$$

where $\mathbf{T} \in \mathbb{R}^{2 \times 2}$ is a matrix containing the eigenvectors of $\mathbf{M}^{-1}\mathbf{K}$, the frequencies $\omega_1, \omega_2 \in \mathbb{R}$ are the square roots of the eigenvalues of $\mathbf{M}^{-1}\mathbf{K}$ that correspond with the suspension frequency and internal resonance frequency, respectively, and $\beta_1, \beta_2 \in \mathbb{R}$ are the relative damping ratios for the eigenmodes. Taking the Laplace transform $\mathcal{L}\{f(t)\}(s) = \int_0^\infty f(t)e^{-st}dt$ for an arbitrarily piecewise continuous function $f(t)$ with $t \geq 0$, (5.28) can be written as

$$(\mathbf{M}s^2 + \mathbf{D}s + \mathbf{K}) \mathbf{X}(s) = (\mathbf{D}_0s + \mathbf{K}_0) X_b(s) + \mathbf{B}_0F_p(s). \quad (5.30)$$

Defining the payload acceleration as output $y := \ddot{x}_p = \mathbf{C}\ddot{\mathbf{x}}$ with $\mathbf{C} = [0, 1]$, and $r := \ddot{x}_b$ and F_p as inputs, input-output relations for the primary path P_1 and the secondary path P_2 of the uncontrolled system are obtained from (5.30) as

$$\begin{aligned} P_1(s) &:= \frac{Y(s)}{R(s)} = \mathbf{C} (\mathbf{M}s^2 + \mathbf{D}s + \mathbf{K})^{-1} (\mathbf{D}_0s + \mathbf{K}_0) \\ &= \frac{ds + k}{s^2 + 2\zeta_1\omega_1s + \omega_1^2} \frac{s^2 + \zeta_{a2}\omega_{a2}s + \omega_{a2}^2}{s^2 + \zeta_2\omega_2s + \omega_2^2}, \end{aligned} \quad (5.31)$$

$$\begin{aligned} P_2(s) &:= \frac{Y(s)}{F_p(s)} = s^2 \mathbf{C} (\mathbf{M}s^2 + \mathbf{D}s + \mathbf{K})^{-1} \mathbf{B}_0 \\ &= \frac{s^2}{s^2 + 2\zeta_1\omega_1s + \omega_1^2} \frac{s^2 + \zeta_{b2}\omega_{b2}s + \omega_{b2}^2}{s^2 + \zeta_2\omega_2s + \omega_2^2}. \end{aligned} \quad (5.32)$$

Values for $d, k, \omega_1, \omega_{a2}, \omega_{b2}, \omega_2, \zeta_{a2}, \zeta_{b2}$ are derived with $\gamma = 1.4$, $g = 9.81 \text{ m/s}^2$, $p_{atm} = 1.0 \times 10^5 \text{ Pa}$, $\rho = 7.0 \text{ kg/m}^3$, $V_{1,0} = V_{2,0} = 4.0 \times 10^{-3} \text{ m}^3$, $A_p = 0.02 \text{ m}^2$, $A_n = 0.001 \text{ m}^2$, $L_n = 0.25 \text{ m}$, $m_p = 1000 \text{ kg}$, $\beta_1 = 0.02$, $\beta_2 = 0.01$. Bode plots for P_1, P_2 are shown in Fig. 5.3, which shows that a pole-zero cancellation occurs in P_2 . Physically this makes sense, because m_p is much larger than m_n , such that the impact of m_n on $\ddot{x}_p$ is negligible compared to F_p . For P_1 , it can be seen that the anti-resonance and resonance frequency do not coincide because the input of P_1 is a prescribed acceleration. The anti-resonance frequency $f_{a2} = \omega_{a2}/(2\pi)$ is approximately given by

$$f_{a2} \approx \frac{c_0}{2\pi} \sqrt{\frac{A_n}{V_{1,0}L_n}}, \quad (5.33)$$

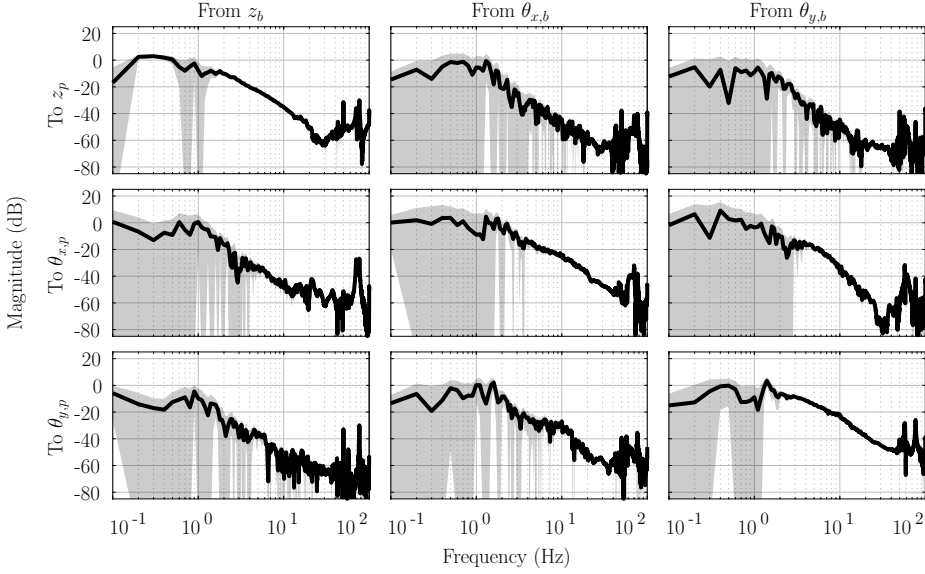


Figure 5.12. FRF Measurement of the primary path $\mathbf{P}_1$, representing the transmissibility function matrix of the vibration isolation system without feed-forward control. The shaded area represents the 95% confidence interval of the FRF measurement. It is observed that the suspension frequencies are around 1 Hz, after which the transmissibility starts to roll off from the 0 dB line. Note that the suspension modes show no resonance peaks because skyhook damping is applied using acceleration feedback.

with c_0 representing the speed of sound in air. At high frequencies, the figure shows that the magnitude of P_1 is increased for frequencies beyond f_{a2} , hence the performance of the passive vibration isolation system is deteriorated due to the Helmholtz effect. Physically this means that the lower air chamber decouples such that the effective air volume decreases and consequently the isolator's stiffness increases.

5.8 Appendix B: System identification

This section presents system identification results for the industrial setup presented in Section 5.2. These results are used to implement the self-tuning controller in Section 5.5.

The primary path $\mathbf{P}_1$ represents the transmissibility matrix of the uncontrolled system, and is measured using the method presented in Chapter 3. In this method, a combination of environmental floor excitations at low frequencies and shaker excitations at high frequencies is used. The shakers are placed in

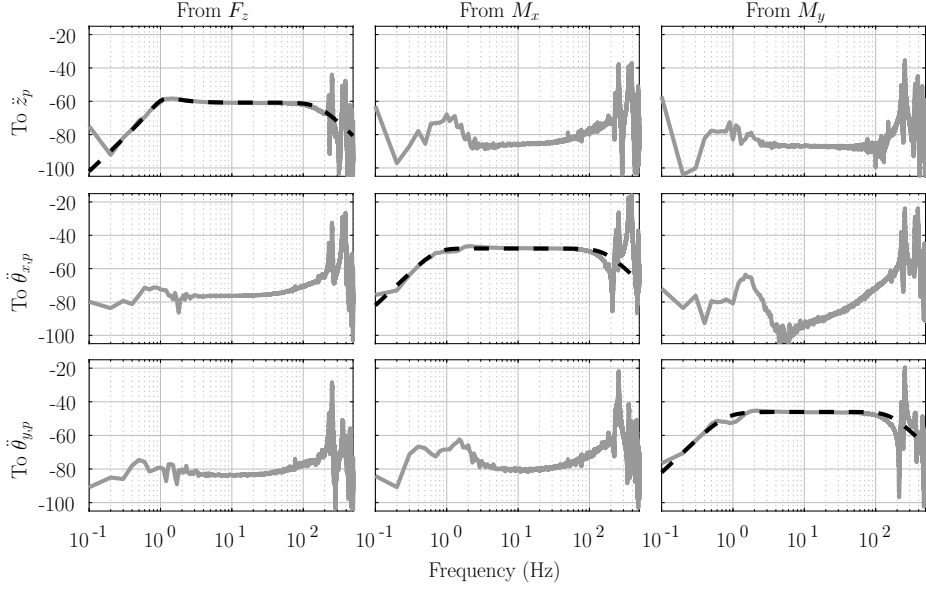


Figure 5.13. Bode magnitude plots of the measured transfer functions $\mathbf{P}_2(j\omega)$ (solid gray) compared to the parametric model $\hat{\mathbf{P}}_2(j\omega)$ (dashed black); the off-diagonal magnitudes are low compared to the main-diagonal magnitudes between 1 and 100 Hz, showing good system decoupling and validating the decentralized structure of the parametric model.

parallel to the base frame support feet, see Figure 5.2. Both floor and shaker excitations are considered as random signals, and a measurement record of 10 minutes is recorded at a fixed sampling frequency of 4 kHz. Spectral analysis is used to estimate the FRF and the 95% confidence regions of the FRF, which are both shown in Figure 5.12.

The FRF of the secondary path $\mathbf{P}_2$ is also obtained using spectral analysis. In this measurement, three random excitation signals are sent simultaneously to all three Lorentz actuators, and a data record of three minutes is recorded. This record is also split in subrecords of 10 seconds, and filtered with a Hanning window with 50% overlap. Figure 5.13 shows the resulting FRF measurements of $\mathbf{P}_2(j\omega)$, and a fitted parametric model which is given by

$$\hat{\mathbf{P}}_2(s) = \underbrace{\begin{bmatrix} \frac{s^2}{ms^2 + d_z s + k_z} & 0 & 0 \\ 0 & \frac{s^2}{J_{xx}s^2 + d_x s + k_x} & 0 \\ 0 & 0 & \frac{s^2}{J_{yy}s^2 + d_y s + k_y} \end{bmatrix}}_{\hat{\mathbf{P}}_{2,m}(s)} \underbrace{\frac{\omega_n^2}{s^2 + 2\zeta_n \omega_n s + \omega_n^2} e^{\tau s} \mathbf{I}_{n_u}}_{\hat{\mathbf{P}}_{2,a}(s)}. \quad (5.34)$$

The parametric model includes the suspended rigid-body behavior of the metrology frame in $\hat{\mathbf{P}}_{\mathbf{2},\mathbf{m}}$, and actuator dynamics described by a second-order low-pass filter at $\omega_n = 2\pi \cdot 160$ rad/s and a time delay of $\tau = 0.001$ s in $\hat{\mathbf{P}}_{\mathbf{2},\mathbf{a}}$. The model parameters are $m = 1100$ kg, $J_{xx} = 250$ kg m², $J_{yy} = 200$ kg m², $k_z = 50$ kN/m, $k_{Rx} = 5$ kN/m, $k_{Ry} = 5$ kN/m, $d_z = 6000$ Ns/m, $d_{Rx} = 1500$ Ns/m, $d_{Ry} = 1500$ Ns/m.

5.9 Appendix C: Proofs for orthonormal basis functions

This appendix contains the proofs related to the orthonormality of basis functions. A proof for uniform convergence speed of all self-tuning parameters is given in Appendix C.1, whereas a sketch of proof for the orthonormality of the n^{th} -order weak integrator is given in Appendix C.2.

5.9.1 Appendix C.1: Proof for uniform convergence speed

This appendix proves uniform convergence speed of all self-tuning parameters. It is shown in Chapter 4 that all eigenvalues of $\mathbb{E} \left[\tilde{\Psi}^T(k) \mathbf{N}^T(q) \mathbf{N}(q) \tilde{\Psi}(k) \right]$ must be identical. In this chapter, $\mathbf{N}(q) = n(q) \mathbf{I}_{(N_p+2)N_u}$ is used, see Assumption 5.2. Then, fast and uniform convergence of the parameters is obtained if all eigenvalues of $\mathbb{E} \left[\tilde{\psi}(k) n^2(q) \tilde{\psi}^T(k) \right]$ are identical, see Chapter 4. Define

$$n(q) \tilde{\psi}(k) = \mathcal{B}(q) \otimes (n(q) \tilde{\mathbf{a}}_{\mathbf{0}}(k)), \quad \mathcal{B}(q) = \begin{bmatrix} \mathcal{B}^{[1]}(q) \\ \vdots \\ \mathcal{B}^{[N_p+2]}(q) \end{bmatrix}. \quad (5.35)$$

Then, using Parseval's rule, it is obtained that

$$\mathbb{E} \left[\left(n(q) \tilde{\psi}(k) \right) \left(n(q) \tilde{\psi}(k) \right)^T \right] = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(n(e^{j\omega}) \tilde{\psi}(e^{j\omega}) \right) \left(n(e^{j\omega}) \tilde{\psi}(e^{j\omega}) \right)^H d\omega \quad (5.36)$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\mathcal{B}(e^{j\omega}) \otimes (n(e^{j\omega}) \tilde{\mathbf{a}}_{\mathbf{0}}(e^{j\omega})) \right) \left(\mathcal{B}(e^{j\omega}) \otimes (n(e^{j\omega}) \tilde{\mathbf{a}}_{\mathbf{0}}(e^{j\omega})) \right)^H d\omega \quad (5.37)$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\mathcal{B}(e^{j\omega}) \otimes (n(e^{j\omega}) \tilde{\mathbf{a}}_{\mathbf{0}}(e^{j\omega})) \right) \left(\mathcal{B}^H(e^{j\omega}) \otimes (n(e^{j\omega}) \tilde{\mathbf{a}}_{\mathbf{0}}(e^{j\omega}))^H \right) d\omega \quad (5.38)$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\mathcal{B}(e^{j\omega}) \mathcal{B}^H(e^{j\omega}) \right) \otimes \mathbf{S}_{\mathbf{a}_{\mathbf{0}}, \mathbf{N} \mathbf{a}_{\mathbf{0}}, \mathbf{N}}(e^{j\omega}) d\omega, \quad (5.39)$$

where the power spectral density matrix of the filtered disturbances $n(q)\tilde{\mathbf{a}}_0$ is given by

$$\mathbf{S}_{\mathbf{a}_0, \mathbf{N} \mathbf{a}_0, \mathbf{N}}(e^{j\omega}) = (n(e^{j\omega})\tilde{\mathbf{a}}_0(e^{j\omega}))(n(e^{j\omega})\tilde{\mathbf{a}}_0(e^{j\omega}))^H. \quad (5.40)$$

In (5.38) it is used that $(A \otimes B)^H = A^H \otimes B^H$, and in (5.39) it is used that $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$. Now, the problem has been reformulated to a frequency-domain design problem of the basis functions in $\mathcal{B}$ such that the matrix resulting from the integral in (5.39) has all identical eigenvalues. There exists no general solution to this design problem for any arbitrary $\mathbf{S}_{\mathbf{a}_0, \mathbf{N} \mathbf{a}_0, \mathbf{N}}(e^{j\omega})$. However, under the assumption

$$\mathbf{S}_{\mathbf{a}_0, \mathbf{N} \mathbf{a}_0, \mathbf{N}}(e^{j\omega}) = \sigma_0^2 \mathbf{U}, \quad (5.41)$$

with unitary matrix $\mathbf{U}$, the disturbances represent (possibly correlated) Gaussian disturbances with zero mean and variance σ_0^2 , and (5.39) reduces to

$$\mathbb{E} \left[\left(n(q)\tilde{\psi}(k) \right) \left(n(q)\tilde{\psi}(k) \right)^T \right] = \underbrace{\frac{\sigma_0^2}{2\pi} \int_{-\pi}^{\pi} \mathcal{B}(e^{j\omega}) \mathcal{B}^H(e^{j\omega}) d\omega}_{\mathbf{A}} \otimes \mathbf{U}. \quad (5.42)$$

In general, the singular values of $(\mathbf{A} \otimes \mathbf{U})$ are given by $\sigma_{\mathbf{A},i} \sigma_{\mathbf{U},j}$, where $\sigma_{\mathbf{A},i}, i \in \{1, \dots, N_p + 2\}$ and $\sigma_{\mathbf{U},j}, j \in \{1, \dots, N_r\}$ represent the singular values of $\mathbf{A}$ and $\mathbf{U}$, respectively. Since $\mathbf{U}$ is unitary, it follows that $\sigma_{\mathbf{U},j} = 1$ for all j . Then, all singular values of (5.40) are identical, i.e. a solution of the basis functions design problem is found, when using orthonormal basis functions [66, 49] as defined in (5.14). Using this design of orthonormal basis functions, the matrix $\mathbf{A}$ in (5.40) reduces to a scaled identity matrix which has eigenvalues $\sigma_{\mathbf{A},i} = \sigma_0^2$ for all i , and which completes the proof.

5.9.2 Appendix C.2: Proof for orthonormality of an n^{th} -order weak integrator

This appendix gives a sketch of proof for the orthonormality of a basis function describing the n^{th} -order weak integrator. To this end, consider a Takenaka-Malmquist basis function given by

$$\mathcal{B}^{[1]} = \frac{(1 - \xi_1^* z)}{(z - \xi_1)} \frac{1}{(z - \xi_2)}, \quad (5.43)$$

And two additional basis functions

$$\mathcal{B}^{[2]} = \frac{(1 - \xi_1^* z)}{(z - \xi_1)} \frac{(1 - \xi_2^* z)}{(z - \xi_2)} \cdot \frac{1}{(z - \alpha)}, \quad (5.44)$$

$$\mathcal{B}^{[3]} = \frac{(1 - \xi_1^* z)}{(z - \xi_1)} \frac{(1 - \xi_2^* z)}{(z - \xi_2)} \cdot \frac{(1 - \alpha z)}{(z - \alpha)} \frac{1}{(z - \alpha)}. \quad (5.45)$$

Two basis functions are said to be orthogonal if

$$\langle \mathcal{B}^{[m]}, \mathcal{B}^{[n]} \rangle := \frac{1}{2\pi i} \oint_{\mathbb{T}} \mathcal{B}^{[m]}(z) \mathcal{B}^{[n]H} \left(\frac{1}{z^H} \right) \frac{1}{z} dz = 0. \quad (5.46)$$

It follows straightforwardly from substitution of (5.43)–(5.45) in (5.46) that $\langle \mathcal{B}^{[1]}, \mathcal{B}^{[2]} \rangle = 0$, $\langle \mathcal{B}^{[1]}, \mathcal{B}^{[3]} \rangle = 0$, and $\langle \mathcal{B}^{[2]}, \mathcal{B}^{[3]} \rangle = 0$. Then, it also holds that $\langle \mathcal{B}^{[1]}, (c_2 \mathcal{B}^{[2]} + c_3 \mathcal{B}^{[3]}) \rangle = 0$ with $c_2, c_3 \in \mathbb{R}$. The latter linear combination can be written as

$$\begin{aligned} c_2 \mathcal{B}^{[2]} + c_3 \mathcal{B}^{[3]} &= \frac{(1 - \xi_1^* z)}{(z - \xi_1)} \frac{(1 - \xi_2^* z)}{(z - \xi_2)} \left[\frac{c_2}{z - \alpha} + \frac{c_3(1 - \alpha z)}{(z - \alpha)^2} \right] \\ &= \frac{(1 - \xi_1^* z)}{(z - \xi_1)} \frac{(1 - \xi_2^* z)}{(z - \xi_2)} \left[\frac{(c_3 - c_4 \alpha)z + (c_4 - c_3 \alpha)}{(z - \alpha)^2} \right] \\ &= \frac{(1 - \xi_1^* z)}{(z - \xi_1)} \frac{(1 - \xi_2^* z)}{(z - \xi_2)} \left[\frac{d_1 z + d_0}{(z - \alpha)^2} \right] \end{aligned} \quad (5.47)$$

with the two variables $d_1 = (c_3 - c_4 \alpha)$, $d_0 = (c_4 - c_3 \alpha)$ which are free to choose since $\langle \mathcal{B}^{[1]}, (c_2 \mathcal{B}^{[2]} + c_3 \mathcal{B}^{[3]}) \rangle = 0$ for any $c_2, c_3 \in \mathbb{R}$. Next, if d_1, d_0 are chosen such that $\left[\frac{d_1 z + d_0}{(z - \alpha)^2} \right]$ describes a weak integrator $H_{(\alpha, n)}(z)$ having order $n = 2$, cut-off frequency α , and a relative degree of one, the linear combination of basis functions in (5.47) can be regarded as a single basis function similar to $\mathcal{B}^{[n_p+2]}$ in (5.18) and orthogonal to the Takenaka-Malmquist basis function in (5.43). This idea can be generalized to any Takenaka-Malmquist basis function, and any arbitrary higher-order weak integrator $H_{(\alpha, n)}(z)$, by extending (5.43)–(5.45) up to n basis functions.

Chapter 6

Combined feedback and feedforward control

This chapter proposes a physics-based approach toward mixed feedback and feedforward control design for active soft-mounted vibration isolators. A model is derived based on measured frequency response data obtained from an industrial vibration isolation system. Using this model, a combination of relative position feedback, acceleration feedback, and disturbance feedforward control is designed and evaluated. By mixing the sensor signals, amplification of sensor noise is minimized while vibration isolation and leveling performance are optimized. It is shown that feedback control greatly enhances the rejection of disturbances that act directly on the isolated metrology frame. In addition, feedforward control improves the rejection of floor vibrations that are transmitted via the vibration isolator. Experimental validations of the feedback and feedforward control strategies show great improvements in vibration isolation performance.¹

¹This chapter is based on:

M.A. Beijen, M.F. Heertjes, H. Butler, and M. Steinbuch, *Mixed feedback and feedforward control for soft-mounted active vibration isolation systems: a loop shaping approach*, in preparation for journal publication.

6.1 Introduction

Vibration isolation systems are widely used in high-precision machines to isolate metrology frames from external disturbances [53, 45]. There are generally two types of disturbances the system has to reject: (a) floor vibrations which enter the metrology frame via the suspension system, and (b) disturbance forces acting directly on the isolated metrology frame. In passive systems, the disturbance rejection properties strongly depend on the suspension stiffness [9]. Systems with a low stiffness (soft mounts) generally achieve a good suppression of floor vibrations but poor rejection of direct disturbances. Systems with a high stiffness (hard mounts) generally achieve a good rejection of direct disturbances but poor rejection of floor vibrations. Hence the existence of an inherent design tradeoff.

In view of this tradeoff, feedback control can be used to reduce the sensitivity for both types of disturbances, see, e.g., [72]. However, high-gain feedback can lead to undesired amplification of sensor noise. To minimize noise amplification, mixed-sensor feedback control strategies are developed that use different types of sensors dedicated to different frequency ranges. A linear mixed-sensor control strategy for soft mounts is presented in [10] where position feedback at low frequencies is combined with acceleration feedback at high frequencies. In [116, 25], sliding mode control strategies are proposed that improve performance by mixing relative position feedback and absolute velocity (or acceleration) feedback. For hard mounts, an example of mixed-sensor feedback control is presented in [90], where a combination of acceleration feedback and force feedback is used. This improves the disturbance rejection properties at lower frequencies while at the same time adding damping to internal modes at higher frequencies. However, this requires high-gain feedback controllers, which can lead to stability problems and inherently induce noise amplification.

Disturbance feedforward control is a promising alternative to feedback control in improving the rejection of floor vibrations, see e.g. [35, 58, 22]. Using feedforward control, the floor vibrations rather than the resulting metrology frame vibrations are measured. This generally improves the signal-to-noise ratio as compared to feedback control [93], because the disturbances to-be-measured are not (mechanically) low-pass filtered by the isolator system. Furthermore, feedforward controllers do not affect the stability properties, because of the absence of a feedback mechanism. However, feedforward control cannot suppress direct disturbances, and it needs an accurate system model which is not always easy to obtain. Therefore, feedforward controllers are generally implemented with self-tuning algorithms to find (online) the performance-relevant controller parameters, as has been discussed in Chapter 4.

In this chapter, a mixed-sensor feedback and feedforward control strategy for active soft mounts is presented. A combination of relative position feedback and absolute acceleration feedback control is designed to maximize performance in terms of leveling and the rejection of direct disturbances. Also, it minimizes the

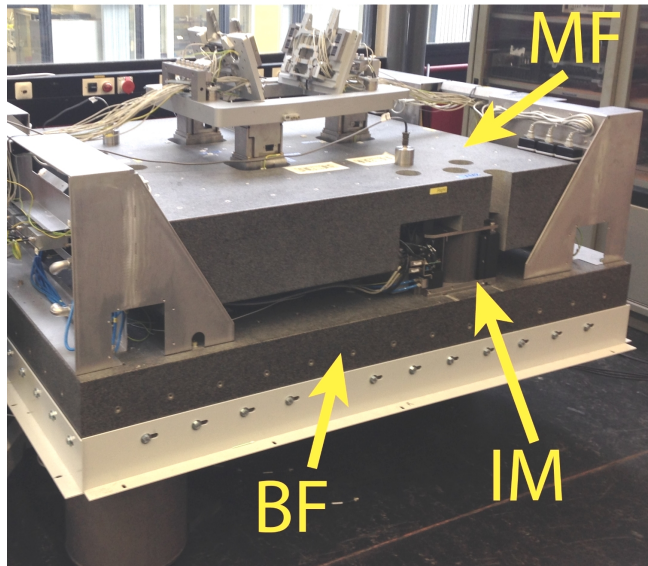


Figure 6.1. Vibration isolation system, basically consisting of a metrology frame (MF), a base frame (BF), and three isolation modules (IMs).

amplification of sensor noise and it guarantees sufficient robust stability properties. In addition, a disturbance feedforward controller is designed to improve the rejection of floor vibrations in a broad frequency range of interest.

This chapter is organized as follows. The vibration isolation test rig and the block diagram for control are presented in Section 6.2. The problem definition is given in Section 6.3. The feedback controller design is given in Section 6.4, while the feedforward controller design is given in Section 6.5. Modeling results of vibration isolation performance in terms of residual metrology frame vibrations are evaluated in Section 6.6. An experimental validation of the proposed control strategies is presented in Section 6.7, and the major conclusions are given in Section 6.8.

6.2 Active vibration isolation system

The system under investigation is the active vibration isolation system shown in Figure 6.1 and consists of the following parts: (a) a six-degree-of-freedom (6-DOF) movable metrology frame (MF), (b) a base frame (BF) that is supported by the floor, and (c) three isolation modules (IM) that are installed between the BF and the MF. Each IM contains a pneumatically controlled air mount that provides gravity compensation, damping, and a low suspension stiffness.

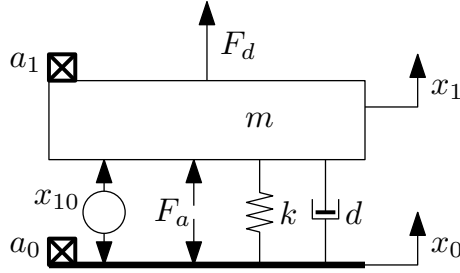


Figure 6.2. Simplified mass-spring-damper model for the vibration isolation test rig in the vertical direction; base frame (BF) motion is represented by x_0 , and metrology frame (MF) motion is represented by x_1 ; the MF has rigid-body mass m ; the isolation module (IM) is modeled as a spring k , a damper d , and an actuator force F_a ; the BF and MF acceleration is measured by the sensors a_0 and a_1 , respectively; sensor x_{10} measures the relative distance between the BF and the MF.

To enable active vibration isolation, each IM is equipped with Lorentz motors, and proximity sensors measuring the relative position between the BF and MF. Furthermore, the BF and MF are equipped with accelerometers to measure the absolute accelerations of both frames in the vertical directions.

The control strategies proposed in this chapter will be applied in three directions, i.e. the coordinates (z, θ_x, θ_y) which are defined with respect to the center of mass of the metrology frame. These coordinates are chosen because they can be calculated from the vertical movements measured at the three isolation modules by applying static coordinate transformations. The principal coordinates (x, y, θ_z) corresponding to horizontal movements are not considered for practical reasons.

6.2.1 Modeling

Measured frequency response functions (FRFs) of the system with actuator forces as the inputs, and measured accelerations as the outputs, are shown in Figure 6.17 in Appendix B in Section 6.10. From this figure, it follows that the system dynamics are strongly decoupled. Therefore, a decentralized plant model will be used for controller design:

$$\mathbf{P}_{\text{acc}}(s) = \begin{bmatrix} \frac{s^2}{ms^2 + d_z s + k_z} & 0 & 0 \\ 0 & \frac{s^2}{J_x s^2 + d_{\theta_x} s + k_{\theta_x}} & 0 \\ 0 & 0 & \frac{s^2}{J_y s^2 + d_{\theta_y} s + k_{\theta_y}} \end{bmatrix}, \quad (6.1)$$

$$\mathbf{P}_{\text{pos}}(s) = \frac{1}{s^2} \mathbf{P}_{\text{acc}}(s),$$

where numerical values for the plant parameters are given in Appendix B in Section 6.10. In fact, the system behaves in each direction as a 1-DOF mass-spring-damper system for which the model is shown in Figure 6.2. The equation of motion (in Laplace domain) for the modeled system is given by

$$(ms^2 + ds + k)X_1(s) = (ds + k)X_0(s) + F_d(s) + F_a(s), \quad (6.2)$$

where s is the Laplace variable, m, d, k are model parameters, X_0 and X_1 represent respectively base frame and metrology frame displacements, and F_a and F_d represent actuator and disturbance forces, respectively.

6.2.2 Disturbance characterization

The metrology frame is basically excited by two types of disturbances, i.e. base frame vibrations $A_0 = s^2 X_0$ and direct disturbance forces F_d . Measured power spectral density (PSD) plots for A_0 and F_d in the z -direction are shown in Figure 6.3. The PSD of A_0 shows increased power below 0.2 Hz due to sensor noise. For frequencies between 2 and 300 Hz, the spectrum is approximately flat. Beyond 450 Hz, the spectrum decreases because the sensor conditioners apply low-pass filtering of the accelerometer signal beyond this frequency. The PSD for F_d shown in Figure 6.3 is based on a measurement of the covariance matrix of the process noise on the metrology frame, i.e. the part of the output power in $\mathbf{A}_1$ that is not correlated with base frame disturbances $\mathbf{A}_0$. Under the assumption that the power of this process noise significantly exceeds the sensor noise $\mathbf{N}_1$, it represents an estimate of the impact of direct disturbances on the metrology frame. If, subsequently, the process noise PSD is filtered with the estimated inverse of the plant model $\hat{P}_{2,acc}$, the PSD of F_d can be retrieved.

The measured sensor noise characteristics are shown in Figure 6.4. These characteristics become important when enabling active vibration isolation, and could induce metrology frame vibrations due to noise amplification. The proximity sensor generally shows a flat frequency spectrum with a magnitude of $10^{-16} \text{ m}^2/\text{Hz}$. At low frequencies it shows a -1 slope which is probably due to pink noise from the electronics. Beyond 100 Hz, the noise spectrum decreases due to low-pass filtering of the sensor signal. The spectrum of the accelerometer noise is approximately flat at the mid- and higher frequencies, but strongly increases at low frequencies. This is typical for many accelerometers. The peak at 850 Hz corresponds to the internal resonance frequency of the sensor [109].

6.2.3 Active vibration isolation

Figure 6.5 shows a block diagram of the active vibration isolation system with feedback and feedforward control. A state-space description for P , representing the mechanical model from Figure 6.2, can be found in Appendix A in Section 6.9. Two external disturbances are considered, i.e. floor vibrations a_0 and

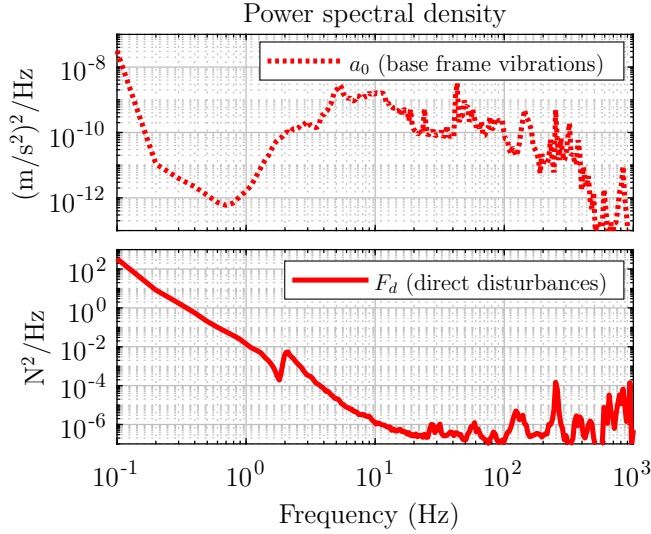


Figure 6.3. Power spectral density plots showing the measured floor vibrations and direct disturbance forces in the z -direction under operating conditions.

direct disturbance forces F_d . The system has three controllers: C_0 provides disturbance feedforward control using base frame accelerations denoted by a_0 , C_1 provides feedback control based on metrology frame accelerations denoted by a_1 , and C_{10} provides feedback control based on the relative position of the two frames which is denoted by x_{10} . Note that a_0 , a_1 , and x_{10} are estimated by measurement through $\tilde{a}_0$, $\tilde{a}_1$, $\tilde{x}_{10}$ which contains the effect of additive sensor noise represented by n_0 , n_1 , and n_{10} , respectively.

Given Figure 6.5, the actuator force F_a can be written in the Laplace domain as

$$F_a(s) = -C_{10}(s)[X_1(s) - X_0(s) + N_{10}(s)] - C_1(s)[A_1(s) + N_1(s)] + C_0(s)[A_0(s) + N_0(s)]. \quad (6.3)$$

Substitution of (6.3) in (6.2), using $A_0(s) = s^2 X_0(s)$, $A_1(s) = s^2 X_1(s)$, and after some rewriting, it is found that

$$A_1(s) = \mathcal{T}(s)A_0(s) + s^2 \mathcal{C}(s)F_d(s) + \mathcal{S}_0(s)N_0(s) + \mathcal{S}_1(s)N_1(s) + \mathcal{S}_{10}(s)N_{10}(s), \quad (6.4)$$

with the performance indicators $\mathcal{T}$, which is the transmissibility function, and

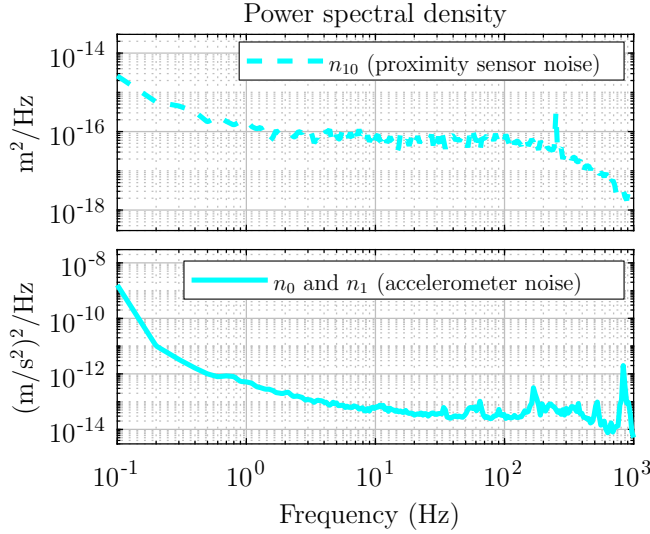


Figure 6.4. Power spectral density plots showing the measured sensor noise characteristics.

$\mathcal{C}$, which is the compliance function, given by

$$\mathcal{T}(s) = \frac{ds + k + C_{10}(s) + s^2 C_0(s)}{ms^2 + ds + k + C_{10}(s) + s^2 C_1(s)}, \quad (6.5)$$

$$\mathcal{C}(s) = \frac{1}{ms^2 + ds + k + C_{10}(s) + s^2 C_1(s)}, \quad (6.6)$$

and the sensor noise sensitivity functions

$$\mathcal{S}_0(s) = \frac{s^2 C_0(s)}{ms^2 + ds + k + C_{10}(s) + s^2 C_1(s)}, \quad (6.7)$$

$$\mathcal{S}_1(s) = \frac{-s^2 C_1(s)}{ms^2 + ds + k + C_{10}(s) + s^2 C_1(s)}, \quad (6.8)$$

$$\mathcal{S}_{10}(s) = \frac{-C_{10}(s)}{ms^2 + ds + k + C_{10}(s) + s^2 C_1(s)}. \quad (6.9)$$

Without control, only floor vibrations and direct disturbances contribute to A_1 which is shown in Figure 6.6. It follows that below 1 Hz and beyond 20 Hz the impact of direct disturbances is dominant. In between these frequencies, the impact of A_0 and F_d are comparable.

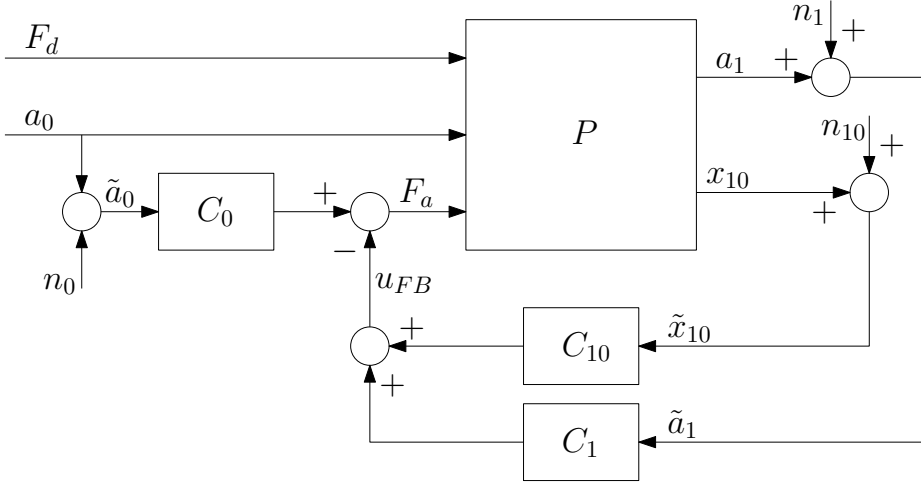


Figure 6.5. Block diagram of the active vibration isolation system in a single direction using feedback plus feedforward control; note that $a_0 = \ddot{x}_0$ and $a_1 = \ddot{x}_1$.

6.3 Problem definition

Reconsider Figure 6.6 regarding the residual power in A_1 without active vibration isolation. The first and main goal in this chapter is to reduce the residual vibrations in the frequency range of interest by properly designing the controllers C_0, C_1, C_{10} . By doing so, a balanced tradeoff is created between the rejection of the physical disturbances a_0, F_d on the one hand, and minimization of noise amplification on the other hand. In case of no sensor noise ($n_0 = n_1 = n_{10} = 0$), the controllers could be designed such that $\mathcal{T}$ and $\mathcal{C}$ have desired magnitudes in the relevant frequency range of interest. In this case, it is (in theory) sufficient to choose $C_{10} = 0$, use C_1 to shape $\mathcal{C}$, and consequently use C_0 to shape $\mathcal{T}$. However, both C_0 and C_1 have measured accelerations as controller input which inherently suffer from low-frequency sensor noise, recall Figure 6.4. To prevent amplification of this noise, the functions $\mathcal{S}_0, \mathcal{S}_1$ may not have a too large gain at low frequencies. Therefore, it is difficult to use C_0 and C_1 for high-gain control at low frequencies without using C_{10} .

The second goal is to obtain leveling, which means that the metrology frame must be maintained at a certain offset distance with respect to the base frame. In general, the goals of leveling and vibration isolation are conflicting, because leveling aims at tracking the base frame motion whereas vibration isolation aims at counteracting it. Fortunately, the interest in leveling is only at very low frequencies (below 0.1 Hz) and therefore does not conflict with our active vibration isolation objectives.

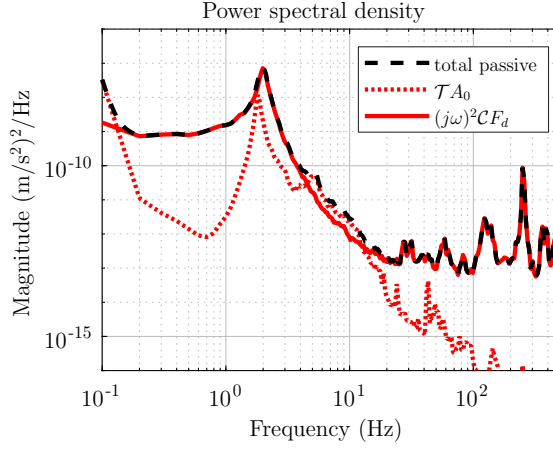


Figure 6.6. Modeled power spectral density plots, showing the contributions of each individual disturbance source to the total residual output power measured at the metrology frame.

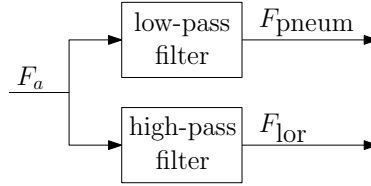


Figure 6.7. Implementation of actuator fusion at the experimental setup; the pneumatic actuator provides static gravity compensation, whereas the Lorentz actuator provides active vibration isolation.

6.3.1 Actuator fusion

In the experimental setup, leveling and gravity compensation are provided by the pneumatic actuators, which can provide large forces but have a slow response due to a large time constant (typically more than 10 s.). In contrast to this, active vibration isolation forces are provided by the Lorentz actuators which can only provide small forces (< 15 N) but with a high accuracy and a fast response thanks to a small time constant (typically less than 0.05 s.). To represent both types of actuators by a single actuator force F_a in Figure 6.2, the concept of actuator fusion is used. This is schematically depicted in Figure 6.7 and combines the attractive properties of large pneumatic actuator forces and fast Lorentz actuator responses. A combination of complementary low-pass and high-pass filters is used with a cut-off frequency of $\omega_a = 2\pi \cdot 0.05$ rad/s. Since the filters are complementary, the total force provided by both actuators is equivalent to

the single force F_a . This enables the possibility to analyze stability in a multi-loop SISO context from open-loop Bode and Nyquist plots based on (measured) FRF data.

6.4 Feedback control

This section proposes a feedback control strategy to improve vibration isolation performance in terms of rejecting direct disturbances, and to obtain leveling of the metrology frame. First, the basic tuning rules are explained to obtain the desired vibration isolation properties using feedback control. Second, the controllers are augmented to provide leveling and robust stability.

6.4.1 Basic tuning rules for active vibration isolation

From a physical point of view, the following tuning rules are used to obtain the desired active vibration isolation properties:

- 1) Design the compliance of soft mounts comparable to that of hard mounts. To this end, relative position feedback control is used to add virtual stiffness, or

$$C_{10}(s) = K_s, \quad (6.10)$$

where K_s has the physical meaning of a stiffness constant corresponding to a virtual spring that is created between the base frame and the metrology frame.

- 2) Virtually increase the mass of the metrology frame, and add skyhook damping to the suspension mode. The addition of virtual mass is necessary to decrease the suspension frequency that otherwise would increase by the addition of virtual stiffness in step 1. Skyhook damping is added to the suspension mode to improve the disturbance rejection properties for frequencies that are close to the suspension frequency. Virtual mass and skyhook damping are added by using proportional and integral acceleration feedback, respectively, or

$$C_1(s) = K_a + \frac{K_v}{s}, \quad (6.11)$$

where gains K_a and K_v have the physical meaning of a virtual mass and a skyhook damper added to the metrology frame [97].

If (6.10) and (6.11) are substituted in (6.5) and (6.6), and feedforward control is neglected ($C_0 = 0$), then after some rewriting, it is obtained that

$$\mathcal{T}(s) = \frac{ds + (k_1 + K_s)}{(m + K_a)s^2 + (d + K_v)s + (k_1 + K_s)}, \quad (6.12)$$

$$\mathcal{C}(s) = \frac{1}{(m + K_a)s^2 + (d + K_v)s + (k_1 + K_s)}. \quad (6.13)$$

The common denominator of $\mathcal{T}$ and $\mathcal{C}$ represents a second-order polynomial, from which a complex pair of poles can be calculated that corresponds to the closed-loop suspension mode. Using (6.13), the following tuning rules are proposed:

- The virtual stiffness is set to $K_s = 100k$, which will make the closed-loop system 100 times stiffer, and virtually increases the suspension frequency from 1.8 Hz to 18 Hz if no virtual mass would be applied.
- To keep the suspension frequency $\sqrt{\frac{k+K_s}{m+K_a}}$ of the closed-loop system at the same value as the suspension frequency $\sqrt{\frac{k}{m}}$ of the uncontrolled system, it follows that $K_a := K_s \frac{m}{k}$; the value $K_v := 2\zeta_r \omega_r (m + K_a) - d$ is based on the desired relative damping ratio ζ_r of the suspension mode of the closed-loop system; a typical value is given by $\zeta_r = 0.7$ which implies that $K_v = 2.10 \cdot 10^6 \text{ Ns/m}$.

The resulting closed-loop transmissibility function $\mathcal{T}$ and compliance function $\mathcal{C}$ are shown in Figure 6.10. It is observed that the suspension frequency of the closed-loop system is maintained at 1.8 Hz and that skyhook damping is added to the suspension mode. At low frequencies, the magnitude of the compliance is decreased with a factor of 100, because the closed-loop stiffness $(K_s + k)$, with $K_s = 100k$, is approximately 100 times larger than the passive stiffness k .

6.4.2 Stability analysis

The stability analysis is performed in the frequency domain by analyzing the transfer function of the open-loop system denoted by L . Using the block diagram in Figure 6.5, L is defined as the transfer function from plant input F_a to controller output u_{FB} under the condition that the feedback loop is opened at u_{FB} . In other words, L is the sum of the separate open-loop transfer functions for relative position feedback and absolute acceleration feedback, or

$$L(s) = L_{\text{pos}}(s) + L_{\text{acc}}(s), \quad (6.14)$$

with $L_{\text{pos}}(s) = C_{10}(s)P_{\text{pos}}(s)$ and $L_{\text{acc}}(s) = C_1(s)P_{\text{acc}}(s)$. The functions $P_{\text{pos}}(s)$ and $P_{\text{acc}}(s)$ are given by the (1,1)-elements of respectively $\mathbf{P}_{\text{pos}}$ and $\mathbf{P}_{\text{acc}}$ in (6.1). Figure 6.8 shows a comparison of the total open-loop system L with the separate open-loop systems L_{pos} , L_{acc} . It is observed that at low frequencies L is dominated by L_{pos} (position feedback), while at high frequencies L is dominated by L_{acc} (acceleration feedback), which is desirable for two reasons. On the one hand, accelerometer measurements are noisy at low frequencies which easily induces drift and actuator saturation problems. On the other hand, absolute

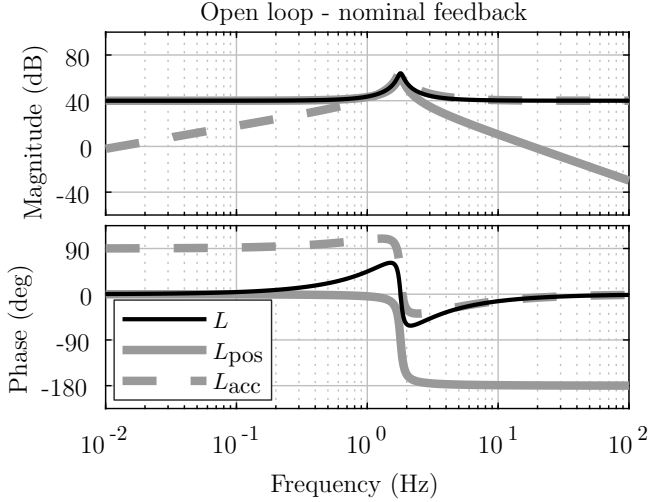


Figure 6.8. Bode plots of the separate open-loop systems L_{pos} (solid gray, relative position feedback) and L_{acc} (dashed gray, absolute acceleration feedback), and the total open loop system $L = L_{\text{pos}} + L_{\text{acc}}$ (black). Note that L still has an infinite bandwidth and needs robustification, see Figure 6.9.

acceleration feedback is desired at higher frequencies, because it has the physical interpretation of adding virtual mass and skyhook damping.

It follows from Figure 6.8 that the closed-loop system has an infinite bandwidth because the magnitude of L never crosses the 0 dB line. In fact the open-loop FRF is positive real, which is rather unrealistic in view of unmodeled higher-order dynamics and actuator delays. Therefore, a robustness filter is designed that renders the closed-loop system robustly stable. This filter consists of a second-order low-pass filter at $\omega_{lp} = 1.8 \cdot 2\pi$ rad/s and $\zeta_{lp} = 0.7$ to limit the controller bandwidth, and a zero at $\omega_z = 2\pi \cdot 16$ rad/s to improve the phase margin around the cross-over frequency, giving

$$H_{\text{rob}}(s) = \frac{\omega_{lp}^2}{s^2 + 2\zeta_{lp}\omega_{lp}s + \omega_{lp}^2} \cdot \frac{s + \omega_z}{\omega_z}. \quad (6.15)$$

The robustness filter will be applied to both C_1 and C_{10} . By doing so, the gain ratio of C_1 and C_{10} will remain constant such that the tuning rules derived to obtain (6.10) and (6.11) remain valid within the restrictions imposed on the realizable bandwidth.

In acceleration feedback, an additional high-pass filter at $\omega_{hp} = 0.1 \cdot 2\pi$ rad/s will be used to obtain a stronger suppression of accelerometer noise at low fre-

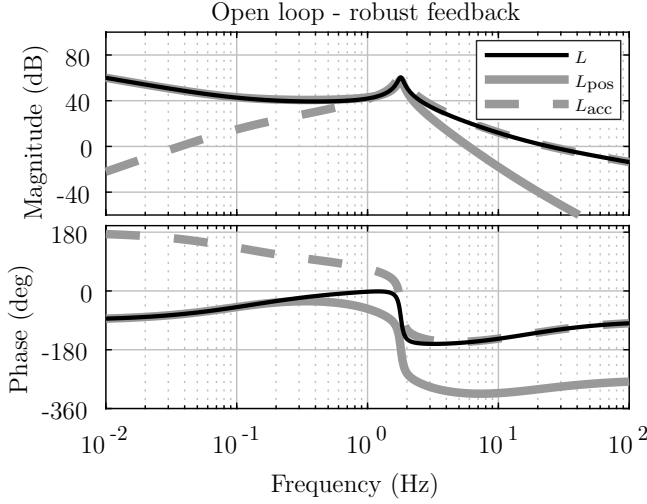


Figure 6.9. Bode plots of the open-loop systems $L_{\text{pos},r}$ (solid gray, relative position feedback), $L_{\text{acc},r}$ (dashed gray, absolute acceleration feedback), and the total open-loop system $L_r = L_{\text{pos},r} + L_{\text{acc},r}$ (black).

quencies:

$$H_{\text{hp}}(s) = \frac{s}{s + \omega_{hp}}. \quad (6.16)$$

6.4.3 Augmented controllers with leveling

The ability to maintain the metrology frame at a desired leveling set-point with respect to the base frame is provided by adding a proportional-integral (PI) action to the position feedback controller,

$$H_{\text{level}}(s) = \frac{\omega_i + s}{s}, \quad (6.17)$$

which provides integral action below $\omega_i = 0.1 \cdot 2\pi$ rad/s and proportional action beyond ω_i .

The augmented feedback controllers are thus given by

$$C_{10,r}(s) = C_{10}(s)H_{\text{rob}}(s)H_{\text{level}}(s), \quad (6.18)$$

$$C_{1,r}(s) = C_1(s)H_{\text{rob}}(s)H_{\text{hp}}(s). \quad (6.19)$$

Figure 6.9 shows Bode plots for the individual open-loop systems $L_{\text{acc},r} = P_{\text{acc}}C_{1,r}$, $L_{\text{pos},r} = P_{\text{pos}}C_{10,r}$, and the total open-loop system $L_r = L_{\text{acc},r} + L_{\text{pos},r}$. Compared to L in Figure 6.8, the cross-over frequency of L_r is limited to 24 Hz

with a sufficiently large phase margin of 58° . The gain margin for L_r in Figure 6.9 is infinite because the line indicating a phase shift of -180° is never crossed. The high-pass filter in (6.16) further decreases $L_{acc,r}$ below 0.1 Hz, while in the same region the magnitude of L_{pos} is increased to obtain the desired leveling properties.

Performance in terms of $\mathcal{T}$ and $\mathcal{C}$ using robust feedback control is shown in Figure 6.10. In terms of $\mathcal{T}$, the performance obtained with robust feedback control is similar to the performance obtained with nominal feedback control from (6.10) and (6.11). A difference, however, is that the robust feedback controller from (6.18) and (6.19) creates a notch around 20 Hz in $\mathcal{T}$ because (6.5) has a zero given the fact that $ds + k + C_{10}(s) = 0$. In terms of $\mathcal{C}$, performance is further improved for frequencies below 0.1 Hz as a result of the leveling integrator included by (6.17). Beyond 20 Hz, performance is no longer improved because the bandwidth is limited by the filter given in (6.15).

6.5 Feedforward control

In Section 6.4 a mixed-sensor feedback controller has been presented that greatly improves $\mathcal{C}$ up to 20 Hz. However, apart from damping the suspension mode, feedback control does not improve performance in terms of $\mathcal{T}$. Therefore, a disturbance feedforward controller C_0 is presented in this section to also enhance $\mathcal{T}$. The considered design method for this feedforward controller is partly adopted from Chapter 4.

In order to design C_0 , recall the expression of $\mathcal{T}$ in (6.5). From this expression, it follows that perfect cancellation of a_0 ($\mathcal{T} = 0$) is obtained if C_0 is designed as

$$C_{0,opt}(s) = -\frac{ds + k + C_{10}(s)}{s^2}. \quad (6.20)$$

However, this optimal controller is not feasible in practice, because it induces drift as a result of using pure integrators in combination with accelerometers. For this reason, it was proposed in Chapter 4 to replace the pure integrators by n^{th} -order weak integrators. That is, the integrating actions are cut off below a certain frequency α . By doing so, a modified feedforward controller is given by

$$C_{0,sub}(s) = -H_{(\alpha,n)}^2(s) \left(ds + k + \frac{sC_{10}(s)}{s + 2\pi 0.1} \right), \quad (6.21)$$

with

$$H_{(\alpha,n)}(s) = \frac{1 - L_{(\alpha,n)}(s)}{s}, \quad (6.22)$$

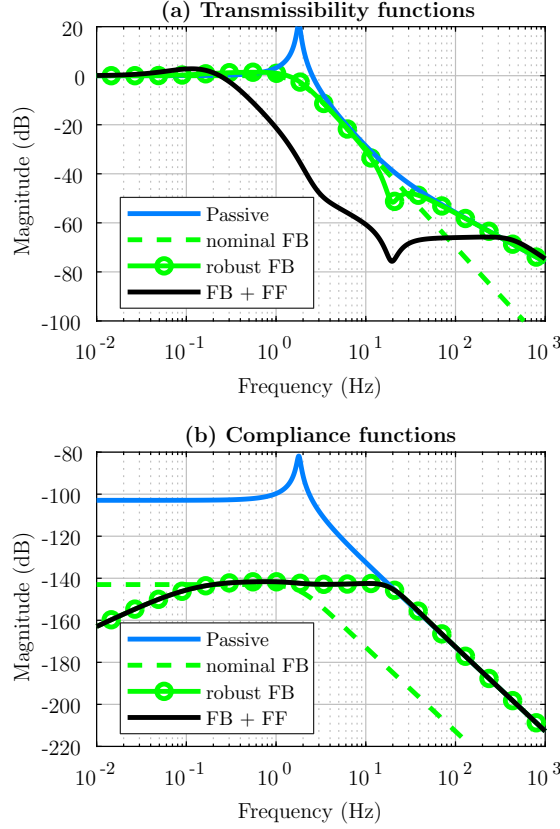


Figure 6.10. Vibration isolation performance in terms of (a) transmissibility, and (b) compliance; performance is shown for the passive and active systems with nominal feedback (FB) control according to (6.10) and (6.11), robust feedback control according to (6.18) and (6.19), and robust feedback plus feedforward (FF) control, the latter given by (6.20); FB control is used to decrease $\mathcal{C}$, whereas FF control is used to decrease $\mathcal{T}$.

describing an n^{th} -order weak integrator, and

$$L_{(\alpha,n)}(s) = \left(\frac{\alpha}{s + \alpha} \right)^n, \quad (6.23)$$

representing an n^{th} -order low-pass filter with cut-off frequency α rad/s. In this chapter, $\alpha = 0.2 \times 2\pi$ rad/s and $n = 2$. In addition, C_{10} is high-pass filtered at 0.1 Hz to cut off the leveling integrator that is included in C_{10} .

The signal conditioners of the accelerometers contain a low-pass filter F_s at

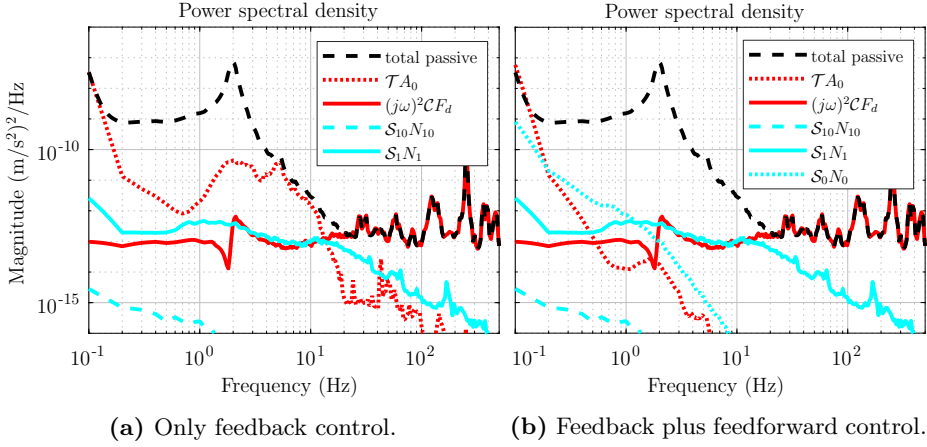


Figure 6.11. Modeled power spectral density plots, showing the contributions of each disturbance and noise source to the total residual vibration power of the metrology frame; compared to Figure 6.6, Figure 6.11a shows that feedback control is used to greatly reduce the impact of direct disturbances; subsequently; Figure 6.11b shows that feedforward control is used to suppress the contribution of base frame vibrations to the output power.

$f_l = 450$ Hz and relative damping $\zeta_l = 0.7$, or

$$F_s(s) = \frac{(2\pi f_l)^2}{s^2 + 2\zeta_l(2\pi f_l)s + (2\pi f_l)^2}. \quad (6.24)$$

This filter limits the frequency range in which disturbance feedforward control can improve performance. To take into account the limiting effect of this filter in the modeling results, it will be regarded as part of the feedforward controller:

$$C_0(s) = -H_{(\alpha,n)}^2(s) \left(ds + k + \frac{sC_{10}(s)}{s + 2\pi 0.1} \right) F_s(s). \quad (6.25)$$

Performance in terms of $\mathcal{T}$ and $\mathcal{C}$ is shown in Figure 6.10 for the system with combined feedback and feedforward control. For frequencies beyond 0.2 Hz, which corresponds to the value for α , the transmissibility is greatly enhanced up to 300 Hz; around the latter frequency performance is limited by the low-pass filter of the accelerometers, see (6.24). Note that improvements up to such high frequencies are generally only feasible with feedforward control, due to stability issues occurring in high-bandwidth feedback control in view of non-modeled system dynamics and actuator delay. For frequencies beyond 300 Hz, performance is slightly deteriorated due to a waterbed effect, as discussed in Chapter 2. For the same reason, performance is also deteriorated for frequencies below α , but this is not a problem because the power of floor vibrations below

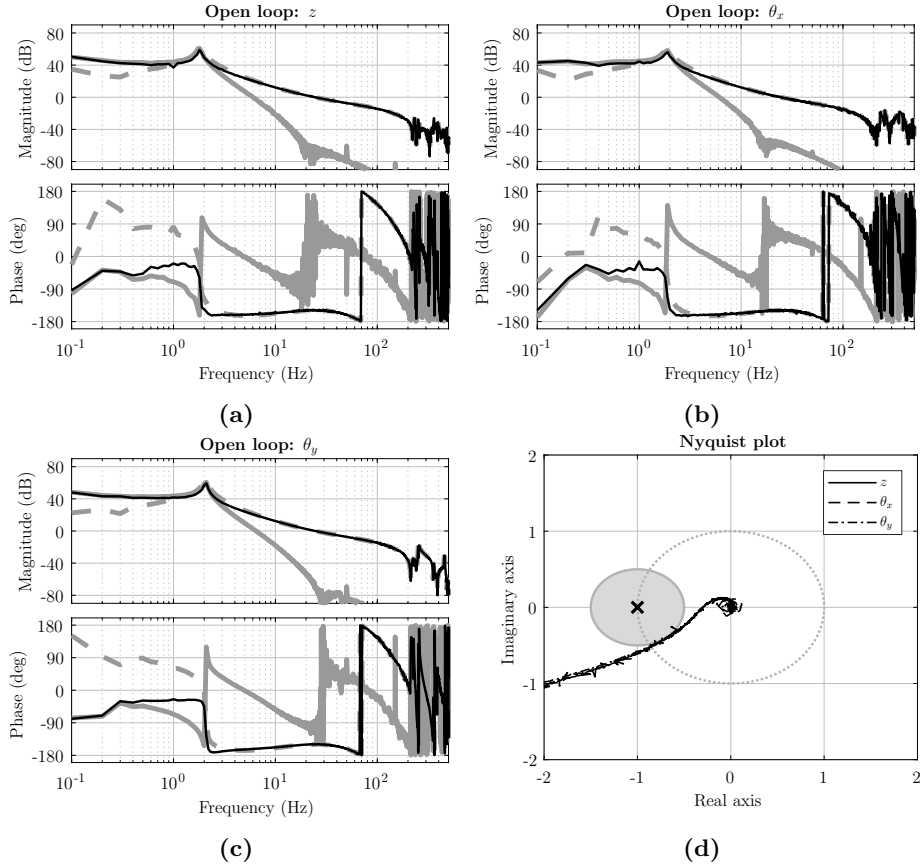


Figure 6.12. (a-c) Bode plots of the open loop systems per direction, showing the position loop (solid gray), the acceleration loop (dashed gray), and the total open loop system (solid black); (d) Nyquist plots of the total open loop systems, showing closed-loop stability of the feedback loops since the point $(-1,0)$ is at the left when the contours approach the origin; note that the contours do not cross the gray disc which indicates a sufficiently large modulus margin of 0.5, hence robust stability.

1 Hz is fairly low, recall Figure 6.3. In the compliance plot, the feedforward curve overlaps the curve obtained with robust feedback control. This is because feedforward control does not affect $\mathcal{C}$, recall (6.6).

6.6 Modeled output power analysis

So far, controllers were designed and evaluated based on $\mathcal{T}$ and $\mathcal{C}$. The effect of sensor noise was only taken into account qualitatively by mentioning that the controllers C_0 and C_1 must have limited gains at low frequencies, because of accelerometer noise. The goal of the current section is to provide a quantitative analysis of the sensor noise contributions to the power in A_1 , because minimization of A_1 is the actual goal in this chapter, recall Section 6.3.

First, the effect of adding feedback control is considered using Figure 6.11a, which presents modeled power spectral density plots for the individual disturbance and noise contributions to A_1 . Compared to the results in Figure 6.6 corresponding to the passive system, the major advantage of feedback control is that the contribution of direct disturbances to A_1 is significantly reduced. This contribution is decreased with a factor of $(100)^2$, which corresponds with the tuning rule in (6.10) to increase the stiffness virtually with a factor of 100. There is no significant reduction of floor vibrations, which is due to the fact that $\mathcal{T}$ is not significantly decreased using feedback control.

Second, the effect of adding feedforward control is considered using the modeled PSD plots shown in Figure 6.11b. Compared to Figure 6.11a, it is observed that adding feedforward control largely decreases the effect of base frame vibrations a_0 beyond 0.2 Hz, but this comes at the cost of amplification of sensor noise n_0 , which becomes the new dominant contribution to A_1 between 0.2 and 1 Hz.

6.7 Experimental validation

This section discusses experimental results of the proposed feedback and feedforward control strategies on the vibration isolation system shown in Figure 6.1 and in three directions, i.e. z , θ_x , and θ_y . First, some practical remarks on implementing the feedback and feedforward controllers are discussed. Afterwards, the closed-loop measurement results are presented.

6.7.1 Practical remarks on controller implementation

For feedback control, the design approach from Section 6.4 is applied in all the three directions. This approach needs the mass, damping, and stiffness parameters of the system which are derived in Appendix B in Section 6.10. Having these parameters, feedback control is used to virtually increase both the mass and the stiffness with a factor of 50, and to obtain a damping ratio of $\zeta_n = 0.7$ for the suspension modes in closed-loop. For each direction, the zero of the integrator from (6.17) that provides leveling is placed at 0.1 Hz. Robustness filters as given in (6.15) are added, where for each direction the low-pass filter frequency ω_{lp} is chosen in line with the suspension frequency of the corresponding

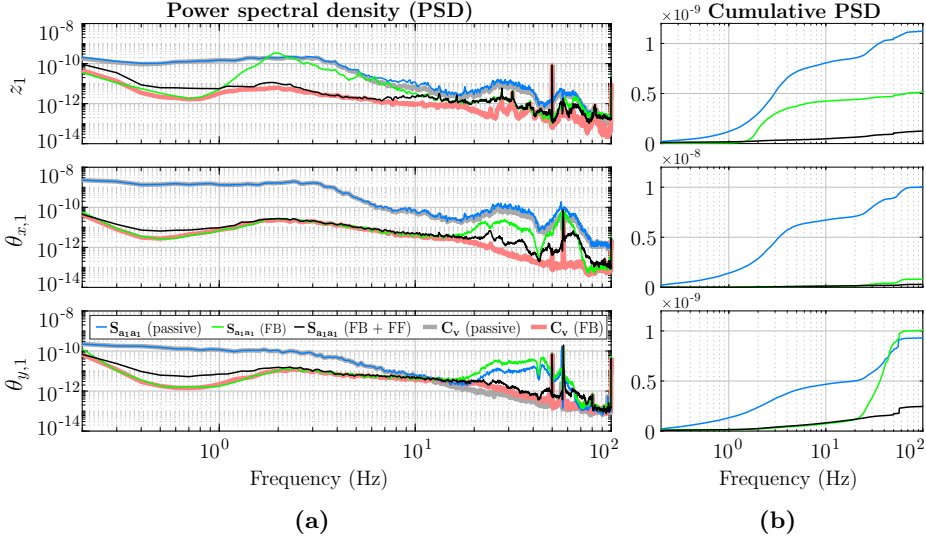


Figure 6.13. This figure shows: (a) measured power spectral density (PSD) plots of the metrology frame accelerations, and (b) measured cumulative PSD plots; the total power in $\mathbf{a}_1$ is represented by $\mathbf{S}_{\mathbf{a}_1 \mathbf{a}_1}$, whereas $\mathbf{C}_v$ represents the fraction of $\mathbf{S}_{\mathbf{a}_1 \mathbf{a}_1}$ that is caused by direct disturbances and sensor noise; feedback control reduces $\mathbf{C}_v$ and partly $\mathbf{S}_{\mathbf{a}_1 \mathbf{a}_1}$; the remaining parts of $\mathbf{S}_{\mathbf{a}_1 \mathbf{a}_1}$ are reduced by feedforward control.

direction (1.8 Hz for z , 2.1 Hz for θ_x , 2.2 Hz for θ_y). The zero from (6.15) that provides phase lead is placed at 13 Hz in all three directions. To ensure robust stability, four notch filters are applied to suppress the peaks corresponding to structural modes of the metrology frame at 228, 255, 355, and 391 Hz. In each of the position loops, an additional second-order low-pass filter at 10 Hz with damping ratio 0.7 is used to compensate for the anti-resonances shown in Figure 6.18. Without this extra filtering, the slope of the position loops would turn with +40 dB/decade after the anti-resonance which is undesired because then effectively too much virtual stiffness is added.

Bode plots of the resulting open-loop systems are given in Figures 6.12a-c. These plots are similar to the modeling results showed in Figure 6.9 where position feedback dominates at low frequencies and acceleration feedback at high frequencies. The cross-over frequencies are encountered around 25 Hz, but have less phase margin compared to the model due to non-modeled actuator delays. Figure 6.12d shows Nyquist plots of the open-loop systems. It is observed that the point $(-1, 0)$ always is at the left when the contours approach the origin, indicating closed-loop stability. Moreover, the contours do not cross the gray disk with a radius of 0.5, which is an indication for sufficient modulus margin,

hence robust stability.

The feedforward controllers are implemented according to (6.21) for all three directions. To reduce parameter uncertainty, the feedforward controller is implemented as a structure with fixed poles and self-tuning zeros to find (online) the controller parameters giving the best performance. A detailed discussion about this self-tuning algorithm can be found in Chapter 4.

6.7.2 Results

Measurement results and performance improvements based on the reduction in output power, transmissibility and compliance functions, and the reduction in leveling errors are discussed in the following. For clarity of presentation, two small modifications to the setup are performed. First, compared to Figure 6.3, the power spectrum of $\mathbf{a}_0$ is artificially increased between 20 and 100 Hz using shakers which are located between the base frame and the floor. Second, a little amount of skyhook damping is applied to the suspension mode of the “passive” system as well to simplify the comparison of the measurements.

Output power

Figure 6.13a shows measured power spectral density (PSD) plots of the metrology frame accelerations $\mathbf{a}_1$. In these plots, the total power in $\mathbf{a}_1$ is represented by $\mathbf{S}_{\mathbf{a}_1\mathbf{a}_1}$, whereas $\mathbf{C}_v$ represents the fraction of $\mathbf{S}_{\mathbf{a}_1\mathbf{a}_1}$ that is not correlated with $\mathbf{a}_0$ but caused by direct disturbances and sensor noise amplification. In other words, $\mathbf{C}_v$ cannot be reduced by feedforward control. It is observed that the passive system generally shows the largest total power which is dominantly caused by $\mathbf{C}_v$. As such, adding feedforward control directly to the passive system will hardly lead to any performance improvement. Therefore, the first step is to apply feedback control to reduce $\mathbf{C}_v$. By doing so, it is observed that both $\mathbf{S}_{\mathbf{a}_1\mathbf{a}_1}$ and $\mathbf{C}_v$ are largely reduced below 2 Hz in the z -direction, and below 20 Hz in the θ_x - and θ_y -directions. In contrast, $\mathbf{S}_{\mathbf{a}_1\mathbf{a}_1}$ is not (largely) decreased between 2 and 20 Hz in the z -direction, and between 20 and 70 Hz in the θ_x - and θ_y -directions. In these regions, base frame vibrations clearly become the dominant source of output power when only feedback control is used. However, this power can be effectively reduced by combining feedback and feedforward control, which is shown in the figure. Note that feedforward control does not reduce $\mathbf{S}_{\mathbf{a}_1\mathbf{a}_1}$ completely up to $\mathbf{C}_v$ due to the limited spring-damper compensation structure of the feedforward controller. It is expected that performance could be further improved by extending the controller structure with basis functions, see Chapter 5.

Figure 6.13b presents cumulative PSD plots showing that the addition of feedback and feedforward control leads to significant performance improvements. Compared to the passive system, the cumulative power at 100 Hz has reduced with 89 % in the z -direction, with 97 % in the θ_x -direction, and with 74 % in the

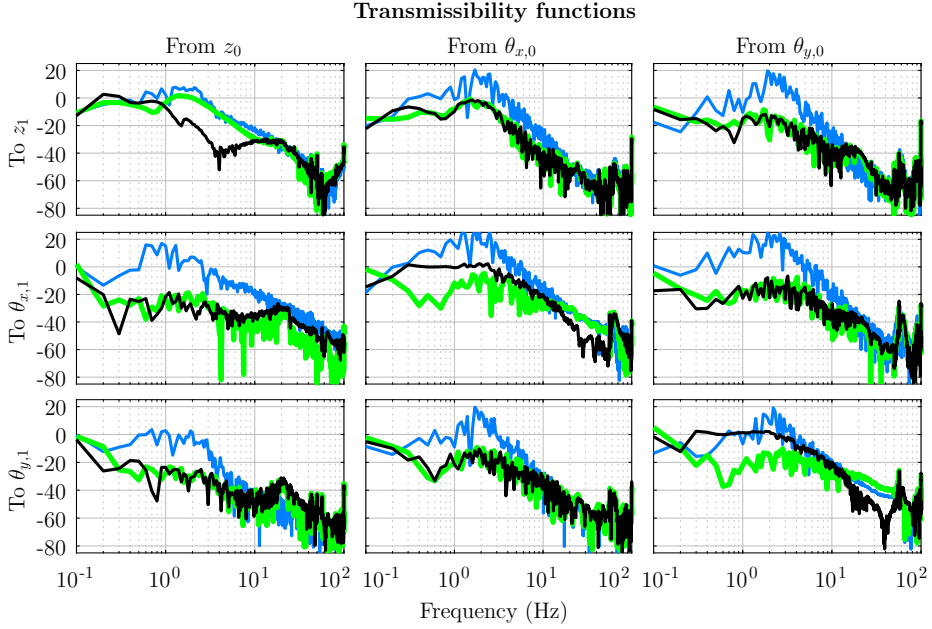


Figure 6.14. Bode magnitude plot of the measured transmissibility functions for the passive system (blue), the system with feedback control (green), and the system with feedback plus feedforward control (black).

θ_y -direction. Observe the remarkable result that in θ_y -direction the system with only feedback control has a larger cumulative power at 100 Hz than the passive system. This is due to the waterbed effect in feedback control which deteriorates performance beyond the bandwidth of 20 Hz, which is the region where shaker-induced disturbances occur. As the figure shows, the problem disappears when also feedforward control is added.

Transmissibility functions

Figure 6.14 shows the measured transmissibility functions. Although it is difficult to sufficiently excite $\mathbf{a}_0$, which makes the measurements noisy, some interesting conclusions can be drawn from the main diagonal entries of the plot. In the z -direction, it is observed that feedback control does not affect the transmissibility function significantly, which is in line with the modeling results in Figure 6.10a. In the same direction, it is shown that feedforward control significantly reduces the transmissibility function at lower frequencies, but not as much as predicted from Figure 6.10a at the mid-frequencies. This is mainly due to the additional dynamics and low-pass filter poles in the position sensor loop and which were not accounted for in the model. In the θ_x - and θ_y -directions, the

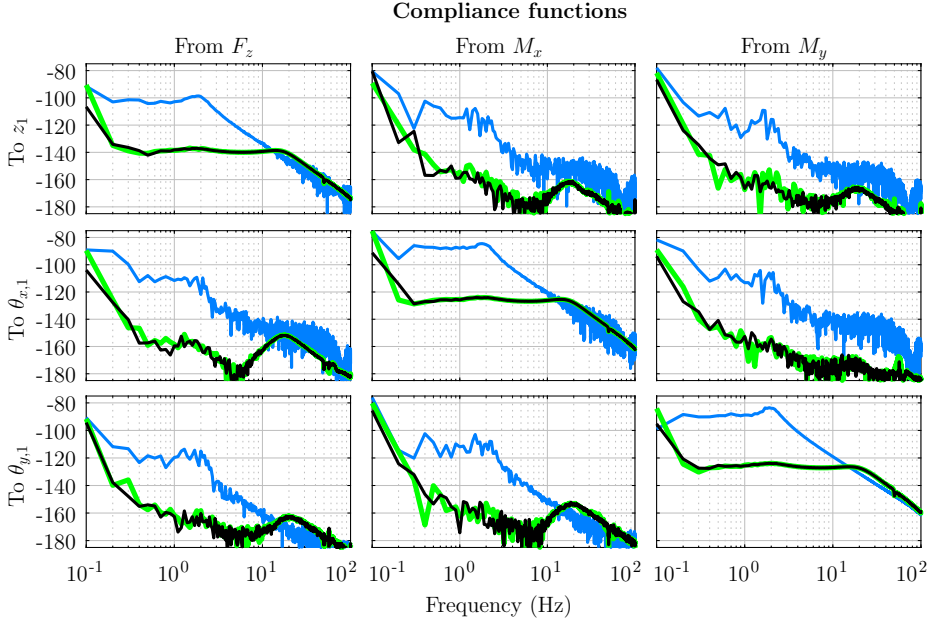


Figure 6.15. Bode magnitude plot of the measured compliance functions for the passive system (blue), the system with feedback control (green), and the system with feedback plus feedforward control (black).

system is only sufficiently excited beyond 20 Hz. Here, the ability of feedforward control is demonstrated to reduce the transmissibility also beyond 20 Hz. This is because feedforward control, in contrast to feedback control, has no bandwidth limitation to guarantee robust stability and it generally induces a better signal-to-noise ratio.

Compliance functions

Figure 6.15 shows the measured compliance functions. It is observed that the main diagonal entries show strong agreements with the modeling results shown in Figure 6.10b. Using feedback control, the magnitudes are lowered up to 34 dB at low frequencies, indicating that the controller indeed renders the closed-loop system about 50 times stiffer. Beyond the bandwidth of 20 Hz, performance is slightly deteriorated due to the waterbed effect. At the off-diagonal entries, the performance improvements are similar but the measurements become noisy because the system dynamics are strongly decoupled. At very low frequencies, the plots show sudden increase because of sensor noise that hampers the measurement. As expected from Figure 6.10b, adding feedforward control does not affect the closed-loop compliance functions.

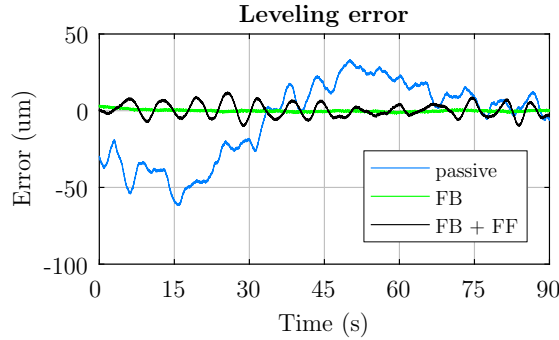


Figure 6.16. Leveling error. The root-mean-square (RMS) leveling errors are $27.0 \mu\text{m}$ with skyhook damping, $1.0 \mu\text{m}$ with feedback control, and $3.4 \mu\text{m}$ with feedback plus feedforward control.

Leveling

Measured performance in terms of leveling is shown in Figure 6.16. The figure only shows performance in the z -direction, but similar results are found for the θ_x - and θ_y -directions. It is observed that feedback control largely reduces the leveling error thanks to the addition of virtual stiffness. Compared to feedback control, the addition of feedforward control slightly deteriorates the leveling performance due to the amplification of low-frequency accelerometer noise $\mathbf{n}_0$, recall Figure 6.11b. However, the root-mean-square (RMS) leveling error is still about 8 times lower compared to the passive system.

6.8 Conclusions

This chapter provides a physics-based control design approach toward feedback and feedforward control design for active soft-mounted vibration isolators in a (multi-loop) SISO context. Relative position feedback is proposed to increase performance in terms of leveling and the compliance functions by adding virtual stiffness. Simultaneously, virtual mass and skyhook damping are added using acceleration feedback to prevent that the transmissibility of floor vibrations is increased. In addition, disturbance feedforward control is proposed to improve the transmissibility function. Experimental results show that the compliance functions are improved up to a factor of 50, and that the RMS leveling error is reduced with a factor of eight, while guaranteeing robust stability via sufficient modulus margin. Application of disturbance feedforward control further reduces the residual output power up to 97 % compared to the passive system, and reduces the transmissibility of floor vibrations in three directions in a broad frequency range.

6.9 Appendix A: State-space model

The mechanical system is described by

$$\begin{bmatrix} x_{10} \\ a_1 \end{bmatrix} = \underbrace{\begin{bmatrix} A & B \\ C & D \end{bmatrix}}_P \begin{bmatrix} a_0 \\ F_d \\ F_a \end{bmatrix} \quad (6.26)$$

with state-space matrices

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{k_1}{m_1} & \frac{d_1}{m_1} & -\frac{k_1}{m_1} & -\frac{d_1}{m_1} \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{1}{m_1} & \frac{1}{m_1} \end{bmatrix},$$

$$C = \begin{bmatrix} -1 & 0 & 1 & 0 \\ \frac{k_1}{m_1} & \frac{d_1}{m_1} & -\frac{k_1}{m_1} & -\frac{d_1}{m_1} \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \frac{1}{m_1} & \frac{1}{m_1} \end{bmatrix}.$$

From this state-space model, the following two transfer functions can be extracted. $P_{pos}(s)$ describes the transfer function from actuator input to relative position output, and $P_{acc}(s)$ describes the transfer function from actuator input to MF acceleration output. From (6.2), it can be derived that

$$P_{pos}(s) := \frac{X_{10}(s)}{F_a(s)} = \frac{1}{ms^2 + ds + k},$$

$$P_{acc}(s) := \frac{A_1(s)}{F_a(s)} = \frac{s^2}{ms^2 + ds + k}. \quad (6.27)$$

These transfer functions are used for stability analysis in Section 6.4.

6.10 Appendix B: System identification

Parametric descriptions for the plant models $\mathbf{P}_{acc}$ and $\mathbf{P}_{pos}$ are given in (6.1). Numerical values for the plant parameters are $m = 1300$ kg, $J_x = 285$ kg m², $J_y = 235$ kg m², $k_z = 170$ kN/m, $k_{\theta_x} = 40$ kNm/rad, $k_{\theta_y} = 40$ kNm/rad, $d_z = 1100$ Ns/m, $d_{\theta_x} = 300$ Nms/rad, $d_{\theta_y} = 170$ Nms/rad.

Figure 6.17 compares the modeled and measured frequency response matrix (FRM) for $\mathbf{P}_{acc}$. Since the system is considered to be decoupled, only FRF data of the main diagonal entries of $\mathbf{P}_{acc}$ are shown. It is observed that in all directions the magnitude response can be accurately described by the model up to 100 Hz. At higher frequencies, the measurement shows additional resonance peaks due to internal modes of the system. At very low frequencies below 0.3 Hz, the measurement becomes unreliable due to accelerometer noise. The measured

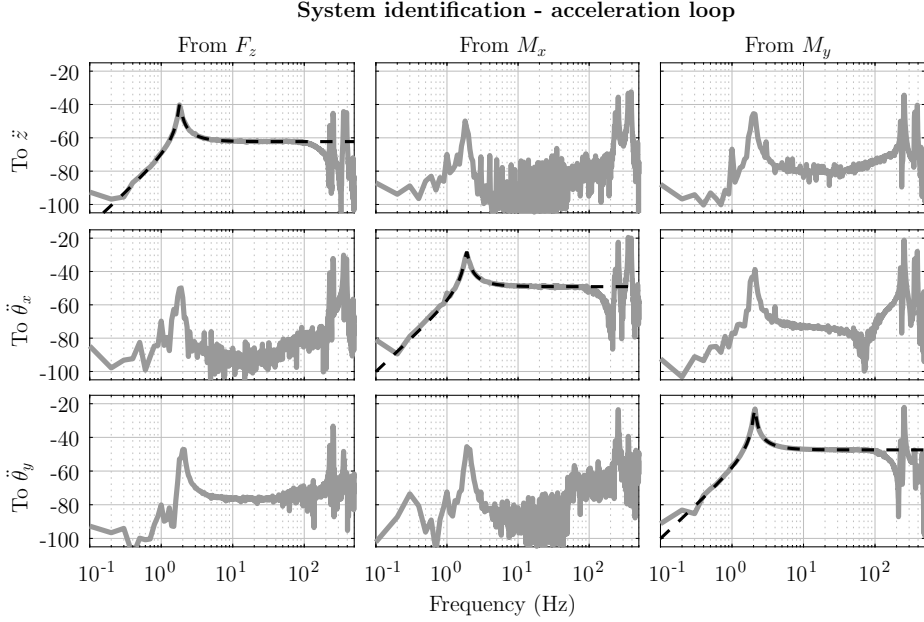


Figure 6.17. Accelerometers - Bode magnitude plots showing the measured FRF matrix $\mathbf{P}_{2,\text{acc}}$ (solid gray) and the parametric fit $\hat{\mathbf{P}}_{2,\text{acc}}$ (dashed black); from actuator input to relative position output.

phase starts to deviate already beyond 10 Hz due to delay in the Lorentz actuators. Note that the suspension frequency is at 1.8 Hz.

Figure 6.18 compares the modeled and measured frequency response matrix (FRM) for $\mathbf{P}_{\text{pos}}$. In contrast to $\mathbf{P}_{\text{acc}}$, the magnitude response of the measured FRF already starts to deviate from the model beyond 15 Hz due to anti-resonances in all three directions. These anti-resonances are caused by a mechanical decoupling of the support plate where the sensors are mounted on.

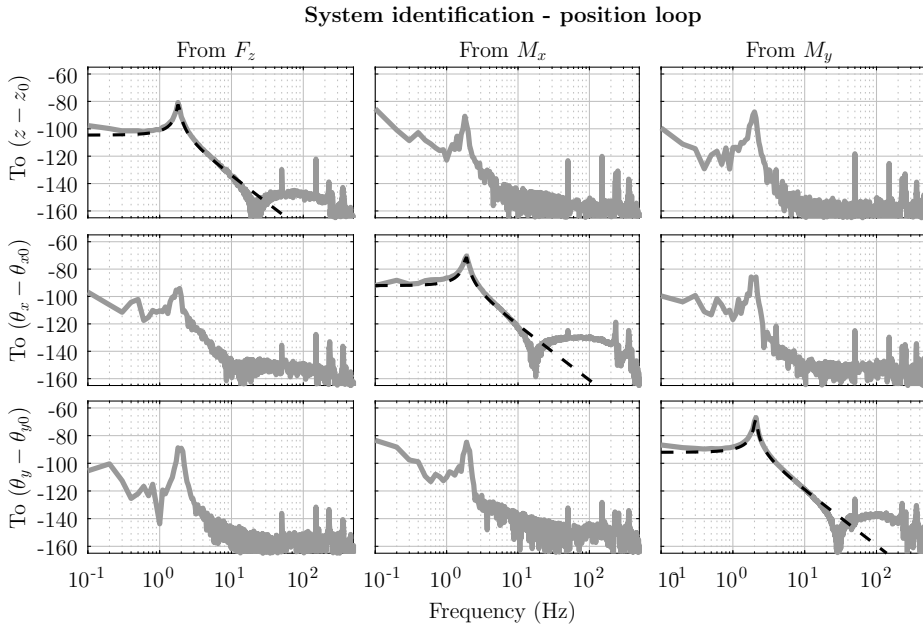


Figure 6.18. Position sensors - Bode magnitude plots showing the measured FRF matrix $\mathbf{P}_{2,\text{pos}}$ (solid gray) and the parametric fit $\hat{\mathbf{P}}_{2,\text{pos}}$ (dashed black); from actuator input to relative position output.

Chapter 7

Conclusions and Recommendations

In this thesis, disturbance feedforward control (DFC) is considered for active vibration isolation systems in high-precision machines. It is shown that DFC can significantly improve performance, despite the fact that DFC inherently comes too late with its counteracting outputs and therefore has a causality problem similar to feedback control. Other key differences are that DFC generally operates under excellent signal-to-noise ratios and that disturbance rejection at high frequencies is not restricted by bandwidth limitations. To obtain performance improvement, design tools for identification and control toward an implementation of DFC in next-generation active vibration isolation systems are provided. These tools have been experimentally validated on both hard- and soft-mounted active vibration isolation systems. Most of the validations are performed on the floating reticle stage test rig, which includes a vibration isolation system that is highly representative for industrial applications in terms of size, weight, and hardware specifications.

7.1 Conclusions

The conclusions in this section are aligned with the research challenges as formulated in Chapter 1.

Inherent design limitations in DFC

Performance tradeoffs are shown to be present in causal feedforward control systems like DFC. The feedforward sensitivity matrix describes performance improvement and deterioration obtained at different frequencies and directions,

and is shown to be constrained by fundamental limitations. More specifically, it is shown that frequency-wise tradeoffs can be exchanged between the principal gains, but not between the principal directions. In view of this, the H_2 control design framework is demonstrated to include an accurate model of the system as well as its disturbance directions to optimally deal with the fundamental limitations. Application to a numerical example of a 2-DOF isolated platform shows how performance is improved by adequate modeling of the disturbances. It also supports the main conclusion, namely that a centralized controller is the optimal solution if the incoming disturbances are correlated, even though the plant dynamics are perfectly decoupled.

Measurement of the transmissibility matrix

A non-parametric identification approach based on spectral analysis is presented to estimate the transmissibility matrix of an industrial active vibration isolation system. Measurements without additional shaker excitation can only provide a good frequency response function for the outputs related to input directions that are sufficiently excited by environmental vibrations from the floor. An analysis method based on principal component analysis is presented to detect the insufficiently excited input directions, for which subsequently the excitation is increased with minimal cost of experiment using an adequate input design of the shaker signals. Experiments with enabled shaker excitation show reliable measurement results for frequencies from 4 to 100 Hz and in three directions. For frequencies below 4 Hz, the measurement results are validated using a parametric model obtained via an indirect measurement. This validation generally shows a good fit of the identified transmissibility matrix with the parametric model at the lower frequencies. This completes the transmissibility matrix estimation as a whole, and leads to the key message: appropriate identification of the transmissibility matrix of industrial vibration isolation systems with minimum experiment cost necessitates the usage of parametric models at low frequencies, and a combination of environmental and shaker excitations at higher frequencies.

Model-based control and data-based self-tuning

A model-based feedforward controller to achieve perfect cancellation of floor vibrations is derived that only depends on the isolator dynamics, i.e. regardless the structural dynamics potentially encountered atop the isolated payload. The isolator dynamics basically consist of parallel spring-damper pairs, possibly augmented with internal isolator dynamics appearing within the frequency range of interest. To prevent problems with drift and actuator saturation, the feedforward controller gain at low frequencies is limited using higher-order weak integrators. Since pure model-based feedforward control is shown to be very sensitive for parameter estimation errors, the controller is implemented as a self-tuning controller for which the poles are fixed in rational basis functions and the zeros are obtained online via self-tuning. Stability and minimization of bias

in the parameter estimates are obtained by using a residual noise shaping filter. Simulations show that the parameters of the self-tuning controller converge to the actual system parameters with (a) predictable bias and (b) in the presence of input sensor noise. A first experimental validation of the self-tuning feedforward control strategy on an active hard-mounted vibration isolation system shows that suppression of floor vibrations is improved up to 40 dB for frequencies between 2 and 300 Hz. A second experimental validation on the floating reticle stage test rig shows a reduction up to 23% of the residual output vibrations under normal operating conditions, and a reduction up to 89% under additional shaker-induced base frame vibrations. The latter being the result of resonances that (in this specific example) were not excited beyond the sensor noise levels given the existing floor vibrations.

Combined feedback and feedforward control

A novel control design approach toward feedback and feedforward control design for active soft-mounted vibration isolators in a multi-loop SISO context is proposed. Nominal feedback control designs are derived from physical insights, whereas robustly stable feedback controllers are obtained by adding robustness and leveling filters to the nominal feedback controllers. Feedback control is used to improve the rejection of direct disturbances up to a factor of 100 within the restricted bandwidth of 25 Hz. Disturbance feedforward control is used to decrease the transmissibility of floor vibrations up to 30 dB for frequencies between 0.2 and 300 Hz. Numerical results show great potential in terms of performance enhancement. Partial experimental validations of the feedback control strategy demonstrates the predicted closed-loop behavior (from simulation) in terms of transmissibility and compliance. The simulations and validations support the key message, i.e. that a breakthrough in active vibration isolation performance enhancement is only possible with the combined effort of both feedback and feedforward control to adequately suppress both direct and indirect disturbances.

7.2 Recommendations

The developments in this thesis show the potential of DFC to improve the rejection of indirect disturbances in industrial high-precision systems. Based on the obtained results, new insights are developed, leading to recommendations for ongoing research.

Mechatronic redesign of vibration isolation systems

Many state-of-the-art vibration isolation systems are designed such that the passive suspension stiffness is low to optimize performance in terms of rejecting floor vibrations. However, this comes with several problems. First, it leads to a poor rejection of direct disturbances, especially at the lower frequency range. Second, it induces internal isolator dynamics at relatively low frequencies which

limit performance and increase the modeling complexity to implement DFC. Third, it leads to long settling times for leveling. A mechatronic redesign with a higher isolator stiffness is therefore recommended, in combination with DFC to maintain the desired suppression of floor vibrations. On the other hand, the stiffness should not be made too high as this might lead to unnecessarily large actuators to provide the forces needed for active vibration isolation. This thesis provides sufficient design tools to support this change of paradigm in active vibration isolation in the context of high-precision machines such as wafer scanners.

Addition of extra sensors

The mixed-sensor feedback and feedforward control approach from Chapter 6 uses relative position feedback, acceleration feedback, and acceleration-based DFC. One of the factors limiting the feedback control bandwidth is the presence of structural modes of the metrology frame, leading to a loss of collocation between sensors and actuators. Force feedback control generally maintains better collocation properties at higher frequencies, enabling a higher control bandwidth. The possibilities of force feedback control in combination with DFC have not been explored in this thesis. Another interesting option to be considered for DFC is the use of geophones, which, compared to accelerometers, have the potential to measure a in broader frequency interval with a similar accuracy.

Improvement of data-based self-tuning

The self-tuning control strategy proposed in this thesis uses Filtered-error Least Mean Squares (FeLMS), showing significant performance improvements with a relatively low computational cost. However, the steady-state variance on the self-tuning parameters generally remains large when the adaptation rate is set too high. To reduce the variance, the adaptation rate needs to be lowered but this significantly reduces convergence time. To improve this, future research should focus on fast and robust data-based tuning. Additionally, current FeLMS algorithms produce updates even when there are no significant input disturbances. In that case, self-tuning should be switched off to prevent the algorithm from learning based on sensor noise alone. A possible solution to this problem is found in switching LMS, that only enables self-tuning when the input disturbances are sufficiently exciting. However, stability and convergence conditions of such algorithms are still partly lacking or too conservative to be useful.

Feedforward control of other disturbances

The current thesis only discusses disturbance feedforward control of base frame vibrations. However, feedforward control of other known or measurable disturbances should be considered to further improve performance. In the context of wafer scanners, for example, dominant sources of base frame vibrations are the self-induced machine vibrations due to aggressive wafer stage setpoints. In

fact, these setpoints are known and its information could be used to improve the feedforward control design. One could even argue that this allows (at least partly) a non-causal implementation that no longer is constrained by the Bode sensitivity integral. As a disadvantage, one could opt that the transmission path $\mathbf{P}_1$ in such a case generally becomes more complex, time-varying, and possibly position-dependent.

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Dankwoord

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Michiel Beijen

List of publications

Peer-reviewed journal articles

- Beijen, M.A., Heertjes, M.F., Van Dijk, J., and Hakvoort, W.B.J. (2018). *Self-tuning MIMO Disturbance Feedforward Control for Active Hard-mounted Vibration Isolators*. Accepted for Control Engineering Practice.
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About the author

Michiel Beijen was born on November 21, 1987 in Borne. In 2006 he finished his secondary education at St. Canisius in Almelo. Subsequently, he studied Mechanical Engineering at the University of Twente, the Netherlands, where he obtained his B.Sc. degree (Cum Laude) in 2009. In 2012, he obtained the M.Sc. degrees in Mechanical Engineering (Cum Laude) and Systems & Control (Cum Laude) from the same university. During his M.Sc. thesis, he worked on two-sensor feedback control strategies for active hard-mounted vibration isolators under the supervision of Dirk Tjepkema and Johannes van Dijk. After finishing his M.Sc. thesis, he worked one year at the University of Twente as a researcher to develop an active vibration isolation control strategy for an industrial measurement device.

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